

ELEMENTARY PARTICLES

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Part I

Particle Physics

1 INTRODUCTION

Elementary or fundamental particles are treated as pointlike objects in particle physics. The spatial size of these particles is not observable yet. All elementary particles of the same type are utterly identical. They can be described as excitations of quantum fields in the framework of quantum field theory (QFT). Elementary particles can be associated with 4 fundamental forces or interactions: gravity, electromagnetism, strong, and weak. The laws of physics between elementary particles and fundamental interactions are formulated in the Standard Model (SM). Figure 1.1 shows the most famous figure (probably) of the SM.

three generations of matter (fermions)					interactions / forces (bosons)	
	I	II	III			
QUARKS	mass charge spin u up	 c charm	 t top	0 0 1 g gluon	 H Higgs	
	 d down	 s strange	 b bottom	0 0 1 γ photon		
	 e electron	 μ muon	 τ tau	 W W boson		
LEPTONS	 ν_e electron neutrino	 ν_μ muon neutrino	 ν_τ tau neutrino	 Z Z boson		

Figure 1.1: The table of the Standard Model. This is definitely not the best figure for particle categorization in the Standard Model. It is misleading that these elementary particles are drawn as spheres while they are commonly known to be pointlike. Also, particles in the same row somehow suggest that they have something in common. This is not the case, as u , c , and t are not more connected to the gluon and Higgs boson than d , s , and b are.

The SM predictions are all confirmed by experiments, but it is not the whole story of physics. For now, gravity is still incompatible with the SM. Moreover, the SM cannot explain dark matter and dark energy.

1.1 Properties of Elementary Particles

Elementary particles can be classified via various properties:

- Mass (m): particles can be massive or massless. Massless particles ($m = 0$) include photons and gluons. Massive particles ($m \neq 0$) include leptons, quarks, the W , Z , and Higgs (H) boson. Electron, muon, and tau are often denoted as ℓ^- altogether, where $\ell = e, \mu, \tau$, and their corresponding neutrinos are denoted as ν_ℓ . Quarks are written as q_i , where $q_i = u, d, s, c, b, t$.
- Spin (J): the spin is an intrinsic angular momentum of a particle that has no classical counterparts. Particles in the SM have spin $J = 0, 1/2$, or 1 .

Spin (J)	0	1/2	1
Particles	Higgs boson	Leptons and quarks	Gauge bosons
Symbols	H	ν_ℓ, ℓ^-, q_i	$\gamma, g, Z, W^\pm$
Fields	Higgs field	Dirac field	Gauge field

- Charge: charges specify the couplings to quantum fields. For example, an electron has the electric charge subject to electromagnetic interactions, quarks have the color charge associated with the strong force, etc. Charges also specify interactions. The logic of reasoning whether a particle experience a certain interaction is the following: What charges does it have? What do these charges couple to? Then What interactions is associated with those charges? For example, quarks have color charge, electric charge, and weak charge. Thus, it can couple to gluons, photons, and W and Z . It experience strong, EM, and weak interactions. Another example is the photon. Photons do not have electric charges, so they *do not couple to other photons*, so photons do not experience EM interactions (with other photons).
- Lifetime (τ): Most particles will decay into other particles after some characteristic time called the mean lifetime. Particle number is not conserved as a consequence.

According to QFT, for each *type* of particle a , there must be a corresponding antiparticle $\bar{a}$. Compared to a , its antiparticle has exactly the same mass, mean lifetime, spin, but opposite charge. For example, the antiparticle of the electron e^- is the positron e^+ .

1.2 Quantum Mechanics + Relativity

When quantum mechanics meets special relativity, they produce relativistic quantum mechanics and later quantum field theory. In standard quantum mechanics, particles are described by the Schrödinger equation:

$$i\hbar \frac{\partial \Psi}{\partial t} = \left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \Psi.$$

One can immediately see that the Schrödinger equation is not complete and is incompatible with special relativity. First, it does not describe particles with $m = 0$ such as photons because the $-\hbar^2/2m$ term blows up. Second, the Schrödinger equation predicts nothing about spins. In non-relativistic quantum mechanics, the spin state of a particle looks like an add-on to its spatial wavefunction. Third, the time-derivative and spatial derivatives are not of the same order. One of the most significant implications of special relativity is that time is treated equally as space, known as the spacetime. If the order of spatial derivatives is 2, then the order of time-derivatives should also be 2.

To reach relativistic quantum mechanics, here is a simple and intuitive modification. Physically, the Schrödinger equation is just the *quantized* version of the classical energy-momentum relation:

$$E = \frac{\mathbf{p}^2}{2m} + V \quad \text{with} \quad \mathbf{p} \rightarrow -i\hbar \nabla, \quad E \rightarrow i\hbar \frac{\partial}{\partial t}.$$

If we stick to free particles for now ($V = 0$), then it just says that $E = \mathbf{p}^2/2m$. In special relativity, a free particle of mass m and 3-momentum $\mathbf{p}$ has the energy-momentum relation

$$E^2 = (mc^2)^2 + (\mathbf{p}c)^2.$$

Substituting $E \rightarrow i\hbar\partial_t$ and $\mathbf{p} \rightarrow -i\hbar\nabla$, one obtain

$$\left(-\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + c^2\nabla^2\right)\phi(\mathbf{x}, t) = \left(\frac{mc^2}{\hbar}\right)^2 \phi(\mathbf{x}, t).$$

Note that we use ϕ , a quantum field, instead of ψ . Dividing by c^2 on both sides gives the **Klein-Gordon equation**.

$$\left(-\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \nabla^2\right)\phi = \left(\frac{mc}{\hbar}\right)^2 \phi.$$

The constant coefficient $(mc/\hbar)$ has units of Length^{-1} . Its inverse is known as the Compton wavelength of a particle,

$$\lambda_C \equiv \frac{\hbar}{mc}.$$

To study or probe the structure of a target particle on a distance scale Δx , the de Broglie wavelength of the probe particle should satisfy $\lambda_{\text{dB}} \sim \hbar/p \leq \Delta x$. The energy of this particle is at least $E \sim pc \geq \hbar c/\Delta x$. If $\Delta x < \lambda_C = \hbar/mc$, the energy of the probe particle is larger than its rest mass energy, $E \geq mc^2$. That is, the energy is enough to create another particle. This means when the length scale is smaller than the Compton wavelength of a particle, particle creation is inevitable—one enters the regime of relativistic quantum mechanics. Elementary particle physics works under this length scale.

1.3 Mass and Energy Units

In particle physics, it is convenient to use natural units, in which fundamental constants c and $\hbar$ are set to 1. In this case, the rest mass energy is the same as mass: $E_{\text{rest}} = mc^2$. The mass unit in particle physics is often expressed in eV/c^2 . For example, $m_p \approx 1.67 \times 10^{-27} \text{ kg} = 0.938 \text{ GeV}/c^2$. This is saying that the rest mass energy of a proton is about 0.938 GeV. The following table lists the energy scale used in physics.

Scale	Value in eV	Physical systems and processes	Examples
1 eV	1 eV	Atomic physics, molecular physics, chemistry.	Ground state energy of a hydrogen atom: about -13.6 eV .
1 keV	10^3 eV	X-rays	Transitions between inner-shell atomic states for high Z atoms.
1 MeV	10^6 eV	Nuclear physics, γ -rays.	Nuclear binding energy per nucleon: $\sim 8 \text{ MeV}$.
1 GeV	10^9 eV	Particle physics	Rest mass energy of a proton: 0.938 GeV .
1 TeV	10^{12} eV	Particle physics	Large hadron collider (LHC) can accelerate particles to 6.5 TeV .
-	10^{28} eV	Quantum gravity	Planck mass: $m_{\text{Pl}} \approx 1.2 \times 10^{19} \text{ GeV}$.

1.4 The History of Elementary Particles

1.4.1 The Classical Era (1897-1932)

Elementary particle physics began with the discovery of electron by J. J. Thomson. Thomson discovered that cathode rays were produced by a hot filament. Using methods in mass spectrometry, Thomson obtained the velocity of the particle and its charge-mass ratio. This ratio was greater than any of the known particles, indicating that it either had a large amount of charge or a very small mass. This particle is later called the **electron**.

Thomson believed that the electron is one component of atoms, but he proposed a wrong model called the **plum-pudding model**, in which electron distributed uniformly in a positively charged paste. Rutherford argued against Thomson's model by his scattering experiment. He showed that the positive charge is concentrated in the **nucleus** at the center of the atom. In his experiment, Rutherford fired an α particle beam to a gold foil. It turned out that most of the α particles passed through the foil while only a few deflected wildly. If the atom was in the plum-pudding model, all α particles should be deflected only a bit. The nucleus of the lightest atom is named as the **proton** by Rutherford.

It is natural to think that for heavier atoms, to support more electrons' orbit, there should be a corresponding number of protons. That is, if there are two electrons, the nucleus should contain two protons and thus be twice as heavier. The truth is helium atom

weighs about 4 times heavier than hydrogen nucleus. This was resolved with the discovery of **neutron** by James Chadwick in 1932, and it marked an end to the classical period of elementary particle physics.

1.4.2 The Photon (1900-1924)

In 1900, Max Planck was trying to explain the **blackbody spectrum** for the electromagnetic radiation emitted by a hot object. He came up with an assumption that the electromagnetic radiation is quantized,

$$E = h\nu,$$

where ν is the frequency of the radiation and $h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$ is the Planck's constant.

In 1905, Einstein used Planck's idea to explain the **photoelectric effect**: electrons were ejected when electromagnetic radiation shines on a metal surface. A surprising result of the photoelectric effect is that the maximum energy of the electron ejected is independent of the intensity of the light, but on its color (i.e. frequency). Einstein argued that electromagnetic radiation is by its nature quantized, but this received harsh criticism. In 1923, A. H. Compton discovered the famous **compton effect**. He found out that the light scattered from a particle at rest is shifted in wavelength and will scatter of with an angle according to the equation

$$\lambda' = \lambda + \lambda_c(1 - \cos \theta),$$

where λ is the incident wavelength, λ' is the scattered wavelength, θ is the scattering angle, and $\lambda_c = h/mc$ is the **Compton wavelength**. This formula can also be derived by treating light as a particle with zero mass, and invoking conservation of relativistic energy and momentum. It is saying that light behaved like a particle, called the **photon**, on the subatomic scale.

1.4.3 Mesons (1934-1947)

After the discovery of protons and neutrons in the nucleus, there appeared a question: what holds these particles together as there should be proton-proton repulsion inside the nucleus. There must be a force stronger than the Coulomb force. Physicists called this force the **strong force**. There appeared another question: if this force is stronger than electromagnetic force, why it does not appear in everyday life? After all, every force we encounter daily is electromagnetic in origin, with gravity the only exception. The answer to this is that the strong force is a short-range force. Typical forces behaves like $e^{-(r/a)}/r^2$, where a is the range. For Coulomb force and gravity, $a = \infty$, but for strong force, $a \sim 10^{-15} \text{ m}$.

The first significant theory of the strong force was proposed by Yukawa in 1934. Before this theory, it was known that some particles served as mediators of forces or fields (e.g. electromagnetic force is mediated by photon) using the uncertainty principle. The range of the force is inversely proportional to the mass of the mediators (e.g. photon is massless so electromagnetic force has an infinite range). Yukawa calculated that the mediator of the strong force should be about 300 times massive than the electron, but this mass is smaller than that of the proton. Such particle is called the **meson**, the middle-weight. By the same token, the electron is called a **lepton**, the light weight, while the proton and the neutron are **baryon**, the heavy-weight.

By 1937, middle-weight particles were detected from the cosmic rays, but those particles appeared lighter than Yukawa's prediction and had a wrong lifetime. In 1946, experiments in Rome showed that these particles interact weakly with atomic nuclei. Everything seemed to be out of order. The resolution came in 1947 when Powell found that there are two middle-weight particles: one is called the **pion**, π , and the other is the **muon**, μ . The correct Yukawa meson is the π

1.4.4 Antiparticles (1930-1956)

The complete nonrelativistic quantum mechanics was invented in 1923-1926. The first significant discovery of relativistic quantum mechanics is the Dirac equation (by Paul Dirac) in 1927. The Dirac equation describes free electrons with energy given by the relativistic formula $E^2 - p^2c^2 = m^2c^4$. Interestingly, there exists two solutions $E = \pm\sqrt{p^2c^2 + m^2c^4}$. To account for the negative solution, Dirac proposed an infinite sea, called the **Dirac sea**, of electrons in negative-energy states. Dirac used the Pauli exclusion principle to explain why all electrons we observed are in positive-energy states (because all negative-energy states are filled). If a negative-energy electron is absence at the expected position in the sea, the "hole" is seen as a particle with a net positive energy and charge. Such particle was confirmed by Anderson's discovery of the **positron**, e^+ , in 1931.

Later in the 1940s, the Feynman-Stueckelberg formulation saw the negative solution as a positive-energy states of a different particle. The dualism in Dirac's equation is a profound and universal feature of quantum field theory: every particle has a corresponding antiparticle, with the same mass and opposite electric charge. The antiparticle is denoted by the symbol of its corresponding particle with a bar. The [antiproton](#), $\bar{p}$, and [antineutron](#), $\bar{n}$, were observed in 1955 and 1956 respectively at Berkeley. Though antineutron is also neutral, it is different from ordinary neutron by other quantum numbers, specifically the change in sign of the baryon number. There also exists a [crossing symmetry](#) for a reaction: for

$$A + B \rightarrow C + D,$$

a particle can be crossed over to the other side by turning it to its antiparticle, e.g.

$$A \rightarrow \bar{B} + C + D.$$

One mystery about antiparticles is the matter-antimatter asymmetry. It is observed that all of the observable universe is made of ordinary matter unless some energetic events occur, like the ones in the particle accelerator.

1.4.5 Neutrinos (1930-1962)

A beta decay is a nuclear reaction that emits an electron,

$$A \rightarrow B + e^-.$$

In 1930, the study of beta decay indicated a missing energy on the right side of the equation. Pauli suggested that a neutral particle was emitted together with the electron, and he called this particle the neutron. In 1932, the actual neutron was discovered and Chadwick adopted this name. The following year Fermi showed that the neutral particle emitted in beta decay must be extremely light, and he called it the [neutrino](#), the little neutral one. In modern terminology, the extra particle emitted in the beta decay is actually the antineutrino,

$$n \rightarrow p^+ + e^- + \bar{\nu}.$$

Neutrinos interact extremely weakly with matter as they can penetrate a thousand light years of lead. In every second, hundreds of billions of neutrinos from the sun pass through a human's body. This makes it difficult to detect neutrinos. By 1950, theoretical framework for the existence of neutrinos was established, but its discovery was in the mid-1950s.

Physicists wanted to study the property distinguishing a neutrino from antineutrino. If neutrino is its own antiparticle, then

$$\nu + n \rightarrow p^+ + e^- \quad \text{and} \quad \bar{\nu} + N \rightarrow p^+ + e^-$$

should have the same footing—they should occur about the same rate. In the late 1950s, Davis and Harmer made an experiment about this and the second reaction did not occur. Hence neutrino and antineutrino are distinct particles. In 1953 Konopinsky and Mahmoud introduced the [lepton number](#), L , and its conservation. They assigned $L = +1$ to electron, the muon, and the neutrino, and $L = -1$ to their corresponding antiparticles. Both sides of the beta decay has total lepton number zero.

Theoretically, the decay of a muon into an electron and a photon follows the conservation of lepton number (and other conservation laws). However, this process was never observed:

$$\mu^- \rightarrow e^- + \gamma.$$

The answer arrived in late 1950s and early 1960s to this is that there should be two different kinds of neutrino, one associated with electron and the other with muon. The [muon number](#) L_μ and the [electron number](#) was invented: $L_\mu = +1$ to μ^- and ν_μ , negative for their corresponding antiparticles, and the same went for electron and electron neutrino ν_e . Originally it was thought that two neutrinos were produced in muon decay,

$$\mu^- \rightarrow e^- + 2\nu.$$

This is true in some sense, but wrong in detail. The corrected version should be

$$\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu.$$

Due to the small mass of neutrinos, in many calculations neutrinos are assumed to be massless. Physically they have mass, but we

do not know what their masses are. Neutrinos belong to the lepton family as they do not participate in strong interactions.

1.4.6 Strange Particles (1947-1960)

In December 1947, Rochester and Butler observed a neutral particle from cosmic rays that decays into a π^+ and a π^- . This new particle is called the kaon K^0 and fell into the meson family. Many more mesons such as the η , the ϕ , the ω , the ρ 's, were discovered. In 1950, Anderson's group at Caltech observed another new particle, now called Λ , that decays into a p^+ and a π^- . Such particle is heavier than proton, so it belongs to the baryon family. Actually, there exists a question in 1938: why is the proton stable? Stückelberg proposed a law of **conservation of baryon number**, which is assigned $A = +1$ to all baryons and $A = -1$ to all antibaryons. Proton is the lightest baryon, so it cannot spontaneously decay into other particles. Hence for the conservation of baryon number, Λ must be in the baryon family. More heavy baryons, such as the Σ 's, the Ξ 's, the Δ 's, were discovered in the next few years.

The new heavy baryons and mesons were collectively known as “strange” particles. Until now, the only source of strange particles are from cosmic rays. In 1952, the first modern particle accelerator called the Brookhaven Cosmotron started operating and it made producing strange particles in the lab possible. The produced strange particles are “strange” because they are produced in $\sim 10^{-23}$ s, but they decay in $\sim 10^{-10}$ s. This suggested another unknown mechanism in decay process. Indeed, the strange particles are produced by the strong force, but the decays are governed by the weak force. Pais theory of decay required that strange particles come in pairs. In 1953 Gell-Mann and Nishijima proposed the **conservation of strangeness** to account for this pair production. They assigned strangeness to each particle, just like baryon numbers. Strangeness is conserved in any strong interaction, but not in weak interactions. For example: the K has strangeness $S = +1$ while the Σ and Λ has $S = -1$ (ordinary particles like proton or pion have $S = 0$),

$$\pi^- + p^+ \rightarrow K^+ + \Sigma^-.$$

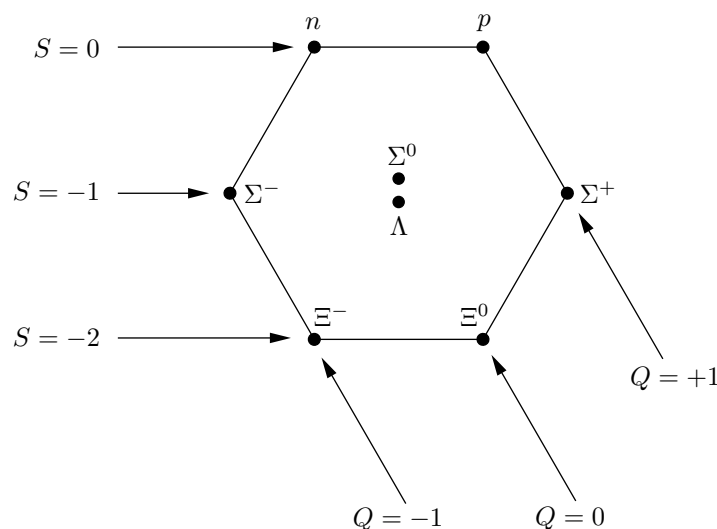
Strangeness is not conserved in decays,

$$\Lambda \rightarrow p^+ + \pi^-.$$

Now there were so many particles discovered, but there is no underlying orders in them. It was just like the situation of chemistry when atoms were found but the periodic table was not invented.

1.4.7 The Eightfold Way (1961-1964)

The “periodic table” called **Eightfold Way** for elementary particles was introduced by Murray Gell-Mann in 1961. The baryons and mesons are arranged in geometric patterns according to their charge and strangeness: the following baryon octet has the eight lightest baryons.



There are also meson octet, baryon decuplet, etc.

1.4.8 The Quark Model (1964)

The Eightfold Way still has a bizarre pattern. There must be a simpler explanation to the existence of different baryons and mesons. In 1964, Gell-Mann and Zweig independently proposed that hadrons are composed of smaller particles. Gell-Mann called these elementary particles **quarks**. There are three types of quarks for now: the u (up) quark with charge $\frac{2}{3}$ and strangeness zero; the d (down) quark with charge $-\frac{1}{3}$ and strangeness zero; the s (strange) quark with charge $-\frac{1}{3}$ and strangeness $S = -1$. Antiquarks have opposite charge and strangeness. In addition, there are two composition rules:

- Every baryon is composed of three quarks, and every antibaryon is composed of three antiquarks.
- Every meson is composed of a quark and an antiquark.

For example, the heavy baryons in the baryon decuplet have quark components

qqq	Q	S	Baryon
uuu	2	0	Δ^{++}
uud	1	0	Δ^+
udd	0	0	Δ^0
ddd	-1	0	Δ^-
uus	1	-1	Σ^{*+}
uds	0	-1	Σ^{*0}
dds	-1	-1	Σ^{*-}
uss	0	-2	Ξ^{*0}
dss	-1	-2	Ξ^{*-}
sss	-1	-3	Ω^-

Table 1.1: The baryon decuplet.

The same quark combination can make different particles, e.g. both the Δ^+ and the proton has uud . This is because quarks can bound in different ways, and their energy difference are very large so that we treat them as different particles.

The quark model is a nice model, but there are some problems with it: there was no observations of an individual quarks. They should be easy to produce by firing particles to protons; they should be easy to identify because of their special fractional charge; and they are easy to store because at least one quark is stable and there is no lighter particle with fractional charge. Yet no one has ever found one. Later physicists who believed in the quark model came up with the **quark confinement**: for some reason quarks are absolutely confined in baryons and mesons. The question becomes: what mechanism does this? Just as what Rutherford did to an atom, experimentalists fired particles to proton. Similarly, most of the incident particles passed through, with only a small fractions scattering sharply. The only difference is that while atoms contains one lumps of charge, the nucleus, the proton contains three.

Consider the particle Δ^{++} with uuu inside. It has three identical particles in it and seems to violate the Pauli exclusion principle. In 1964, O. W. Greenberg suggested that quarks are characterized by another charge called the **color** (red, green, and blue). Then the three u 's are no longer identical. Moreover, the concept of color has another nice feature: all naturally occurring particles are colorless. The word “colorless” means that either the total amount of each color is zero or all three colors are present in equal amounts.

1.4.9 The November Revolution and Its Aftermath (1974-1983 and 1995)

The quark model was not rescued by color, quark confinement, or anything similar above. It was the discovery of the ψ meson by C. C. Ting's group and Burton Richter's group both independently in 1974 that did the job. The ψ was a neutral and heavy meson which weighed over 3 times the mass of the proton (since many mesons appeared to be heavier than the proton, the “middle-weight” meaning was no longer available). This particle has an extremely long life time ($\sim 10^{20}$ s) as other particle with the same mass has typical lifetime $\sim 10^{-23}$ s. The discovery of the ψ marked the start of the **November Revolution**.

The quark model was the ultimate winner of this discovery: ψ is a bound state of a new quark called the c , **charm** quark, and its antiquark, $\psi = (c\bar{c})$. Indeed, Bjorken and Glashow proposed the idea of the fourth quark many years ago because they believed if there were four leptons, there should also be four quarks. In other words, the quark model was waiting for the discovery of the ψ and to provide an explanation. As there are now four quarks, new baryons and mesons carrying different amounts of charm should emerge. The charm of the ψ is hidden as it has a charm quark and an anticharm quark. The first charmed baryons ($\Lambda_c^+ = udc$ and $\Sigma_c^{++} = uuc$) were discovered in 1975. The first charmed mesons ($D^0 = c\bar{u}$ and $D^+ = c\bar{d}$) appeared in 1976, and charmed strange meson $D_s^+ = c\bar{s}$ in 1977.

Aside from the charm quark, a new lepton called the **tau** and its neutrino were found in 1975. Glashow's lepton and quark symmetry seem to be spoiled. However, in 1977 there appeared a new heavy meson $\Upsilon = b\bar{b}$ carrying a fifth quark called b , the **bottom**. Later in the 1980s the first bottom baryon, $\Lambda_b^0 = udb$ were found; the first bottom meson ($\bar{B}^0 = b\bar{d}$ and $B^- = b\bar{u}$) were observed in 1983. The prediction of the sixth quark should be a natural thing here, for there was already six leptons. The t , the **top** quark, appeared to be extraordinarily heavy ($174 \text{ GeV}/c^2$) and its lifetime was too short to form bound states. In 1995, accelerator Tevatron provided the observed reaction $u + \bar{u} \rightarrow t + \bar{t}$, from which the top and anti-top decayed immediately.

1.4.10 Intermediate Vector Bosons (1983)

In the theory of beta decay, Fermi thought of the process as a contact interaction. He did a great approximation because the **weak force** was a short-range force. The mediator was named **intermediate vector boson**. The weak force was too weak to bind particles together, so we cannot directly measure the range of it. Later, Glashow, Weinberg, and Salam came up with the electroweak theory. They believed there should be three intermediate vector bosons, $W^\pm$, and Z . Their calculated masses were

$$M_W = (82 \pm 2) \text{ GeV}/c^2, \quad M_Z = (92 \pm 2) \text{ GeV}/c^2.$$

In 1983, Carlo Rubbia's group discovered the W and the Z and they measured the masses to be

$$M_W = (80.403 \pm 0.029) \text{ GeV}/c^2, \quad M_Z = (91.188 \pm 0.002) \text{ GeV}/c^2.$$

1.4.11 The Standard Model (1978-?)

Now, all matter is constituted of three categories of particles: leptons, quarks, and mediators. The leptons fall into three generations and are classified according to their charge, L_e , L_μ , and L_τ :

ℓ	Q	L_e	L_μ	L_τ
e	-1	1	0	0
ν_e	0	1	0	0
μ	-1	0	1	0
ν_μ	0	0	1	0
τ	-1	0	0	1
ν_τ	0	0	0	1

Table 1.2: Lepton classification

There are also 6 antileptons, which give 12 leptons in total.

There are six flavors of quarks: up, down, strange, charm, bottom, and top, each with associated conservation. Similar to leptons, they fall into three generations (and have corresponding antiquarks):

q	Q	D	U	S	C	B	T
d	$-1/3$	-1	0	0	0	0	0
u	$2/3$	0	1	0	0	0	0
s	$-1/3$	0	0	-1	0	0	0
c	$2/3$	0	0	0	1	0	0
b	$-1/3$	0	0	0	0	-1	0
t	$2/3$	0	0	0	0	0	1

Table 1.3: Quark classification

Finally, there are mediators with corresponding interactions: the photon for the electromagnetic force, the W 's and the Z for the weak force, and the (hypothetical) graviton for gravity. There is some problem with the strong force. In Yukawa's theory the mediator was the pion, but now there exists more heavy mesons that could act as mediators. Moreover, with the quark model, protons, neutrons, and mesons are just composites of quarks. Then what is the exchange particle between quarks? The answer is the **gluon**, and there are eight of them in the Standard Model. Just like quarks, gluons have carry color and cannot exist as lone particles.

In all, there are 60 elementary particles: 12 leptons, 36 quarks, 12 mediators (graviton is not counted in the Standard Model). The Glashow-Weinberg-Salam theory calls for one additional particle called the **Higgs boson**, which is discovered recently at CERN in 2012. One shall ask then, who needs the two "extra" generations of particles when ordinary matter is made up of up and down quarks and electrons? Surprisingly, this has to do with the matter-antimatter asymmetry: this asymmetry can be explained by the Standard Model only if there are at least three generations. Nonetheless, even with the Standard Model, there are still many questions for particle physics:

- Why are there only three generations?
- All quark and lepton masses were determined by experiments. How do we calculate their masses, as a mature theory always do?
- The Grand Unified Theories (GUTs) that unify strong, electromagnetic, and weak interactions?
- Supersymmetry: every fermion has an associated boson?
- What about the graviton?

2 SPECIAL RELATIVITY

There are two postulates in special relativity:

1. The laws of physics take the same form in all inertial reference frames. All inertial frames are fundamentally indistinguishable based on experiments in that frame only.
2. The speed of light (and other massless particles) is the maximum speed in the universe and has the same value in all inertial reference frames,

$$c = 2.99792458 \times 10^8 \text{ m/s}.$$

Note that this number is exact because modern physics define the “meter” using the speed of light in vacuum. With these assumptions, many odd things would occur: Galilean transformation is no longer true, clocks do not run universally, etc. In this chapter, we will review special relativity.

2.1 Spacetime Geometry

In special relativity, time and space are somewhat mixed up in coordinate transformations. Thus, they are unified into a four-dimensional entity called **spacetime**. To be precise, let’s list properties of an inertial frame in special relativity:

1. Clocks that have been synchronized at some time will remain synchronized.
2. The properties of points in *space* is described by Euclidean geometry.
3. The spatial separation between points in space with specified coordinates does not change with time.

When one transform one inertial frame to another, simultaneous events in one inertial reference frame need not be simultaneous in other frames. Events that occur at the same place need not be at the same place in other frames.

An event in an inertial frame can be labeled with four coordinates known as **Minkowski coordinates**: (ct, x, y, z) . (In spacetime coordinates, time is conventionally the zeroth coordinate and is multiplied by a “ c ” to ensure the dimension is right. Later, we will set $c = 1$ for convenience so that the time coordinate is just t , but for now let’s stick with ct .) The geometry of spacetime in an inertial frame is given by the line element

$$ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2. \quad (2.1)$$

(We will prove it later using Lorentz transformations.) It is the square of the **spacetime interval** ds between two infinitesimally separated events. This spacetime is called **Minkowski spacetime**. The line element is a scalar quantity, invariant under any coordinate transformations. In another frame with Minkowski coordinates (ct', x', y', z') , the line element has exactly the form

$$ds^2 = c^2 dt'^2 - dx'^2 - dy'^2 - dz'^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2.$$

Note the minus sign in front of $d\mathbf{x}^2$. This indicates that the spacetime is flat, but non-Euclidean. The line element ds^2 can be positive, negative, or zero. We can put any two infinitesimally separated events into these three categories:

1. $ds^2 > 0$: the two events are **timelike** separated. The time separation dominates over the distance separation. For example, setting $dx = dy = dz = 0$ makes two events occur at the same point in space. Any Lorentz transformation cannot reverse the time ordering of the two events.
2. $ds^2 < 0$: the two events are **spacelike** separated. The distance separation dominates over the time separation. For example, setting $dt = 0$ makes two events occur at the same time, or simultaneously. A Lorentz transformation can reverse the time ordering of the two events, so these events cannot be causal.
3. $ds^2 = 0$: in this case, $c^2 dt^2 = dx^2 + dy^2 + dz^2$, or $d\mathbf{x}^2/dt^2 = c^2$. This describes a photon, so such events are called **lightlike** separated or **null** separated.

Since ds^2 is (Lorentz) invariant, these qualitative result (spacelike, null, or timelike) are also independent of reference frame or coordinate system. In otherwords, if two events are spacelike separated (not necessarily occur at the same time), one can always find a reference frame such that these two events occur simultaneously. Similarly, if two events are timelike separated, one can always find a reference frame such that they occur at the same place.

2.2 The Lorentz Transformation

In Newtonian mechanics, suppose an observer in frame S assigns an event A with coordinates (t, x, y, z) and another observer in a frame S' that has a relative velocity v in x -direction to S . Assume both observers are in inertial frames, and their x -axes are aligned. Also, assume that $t = 0$ represents the point when the origins of S and S' (denoted by $\mathcal{O}$ and $\mathcal{O}'$ respectively) overlap. Then according to Galilean transformation, the coordinates (t', x', y', z') of the event A in S' frame will be

$$\begin{aligned} t' &= t, \\ x' &= x - vt, \\ y' &= y, \\ z' &= z. \end{aligned}$$

This is the usual coordinate transformation between inertial frames in classical mechanics. In special relativity, S' will assign the coordinates of A according to the **Lorentz transformation** from S to S' ,

$$\begin{aligned} ct' &= \gamma(ct - \beta x), \\ x' &= \gamma(x - \beta ct), \\ y' &= y, \\ z' &= z, \end{aligned} \tag{2.2}$$

where c is the speed of light, $\beta = v/c$, and $\gamma = 1/\sqrt{1 - \beta^2}$ is the Lorentz factor. A detailed derivation of the Lorentz transformation is given in Appendix A.1.1. Taking the non-relativistic limit ($v/c \rightarrow 0$) reduces the Lorentz transformation to the Galilean transformation. One can show that the line element satisfies $ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2 = c^2 dt'^2 - dx'^2 - dy'^2 - dz'^2$. The inverse Lorentz transformation (from S' to S) can be obtained by replacing β with $-\beta$ (reversing the velocity).

The **proper time** of an object τ is the time measured by a clock that always travels with that object. In that object's own frame, the proper time is its time coordinate, and its spatial coordinates are all zero. Thus, the proper time can be calculated by the line element

$$c^2 d\tau^2 = ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2$$

in some other frame S . In other words, the proper time is the shortest time measured between two events in any frame.

Example 2.1. Time dilation and muon decay

The proper time τ of a clock fixed at $\mathcal{O}'$ is just the time t' measured by that clock in S' frame,

$$t' = \gamma \left(t - \frac{vx}{c^2} \right) \equiv \tau.$$

In S frame, the clock is moving at a velocity of v , so its position is $x = vt$. Hence the proper time τ of the clock is related to t by

$$\tau = \gamma \left(t - \frac{v^2 t}{c^2} \right) = \gamma \left(1 - \frac{v^2}{c^2} \right) t = \frac{t}{\gamma}.$$

As $\gamma \geq 1$, moving clocks always run slower ($\tau < t$). Time dilation plays a very important role in particle physics. A typical example is the decay of muon. Muons have a mean lifetime (this is the proper time) of $\tau = 2.2 \mu\text{s}$. Suppose a muon travels near the speed of light, $v \lesssim c$. If we ignore relativity, then the mean distance it travels before it decay is

$$\langle \Delta x \rangle < c\tau = 6.6 \times 10^2 \text{ m}.$$

However, we see muons produced by cosmic-rays protons at an altitude of ~ 15 km. Muons typically have $\gamma \approx 40$, and $\gamma\langle\Delta x\rangle > 15$ km, so most muons do reach the Earth.

From now on, we will in general use the natural units $c = 1$. The Lorentz transformation is given by

$$t' = \gamma(t - vx), \quad x' = \gamma(x - vt), \quad y' = y, \quad z' = z. \quad (2.3)$$

It looks more symmetric for t' and x' . Also, the line element and proper time are essentially the same,

$$ds^2 = dt^2 - dx^2 - dy^2 - dz^2 = d\tau^2. \quad (2.4)$$

To return to ordinary units, we shall use dimensional analysis to find out where should the c go. For now, let's develop some math tools that are useful for relativity.

2.2.1 Addition of Velocities

Suppose in frame S , an object is traveling at speed u_x in the x -direction. Then in frame S' with a velocity of v relative to S , the object is also traveling in the x -direction, but certainly with some other velocity u'_x . To determine u'_x , we use the definition of velocity and the Lorentz transformation (2.3),

$$u'_x = \frac{dx'}{dt'} = \frac{\gamma(dx - v dt)}{\gamma(dt - v dx)} = \frac{dx/dt - v}{1 - v dx/dt} = \frac{u_x - v}{1 - u_x v}.$$

This is the velocity addition formula. To check that it is valid, we will first consider the classical limit where $u_x, v \ll 1$. Then by Taylor expanding the denominator and keeping the first order in velocities,

$$u'_x \approx (u_x - v)(1 + u_x v) \approx u_x - v,$$

This is the classical Galilean velocity addition. Next, let's check the constancy of light. If $u_x = 1$ (a photon), then the velocity addition formula says

$$u'_x = \frac{1 - v}{1 - v} = 1.$$

In frame S' , the photon also travels at the speed of light as expected.

2.2.2 Relativistic Doppler Shift

Consider a sinusoidal electromagnetic wave of frequency ν in frame S . Its wavelength is given by $\lambda = 1/\nu = T$, where T is the period of this EM wave. In frame S , if we focus on the same crest between a time interval $\Delta t = T$, it will travel by a distance of one wavelength $\Delta x = \lambda = T$. In frame S' that travels along the wave direction with a velocity v will measure a period T' according to the Lorentz transformation (2.3):

$$T' = \gamma(T - vT) = T \frac{1 - v}{\sqrt{1 - v^2}} = T \sqrt{\frac{1 - v}{1 + v}}.$$

The relativistic Doppler shift is then

$$\nu' = \nu \sqrt{\frac{1 + v}{1 - v}}. \quad (2.5)$$

2.3 Vector and Tensor Algebra

2.3.1 4-Vectors

In special relativity, we often use Minkowski coordinates with basis vectors $\{e_\mu\} = \{e_t, e_x, e_y, e_z\}$. The basis vectors always have *subscript* indices. Conventionally, Greek indices represent numbers $(0, 1, 2, 3) \rightarrow (t, x, y, z)$, while Latin indices represent $(1, 2, 3) \rightarrow$

(x, y, z) . The vector pointing from the origin to an event in S frame can be written as:

$$\mathbf{x} = x^0 \mathbf{e}_0 + x^1 \mathbf{e}_1 + x^2 \mathbf{e}_2 + x^3 \mathbf{e}_3 = \sum_{\mu=0}^3 x^\mu \mathbf{e}_\mu = (t, x, y, z).$$

Note that all components $\{x^\mu\}$ are labeled with *superscript* indices. These vectors are called **4-vectors**. To avoid writing the summation symbol everywhere, we will adopt the **Einstein summation convention**: repeated subscript and superscript indices are implicitly summed over. They are called dummy indices. Indices that do not repeat are known as free indices. In this notes, I will write 4-vectors bolded-italic (e.g. $\mathbf{x}, \mathbf{p}, \mathbf{u}$), or sometimes just italic (x, p, u) , to distinguish with bolded three-vectors (e.g. $\mathbf{x}, \mathbf{p}, \mathbf{u}$). Therefore, a 4-vector can be written compactly as

$$\mathbf{x} = x^\mu \mathbf{e}_\mu.$$

By definition, the component of any 4-vector $\mathbf{a}$ (not just position) transform from one inertial frame to another according to the Lorentz transformation:

$$\begin{aligned} a'^0 &= \gamma(a^0 - va^1), \\ a'^1 &= \gamma(a^1 - va^0), \\ a'^2 &= a^2, \\ a'^3 &= a^3. \end{aligned}$$

A 4-vector have a “length”, or **magnitude**, defined by

$$\mathbf{a} \cdot \mathbf{a} = a^0 a^0 - a^1 a^1 - a^2 a^2 - a^3 a^3 \begin{cases} < 0, & \implies \text{timelike vector,} \\ = 0, & \implies \text{null/lightlike vector,} \\ > 0, & \implies \text{spacelike vector.} \end{cases}$$

The magnitude is an invariant quantity, just like the line element. (However, the position or displacement 4-vectors can only be defined in flat spacetime. In curved spacetime, 4-vectors need to be defined in a local, flat, tangent space to the curved spacetime.) The definition of a magnitude can be generalized to a **scalar product** (or inner product) of 4-vectors:

$$\mathbf{a} \cdot \mathbf{b} = a^0 b^0 - a^1 b^1 - a^2 b^2 - a^3 b^3 = \mathbf{b} \cdot \mathbf{a} \quad \text{or} \quad \mathbf{a} \cdot \mathbf{b} = \eta_{\mu\nu} a^\mu b^\nu.$$

Here $\eta_{\mu\nu}$ is called the **Minkowski metric**. The $\{\eta_{\mu\nu}\}$ are components of the **metric tensor** in the (t, x, y, z) Minkowski coordinate basis of an inertial frame in special relativity. It can be represented by a diagonal matrix:

$$[\eta_{\mu\nu}] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}. \quad (2.6)$$

Note that $\eta^{\mu\alpha} \eta_{\alpha\nu} = \delta^\mu_\nu$, the identity matrix. The inner product of four vectors can also be written as

$$\mathbf{a} \cdot \mathbf{b} = a_\nu b^\nu = a^\mu b_\mu$$

where $a_\nu = \eta_{\mu\nu} a^\mu$ and $b_\mu = \eta_{\mu\nu} b^\nu$. Components of vectors with subscript indices, $\{a_\mu\}$, are called **covariant components**. Components of vectors with superscript indices, $\{a^\mu\}$, are called **contravariant components**. In special relativity, they are related by $a^0 = a_0$, and $a^i = -a_i$. Note that now the inner product is explicitly

$$\mathbf{a} \cdot \mathbf{b} = a^0 b^0 + (-a^1) b^1 + (-a^2) b^2 + (-a^3) b^3 = a^0 b^0 + a^1 (-b^1) + a^2 (-b^2) + a^3 (-b^3).$$

The metric tensor $\eta_{\mu\nu}$ directly related to basis vectors. We can write $\mathbf{a} \cdot \mathbf{b} = (a^\mu \mathbf{e}_\mu) \cdot (b^\nu \mathbf{e}_\nu) = a^\mu b^\nu (\mathbf{e}_\mu \cdot \mathbf{e}_\nu)$. Hence

$$\eta_{\mu\nu} = \mathbf{e}_\mu \cdot \mathbf{e}_\nu.$$

This implies that the Minkowski coordinate basis vectors satisfy $\mathbf{e}_\mu \cdot \mathbf{e}_\nu = 0$ if $\mu \neq \nu$ —it is an orthogonal basis. Furthermore, they are also unit vectors because all nonzero components of $\eta_{\mu\nu}$ are only -1 or 1 . In conclusion, the Minkowski coordinate basis vectors form an *orthonormal* set of basis vectors. Now with the metric, we can write down the most usual notation of the line element:

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu.$$

This is why $\eta_{\mu\nu}$ is called the metric tensor. It enables one to compute the infinitesimal spacetime interval ds between events with infinitesimal differences in coordinates.

2.3.2 The 4-Derivative

In special relativity, time and space are unified. One may suggest that the derivatives $\partial/\partial t$ and ∇ should also be unified as a 4-vector. This section studies Lorentz-transformation properties of the object $\partial_\mu = \partial/\partial x^\mu$, or explicitly,

$$\{\partial_\mu\} = \left(\frac{\partial}{\partial x^0}, \frac{\partial}{\partial x^1}, \frac{\partial}{\partial x^2}, \frac{\partial}{\partial x^3} \right) = \left(\frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right).$$

Later we will know why ∂_μ has a subscript instead of a superscript. First, looking at the Lorentz transformation equations (2.3),

$$x'^0 = \gamma(x^0 - vx^1), \quad x'^1 = \gamma(x^1 - vx^0), \quad x'^2 = x^2, \quad x'^3 = x^3,$$

we can immediately summarize the four equations into a compact one:

$$x'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} x^\nu.$$

This is because the Lorentz transformation is linear. Now we know that partial derivatives pick out coefficients before x^μ , we can apply this to the chain rule and obtain ∂_μ :

$$\begin{aligned} \frac{\partial}{\partial t} &= \frac{\partial t'}{\partial t} \frac{\partial}{\partial t'} + \frac{\partial x'}{\partial t} \frac{\partial}{\partial x'} + \frac{\partial y'}{\partial t} \frac{\partial}{\partial y'} + \frac{\partial z'}{\partial t} \frac{\partial}{\partial z'}, \\ \frac{\partial}{\partial x} &= \frac{\partial t'}{\partial x} \frac{\partial}{\partial t'} + \frac{\partial x'}{\partial x} \frac{\partial}{\partial x'} + \frac{\partial y'}{\partial x} \frac{\partial}{\partial y'} + \frac{\partial z'}{\partial x} \frac{\partial}{\partial z'}, \\ \frac{\partial}{\partial y} &= \frac{\partial t'}{\partial y} \frac{\partial}{\partial t'} + \frac{\partial x'}{\partial y} \frac{\partial}{\partial x'} + \frac{\partial y'}{\partial y} \frac{\partial}{\partial y'} + \frac{\partial z'}{\partial y} \frac{\partial}{\partial z'}, \\ \frac{\partial}{\partial z} &= \frac{\partial t'}{\partial z} \frac{\partial}{\partial t'} + \frac{\partial x'}{\partial z} \frac{\partial}{\partial x'} + \frac{\partial y'}{\partial z} \frac{\partial}{\partial y'} + \frac{\partial z'}{\partial z} \frac{\partial}{\partial z'}. \end{aligned}$$

Extracting all partial derivatives from the Lorentz transformation:

$$\begin{array}{ccc} \partial_t = \gamma \partial_{t'} - \gamma v \partial_{x'}, & & t = \gamma t' + \gamma v x', \\ \partial_x = -\gamma v \partial_{t'} + \gamma \partial_{x'}, & \text{Compare} & x = \gamma v t' + \gamma x', \\ \partial_y = \partial_{y'}, & \Longleftrightarrow & y = y', \\ \partial_z = \partial_{z'}. & & z = z'. \end{array}$$

The LHS is the transformation of derivatives from S' to S , while the RHS is the *inverse* Lorentz transformation, from S' to S . This tells that the derivatives ∂_μ transform according to the inverse Lorentz transformation. They transform the same way as x_μ does but not x^μ . The derivatives ∂^μ (with superscript) can be obtained by $\eta^{\mu\nu} \partial_\nu$, resulting in $\{\partial^\mu\} = (\partial_t, -\nabla)$. There is a Lorentz-invariant operator, called the **wave operator** or **d'Alembert operator**, denoted as

$$\square = \partial_\mu \partial^\mu = \frac{\partial}{\partial x^\mu} \frac{\partial}{\partial x_\mu} = \frac{\partial^2}{\partial t^2} - \nabla^2. \quad (2.7)$$

Theorem 2.1. Lorentz Transformation Rule

Contravariant components of any 4-vector transform the way the coordinates do:

$$dx'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} dx^\nu \quad \Longleftrightarrow \quad a'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} a^\nu.$$

Covariant components of any 4-vector transform the way the derivatives of the contravariant coordinates do:

$$\frac{\partial}{\partial x^\mu} = \frac{\partial x'^\nu}{\partial x^\mu} \frac{\partial}{\partial x'^\nu} \quad \Longleftrightarrow \quad a_\mu = \frac{\partial x'^\nu}{\partial x^\mu} a'_\nu \quad \text{or} \quad a'_\mu = \frac{\partial x^\nu}{\partial x'^\mu} a_\nu.$$

These rules are how we define a 4-vector. A way to incorporate all these ideas is to verify the invariant inner product,

$$a'^{\mu}b'_{\mu} = \frac{\partial x'^{\mu}}{\partial x^{\nu}}a^{\nu}\frac{\partial x^{\alpha}}{\partial x'^{\mu}}b_{\alpha} = \frac{\partial x'^{\mu}}{\partial x^{\nu}}\frac{\partial x^{\alpha}}{\partial x'^{\mu}}a^{\nu}b_{\alpha} = \frac{\partial x^{\alpha}}{\partial x^{\nu}}a^{\nu}b_{\alpha} = \delta^{\alpha}_{\nu}a^{\nu}b_{\alpha} = a^{\nu}b_{\nu}.$$

Another way is to note that the *divergence*:

$$\partial_{\mu}a^{\mu} = \partial_t a^t + \partial_x a^x + \partial_y a^y + \partial_z a^z = \partial_t a^t + \nabla \cdot \mathbf{a}.$$

It does not have a minus sign before ∇ !

2.3.3 The Lorentz Group

A **group** G is a set of **group elements**, $\{g_a\}$. The elements can be composed together, or multiplied together, by a **composition**. The composition is usually indicated by a dot: $g_a \cdot g_b$. A composition satisfies four axioms:

1. Closure: given any two elements g_a and g_b in a group G , their product $g_a \cdot g_b = g_c$ is also in G .
2. Associativity: composition is associative, $(g_a \cdot g_b) \cdot g_c = g_a \cdot (g_b \cdot g_c)$.
3. Existence of the identity: there exists a group element called the **identity**, I , such that $I \cdot g_a = g_a \cdot I = g_a$.
4. Existence of the inverse: for every group element g_a , there exists a unique group element called the **inverse** of g_a , denoted by g_a^{-1} , such that $g_a^{-1} \cdot g_a = g_a \cdot g_a^{-1} = I$.

The group theory is the key mathematical tool to study symmetries. A **symmetry** is a feature of a system that is invariant under some transformation. The postulates of special relativity already suggest some symmetry: the laws of physics take the same form in all inertial frames; the spacetime interval Δs^2 between two events is invariant under Lorentz transformations.

The **Lorentz group** is a set of transformations that leave $ds^2 = \eta_{\mu\nu}dx^{\mu}dx^{\nu}$ invariant. In general, we want to find matrix representations of a group to study its properties. The Lorentz transformation (2.3) can be expressed in the matrix form:

$$\begin{bmatrix} t' \\ x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \gamma & -v\gamma & 0 & 0 \\ -v\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} t \\ x \\ y \\ z \end{bmatrix}. \quad (2.8)$$

The transformation matrix is denoted as Λ . It is specified by the continuous variable v . We can check whether Lorentz transformations satisfy all group axioms:

1. Closure: if Λ_1 and Λ_2 are two Lorentz transformations along the x -axis, then $\Lambda_1 \cdot \Lambda_2$ is also a Lorentz transformation along the x -axis. However, the product of two Lorentz transformations along different axes is not a pure Lorentz transformation in general, but a combination of a rotation and a Lorentz transformation. Thus, the Lorentz group should also include rotations.
2. Associativity: matrix multiplication is associative.
3. Identity: the identity matrix which has $v = 0$.
4. Inverse: the inverse of a Lorentz transformation with argument v is a Lorentz transformation with argument $-v$. The existence of inverse requires $\det \Lambda \neq 0$.

As we have mentioned, Lorentz transformations are linear. The matrix element of Λ are denoted as

$$\{\Lambda^{\mu}_{\nu}\} \equiv \left\{ \frac{\partial x'^{\mu}}{\partial x^{\nu}} \right\} \implies x'^{\mu} = \Lambda^{\mu}_{\nu}x^{\nu}. \quad (2.9)$$

Note that the superscript index refers to frame S' while the subscript index refers to frame S . To find out what other transformations are in the Lorentz group, we invoke our invariant line element ds^2 :

$$ds^2 = \eta'_{\mu\nu}dx'^{\mu}dx'^{\nu} = \eta_{\mu\nu}dx^{\mu}dx^{\nu},$$

where $\eta'_{\mu\nu} = \eta_{\mu\nu}$. The line element in S' frame can be written as $ds^2 = \eta'_{\mu\nu} dx'^{\mu} dx'^{\nu} = \eta'_{\mu\nu} (\Lambda^{\mu}_{\alpha} dx^{\alpha}) (\Lambda^{\nu}_{\beta} dx^{\beta})$, so

$$\eta'_{\mu\nu} (\Lambda^{\mu}_{\alpha} dx^{\alpha}) (\Lambda^{\nu}_{\beta} dx^{\beta}) = \eta_{\mu\nu} dx^{\mu} dx^{\nu} = \eta_{\alpha\beta} dx^{\alpha} dx^{\beta} \implies (\eta'_{\mu\nu} \Lambda^{\mu}_{\alpha} \Lambda^{\nu}_{\beta} - \eta_{\alpha\beta}) dx^{\alpha} dx^{\beta} = 0.$$

In the last step of the LHS equation, we change the dummy indices (μ, ν) to (α, β) . Since $dx^{\alpha} dx^{\beta}$ is arbitrary, the terms in the bracket must vanish,

$$\eta_{\alpha\beta} = \eta'_{\mu\nu} \Lambda^{\mu}_{\alpha} \Lambda^{\nu}_{\beta}. \quad (2.10)$$

This is the defining equation for all matrices Λ that make up the Lorentz group. Note that all of $\eta_{\mu\nu}$, Λ^{μ}_{α} and Λ^{ν}_{β} are just numbers, so they commute and we can write

$$\eta_{\alpha\beta} = \Lambda^{\mu}_{\alpha} \eta_{\mu\nu} \Lambda^{\nu}_{\beta} \iff \eta = \Lambda^T \eta \Lambda,$$

where Λ^T is the transpose of the matrix Λ . The above two equations are completely equivalent if you match indices according to matrix multiplication $(\eta \Lambda)_{\mu\beta} = \eta_{\mu\nu} \Lambda^{\nu}_{\beta}$. The first index represents rows, while the second represents columns. In matrix multiplication, the second index of the first matrix always matches the first index of the second matrix. If they do not match but still have the same index, such as $\Lambda^{\mu}_{\alpha} \eta_{\mu\nu}$, then taking the transpose of Λ interchanges μ and α . In this case, the matrix multiplication is valid.

Taking the determinant,

$$\det \eta = \det(\Lambda^T \eta \Lambda) = \det \Lambda^T \det \eta \det \Lambda = (\det \Lambda)^2 \det \eta.$$

Since $\det \eta = -1 \neq 0$, we must have $(\det \Lambda)^2 = 1$, or $\det \Lambda = \pm 1$. In conclusion, to satisfy the invariance of ds^2 , each element Λ in the Lorentz group should have the property $\det \Lambda = \pm 1$. There are four types of transformations that satisfy such a property:

1. Lorentz transformations (boosts): transformations between inertial frames with different velocities, specified by three real parameters: $\mathbf{v} = (v_x, v_y, v_z)$.
2. Rotations: transformations describing rotations of spatial coordinates (time is unaffected), specified by three Euler angles (α, β, γ) about three axes of rotation. Since time is unchanged, rotations preserve $d\ell^2 = dx^2 + dy^2 + dz^2$.
3. The time-reversal transformation T : a discrete transformation that transforms $(t, x, y, z) \rightarrow (-t, x, y, z)$. Its matrix form is $\Lambda_T = \text{diag}(-1, 1, 1, 1)$.
4. The parity operation P : a discrete transformation that transforms $(t, x, y, z) \rightarrow (t, -x, -y, -z)$. Its matrix form is $\Lambda_P = \text{diag}(1, -1, -1, -1)$.

Lorentz boosts and rotations are continuous transformations with determinant $+1$. The time-reversal transformation and parity are discrete transformations with determinant -1 . The latter two will play important roles in understanding the nature of interactions.

2.3.4 Tensors

So far we have encountered four types of mathematical objects in special relativity: scalars, 4-vectors, the metric tensor, and Lorentz transformation matrices. The first three are tensors in special relativity, but the last one is not. A tensor can be specified by its rank: a rank $\binom{m}{n}$ tensor has m contravariant indices and n covariant indices. It transform under the rule

$$(T')^{\alpha_1 \alpha_2 \dots \alpha_m}_{\beta_1 \beta_2 \dots \beta_n} = (\Lambda^{\alpha_1}_{\mu_1} \Lambda^{\alpha_2}_{\mu_2} \dots \Lambda^{\alpha_m}_{\mu_m}) (\Lambda^{\nu_1}_{\beta_1} \Lambda^{\nu_2}_{\beta_2} \dots \Lambda^{\nu_n}_{\beta_n}) T^{\mu_1 \mu_2 \dots \mu_m}_{\nu_1 \nu_2 \dots \nu_n}. \quad (2.11)$$

Each upper index is associated with a contravariant Lorentz transformation, and each lower index is associated with a covariant Lorentz transformation.

Example 2.2. A 4-vector

A 4-vector is a tensor. The transformation rule of a 4-vector is

$$(A')^{\alpha} = \Lambda^{\alpha}_{\mu} A^{\mu},$$

where the prime is to be written out for clarity. It satisfies (2.9) and (2.11) if we replace T by A .

Example 2.3. The metric tensor

The metric tensor $\eta_{\mu\nu}$ is another good example of a tensor. It has two covariant indices. From (2.10), it follows the rule

$$\eta_{\alpha\beta} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta \eta'_{\mu\nu}.$$

But the prime index is on the wrong side of the equation comparing to (2.11). Recall that $\Lambda^\mu{}_\alpha$ and $\Lambda^\nu{}_\beta$ comes from $dx'^\mu = \Lambda^\mu{}_\alpha dx^\alpha$ and $dx'^\nu = \Lambda^\nu{}_\beta dx^\beta$. These two Λ 's are for contravariant Lorentz transformation because dx'^μ , dx^α , dx'^ν , and dx^β are contravariant components (they have upper indices). We know that the Lorentz transformation for contravariant and covariant components are inverse of each other. Now we can see why $\eta_{\alpha\beta} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta \eta'_{\mu\nu}$ has the prime index on the opposite side. $\Lambda^\mu{}_\alpha$ and $\Lambda^\nu{}_\beta$ are contravariant transformation, the inverse of covariant transformation, so it is transforming back from the prime coordinates to unprimed coordinates. If $\Lambda^\mu{}_\alpha$ and $\Lambda^\nu{}_\beta$ were covariant transformation, then we will have (2.11):

$$\eta'_{\alpha\beta} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta \eta_{\mu\nu}, \quad [\Lambda \text{ covariant trans.}] \iff \eta_{\alpha\beta} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta \eta'_{\mu\nu}, \quad [\Lambda \text{ contravariant trans.}]$$

The metric tensor is not only a tensor. It is an **invariant tensor** because $\{\eta'_{\mu\nu}\} = \{\eta_{\mu\nu}\}$, i.e. the matrix elements are the same.

One thing to notice is that a tensor must have all indices refer to components measured in one inertial frame. The indices on $\Lambda^\mu{}_\nu = \partial x'^\mu / \partial x^\nu$ are different: the upper index is for one frame, and the lower is for the other. Therefore, Lorentz transformation matrices are not tensors.

2.4 Relativistic Dynamics

2.4.1 Proper Time and 4-Velocity

The trajectory of a particle in spacetime is called its **world line**. In general, it can be parametrized by a smooth and differentiable parameter λ ,

$$t = t(\lambda), \quad x = x(\lambda), \quad y = y(\lambda), \quad z = z(\lambda).$$

(It is similar to how we parametrize a curve in three-space by a parameter t . But t is now a coordinate, so it is better to use a fifth parameter λ for generality.) Recall that the infinitesimal proper time $d\tau$ is defined by (2.4). We can now write

$$d\tau = \sqrt{\eta_{\mu\nu} dx^\mu dx^\nu} = \sqrt{(dt)^2 - dx^2 - dy^2 - dz^2}.$$

The proper time of a matter particle is guaranteed to be real because all particles will travel below the speed of light, $dt^2 > dx^2 + dy^2 + dz^2$. The proper time of a particle along the world line is the integral

$$\tau = \int d\tau = \int_{\lambda_{\min}}^{\lambda_{\max}} d\lambda \sqrt{\left(\frac{dt}{d\lambda}\right)^2 - \left(\frac{dx}{d\lambda}\right)^2 + \left(\frac{dy}{d\lambda}\right)^2 + \left(\frac{dz}{d\lambda}\right)^2} = \int_{\lambda_{\min}}^{\lambda_{\max}} d\lambda \sqrt{\eta_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}}.$$

Example 2.4. Muon in a ring

Consider a muon moving with speed $v = \omega R$ on a circular ring, where ω is the angular frequency and R is the radius of the ring. For convenience, we will parametrize the motion by the time t measured in the lab frame:

$$t = t, \quad x = R \cos \omega t, \quad y = R \sin \omega t, \quad z = z_0.$$

We are interested in how much time has passed in one revolution in muon's frame (assuming that it does not decay). The proper time differential is

$$d\tau = \sqrt{dt^2 - dx^2 - dy^2 - dz^2} = \sqrt{dt^2 - (\omega R)^2 dt^2} = dt \sqrt{1 - (\omega R)^2}.$$

The total proper time is

$$\tau = \int_0^{2\pi R/v} dt \sqrt{1 - (\omega R)^2} = \frac{2\pi R}{v} \sqrt{1 - v^2} = \frac{t_{\text{lab}}}{\gamma},$$

as expected by time dilation.

The most prototypical 4-vector is the differential displacement $\{dx^\mu\} = (dt, dx, dy, dz)$. From this we are able to generalize other classical three-vectors to their counterparts in special relativity. The first one is the velocity:

$$\mathbf{v} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right) \xrightarrow{?} \left\{ \frac{dx^\mu}{dt} \right\} = \left(\frac{dt}{dt}, \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right).$$

Though (dt, dx, dy, dz) is a 4-vector, when they are divided by dt , the resulting object is no longer a 4-vector because dt is one of the components, and it is *not* Lorentz invariant. To preserve Lorentz invariance of (dt, dx, dy, dz) , we want to divide a $d\lambda$ that is also Lorentz invariant. Meanwhile, $d\lambda$ should have units of time, since we want a velocity object. The only Lorentz-invariant measure of time we know is $d\tau$. Thus, we define the **4-velocity** of a particle to be

$$u \equiv \{u^\mu\} = \left\{ \frac{dx^\mu}{d\tau} \right\} = \left(\frac{dt}{d\tau}, \frac{dx}{d\tau}, \frac{dy}{d\tau}, \frac{dz}{d\tau} \right). \quad (2.12)$$

Using $dt/d\tau = \gamma$, we can relate the 4-velocity to ordinary velocity (v_x, v_y, v_z) ,

$$u = \left(\frac{dt}{d\tau}, \frac{dx}{dt} \frac{dt}{d\tau}, \frac{dy}{dt} \frac{dt}{d\tau}, \frac{dz}{dt} \frac{dt}{d\tau} \right) = (\gamma, \gamma v_x, \gamma v_y, \gamma v_z).$$

The magnitude of the 4-velocity must be invariant to be a 4-vector. Indeed,

$$u \cdot u = \eta_{\mu\nu} u^\mu u^\nu = \eta_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = \frac{d\tau^2}{d\tau^2} = 1.$$

This means in ordinary units, $u \cdot u = c^2$, which is of course Lorentz invariant. Interestingly, the result above shows that the 4-velocity is a timelike unit vector. This can be seen in a particle's rest frame, where its spatial velocity is zero. The only nonzero component of the 4-velocity is then its time component, solely along its time-axis. As a consequence, the 4-velocity is always tangent to the world line (because it represents the time-axis) viewed other frame.

2.4.2 4-Momentum and the Invariant Mass

Consider a free particle with $m > 0$ for now. The **4-momentum** of this particle is defined as the product of its 4-velocity and its mass,

$$p \equiv \{p^\mu\} = (E, p_x, p_y, p_z) = (\gamma m, \gamma m v_x, \gamma m v_y, \gamma m v_z).$$

The components are relativistic energy and momentum. In ordinary units, the 4-momentum is

$$p = (\gamma m c^2, \gamma m v_x c, \gamma m v_y c, \gamma m v_z c).$$

We can check both energy and momentum in the non-relativistic limit $v \ll 1$,

$$E = \gamma m = \frac{m}{\sqrt{1 - v^2}} = \left(1 + \frac{1}{2} v^2 + \dots \right) m = m + \frac{1}{2} m v^2 + \mathcal{O}(v^4),$$

$$p_x = \gamma m v = \frac{m v}{\sqrt{1 - v^2}} = \left(1 + \frac{1}{2} v^2 + \dots \right) m v = m v + \mathcal{O}(v^3).$$

The total energy is the rest mass energy plus the classical kinetic energy, and the momentum is the mass times velocity. In the non-relativistic limit, $v \rightarrow 0$, we have $E \rightarrow m$ and $|\mathbf{p}| \rightarrow 0$. In the relativistic limit, $v \rightarrow 1$ and $|\mathbf{p}| \rightarrow E$.

The associated Lorentz-invariant quantity of the 4-momentum is the invariant mass:

$$p \cdot p = \eta_{\mu\nu} p^\mu p^\nu = m^2 \eta_{\mu\nu} u^\mu u^\nu = m^2. \quad (2.13)$$

In terms of energy and spatial momentum, this is saying

$$p_\mu p^\mu = E^2 - \mathbf{p}^2 = m^2 \quad \text{or} \quad E^2 = \mathbf{p}^2 c^2 + m^2 c^4.$$

the famous relation in relativity. Like the three-momentum in classical mechanics, 4-momentum is conserved for a free particle (or a system of free particles). The conservation of 4-momentum plays an important role in particle physics. It represents four conserved quantities: one for conserved energy, and three for conserved spatial momentum. These conservation laws determine whether a decay/scattering is kinematically forbidden or not.

2.4.3 Action and Lagrangian for a Free Particle

Some review: in classical mechanics, we define the Lagrangian to be $L = T - V$, where T and V are the kinetic and potential energy, respectively, of the system. The action along a trajectory from point a to point b is defined as

$$S[q_i(t)] = \int_{t_a}^{t_b} dt L(q_i, \dot{q}_i),$$

where q_i is some generalized coordinates that specify every point on the trajectory. By Hamilton's principle (or principle of stationary action), the action of a actual physical trajectory is extremized among all possible trajectories going from a to b . We then extremize the action using calculus of variations. It helps obtain the equation of motion for each $q_i(t)$, which is the Euler-Lagrange equation

$$\frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} = 0.$$

For a non-relativistic free particle of mass m , if q_i represents (x, y, z) , then

$$T = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2), \quad V = 0.$$

The Euler-Lagrange equation says that

$$0 = \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} = \frac{d}{dt} (m \dot{q}_i) = m \ddot{q}_i.$$

This is just Newton's second law for a free particle $m\mathbf{a} = 0$. It says that a free particle should travel at constant velocity in space.

The Lagrangian/action is the most fundamental one among all physical quantities. It tells which physical path a particle will take. Even though observers in each frame can use different coordinates to represent the trajectory, every one should agree with the actual physical trajectory it takes. Hence the Lagrangian/action should be the "most" invariant quantity, so it needs to be Lorentz invariant in special relativity.

The goal now is to find L for a relativistic particle. The classical Lagrangian cannot do the job because: 1. it has no c and hence no speed limit; 2. it is not Lorentz invariant. Note that the action has dimensions of angular momentum, $[S] = [E \cdot T]$. We want to have some Lorentz-invariant quantities that make the units right. There are three fundamental constants we can use: the speed of light c , Planck constant $\hbar$, and gravitational constant G . (In this section we will write c 's out explicitly.) Planck constant $\hbar$ has the unit of angular momentum, but we cannot use it because there is no quantum mechanics going on yet. Of course G is not useful since there is no gravity. The only available constants now are c and m . We have no choice but to set $L \propto mc^2$. To get dimensions of T , the only Lorentz-invariant candidate is the proper time τ . In conclusion, the action for a relativistic free particle must be of the form

$$S \propto mc^2 \int_{\text{WL}} d\tau,$$

where WL stands for world line. For future convenience, we will choose the constant of proportionality to be -1 . To make it an integral over dt in general reference frame, use the relation $d\tau = dt/\gamma$. Then the action along the world line is

$$S = -mc^2 \int_{t_a}^{t_b} \frac{dt}{\gamma} = -mc^2 \sqrt{1 - \beta^2} \int_{t_a}^{t_b} dt,$$

where $\beta = v/c$ indicates the speed of this free particle in some inertial frame. We can identify the Lagrangian to be

$$L = -mc^2 \sqrt{1 - \beta^2}. \quad (2.14)$$

Checking the non-relativistic limit:

$$L = -mc^2 \left(1 - \frac{1}{2}\beta^2 + \dots \right) = -mc^2 + \frac{1}{2}mv^2 + \mathcal{O}(v^4).$$

Comparing to $L = T - V$ in classical mechanics, the rest mass energy acts like a potential energy of the particle.

In the last section we stated the definition of 4-momentum without a reason, and we did not prove its conservation either. Now with the Lagrangian, it is time to compute the conserved quantities. We know that if a coordinate q_i is cyclic (which means it does not appear in the Lagrangian), then its conjugate generalized momentum $\partial L / \partial \dot{q}_i$ is conserved. In the free-particle case, there is no explicit x, y, z . The conserved momentum in x is

$$p_x = \frac{\partial L}{\partial \dot{x}} = -mc^2 \frac{\partial}{\partial \dot{x}} \left(1 - \frac{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}{c^2} \right)^{-1/2} = -mc^2 \left(\frac{\dot{x}/c^2}{\sqrt{1 - (v/c)^2}} \right) = mc \frac{\dot{x}/c}{\sqrt{1 - (v/c)^2}} = \gamma mc \beta_x.$$

The same holds for p_y and p_z . Note that the Lagrangian has no explicit time-dependence, which means the Hamiltonian/total energy is also conserved. The Hamiltonian is defined as

$$\begin{aligned} H &= \sum_i p_i \dot{q}_i - L \\ &= (mc\beta\gamma)(c\beta) - (-mc^2\sqrt{1 - \beta^2}) \\ &= mc^2 (\gamma\beta^2 + \sqrt{1 - \beta^2}) \\ &= mc^2 \left[\frac{\beta^2}{\sqrt{1 - \beta^2}} + \frac{1 - \beta^2}{\sqrt{1 - \beta^2}} \right] \\ &= \gamma mc^2. \end{aligned}$$

This is the conserved relativistic energy. We have proved almost everything. The only problem now is the massless particle, for which $L = 0$. By the invariant mass (2.13),

$$p_\mu p^\mu = E^2 - p^2 = m^2 = 0.$$

This forces $E = p$ for a massless particle. Then how do we know E , for example, for a photon? It is given by Planck's formula $E = \hbar\omega = h\nu$.

2.4.4 Decay and Scattering

In this section, we will look at kinematics behind important processes in particle physics using conservation of 4-momentum. The examples include: two-body decay, ordinary scattering, and Compton scattering.

Two-Body Decay

A decay process involves a parent particle A and more than one daughter particles: $A \rightarrow 1 + 2 + \dots + N$. To analyze a decay, we focus on several questions: Is a given process kinematically allowed or forbidden? What are the energies of the daughter particles? Does the decay violate other conservation laws, such as charge, baryon number, angular momentum, parity, etc.? What is the decay rate?

The simplest decay process is a two-body decay: $A \rightarrow 1 + 2$, where A is a parent particle with mass m_A , and 1 and 2 are two daughter particles with mass m_1 and m_2 , respectively. Let the particles A , 1, and 2 have 4-momentum p_A , p_1 , and p_2 , respectively. The invariant mass says that

$$p_A^2 = m_A^2, \quad p_1^2 = m_1^2, \quad p_2^2 = m_2^2.$$

The general conservation of 4-momentum reads

$$p_A^\mu = p_1^\mu + p_2^\mu \quad \Longleftrightarrow \quad p_A = p_1 + p_2.$$

which represents four independent equations. It is most convenient to analyze the problem in the **center-of-momentum** (CM) frame of A . In this frame, the 4-momentum of A is $p_A = (m_A, 0, 0, 0)$. Due to conservation of 3-momentum, this frame forces the

daughter particles to travel in opposite direction. Now we will try to determine the energies of the daughter particles. It turns out that we don't even need to worry about the index μ because the invariant mass will do all the job.

First, rewrite the conservation of 4-momentum as

$$p_A - p_1 = p_2.$$

Squaring both sides,

$$p_A^2 + p_1^2 - 2p_A \cdot p_1 = p_2^2.$$

(By squaring a 4-vector p we mean to compute the Lorentz-invariant inner product with itself $p_\mu p^\mu$.) Now using the invariant mass,

$$m_A^2 + m_1^2 - 2p_A \cdot p_1 = m_2^2.$$

In the CM frame, the dot product $p_A \cdot p_1 = (m_A, 0, 0, 0) \cdot (E_1, \mathbf{p}_1) = m_A E_1$. This is why we choose the CM frame, the zeros take care of the unknown $\mathbf{p}_1$. Putting this back into the equation, we have the energy:

$$m_A^2 + m_1^2 - 2m_A E_1 = m_2^2 \implies E_1 = \frac{m_A^2 + m_1^2 - m_2^2}{2m_A}. \quad (2.15)$$

Interchanging $1 \leftrightarrow 2$ or using $E_2 = m_A - E_1$ gives

$$E_2 = \frac{m_A^2 + m_2^2 - m_1^2}{2m_A}.$$

After knowing the energies and masses, we can compute the spatial momentum:

$$\begin{aligned} |\mathbf{p}_1| = |\mathbf{p}_2| &= \sqrt{E_1^2 - m_1^2} = \sqrt{\frac{(m_A^2 + m_1^2 - m_2^2)^2 - 4m_A^2 m_1^2}{4m_A^2}} \\ &= \frac{\sqrt{(m_A^2 + m_1^2 - m_2^2 + 2m_A m_1)(m_A^2 + m_1^2 - m_2^2 - 2m_A m_1)}}{2m_A} \\ &= \frac{\sqrt{[(m_A + m_1)^2 - m_2^2][(m_A - m_1)^2 - m_2^2]}}{2m_A} \\ &= \frac{\sqrt{[m_A^2 - (m_1 + m_2)^2][m_A^2 - (m_1 - m_2)^2]}}{2m_A}. \end{aligned}$$

The most crucial conclusion from this two-body decay is that the energies of the decay products are uniquely determined by the masses. We say that the decay products are **mono-energetic**. The decay products of a three-body decay is not mono-energetic. For instance, in the beta decay $n \rightarrow pe^- \bar{\nu}_e$ (this notation is no different from $n \rightarrow p + e^- + \bar{\nu}_e$), the electron in the beta decay has a spectrum of energy. In fact, it was this mysterious spectrum that inspired Wolfgang Pauli to propose that there must be an invisible particle—a neutrino—along with a proton and an electron.

Example 2.5. Kinematically forbidden pair production

The pair production from one photon to an electron-positron pair, $\gamma \rightarrow e^+ e^-$, is kinematically not allowed. In principle, we can always find a frame such that the total 3-momentum of the electron-positron pair is zero. But there is no such frame for *one* photon: a photon has a nonzero energy, and its magnitude of momentum is exactly equal to the energy. In other words, the magnitude of its 4-momentum is $p_\gamma^2 = E_\gamma^2 - \mathbf{p}_\gamma^2 = 0$. To have a kinematically allowed pair production, one needs at least two photons.

Scattering Processes

A scattering process is a process in which two particles collide/interact to produce two or more particles: $A + B \rightarrow 1 + 2 + \dots + N$. In the CM frame, particles A and B have 4-momenta

$$p_A = (E_A, \mathbf{p}_A) = \left(\sqrt{m_A^2 + \mathbf{p}_A^2}, \mathbf{p}_A \right), \quad p_B = (E_B, \mathbf{p}_B) = \left(\sqrt{m_B^2 + \mathbf{p}_B^2}, \mathbf{p}_B \right).$$

The total 4-momentum is then

$$p_{\text{tot}} = p_A + p_B = (E_A + E_B, 0, 0, 0).$$

Here we can compute the invariant quantity $\sqrt{s}$:

$$\sqrt{s} \equiv \sqrt{(p_A + p_B)^2} = E_A + E_B = E_{\text{CM}}.$$

Once we know $\sqrt{s}$ in the CM frame, it is the same in every other inertial frames. In experiments with colliding beams, the particles usually have the same mass, e.g. $e^+ + e^- \rightarrow 1 + 2 + \dots + n$. In these cases, $\sqrt{s} = 2E$. We need to note that $\sqrt{s} = E_{\text{CM}}$ is linearly proportional to the beam energy E .

Particle physics always works with E_{CM} because it determines whether a process is allowed. The useful energy for creating particles comes from E_{CM} . As long as E_{CM} exceeds the total rest mass $m_1 + m_2 + \dots + m_N$, the scattering process is energetically allowed.

Now consider a fixed target scattering, where the particle B is at rest in the lab frame. This lab frame is not the CM frame. As usual, first list the 4-momenta of particles A and B :

$$p_A = (E_A, \mathbf{p}_A) = \left(\sqrt{m_A^2 + \mathbf{p}_A^2}, \mathbf{p}_A \right), \quad p_B = (E_B, \mathbf{p}_B) = (m_B, 0, 0, 0).$$

The total 4-momentum is

$$p_{\text{tot}} = p_A + p_B = (E_A + m_B, \mathbf{p}_A).$$

The E_{CM} is given by

$$E_{\text{CM}} = \sqrt{s} = \sqrt{p_A^2 + p_B^2 + 2p_A \cdot p_B} = \sqrt{m_A^2 + m_B^2 + 2E_A m_B}. \quad (2.16)$$

The beam energy E_A is often much larger than the rest mass energy of A or B , so $E_{\text{CM}} \simeq \sqrt{2E_A m_B}$. The total energy in the CM frame is proportional to the *square root* of the beam energy E_A in a fixed target experiment. This is big trouble: much of the energy in the lab frame is used to satisfy conservation of linear momentum. The final-state particles acquire kinetic energies, so there is not enough energy to create additional particles or massive particles we want.

Example 2.6. The trouble

In the LHC, the lab frame is the CM frame of two proton beams, in which each proton has energy $E_p = 6.5 \text{ TeV}$. The total energy is $E_{\text{CM}} = 2E_p = 13 \text{ TeV}$. Let's calculate how much energy E_p^{FT} is required for the proton beam to hit a stationary proton target to reach $E_{\text{CM}} = 13 \text{ TeV}$. Using $E_{\text{CM}} \simeq \sqrt{2E_A m_B}$, where $E_A = E_p^{\text{FT}}$ now,

$$E_p^{\text{FT}} = \frac{E_{\text{CM}}^2}{2m_p} = \frac{(13 \times 10^3 \text{ GeV})^2}{2 \times 0.938 \text{ GeV}} \simeq 10^5 \text{ TeV}.$$

Evidently, this is far higher than the energy in the LHC. This tells us it is much better to collide two beams with each other than a to collide with a fixed target.

Sometimes one would like to produce a particular set of final-state particles in a fixed-target experiment. The goal is to find the minimum beam energy needed. The conservation of 4-momentum says

$$p_A + p_B = p_1 + p_2 + \dots + p_N.$$

In the CM frame (primed frame),

$$p'_1 + p'_2 + \dots + p'_N = (E'_1 + E'_2 + \dots + E'_N, 0, 0, 0) = \left(\sqrt{\mathbf{p}_1'^2 + m_1^2} + \sqrt{\mathbf{p}_2'^2 + m_2^2} + \dots + \sqrt{\mathbf{p}_N'^2 + m_N^2}, 0, 0, 0 \right).$$

By (2.16), minimizing E_{CM} means minimizing E_A . Remember that $E_{\text{CM}} = \sqrt{s}$ is frame invariant, which means it can also be written as $E_{\text{CM}} = \sqrt{(p'_A + p'_B)^2} = \sqrt{(p'_1 + p'_2 + \dots + p'_N)^2}$. To minimize E_{CM} , we just need all spatial momenta of final-state particles to vanish so that there is no kinetic energy required. Then all energies are for creating particles' rest mass,

$$E_{\text{CM}}^{\text{min}} = m_1 + m_2 + \dots + m_N.$$

Setting this equal to (2.16),

$$m_A^2 + m_B^2 + 2E_A^{\text{min}} = (m_1 + m_2 + \dots + m_N)^2.$$

The minimum beam energy required is

$$E_A^{\min} = \frac{(m_1 + m_2 + \cdots + m_N)^2 - m_A^2 - m_B^2}{2m_B}. \quad (2.17)$$

4-Momentum Transfer

A scattering process may have an initial-state particle that survives to the final state. Suppose such a particle has mass m , initial 4-momentum $p_i = (E_i, \mathbf{p}_i)$, and final 4-momentum $p_f = (E_f, \mathbf{p}_f)$. The change in its 4-momentum is

$$q = p_f - p_i = (E_f - E_i, \mathbf{p}_f - \mathbf{p}_i).$$

Since q is also a 4-momentum, its magnitude is Lorentz-invariant,

$$q^2 = (p_f - p_i)^2 = p_f^2 + p_i^2 - 2p_i \cdot p_f = 2m^2 - 2(E_i E_f - \mathbf{p}_i \cdot \mathbf{p}_f).$$

This change in momentum is called the **4-momentum transfer**.

Compton Scattering

Compton scattering is a photon-electron scattering process:

$$\gamma + e^- \rightarrow \gamma + e^-.$$

A similar process can be $g + q \rightarrow g + q$, where g is a gluon and q is a quark. In Compton's original scattering experiment, the initial electron is actually not a free particle, but instead bounded to an atom. However, the binding energy of the electron $E_{\text{bind}} \ll E_\gamma$ for energetic photons such as X-rays, so the electron can be treated as a free electron. Compton found that when a photon scatters off an electron, its wavelength λ increases by

$$\Delta\lambda = \frac{2\pi\hbar}{m_e c} (1 - \cos\theta),$$

where θ is the scattering angle (the angle between the final 3-momentum and initial 3-momentum of the photon). We will show this by conservation of momentum.

$$p_e + p_\gamma = p'_e + p'_\gamma.$$

where the primes indicate final-state quantities. Since we do not know anything about the final 4-momentum p'_e of the electron, we will move it to one side and all other quantities to the other side:

$$p_e + p_\gamma - p'_\gamma = p'_e,$$

Squaring both sides gives

$$p_e^2 + p_\gamma^2 + p'^2_\gamma + p_e \cdot p_\gamma - p_e \cdot p'_\gamma - p_\gamma \cdot p'_\gamma = p'^2_e.$$

By the invariant mass, $p_e^2 = p'^2_e = m_e^2$ and $p_\gamma^2 = p'^2_\gamma = 0$. We see that p'^2_e is eliminated by the invariant mass and we don't have to worry about it anymore:

$$\cancel{p_e^2} + p_e \cdot (p_\gamma - p'_\gamma) - p_\gamma \cdot p'_\gamma = \cancel{p'^2_e}.$$

Now using the fact that the electron is initially at rest, so $p_e = (m_e, 0, 0, 0)$. Its dot product with $p_\gamma - p'_\gamma$ is just $m_e(E_\gamma - E'_\gamma)$. Meanwhile,

$$p_\gamma \cdot p'_\gamma = E_\gamma E'_\gamma - |\mathbf{p}_\gamma| |\mathbf{p}'_\gamma| \cos\theta = E_\gamma E'_\gamma (1 - \cos\theta),$$

Putting these back

$$m_e(E - E') - E_\gamma E'_\gamma (1 - \cos\theta) = 0 \implies m_e(E - E') = E_\gamma E'_\gamma (1 - \cos\theta).$$

Dividing both sides by $m_e E_\gamma E'_\gamma$, we get the Compton scattering formula:

$$\boxed{\frac{1}{E'_\gamma} - \frac{1}{E_\gamma} = \frac{1}{m_e} (1 - \cos\theta)}. \quad (2.18)$$

It related the final energy of the photon E'_γ to the scattering angle θ . Using Planck's formula to connect energy to wavelength,

$$E_\gamma = \frac{hc}{\lambda} = \frac{2\pi\hbar c}{\lambda},$$

the Compton scattering formula becomes

$$\frac{\lambda'}{2\pi\hbar c} - \frac{\lambda}{2\pi\hbar c} = \frac{1}{m_e c^2}(1 - \cos\theta).$$

In terms of $\Delta\lambda = \lambda' - \lambda$, it is

$$\boxed{\Delta\lambda = \frac{2\pi\hbar}{m_e c}(1 - \cos\theta).} \quad (2.19)$$

If $\theta \neq 0$ (the trivial scattering) in the Compton scattering process, the photon wavelength increases because $\cos\theta < 1$, or $1 - \cos\theta > 0$. As a result, the photon frequency ν must decrease because of the inverse relation $\nu = c/\lambda$, and so does the energy.

3 OBSERVABLES

Particle physics require observations. There are many observables and measurements that connects experiments to theory. Most of the measurements serve the following goals:

1. **Determination of the properties of particles.** These properties include masses, lifetimes, spins, charges, magnetic moments, decay rates, branching fractions, etc. Because most particles have finite lifetimes, there appear different methods to measure these quantities experimentally.
2. **Searches for new particles.** Throughout the history, particle physicists are always searching for new particles (e^+ , ν 's, $W^\pm$ and Z , the Higgs boson H , etc.) to perfect the standard model. There are also searches that seek particles outside the standard model, such as supersymmetric particles and dark matter.
3. **Measurement of cross sections.** The cross sections are relevant to scattering processes. They are quantities that help study how particles interact.

Natural units are adopted, $\hbar = 1$, $c = 1$, except when we are dealing with numbers.

3.1 Observables

3.1.1 Why do Particles Decay?

When studying quantum mechanics, we put efforts finding the eigenstates and eigenvalues of the Hamiltonian. These states have perfectly defined energies. For example, the energy levels of the hydrogen atom is

$$E_n = -\frac{13.6 \text{ eV}}{n^2},$$

where n is the principal quantum number, determined by the radial equation of the Schrödinger equation. The time dependence of energy eigenstates are described by a oscillatory phase factor e^{-iEt} ,

$$\psi(\mathbf{x}, t) = \psi_{n\ell m}(\mathbf{x})e^{-iE_n t},$$

where $\ell = 0, 1, \dots, n-1$ is the azimuthal/orbital quantum number, and $m = -\ell, -\ell+1, \dots, -1, 0, 1, \dots, \ell-1, \ell$ is the magnetic quantum number. The $\psi(\mathbf{x}, t)$ above oscillates forever and hence cannot undergo decay. But we know that excited states of the hydrogen atom *do* decay—they spontaneously emit one or more photons and return to the ground state in a finite amount of time. There appears a contradiction between theory and observations. The resolution is that the Hamiltonian in the non-relativistic Schrödinger equation of the hydrogen atom is not the full Hamiltonian. The eigenstates $\psi_{n\ell m}$ obtained in quantum mechanics are not the “true eigenstates”. In fact, for any quantum state that survives for a finite interval Δt , such a state must be made up of a superposition of energy eigenstates with a spread of energy ΔE . These uncertainties are related by the uncertainty principle

$$\Delta E \Delta t \geq \frac{\hbar}{2}.$$

3.1.2 The Exponential Law and Decay observables

The decay of particles follow the exponential decay law. Let $P(t)$ be the probability to survive until time t ,

$$P(t) = 1 - \int_0^t \mathcal{P}(t') dt',$$

where $\mathcal{P}(t)$ is the probability density to decay at t . Then the probability to survive until $t + dt$ is

$$\begin{aligned} P(t + dt) &= (\text{prob. to survive until } t)(\text{prob. to then survive from } t \text{ to } t + dt) \\ &= (\text{prob. to survive until } t)(1 - \text{prob. to decay during } dt) \\ &= P(t)(1 - \mathcal{P}(t) dt). \end{aligned}$$

For unstable particles, the probability to decay during dt is a constant $\mathcal{P} = K$ that is independent of t . That is, the probability to decay in the next dt does not depend on how long the particle has lived. Then we can write

$$P(t + dt) = P(t)(1 - K dt).$$

Rearranging some terms,

$$\frac{P(t + dt) - P(t)}{dt} = -KP(t) \implies \frac{dP}{dt} = -KP(t).$$

This gives the exponential decay law,

$$P(t) = e^{-Kt}.$$

We define the constant K to be the **total decay rate**,

$$K = R_{\text{tot}} = \sum_i R_i, \quad (3.1)$$

where R_i is the decay rate of the i th decay mode. The **mean proper lifetime** (or mean lifetime or just lifetime), τ , is the inverse of the decay rate

$$\tau \equiv \frac{1}{K} = \frac{1}{R_{\text{tot}}}. \quad (3.2)$$

The exponential decay law can be written in as

$$P(t) = e^{-t/\tau}. \quad (3.3)$$

At each time, there is a probability density $\mathcal{P}(t) = P(t)$ that determines the decay probability within time t and $t + dt$, defined as

$$P(t) = 1 - \int_0^t \mathcal{P}(t') dt'.$$

Taking the time-derivative on both sides gives

$$\boxed{\mathcal{P}(t) = \frac{1}{\tau} e^{-t/\tau}}. \quad (3.4)$$

This ensures that the probability density $\mathcal{P}(t)$ is normalized. This probability density should not be confused with the decay rate K : $\mathcal{P}(t) dt$ is the probability to first survive to t and decay within time t and $t + dt$. It incorporates the probability of a particle survival to t after it was created. On the other hand, K is the probability to decay within any dt , given that the particle has not decay. It does not care about how much time the particle has lived.

The reason why τ is called the mean lifetime is obvious. It is the *mean* lifetime of a particle in probabilistic sense,

$$\langle t \rangle = \int_0^\infty dt t \cdot \mathcal{P}(t) = \int_0^\infty dt t \left(\frac{1}{\tau} e^{-t/\tau} \right) = \tau \int_0^\infty \frac{dt}{\tau} \frac{t}{\tau} e^{-t/\tau} = \tau \int_0^\infty dx x e^{-x} = \tau.$$

Back to the decay rates R . We can write (3.1) as

$$1 = \sum_i \frac{R_i}{R} = \sum_i B_i \quad \text{where} \quad B_i \equiv \frac{R_i}{R} \quad (3.5)$$

is called the **branching fraction** of the i th decay mode. A branching fraction can be determined by measuring the fractions of decays that lead to a specific decay final state. When one obtain all R_i 's, the branching fraction B_i can be easily calculated. Conversely, obtaining all B_i 's does not mean knowing decay rates, because we have lost some information. The *rates* is lost in the fraction R_i/R , where s^{-1} are canceled. Hence measuring all branching fraction cannot tell the mean lifetime either.

It is convenient to express the decay rate in energy units, so we define the **decay width** Γ to be

$$\Gamma_i = \hbar R_i, \quad \Gamma = \sum_i \Gamma_i = \sum_i \hbar R_i. \quad (3.6)$$

The quantities Γ_i are called **partial decay widths** or simply partial widths, and Γ is the **total decay width**, or the total width of the particle. The decay width is connected to everything: the decay rate, branching fractions, and mean lifetime. In particular, the total width is related to the mean lifetime by

$$\tau = \frac{\hbar}{R} = \frac{1}{\sum_i R_i} = \frac{\hbar}{\sum_i \Gamma_i} = \frac{\hbar}{\Gamma},$$

and the branching fraction by

$$\Gamma_i = \Gamma \cdot \frac{\hbar \Gamma_i}{\hbar \Gamma} = \Gamma \frac{R_i}{R} = \frac{\hbar}{\tau} B_i.$$

In natural units, $\Gamma_i = R_i$ and $\Gamma = 1/\tau$. This will be useful when we derive the Breit-Wigner line shape.

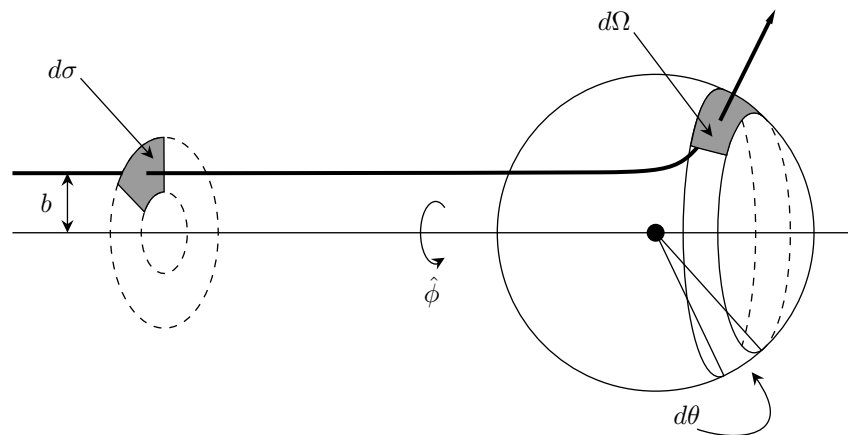
All above are referred to the rest frame of the particle. In the lab frame, particles are moving at relativistic speeds, so quantities such as the observed mean lifetime will be different from the mean proper lifetime.

3.1.3 Angular Measurements

The parameter to describe a scattering process is the cross section σ of the incident beam. The cross section σ has dimensions of an area A . We are interested in the total cross section

$$\sigma_{\text{tot}} = \sum_{i=1}^n \sigma_i.$$

A scattering can be described as the following:



Suppose a particle come across a potential and scatters of at the **scattering angle** θ . The scattering angle is a function of the **impact parameter** b , the distance by which the incident particle would have missed the scattering center. If the incident beam passes through an infinitesimal cross section $d\sigma$, it will scatter into a solid angle $d\Omega$, and

$$d\sigma = D(\theta) d\Omega,$$

where $D(\theta)$ is called the **differential cross section**. The figure shows that

$$d\sigma = |b db d\phi| \quad \text{and} \quad d\Omega = |\sin \theta d\theta d\phi|.$$

Hence the differential cross section can be expressed as

$$D(\theta) = \frac{d\sigma}{d\Omega} = \left| \frac{b}{\sin \theta} \left(\frac{db}{d\theta} \right) \right|.$$

The total cross section is evidently

$$\sigma = \int d\sigma = \int D(\theta) d\Omega.$$

Experimentally, we can control the **luminosity** L of a beam of incident particles. L is the number of particles passing down the line per unit time, per unit area. $dN = L d\sigma$ is the number of particles per unit time passing through area $d\sigma$, and the number per unit time scattered into the solid angle $d\Omega$:

$$dN = L d\sigma = L D(\theta) d\Omega.$$

If the luminosity is known (controlled), we can measure the number of particles entering the detector over a solid angle, and subsequently determine the differential cross section,

$$D(\theta) = \frac{dN}{L d\Omega}.$$

3.1.4 Breit-Wigner Line Shape and the Decay Width

In non-relativistic quantum mechanics, the time evolution of an eigenstate can be described by

$$|\psi(t)\rangle = e^{-iEt} |\psi(0)\rangle.$$

In the particle's rest frame, $E = M$, and let $|\psi(0)\rangle$ be normalized,

$$|\psi(t)\rangle = e^{-iMt} |\psi(0)\rangle \quad \text{with} \quad \langle\psi(0)|\psi(0)\rangle = 1.$$

We see that once $|\psi\rangle$ is normalized at time $t = 0$, it is normalized forever,

$$\langle\psi(t)|\psi(t)\rangle = \langle\psi(0)|(e^{iEt}e^{-iEt})|\psi(0)\rangle = \langle\psi(0)|\psi(0)\rangle = 1.$$

Physically, an eigenstate has a definite energy and it lives forever. This does not describe particles that can decay. For an unstable particle, the probability density to decay at time t is given by (3.4),

$$\mathcal{P}(t) = \langle\psi(t)|\psi(t)\rangle = \frac{1}{\tau} e^{-t/\tau},$$

where τ is the mean lifetime. This suggests that if we replace $M \rightarrow M - i\Gamma/2$ and multiply a prefactor $\sqrt{\Gamma}$ to $|\psi(0)\rangle$,

$$|\psi(t)\rangle = \sqrt{\Gamma} |\psi(0)\rangle e^{-i(M-i\Gamma/2)t} \implies \mathcal{P}(t) = \langle\psi(t)|\psi(t)\rangle = \langle\psi(0)|\psi(0)\rangle \Gamma e^{-\Gamma t} = e^{-\Gamma t},$$

This reproduces $\tau = 1/\Gamma$. The time and energy are conjugate variables. We can Fourier transform $|\psi(t)\rangle$ to the energy space $|\tilde{\psi}(E)\rangle$, starting at $t = 0$,

$$|\tilde{\psi}(E)\rangle = \frac{1}{\sqrt{2\pi}} \int_0^\infty dt e^{iEt} |\psi(t)\rangle = \sqrt{\frac{\Gamma}{2\pi}} \int_0^\infty dt e^{[i(E-M)-\Gamma/2]t} |\psi(0)\rangle = \sqrt{\frac{\Gamma}{2\pi}} \frac{e^{[i(E-M)-\Gamma/2]t}}{i(E-M)-\Gamma/2} \Big|_0^\infty |\psi(0)\rangle.$$

(Here E is not exactly equal to M because there will be a spread in E , resulting from a finite lifetime of the particle.) The evaluation at ∞ vanishes because of the exponential decay term $e^{-\Gamma t/2}$, so

$$|\tilde{\psi}(E)\rangle = -\sqrt{\frac{\Gamma}{2\pi}} \frac{1}{i(E-M)-\Gamma/2} |\psi(0)\rangle.$$

The probability density of finding the particle with energy E is

$$\mathcal{P}(E) = \langle\tilde{\psi}(E)|\tilde{\psi}(E)\rangle = \frac{\Gamma}{2\pi} \frac{1}{(E-M)^2 + (\Gamma/2)^2} \langle\psi(0)|\psi(0)\rangle = \frac{1}{2\pi} \frac{\Gamma}{(E-M)^2 + (\Gamma/2)^2}.$$

This probability density is a function of E , with parameters M and Γ . Meanwhile, E represents the rest mass energy m of a state. In other words,

$$\boxed{\mathcal{P}(m) = \frac{1}{2\pi} \frac{\Gamma}{(m-M)^2 + (\Gamma/2)^2}} \quad (3.7)$$

This is the **Breit-Wigner line shape**, describing the spread of the observed mass m around the most probable mass M . The spread is characterized by the width Γ . Evidently, the maximum of $\mathcal{P}$ is at $m = M$, and $\mathcal{P}(m)$ is symmetric about $m = M$. There exist two points of m at which $\mathcal{P}(m)/\mathcal{P}(M) = 1/2$ (half maximum). One can easily calculate that

$$m_{\text{half max}} = M \pm \frac{1}{2}\Gamma.$$

This is why Γ is called the decay *width*. It is the full width at half maximum of the probability density. If we obtain the distribution of m in an experiment, we can calculate Γ as full width at half maximum. Quantum field theory will improve the formula for the Breit-Wigner line shape, but we shall stick with (3.7) now.

3.2 Particle Detection

3.2.1 Fundamental Principles of Particle Detection

There are three fundamental principles of *direct* particle detection (not decay products).

- **Principle I.** To detect a given particle A , the detection process must involve a type of fundamental interaction that couples to A . The detection process must also involve a type of fundamental interaction that couples to the matter or fields in the detector.

There is no universal interaction that all particles must possess, except gravity. However, gravity does not play a major role in the Standard Model, so the detection of strong, electromagnetic, and weak interactions should be viewed differently. For example, to detect neutrinos, the detection process must involve weak interaction. This results in a low detection probability of neutrinos. Each interaction process is associated with a cross section σ that characterizes the strength and range of an interaction between A and a target particle. The probability for A to undergo an interaction also depends on the number density n of target particles. Combining the two gives a characteristic interaction length for each type of interaction process i ,

$$\lambda_i = \frac{1}{n_i \sigma_i}.$$

The probability for A to interact with the detector by a certain process when traveling a distance Δx through the detector is

$$P(\Delta x) = 1 - e^{-\Delta x / \lambda_i} = 1 - e^{-n_i \sigma_i \Delta x}.$$

If $\Delta x / \lambda_i \ll 1$, then $P(\Delta x) \simeq \Delta x / \lambda_i$.

- **Principle II.** The particle interacting with the detector must transfer an amount of energy that is much larger than the background noise in the system. The electronics that read out a detector should have a threshold criterion. This criterion varies from particles to particles, depending on the amount of energy transferred and the form of this energy. For example, a key energy transfer process in charged particle is ionization. It allows us to track the trajectory of a charged particle without much perturbation to its path. On the contrary, the detection of electrically neutral particles are often “destructive”—the particle is typically absorbed in the detection process.
- **Principle III.** A detector’s properties and performance must be well understood. The resolution for a certain quantity should be known; all parts of the detector should work properly. This is to ensure that no event is mis-measured.

3.2.2 Classification of Particles by Stability and Interactions

The classification of particles is based on their mean lifetime and the types of interactions that they experiences. There are hundreds of particles in the Standard Model, but only a few of them can be detected directly. Most particles decay so fast that only their decay products are detectable. Almost all particles that decay via strong or electromagnetic interactions can only be detected indirectly. (There are also weak decay particles that are not directly detectable, such as W , Z , H , and the top quark t , mainly because of their very large masses.) To see why, let’s calculate how far they can travel in a detector.

Example 3.1. A typical strong interaction decay

Consider a particle with decay width Γ of the order $10 - 100$ MeV (a typical decay via strong interactions). Their mean lifetime, according to (3.6), is about

$$\tau = \frac{\hbar}{\Gamma} \simeq \frac{66 \times 10^{-23} \text{ MeV s}}{\Gamma \text{ (MeV)}} \sim 10^{-23} \text{ s}.$$

This is the mean proper lifetime in the particle’s rest frame. In the lab frame, there will be relativistic time dilation. The distance it can travel before decay at time Δt_{rest} is

$$\Delta x = v \Delta t = \gamma \beta c \Delta t_{\text{rest}},$$

where $\beta = v/c$. The mean distance traveled is

$$\langle \Delta x \rangle = \gamma \beta c \langle \Delta t_{\text{rest}} \rangle = \gamma \beta c \tau.$$

Recall that the relativistic 3-momentum is given by $|\mathbf{p}| = \gamma m v = \gamma \beta m c$. To achieve the maximum mean distance traveled, we want the largest value of $\gamma \beta = |\mathbf{p}|c/mc^2$. In a typical high energy experiment, a hadron with mass $mc^2 \sim 1 \text{ GeV}$ has momentum $|\mathbf{p}|c \sim 10 \text{ GeV}$, so the mean distance traveled is

$$\Delta x = (10c) \times 10^{-23} \text{ s} = 3 \times 10^{-14} \text{ m}.$$

This is only 10 times the size of an atomic nucleus, and is much smaller than the distance of an atom. There is no way to detect it directly with current technologies.

Here we will classify particles into three categories:

- **Detector-stable particles.** These particles usually traverse all the way through a detector. Some of them are *completely* stable (e.g. γ , p , $e^\pm$), and some are not (e.g. n , $\mu^\pm$, $\pi^\pm$).
- **Detector quasi-stable particles.** These particles travel a measurable distance from their production point before decaying via weak interactions. They can produce decay **vertices**, points in space where the measured trajectories of their daughter particles intersect. These vertices are distinctive and can be easily measured.
- **Detector-unstable particles.** These particles decay before traveling any measurable distance from their production point. Most particles are in this category. They can be detected indirectly using the conservation of 4-momentum.

Table 3.1 and 3.2 lists all detector-stable and detector quasi-stable particles, respectively.

Particle	Mass (GeV)	J	τ	$c\tau$ (m)	Ionization track?	EM casc. shower?	Strong casc. shower?	Penetrate had. absorber?
γ	0	1	stable	stable	-	✓	-	-
$e^\pm$	0.00511	1/2	stable	stable	✓	✓	-	-
$\mu^\pm$	0.106	1/2	$2.2 \mu\text{s}$	659	✓	-	-	✓
$\nu_\ell, \bar{\nu}_\ell$	~ 0	1/2	stable	stable	-	-	-	-
$\pi^\pm$	0.140	0	26 ns	7.8	✓	-	✓	-
$K^\pm$	0.494	0	12 ns	3.7	✓	-	✓	-
K_L^0	0.498	0	0.51 ns	0.153	-	-	✓	-
$p, \bar{p}$	0.938	1/2	stable	stable	✓	-	✓	-
$n, \bar{n}$	0.940	1/2	879 s	2.6×10^{11}	-	-	✓	-
α	3.727	0	stable	stable	✓	-	✓	-

Table 3.1: A list of all detector-stable particles.

Particle	Mass (GeV)	J	τ	$c\tau$	Typical decay location
$\tau^\pm$	1.777	1/2	0.3 ps	$87 \mu\text{m}$	inside beam pipe
K_S^0	0.498*	0	0.09 ns	2.7 cm	outside beam pipe
$\Lambda, \bar{\Lambda}$	1.116	1/2	0.26 ns	7.9 cm	outside beam pipe
$D^0, \bar{D}^0$	1.865	0	0.4 ps	$123 \mu\text{m}$	inside beam pipe
$D^\pm$	1.870	0	1.0 ps	$312 \mu\text{m}$	inside beam pipe
$D_s^\pm$	1.968	0	0.5 ps	$151 \mu\text{m}$	inside beam pipe
$B^\pm$	5.279	0	1.6 ps	$491 \mu\text{m}$	inside beam pipe
$B^0, \bar{B}^0$	5.280	0	1.5 ps	$455 \mu\text{m}$	inside beam pipe
$B_s^0, \bar{B}_s^0$	5.367	0	1.5 ps	$454 \mu\text{m}$	inside beam pipe

Table 3.2: A list of all detector quasi-stable particles.

Table 3.1 also lists the signatures for detector-stable particles. These signatures are detected in ionization tracking detectors, electromagnetic calorimeters, hadronic calorimeters, and absorber penetration detectors for muons. Charged particles produce an ionization track from EM interactions with atoms. The photon and electron produce EM cascade showers in dense matter. Detector-stable hadrons produce strong cascade showers in dense matter. The α -particle (helium-4 nucleus) is included because it is important in nuclear transitions. Whatever particles that make up dark matter is not included in Table 3.1, but they are stable particles. They exist in galaxies since the early universe, but are hard to detect.

Table 3.2 consists of all detector quasi-stable particles. All these particles except the τ -lepton are ground-state hadrons containing s , c , or b quarks. They cannot decay via strong or electromagnetic interactions because there are no lighter hadrons containing the same quark content. These particles decay either inside or outside around the beam pipe in a colliding beam experiment.

This kind of classification certainly depends on the detector, and it also depends on the momentum of the particle. (If the particle is produced at rest in the lab frame, we will observe it to decay very quickly.) Nonetheless, they work very well for most particle physics experiments.

3.2.3 Signatures of Particles in Detectors

4 FEYNMAN DIAGRAMS

This chapter is an introduction to elementary particle interactions and the **Feynman diagrams** that represent the allowed interactions. It is a qualitative chapter that shows how each of the relevant forces acts on quarks and leptons. A more quantitative treatment is discussed in Chapter 8.

To be written.

4.1 The Four Interactions

There are four fundamental forces/interactions

Force	Strength	Theory	Mediator
Strong	1	Chromodynamics	Gluon g
Electromagnetic	10^{-2}	Electrodynamics	Photon γ
Weak	10^{-13}	Flavordynamics	W and Z
Gravitational	10^{-42}	Geometrodynamics/Relativity	Graviton (?)

Table 4.1: Four fundamental forces/interactions. The numbers in this table (especially the one for the weak force) should not be taken seriously because the forces depends on the sources and how far away they are.

The classical theory of gravity follows Newton's law of universal gravitation. The relativistic version is the general relativity by Einstein. Yet, there is still no satisfactory quantum theory of gravity. The classical electrodynamics is formulated by Maxwell and is consistent with special relativity. The quantum electrodynamics is perfected by Tomonaga, Feynman, and Schwinger in the 1940s. The first theory of weak forces is by Fermi in 1933, later refined by Lee and Yang, Feynman and Gell-Mann, and put into the modern form by Glashow, Weinberg, and Salam in the 1960s. In this notes, the theory of weak forces is referred to as the Glashow-Weinberg-Salam (GWS) theory. The strong force theory was first presented by Yukawa in 1934, and the chromodynamics appeared in the 1970s.

The hierarchy of interaction strengths is typically Strong $\gg$ Electromagnetic $\gg$ Weak. This hierarchy holds true when comparing cross sections and decay rates, and its inverse is true when comparing lifetimes. If the CM-frame energy $\sqrt{s}$ is comparable to or larger than the mass of the W and Z , then the weak interaction can have strength comparable to the one of electromagnetism:

$$\begin{aligned} \text{Strong} &\gg \text{Electromagnetic} \gg \text{Weak}(\sqrt{s} \ll m_W, m_Z), \\ \text{Strong} &\gg \text{Electromagnetic} \sim \text{Weak}(\sqrt{s} \geq m_W, m_Z). \end{aligned}$$

4.1.1 Scattering Amplitudes

In non-relativistic quantum mechanics, a system of particles at an initial time t_0 is a state $|\psi_i\rangle = |\psi(t_0)\rangle$. Its time evolution is governed by the Schrödinger equation, but here we will use another approach. Define the **time-evolution operator** $U(t, t_0)$ such that

$$|\psi(t)\rangle = U(t, t_0) |\psi(t_0)\rangle.$$

The time-evolution operator is usually a function of the Hamiltonian H . The **amplitude** of a final state $\psi(t)_f$ at time t is given by the inner product

$$\mathcal{A}_{fi}(t, t_0) = \langle \psi_f(t) | \psi(t) \rangle = \langle \psi_f | U(t, t_0) | \psi_i \rangle.$$

It is simply a projection of the true state $|\psi(t)\rangle$ at time t into the particular final state, $|\psi_f(t)\rangle$, because the true state is often a superposition of $|\psi_f(t)\rangle$ with different f . The square modulus of the amplitude $\mathcal{A}^2$ is the transition probability. To determine $U(t, t_0)$,

we write the (total) Hamiltonian as

$$H = H_0 + H',$$

where H_0 is a free-particle Hamiltonian and H' describes the interactions. In many cases, the interaction Hamiltonian H' can be treated as a perturbation, but there are situations where it cannot.

4.1.2 The S -matrix

The study of particle physics requires relativistic quantum mechanics. The free-particle states in relativistic quantum mechanics depend on the spin of the particles. We will focus on spin-0, spin-1/2, and spin-1 particles. Elementary particles in the Standard Model all falls within one of these categories. The particles of those spins are described by the Klein-Gordon, Dirac, and Maxwell-like equations, respectively. For particle interaction processes (such as scattering), in order to have initial-state particles being free particles, they must be far apart so that their interactions are negligible ($H' = 0$) at $t_0 \rightarrow -\infty$. Similarly, at $t \rightarrow \infty$, the final-state particles must be far apart to be free particles. We assume that both conditions will be true when $t_0 \rightarrow -\infty$ and $t \rightarrow \infty$. The amplitude is then given by the S -matrix,

$$S_{fi} \equiv \lim_{\substack{t \rightarrow +\infty \\ t_0 \rightarrow -\infty}} \mathcal{A}_{fi}(t, t_0) = \lim_{\substack{t \rightarrow +\infty \\ t_0 \rightarrow -\infty}} \langle \psi_f | U(t, t_0) | \psi_i \rangle. \quad (4.1)$$

Both the initial state $|\psi_i\rangle = |\psi(t_0)\rangle$ and final state $|\psi_f(t)\rangle$ are eigenstates of H_0 —they are free-particle states. In general, the S -matrix elements are complex-valued function of the particle properties, including masses, spins, charges, 4-momenta, even orbital angular momenta. They also involve quantities that characterize any intermediate-state particles before they evolve into the final-state particles. These intermediate-state particles are usually **virtual particles**. There can be multiple allowed transition pathways through intermediate states as we will see in Feynman diagrams. This is how we define a *process*: a process should have the same initial state and the same final state, but it does not care about how the initial state become the final state. For example, $e^+ + e^- \rightarrow \gamma^* \rightarrow \mu^+ + \mu^-$ and $e^+ + e^- \rightarrow Z^* \rightarrow \mu^+ + \mu^-$ count as the same process. It does not care about intermediate particles, whether it is a photon or a Z -boson. The amplitude should be summed up before squaring to get the probability or probability density,

$$\mathcal{P}_{fi} \propto |\mathcal{A}_{fi, \text{tot}}|^2 = \left| \sum_{j=1}^n \mathcal{A}_{fi, j} \right|^2.$$

Note that we do not calculate individual probabilities first and sum them up because the amplitudes interfere with each other, so their relative phases do matter. If the set of final-state particles (or initial-state particles) are different, we should sum up the probabilities as they represent completely different processes. Summing up the amplitude of different processes are meaningless in this case.

4.1.3 Feynman Diagrams

A Feynman diagram represents a scattering amplitude of a process, with the intermediate state fully specified. A set of Feynman diagrams describe different scattering amplitudes order-by-order in perturbation theory. Though Feynman diagrams look like cartoons, they are extremely precise and rigorous. They are derived from the Lagrangian of the Standard Model

$$L_{\text{SM}} = L_{\text{strong}} + L_{\text{EM}} + L_{\text{weak}}.$$

The rules to draw Feynman diagrams are different for each type of interactions as we will discuss in the following sections.

A Feynman diagram consists of three components: external lines, internal lines, and fundamental interaction vertices. The external lines represent free-particle momentum eigenstates. The internal lines represent intermediate-state particles. In some processes in which no internal lines are present, there is a direct interaction between the initial-state particles and final-state particles. The interaction vertices specify a particular interaction between particles. In each vertex, the 4-momentum is strictly conserved. We will see many examples below.

There are several conventions to draw Feynman diagrams. In this notes and in most of the books, the time axis runs from left to right. The line-style conventions are listed in Table 4.2. Note that Feynman diagrams with precise Feynman rules work only for elementary particles in SM. Hadrons are not elementary particles as they are composed of quarks and gluons. Thus, they cannot be represented by a single line. Sometimes a hadron is represented by explicitly drawing all the quark lines in Feynman diagrams, but because these quarks are confined (they are not free particles), Feynman rules for them should be modified.

Line style	Description	Particles Represented
	Solid line with no arrow.	Fermion or antifermion according to label.
	Solid line with right arrow	Fermion traveling to the right in time.
	Solid line with left arrow	Antifermion traveling to the right in time.
	Wiggly line	$J = 1$ colorless boson: $\gamma, Z, W^\pm$ in SM.
	Spring	$J = 1$ colored boson: g in SM.
	Dashed line	$J = 0$ boson: H in SM.

Table 4.2: Line-style conventions commonly used for Feynman diagrams.

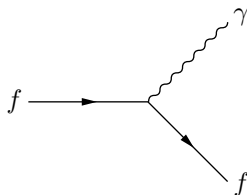
In the following sections, we will discuss how to draw Feynman diagrams in theories describing particular interactions: quantum electrodynamics (QED), quantum chromodynamics (QCD), and weak/GWS theory.

4.2 Quantum Electrodynamics (QED)

In quantum electrodynamics (QED), photons interact with electrically charged particles. This section mainly talks about interactions between photons and spin-1/2 fermions of the SM, i.e. charged leptons and quarks. The charged spin-1/2 fermions are denoted as f , and their antiparticles as $\bar{f}$.

4.2.1 Fundamental Vertices in QED

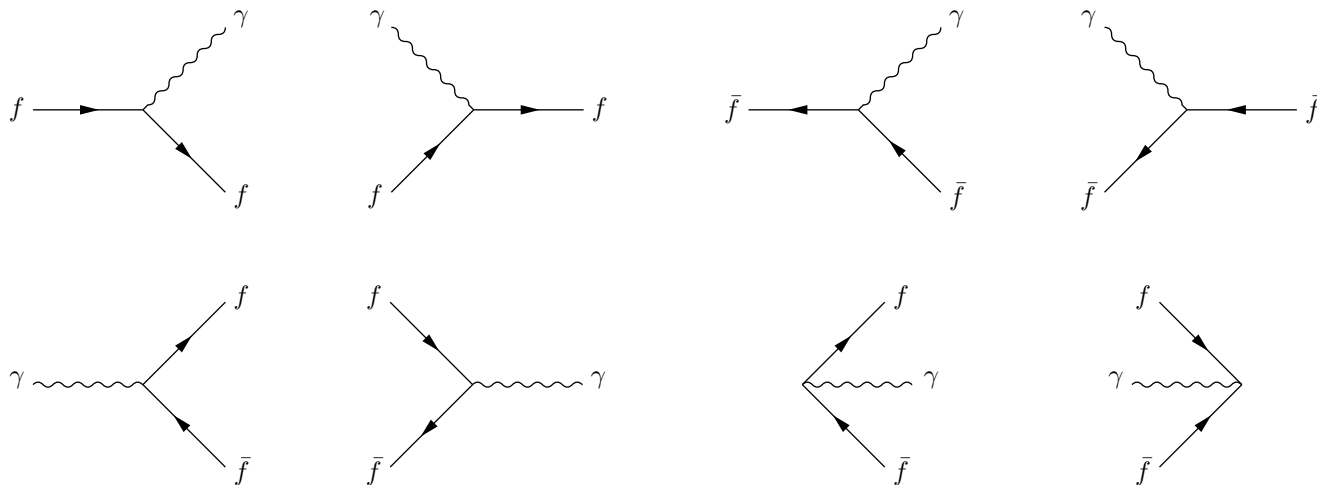
All electromagnetic phenomena are reducible to the following process:



There are several features on this single vertex.

- A photon is involved, so it must be an electromagnetic process.
- The separation of lines in a Feynman diagram does not represent physical separation. The angle between the lines do not matter; they can be whatever they want as long as the time ordering is unchanged.
- The fermion on the LHS of the vertex is an incoming particle, while the fermion and the photon are outgoing particles. This vertex can be written as $f \rightarrow f\gamma$, where f is a spin-1/2 fermion of the *same* type. Thus, the process $c \rightarrow u\gamma$ is not allowed, even though it conserves electric charge. The electromagnetic vertex simply does not allow the fermion to change type in the process. On the contrary, the 4-momentum and the spin direction of the fermion can change.
- The mathematical factor associated with an electromagnetic vertex is linear in the fermion's charge. In other words, the amplitude for emitting or absorbing a photon is proportional to the charge of the particle. For example, the strange quark has charge $-\frac{1}{3}e$, while the electron has $-e$. The ratio of amplitude for emitting or absorbing a photon is 1/3, giving a probability of 1/9.
- The 4-momentum is conserved at every vertex in a Feynman diagram. This is a subtle point: if a process only involve one vertex, then the 4-momentum is not conserved. That is, the diagram above is not a physical process. If the 4-momentum were conserved in this process, $p_f = p'_f + p_\gamma$, then squaring both sides gives $m_e^2 = m_e^2 + 2p_\gamma \cdot p'_f$, or $p_\gamma \cdot p'_f$. Let the energy of the photon be E_γ , so $E_\gamma E'_f - E_\gamma \sqrt{E_f'^2 - m_e^2} \cos \theta = 0$, where θ is the angle between the final electron and the photon. The LHS is greater than zero; 4-momentum is not conserved in this process. The resolution is that this vertex is only a part of a Feynman diagram. Later we will see that if this photon is an intermediate particle, then it is in principle not observable.

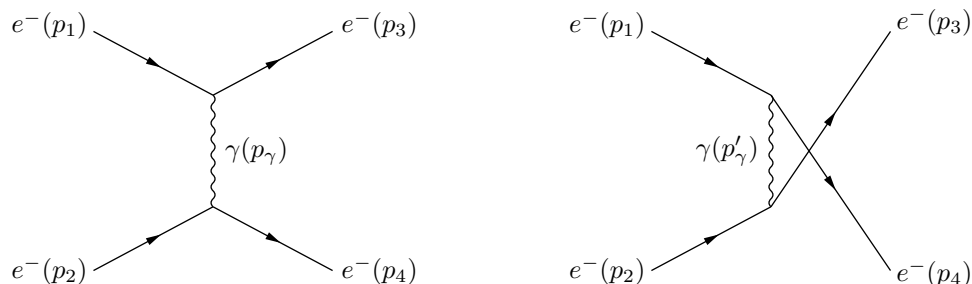
A way to generate alternative vertices that involve antiparticles is known as **crossing**. The rule is that if one switch an incoming particle to an outgoing one, then it changes into its antiparticle. Crossing symmetry can be written compactly: $A + B \rightarrow C + D$ implies that $A \rightarrow \bar{B} + C + D$ is allowed. There are in total eight possibilities to draw a fundamental vertex:



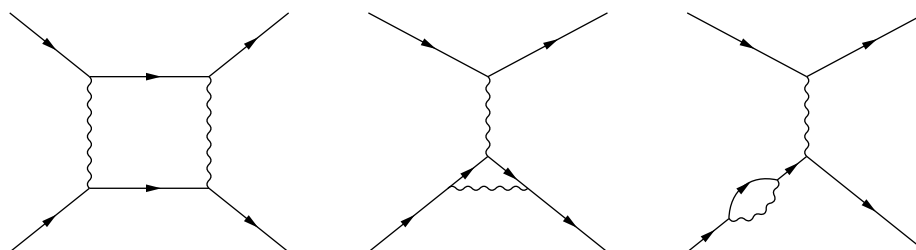
Evidently, the last two are not possible because they either have no incoming or outgoing particles. All the other six vertices are valid. One can quickly find a common feature of these vertices: the arrows on the lines cannot point towards the same vertex. All electromagnetic processes can be represented by combinations of those vertices. The next section introduces some examples.

4.2.2 Feynman Diagrams in QED

Møller Scattering. The first and probably the simplest process is called **Møller scattering**: $e^- + e^- \rightarrow e^- + e^-$. It is just an electron scattering from an electron.

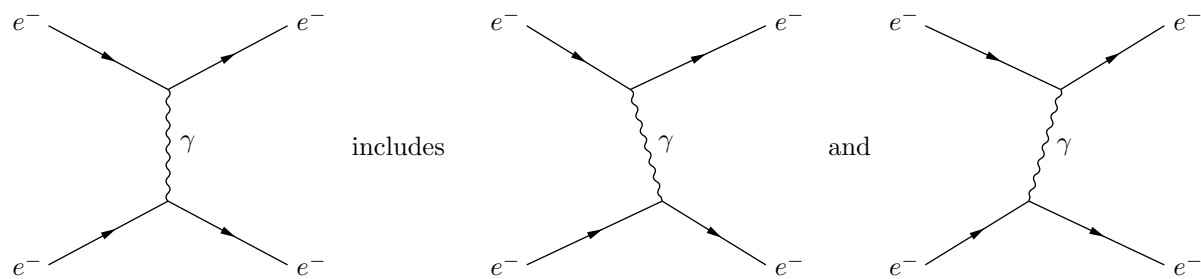


The two Feynman diagrams above are the lowest order diagrams in EM coupling for Møller scattering. The lowest order means that it contains the fewest number of interaction vertices (2 in this case). The Feynman diagram with topology the same as the left one is called a **t-channel**. The right one is called a **u-channel**. They are more related to physical scattering of particles. Because electrons are identical particles, it is in principle impossible to distinguish the two identical final states above (if their spin configurations are also the same). However, the intermediate photons have distinct 4-momenta: the left one has $p_\gamma = p_1 - p_3$, but the right one has $p'_\gamma = p_1 - p_4$. Hence they contribute to two distinct amplitudes. This happens when we have two identical outgoing particles. Higher order diagrams for Møller scattering can be

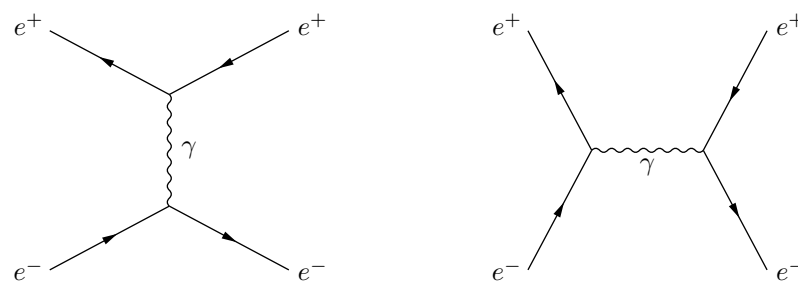


They have more vertices so they are less probable. In most cases, the lowest order Feynman diagrams are good enough. Moreover, we are not considering Feynman diagrams with loops. They are more complicated and advanced diagrams. All we will draw are tree diagrams.

Another thing to notice is that the photon is drawn vertically in these Feynman diagrams. For Feynman diagrams with intermediate particles drawn vertically, it means that both time orderings are included:

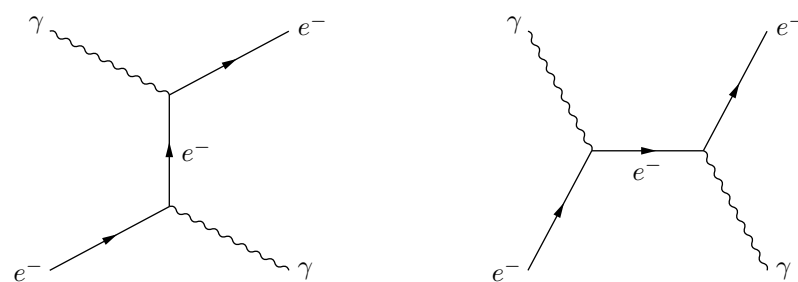


Bhabha Scattering. The second Feynman diagram to study is **Bhabha scattering**, $e^+ + e^- \rightarrow e^+ + e^-$, or electron-positron scattering. The lowest order diagrams are



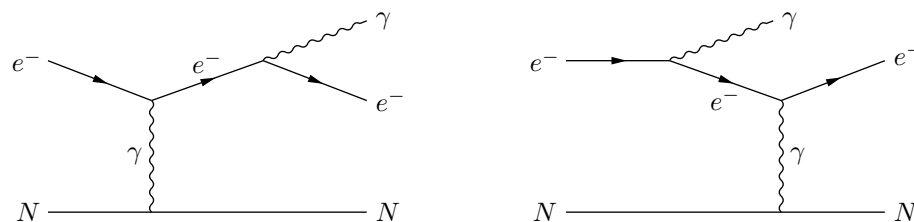
The Bhabha scattering is one of the most important processes at an e^+e^- collider. The left diagram above is called a **s-channel**, which involves an annihilation of initial particles.

Compton Scattering. The Compton scattering is $\gamma + e^- \rightarrow \gamma + e^-$. The lowest order EM coupling diagrams are



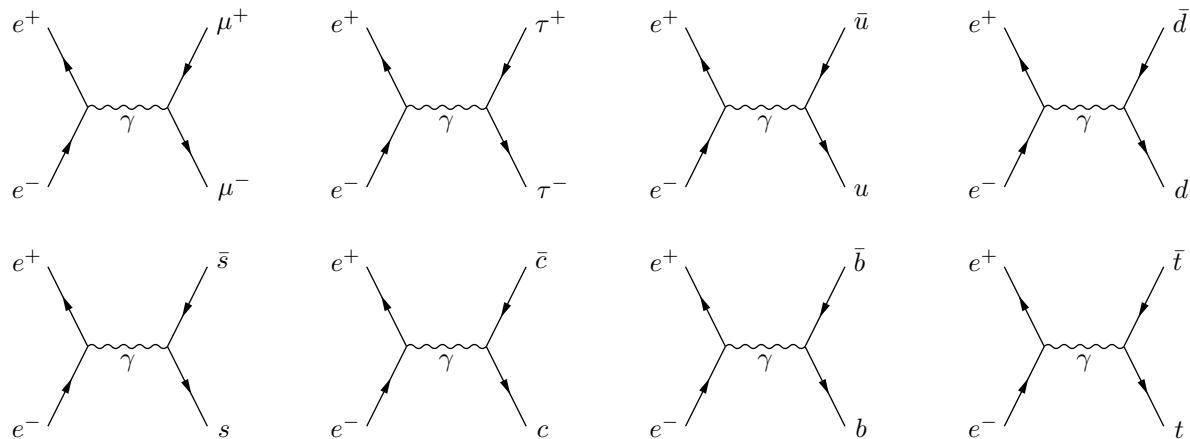
In compton scattering, the intermediate-state particle can no longer be a photon because that will violate charge conservation. The intermediate state for a QED Compton scattering process will be an electron.

Bremsstrahlung. The word **bremsstrahlung** means “braking radiation” in German. We know that accelerating a charged particle will make it radiate. Accelerating an electron can easily achieve such an effect because electrons are very light. The process is described as $e^- + N \rightarrow e^- + \gamma + N$, where N is a nucleus with charge $+Ze$. (Here Z is the atomic number, not the Z -boson.) An electron is stopped by a nucleus so it radiates energetic photons. There are two lowest order diagrams for Bremsstrahlung in QED:



The intermediate process are distinct. It is straight forward to find that the 4-momenta of the intermediate electron in the two diagrams are different.

Pair Production and Annihilation If the energy of a e^+e^- pair is high enough, they can produce a pair of charged fermions. The charged fermions need not be leptons; they can also be quarks. They cannot be neutrinos because neutrinos do not couple to photons as they don't have electric charge. In QED, the pair production process for one species of particle in the final state has unique lowest order Feynman diagram (except electron pair production which is mentioned in Bhabha scattering). Here are possible outcomes of e^+e^- annihilation to pair production:

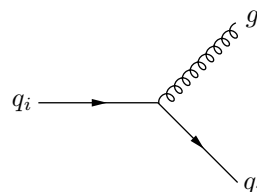


Charges of all kinds are strictly conserved in these processes. All vertices have net charge zero.

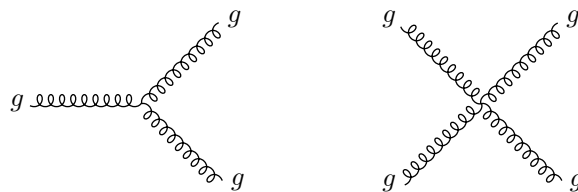
4.3 Quantum Chromodynamics (QCD)

4.3.1 Fundamental Vertices in QCD

Color is the “charge” in chromodynamics, and the fundamental interaction is between quarks and gluons:

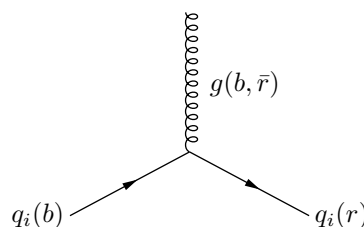


where q_i represents any quark (u, d, s, c, b, t). The force between two quarks is mediated by the exchange of gluons. In QED, photons cannot couple to other photons because they don't possess electric charge. However, in QCD, gluons are able to couple to other gluons because they have color charge. Hence there are two more fundamental vertices involving only gluons:



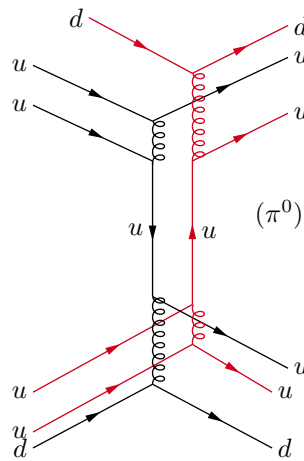
QCD has three differences from QED:

- There are three kinds of color (red, green, blue). The color (not flavor) of the quark may change, but they are always conserved. The gluon must carry away the color difference:

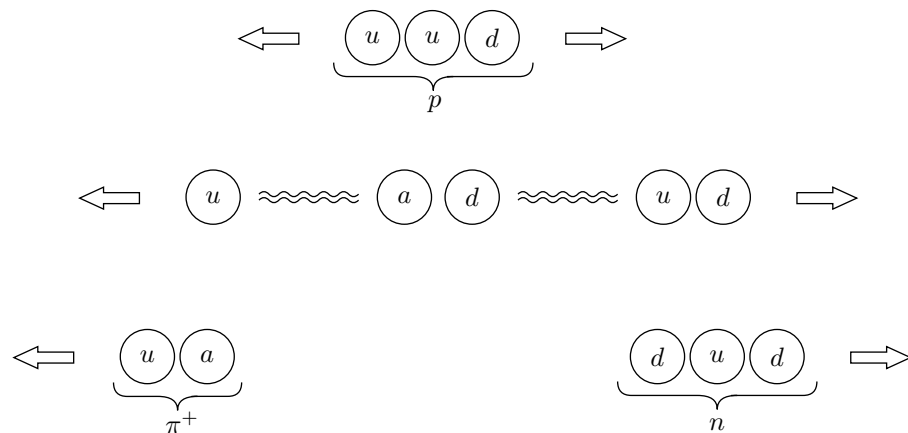


Here b stands for blue, r stands for red, and $\bar{r}$ is anti-red.

- QCD and QED differs from the coupling constant. Recall that in QED, each vertex contribute a factor of $\alpha = 1/137$. However, the coupling constant α_s in QCD is greater than 1. Hence Feynman rules do not work in QCD. In fact, the coupling “constant” depends on the separation distance between particles. At a distance less than the size of a proton the coupling constant becomes small. This phenomenon is called the **asymptotic freedom**.
- The last difference is that while many particles carry charges, no natural particles carry color. The strong force between natural particles is a more complicated process. For example: an interaction between two protons can be



This is the exchange of the pion between protons in Yukawa's theory. If QCD is correct, then it must account for quark confinement. One possible explanation is that the energy required to separate quarks apart is enough to create a new quark-antiquark pairs:



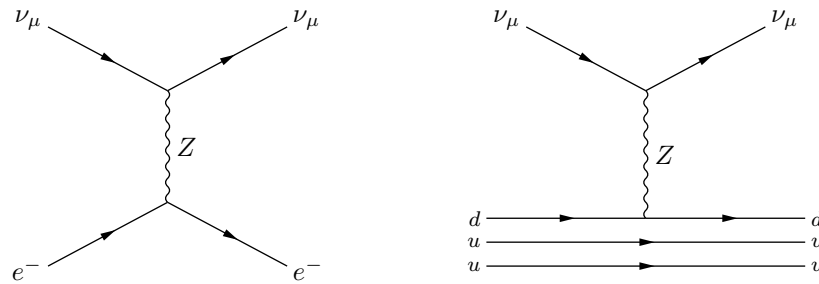
Confinement involves long-range behavior of strong force, and its explanation is hard because this is where Feynman rules fail.

4.4 Weak Interactions

There is no “charge” that produce weak forces. All quarks and leptons can have weak interactions. There are two kinds of weak interactions: the charged mediated by the $W^\pm$, and neutral mediated by the Z . There are many fundamental vertices that are associated with the W and the Z . We will put them in Section 4.5.2. Here we will discuss some examples.

4.4.1 Neutral

Neutrino-electron scattering and neutrino-proton scattering:

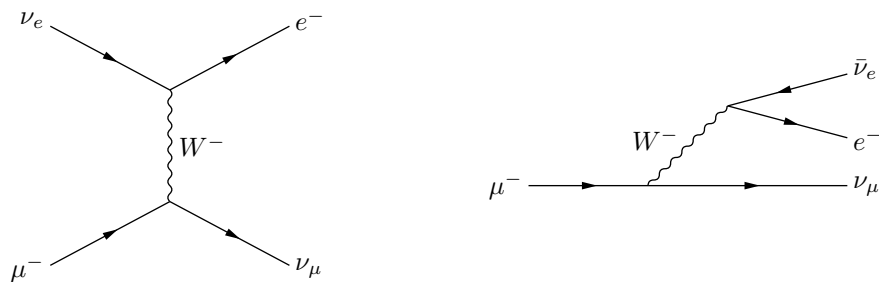


Any process mediated by the photon can also be mediated by the Z . The photon process dominates, but we can still trace the weak process because it violates the conservation of parity. A pure neutral weak interaction occurs only in neutrino scattering.

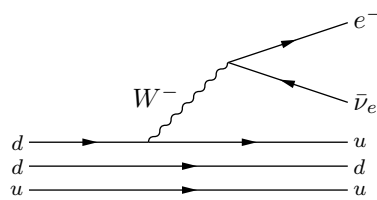
4.4.2 Charged

In strong, electromagnetic, and neutral weak interactions, same quarks or leptons come out as they went in (quarks may differ in color but never in flavor). Only the charged weak interaction can change flavor.

For leptonic processes, a lepton is converted into its corresponding neutrino by emitting a W^- or absorbing a W^+ . Here is an example of a process with two vertices:



In quark weak interactions, the quark going in must have the same color as the one going out, but a different flavor. At the other end of the W^- line can be leptons, which is called a **semileptonic process**, or it can be quarks, the pure **hadronic process**. Here is a classic example of semileptonic process, the beta decay:



W and Z can directly couple to each other just like gluon-gluon couplings. Additionally, W is charged so it can also couple to the photon.

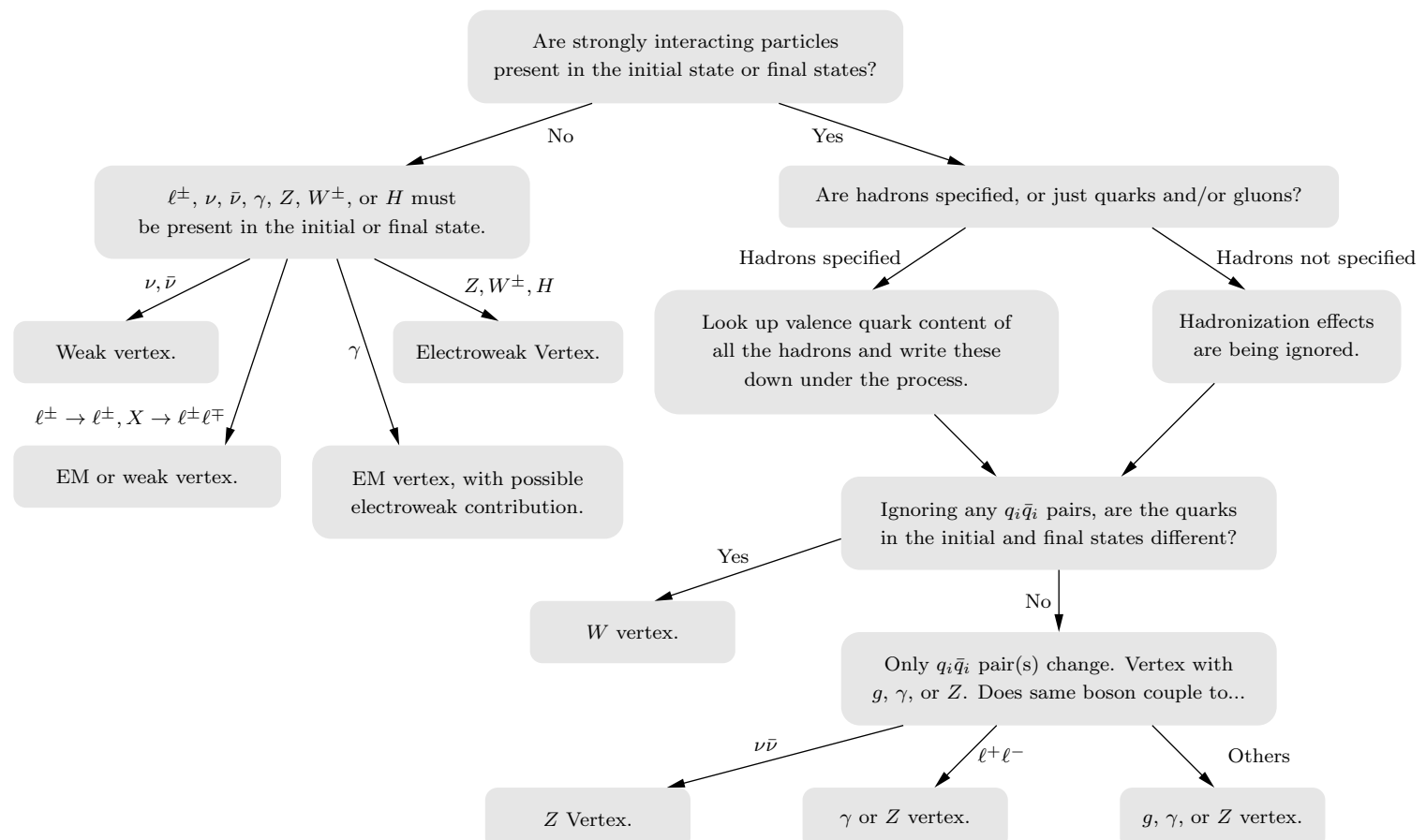
4.5 Algorithm to Draw Feynman Diagrams

4.5.1 Conservation Laws

This section summarizes an algorithm and several tips to draw Feynman diagrams. We focus mainly on conservation laws. Some conservation laws are universal, such as 4-momentum conservation and charge conservation. Some are true in the Standard Model, but are violated in physics beyond the Standard model, such as baryon number conservation. Some are true for specific interactions or situations, such as lepton number conservation and flavor conservation.

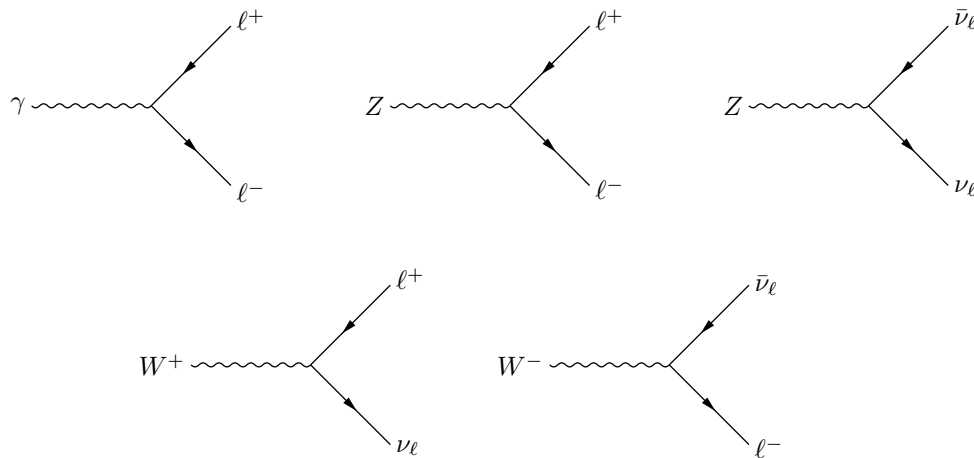
1. **4-momentum conservation.** Before drawing a Feynman diagram, classify the process as a decay or a scattering. For a decay process, check if the mass of the parent particle is higher than the total mass of the daughter particles. If not, the process is forbidden by conservation of energy. For a scattering process, unless $\sqrt{s}$ is specified, one may assume that $\sqrt{s}$ is enough to produce final-state particles. If angular momentum are specified, always check whether it is conserved.
2. **Charge conservation.** Electric charge is conserved locally at every SM vertex and in the overall process. In general, we will not worry about color charge in a Feynman diagram.
3. **Baryon number conservation.** The baryon number B of a particle is defined as $B = N_{\text{quark}} - N_{\text{antiquark}}$. Thus, all baryons (which have 3 quarks) have baryon number 1, and all mesons (which have 1 quark and 1 antiquark) have baryon number 0. Leptons, gauge bosons, and Higgs boson have baryon number 0.
4. **Lepton number conservation.** The ℓ -lepton number L_ℓ (where $\ell = e, \mu, \tau$) is defined as $L_\ell = N(\ell^-, \nu_\ell) - N(\ell^+, \bar{\nu}_\ell)$. For example, e^- has electron-lepton number 1; $\bar{\nu}_\tau$ has τ -lepton number -1 . The lepton number of three families are separately conserved. The conservation of lepton number is true in most cases, but neutrino oscillations (such as $\nu_e \leftrightarrow \nu_\mu$) violate such conservations.
5. **Flavor.** The conservation of flavor works in strong and electromagnetic interactions, but not in weak ones. If there is a flavor change, then there must be a W involved in the process.
6. **Other conservations.** A decay is electromagnetic if a photon comes out; it is weak if a neutrino comes out; it is strong if a gluon comes out. If there are only quarks and leptons, other conservation rules should be taking into consideration. These conservation rules include parity, charge conjugation, CP, isospin, etc.
7. **The OZI rule.** as we have mentioned in Chapter 1, the J/ψ ($c\bar{c}$) has a 1000 times longer lifetime than typical strong decays. The explanation is the following: Okubo, Zweig, and Iizuka came up with the **OZI rule** to account for the branching fraction of the ϕ ($s\bar{s}$). The ϕ has two decay modes, two K 's decay and three π 's decay where the latter is energetically favored because π weights much lower. The puzzle is that the branching fraction is much higher for two K 's decay. The OZI rule says that the three π 's decay is suppressed because the gluons have high energy in this process. The asymptotic freedom makes the gluons couple weakly at high energy, so the ϕ is more likely to go the two K 's path. The OZI rule work for the J/ψ : the J/ψ can decay into three π 's or two D 's. Again the three π 's path is suppressed, but meanwhile, the two D 's path is forbidden because they weigh more than the J/ψ , which is kinematically forbidden. Hence the J/ψ will live unusually longer.

Here is a flow chart to draw Feynman diagrams.

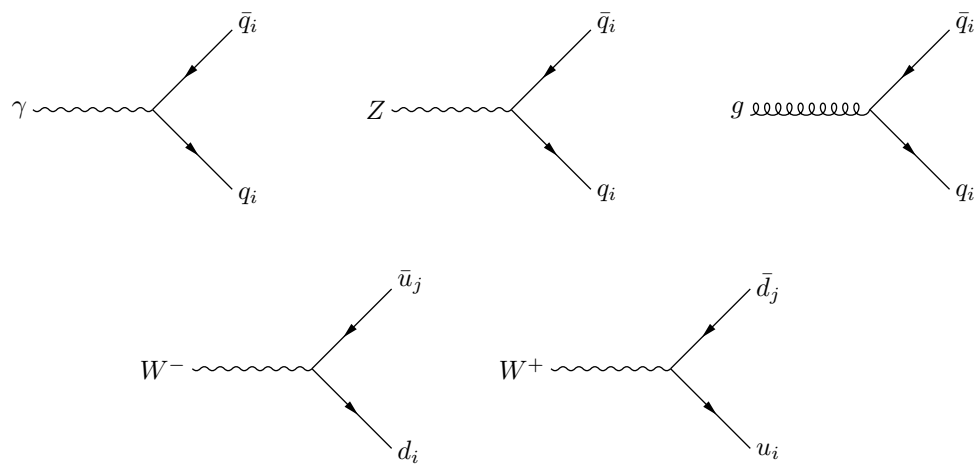


4.5.2 Allowed Vertices

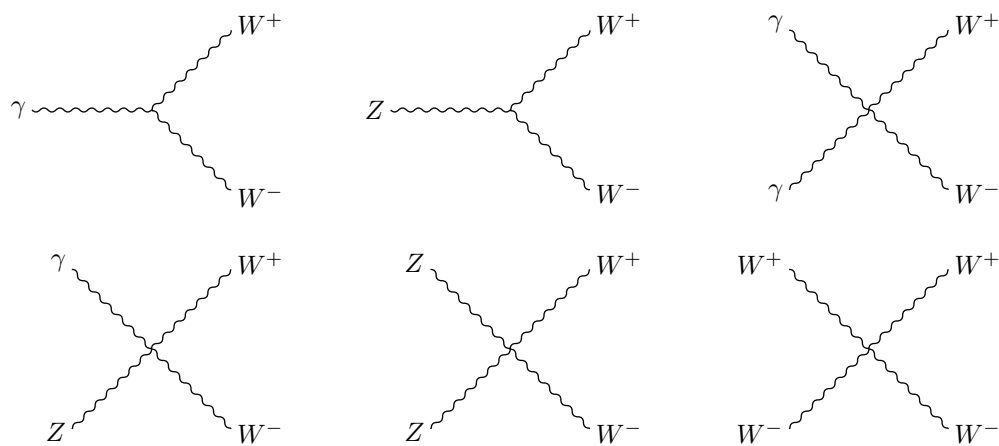
Vertices involving gauge bosons and leptons: $\ell = e, \mu, \tau$.



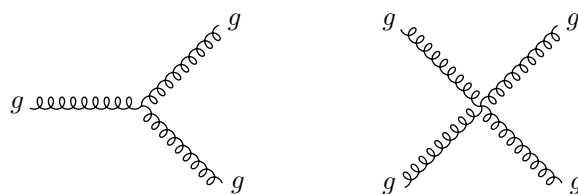
Vertices involving gauge bosons and quarks: $q_i = u, d, s, c, b, t$, $\bar{q}_i = \bar{u}, \bar{d}, \bar{s}, \bar{c}, \bar{b}, \bar{t}$, $u_i = u, c, t$, $d_i = d, s, b$.



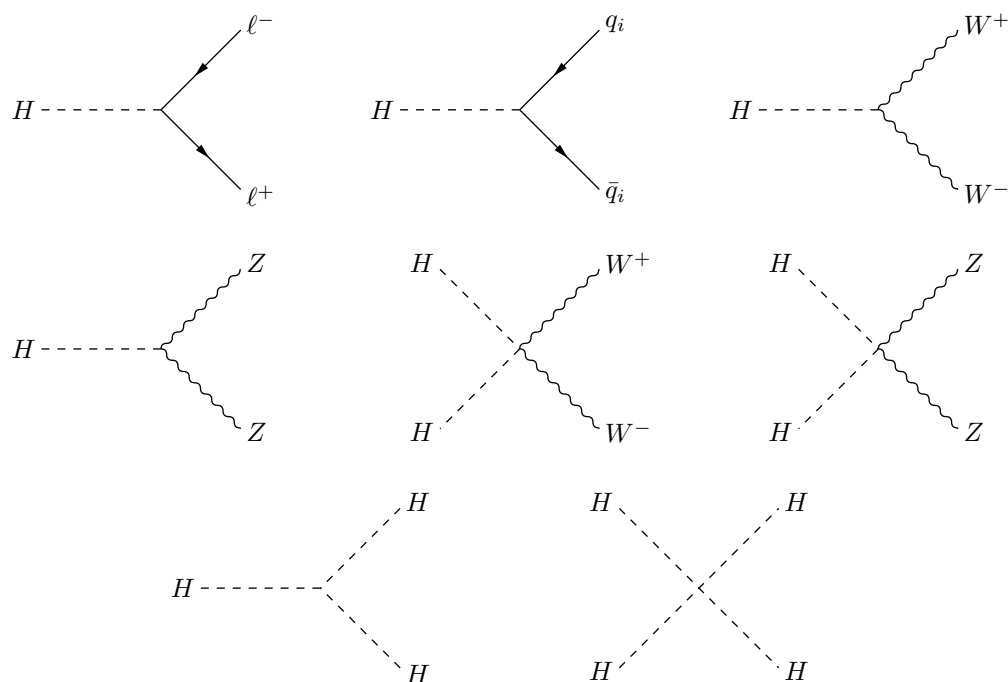
Vertices involving only $J = 1$ bosons (electroweak):



Vertices involving only $J = 1$ bosons (strong):



Vertices involving Higgs bosons:



Each fundamental vertices are not alone. One can always invoke crossing symmetry to produce other vertices, just like we did in Section 4.2.1, as long as 4-momentum is conserved. Many common Feynman diagrams are recorded in Appendix B for reference.

4.6 Phenomenology

In physics, phenomenology is the use of theory to predict results of experiment. In this section, we will talk about how to predict cross sections and decay rate from Feynman diagrams and some simple results from quantum field theory. The key ideas and methods include:

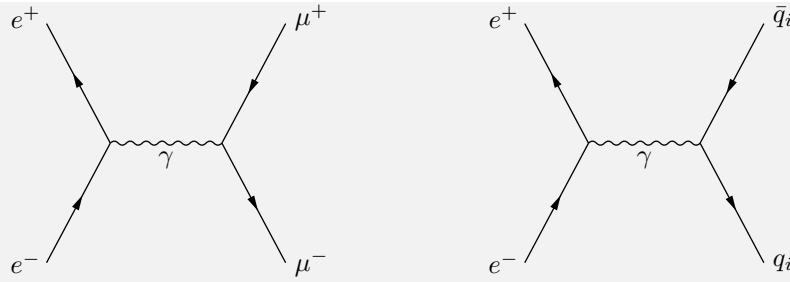
- Drawing relevant Feynman diagrams.
- Identifying relevant energies and masses, and making possible approximations.
- Considering effects of vertex factors on amplitude $\mathcal{A}$ and observables $|\mathcal{A}|^2$.
- Considering effects of propagator factors on amplitudes and observables.
- Using dimensional analysis in the framework of natural units.

4.6.1 Vertex Factors

We know that the amplitude of Feynman diagrams with more vertices are more suppressed. However, each vertex does not contribute to the same “suppression factor”, but according to the **vertex factor**. For an electromagnetic process, the vertex factor is proportional to the electric charge of the particle produced. For example, the vertex $e^+e^- \rightarrow \gamma$ contributes a factor of $q_e = -e$, while the vertex $\gamma \rightarrow q_i\bar{q}_i$ contributes a factor of q_{q_i} , which can be $\frac{2}{3}e$ or $-\frac{1}{3}e$.

Example 4.1. Cross section of e^+e^- collisions

We would like to study the cross sections of $e^+e^- \rightarrow \mu^+\mu^-$ and $e^+e^- \rightarrow q_i\bar{q}_i$, and their ratio $R(\sqrt{s}) = \sigma_{\text{hadrons}}(\sqrt{s})/\sigma_{\mu\mu}(\sqrt{s})$. Following the steps above, we should first draw Feynman diagrams for the two processes:



The second step is to list out the energies and masses. Suppose we know the center of momentum energy is $\sqrt{s} \ll m_Z$ so that the processes are dominated by electromagnetic interaction. The masses involved in $e^+e^- \rightarrow \mu^+\mu^-$ are m_e and m_μ . If we take $\sqrt{s} \gg m_e, m_\mu$, the only energy or mass scale relevant in this process is $\sqrt{s}$. By dimensional analysis (in natural units), the dimensions of a cross section σ is

$$[\sigma] = L^2 = \left[\frac{\hbar c}{E} \right] \rightarrow \frac{1}{E^2} = \frac{1}{M^2}.$$

We arrive at a very important conclusion: the cross section goes down with increasing $\sqrt{s}$

$$\sigma \propto \frac{1}{E_{\text{CM}}^2} \sim \frac{1}{\sqrt{s}} = \frac{1}{s}.$$

Though we used a lot of approximations, the result is perfectly consistent with experimental results. Now consider the second process $e^+e^- \rightarrow q_i\bar{q}_i$. In most cases, the quark and antiquark produced form hadrons with other quarks. However, at resonance $\sqrt{s}$ close to meson with content $(q_i\bar{q}_i)$, they form a meson themselves, such as a $J/\psi(c\bar{c})$ when $\sqrt{s} \approx m_{J/\psi}$. We make the same assumption that $\sqrt{s} \gg 2m_{q_i}$ so that the only relevant energy in the problem is $\sqrt{s}$. Also, we will neglect resonance for simplicity now.

The ratio of cross sections can be calculated as

$$R(\sqrt{s}) \equiv \frac{\sum_i \sigma(e^+e^- \rightarrow q_i\bar{q}_i, \sqrt{s})}{\sigma(e^+e^- \rightarrow \mu^+\mu^-, \sqrt{s})} \simeq \frac{\sum_i (E_{\text{CM}})^{-2} \cdot 3 \cdot [(-e) \cdot q_i]^2}{(E_{\text{CM}})^{-2} (-e)^2} \simeq 3 \sum_i q_i^2,$$

where i runs over all kinematically accessible quarks. The factor of 3 counts for color charge (R,G,B) the quark can have. That is, $\sigma(e^+e^- \rightarrow q_R\bar{q}_R) = \sigma(e^+e^- \rightarrow q_G\bar{q}_G) = \sigma(e^+e^- \rightarrow q_B\bar{q}_B)$. One important thing to notice is that different q_i and colors count for different final states. Their amplitudes need to be square and then add up. Only when the final state is the same but with different intermediate states, the amplitude first add up and then squared.

For concreteness, consider $\sqrt{s} = 27 \text{ GeV}$ as an example. This energy is enough to produce 2 bottom quarks, then

$$R(\sqrt{s} \sim 27 \text{ GeV}) \simeq 3 \left[\left(\frac{2}{3} \right)^2 + \left(\frac{1}{3} \right)^2 + \left(\frac{1}{3} \right)^2 + \left(\frac{2}{3} \right)^2 + \left(\frac{1}{3} \right)^2 \right] = \frac{11}{3}.$$

In real measurements, when we increase $\sqrt{s}$, we can observe “steps” in $R(\sqrt{s})$. The quark mass threshold perfectly explain this because the amplitudes add discrete steps except at resonance masses of particles like J/ψ , Υ , Z , etc.

For vertices that change the flavor of quarks, the magnitude of each vertex factor is proportional to gV_{ij} . Here i refers to quarks that have charge $\pm 2/3$ and j refers to quarks that have charge $\mp 1/3$. For example, the process $W^- \rightarrow u\bar{d}$ has a vertex factor gV_{ud} . The quantities V_{ij} are elements of the 3×3 **Cabibbo-Kobayashi-Maskawa (CKM) quark mixing matrix** of the Standard Model. For most purposes, we only need to know the magnitudes of V_{ij} :

$$\begin{bmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{bmatrix} = \begin{bmatrix} 0.974 & 0.225 & 0.004 \\ 0.23 & 0.96 & 0.041 \\ 0.008 & 0.039 & 1.0 \end{bmatrix}.$$

4.6.2 Propagators

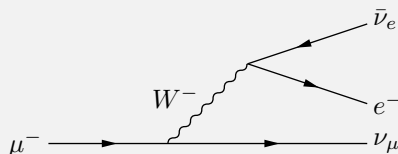
Another factor that contributes to the amplitude of a Feynman diagram is the **propagator**:

$$\Delta = \frac{i}{p^2 - m^2 + i\epsilon}. \quad (4.2)$$

Here p is the momentum of the intermediate particle, which may not equal to m^2 . The quantity p^2 is the *off-shell* mass of the intermediate particle, while m^2 is the *on-shell* mass. In general, the term $i\epsilon$ can be neglected unless $p^2 = m^2$.

Example 4.2. Muon decay

Consider a simple decay mediated by a W -boson. $\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$. As usual, first draw the Feynman diagram for this process.



Since this is a weak process, the vertex factor is no longer the charge of particles, but is related to a weak coupling constant g . Considering both vertex factors and the propagator, the amplitude is proportional to

$$\mathcal{A}(\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu) \propto \frac{g^2}{p^2 - m_W^2}.$$

The decay rate or the width Γ is proportional to the amplitude squared,

$$\Gamma(\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu) \propto |\mathcal{A}|^2 \propto \frac{g^4}{(p^2 - m_W^2)^2}.$$

Here are some reasonings we can make. First, Γ has dimensions of M , but the RHS has dimensions M^{-4} , so the proportionality factor should have dimensions M^5 . The mass of the W -boson does not play a role in the proportionality factor; it only affects the propagator. Hence we need some other mass. The momentum of the intermediate W is negligible compared to m_W . This is because in the rest frame of the initial muon, its 4-momentum is $(m_\mu, 0, 0, 0)$. The transferred momentum to the W is

$$p^2 = (p_\mu^2 - p_{\nu_\mu}^2) = m_\mu^2 + m_{\nu_\mu}^2 - 2m_\mu E_{\nu_\mu}.$$

The maximum p^2 the W can have is $p_{\text{max}}^2 = m_\mu^2 + m_{\nu_\mu}^2 - 2m_\mu m_{\nu_\mu} = (m_\mu - m_{\nu_\mu})^2 < m_\mu^2 \ll m_W^2$ because $m_W = 80.37 \text{ GeV}$! Therefore, p^2 is negligible in the amplitude. Finally, notice that $m_e, m_{\bar{\nu}_e}, m_{\nu_\mu} \ll m_\mu$, we may guess that the proportionality factor depends only on m_μ effectively, so

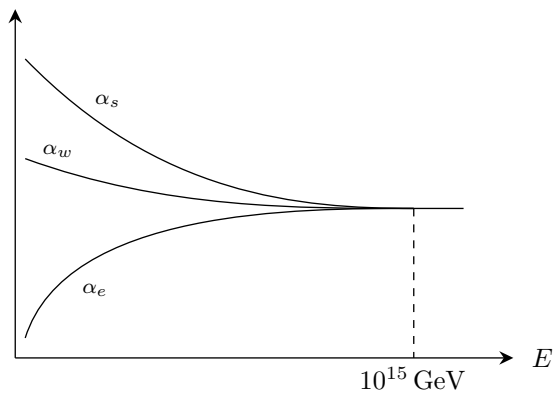
$$\Gamma(\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu) \propto \frac{m_\mu^5 g^4}{m_W^4}.$$

It turns out that this method can be generalized to other processes mediated by a W -boson, so our reasoning works.

4.7 Unification Schemes

Historically, electricity and magnetism were used to be two separate subjects. Maxwell unified them into electromagnetism. Einstein wanted to combine gravity with electromagnetism but he failed. On the other way, Glashow, Weinberg, and Salam combined the weak and electromagnetism into the electroweak theory. The theory begins with four massless mediators, but three of them (the $W^\pm$ and the Z) acquire mass via **Higgs mechanism**, the remaining massless one is the photon.

Started from the 1970s, people worked to unify chromodynamics with the electroweak theory into the **Grand Unified Theory** (GUT). This is possible, as the strong coupling constant α_s decreases at short distance and so does the weak α_w , but with a slower rate. The electromagnetic coupling constant α_e (smallest among the three) increases with shorting distance. They may converge to a limiting value at extraordinary high energy:



Another prediction of the GUT is that proton can decay, so that the baryon number and lepton number is no longer conserved:

$$p^+ \rightarrow e^+ + \pi^0 \quad \text{or} \quad p^+ \rightarrow \bar{\nu}_\mu + \pi^+.$$

Its half-life is at least 10^{19} times the age of the universe, so protons are still very stable. After all the unification of the strong, electromagnetic, and weak forces, the final step is to incorporate gravity.

5 SYMMETRIES

A **symmetry** is an operation you can perform on a system that leaves it invariant. Symmetries can be classified into two main types:

- **Continuous symmetries** include transformations defined by a continuous parameter. The groups associated with continuous symmetries are **Lie groups**. For example, the Lorentz group includes translation in spacetime (parametrized by x^μ), spatial rotations (parametrized by 3 angles θ_i), and boosts (parametrized by velocity v_i).
- **Discrete symmetries** include discrete transformations. They are very important to quantum mechanics and particle physics. In ordinary quantum mechanics, we encounter parity P , changing $\mathbf{x} \rightarrow -\mathbf{x}$, and time-reversal T , changing $t \rightarrow -t$. In particle physics, we will encounter charge conjugation C .

In this chapter, we first review symmetries in field theory. Then, we discuss some basics of group theory, in particular Lie groups and their application in strong isospin and other continuous symmetries. Finally, we turn to discrete symmetries: parity, charge conjugation, time-reversal, and their application in interactions.

5.1 Symmetries in Field Theory

5.1.1 The Lagrangian and Hamiltonian Formalism

A classical scalar field ϕ is a continuous function of x^μ , which we shall write it as $\phi(x)$ or $\phi(\mathbf{x}, t)$. The Lagrangian is now a function of the field ϕ , its spatial derivatives $\nabla\phi$, and its time-derivative $\dot{\phi}$,

$$L = L(\phi, \partial_\mu \phi).$$

There could be higher derivatives, but we are not considering them for now. We usually formulate the field theory using the **Lagrangian density** $\mathcal{L}$, defined by

$$L = \int d^3x \mathcal{L},$$

where L is the usual Lagrangian. Because we will nearly always deal with the Lagrangian density in field theory, it is more convenient to call the Lagrangian density just the “Lagrangian”. The action S is the same as the one in classical mechanics: it is the integral of the Lagrangian over time,

$$S = \int dt L = \int dt d^3x \mathcal{L} \equiv \int d^4x \mathcal{L}.$$

The fundamental principle, the **principle of stationary action** (or **Hamilton’s principle**) still holds,

$$\delta S = 0.$$

Any dynamics of the field ϕ follows a stationary action. By varying the action with respect to ϕ , we get

$$\begin{aligned} \delta S &= \int d^4x \mathcal{L}(\phi + \delta\phi, \partial_\mu\phi + \partial_\mu(\delta\phi)) - \int d^4x \mathcal{L}(\phi, \partial_\mu\phi) \\ &= \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)} \partial_\mu(\delta\phi) + \mathcal{O}(\delta\phi^2) \right] \\ &= \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)} \delta\phi \right) - \delta\phi \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)} \right) \right]. \end{aligned}$$

Just like we require the end points of paths to be fixed in classical mechanics, in field theory, we require the “surface term” to vanish. That sets

$$\int d^4x \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)} \delta\phi \right) = 0.$$

We are left with

$$\delta S = \int d^4x \delta\phi \left[\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \right) \right] = 0.$$

Since $\delta\phi$ are arbitrary variations on the field, the Euler-Lagrange equation in field theory is

$$\boxed{\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \right) = 0.} \quad (5.1)$$

Note the implicit sum in the index μ .

Since we are interested to apply field theory to quantum systems, it is more useful to find the Hamiltonian for a field. As usual, we first write the Hamiltonian as the **Hamiltonian density** $\mathcal{H}$,

$$H = \int d^3x \mathcal{H},$$

where H is the usual Hamiltonian. In classical mechanics, the Hamiltonian is defined by the generalized coordinates and their conjugate momenta

$$H(p_i, q_i) = \sum_i p_i \dot{q}_i - L, \quad \text{where } p_i \equiv \frac{\partial L}{\partial \dot{q}_i}.$$

In field theory, we define the **momentum field** $\pi(x)$ to be the conjugate momentum of field $\phi(x)$. Thus, the Hamiltonian density in analogy to the Hamiltonian in classical mechanics is defined as

$$\mathcal{H}(\phi, \pi) = \sum_i \pi_i(x) \dot{\phi}_i(x) - \mathcal{L}, \quad \text{where } \pi_i \equiv \frac{\partial\mathcal{L}}{\partial\dot{\phi}_i}. \quad (5.2)$$

5.1.2 Symmetries and Conserved Currents

In particle physics, we are interested in symmetries inside the Lagrangian $\mathcal{L}$ because they give rise to conserved currents. To see this, consider an infinitesimal transformation of the fields

$$\phi_i(x) \rightarrow \phi'_i(x) = \phi_i(x) + \delta\phi_i(x),$$

where $\delta\phi_i$ is small. The variation in the Lagrangian is given by

$$\delta\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\phi_i} \delta\phi_i + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)} \delta(\partial_\mu\phi_i) = \frac{\partial\mathcal{L}}{\partial\phi_i} \delta\phi_i + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)} \partial_\mu(\delta\phi_i),$$

where i is implicitly summed over. This statement is true for any fields. In physics, the fields should of course be physical, so they need to obey the Euler-Lagrange equation (5.1). Then $\delta\mathcal{L}$ can be written as

$$\delta\mathcal{L} = \partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)} \right) \delta\phi_i + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)} \partial_\mu(\delta\phi_i) = \partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)} \delta\phi_i \right).$$

The change in the Lagrangian is in the form of a divergence. Suppose $\mathcal{L}$ is invariant under the transformation $\phi_i \rightarrow \phi_i + \delta\phi_i$, i.e. a symmetry transformation, then the divergence is zero. This give rise to the conserved 4-current J^μ ,

$$0 = \partial_\mu J^\mu, \quad \text{where } J^\mu \propto \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)} \delta\phi_i.$$

5.2 Symmetries in Quantum Mechanics

5.2.1 Symmetry Operators

The role of symmetries is extended in quantum mechanics. We shall focus on the Hamiltonian formalation of quantum mechanics, where the governing equation is the Schrödinger equation. Almost all symmetry operators in quantum mechanics are unitary (*almost* because the time-reversal operator is an exception). There is a specific reason for this: we expect the symmetry operators to preserve

the behavior of the system. In particular, we want to preserve normalization. Consider a symmetry operator U that transforms a normalized state $|\psi\rangle$ by

$$|\psi\rangle \rightarrow |\psi'\rangle = U|\psi\rangle.$$

Then we want

$$\langle\psi'|\psi'\rangle = \langle\psi|U^\dagger U|\psi\rangle = \langle\psi|\psi\rangle = 1.$$

This implies that $U^\dagger U = 1$, so U is a unitary operator. Next, we say that a unitary operator U is a symmetry of the Hamiltonian if they commute:

$$[U, H] = 0.$$

There are two immediate consequences:

1. Suppose $|\psi(t)\rangle$ is a solution to the Schrödinger equation,

$$H|\psi(t)\rangle = i\frac{d}{dt}|\psi(t)\rangle.$$

Let U be time-independent and linear (which are almost always true). Apply the symmetry operator on both sides of the Schrödinger equation,

$$UH|\psi(t)\rangle = U\left(i\frac{d}{dt}\right)|\psi(t)\rangle.$$

Since U is time-independent and commutes with H ,

$$H(U|\psi(t)\rangle) = i\frac{d}{dt}(U|\psi(t)\rangle) \implies H|\psi'(t)\rangle = i\frac{d}{dt}|\psi'(t)\rangle.$$

This means if $|\psi\rangle$ is a solution to the Schrödinger equation, then so is $|\psi'(t)\rangle = U|\psi(t)\rangle$.

2. The mean energy of a system in the state $|\psi\rangle$ is given by $\langle E \rangle = \langle\psi|H|\psi\rangle$. We can insert an identity operator on the RHS,

$$E = \langle\psi|U^\dagger UH|\psi\rangle = \langle\psi|U^\dagger H U|\psi\rangle = \langle\psi'|H|\psi'\rangle.$$

In other words, the mean energy of $|\psi'\rangle$ is the same as that of $|\psi\rangle$.

5.2.2 Introduction to Lie Groups

The set of all symmetry operations R on a particular system has the following properties:

- Closure: if R_i and R_j are in the set, then $R_i R_j$ (first perform R_j and then perform R_i) is also in the set.
- Identity: there exists an identity element 1 such that $1R_i = R_i1 = R_i$ for all elements R_i .
- Inverse: for every element R_i there is an inverse, R_i^{-1} , such that $R_i R_i^{-1} = R_i^{-1} R_i = 1$.
- Associativity: $R_i(R_j R_k) = (R_i R_j)R_k$.

Mathematically, this is precisely the definition of a **group**, so group theory studies symmetries. If all elements in a group commute, then this group is called **abelian**. For example, translations in space and time are abelian; rotations in 2 dimensions is abelian, but rotations in 3 dimensions is not. The order of a group is the number of elements in the group. If the order of a group is infinite, then we call it an infinite group. Otherwise, the group is called a finite group.

In particle physics, infinite groups are usually labeled by one or more continuous, real parameters. For example, consider the rotation operator $R(\theta, \hat{\mathbf{n}}) = e^{i\hat{\mathbf{n}} \cdot \mathbf{J}\theta}$. It is labeled by a continuous parameter θ , specifying the angle of rotation, and $\hat{\mathbf{n}}$ is the rotation axis. Let's see how group properties are satisfied when the group elements are labeled by a continuous, real parameter. Consider a group G . The group element $a \in G$ can be written as $a(\theta)$, where θ is a continuous, real parameter. For convenience, define θ such that let $a(0) = 1$. The group properties require:

- Closure: for any parameters θ, ϕ in the parameter space, there is another parameter ξ such that $a(\theta)a(\phi) = a(\xi)$. There is some function $f(\theta, \phi) = \xi$ that maps the parameter space into itself.

- Identity: there is an identity element $a(0) = 1$, so

$$a(0)a(\theta) = a(\theta)a(0) = a(\theta).$$

The map therefore satisfy $f(0, \theta) = f(\theta, 0) = \theta$.

- Inverse: the existence of an inverse for each group element says that there is some $a^{-1}(\theta)$ such that $a(\theta)a^{-1}(\theta) = 1$. Suppose $a(\theta') = a^{-1}(\theta)$. Then the map must satisfy $f(\theta, \theta') = 0$.
- Associativity: the associativity $[a(\theta)a(\phi)]a(\xi) = a(\theta)[a(\phi)a(\xi)]$ implies that $f(f(\theta, \phi), \xi) = f(\theta, f(\phi, \xi))$.

The precise mathematical definition of a Lie group requires some topology and differential geometry that are not very relevant with particle physics. Therefore, we adopt a simpler yet sufficient physicists' definition: a group G is a **Lie group** if the function $f(\theta, \phi)$ defined above is continuously differentiable. In particle physics, we look for unitary matrix representations of the group elements in a Lie group.

5.2.3 Generators of Lie Groups

We begin with an example of the group of translation operators in ordinary quantum mechanics. Consider a system with a one-dimensional (differentiable) wave function $\psi(x)$. We perform an infinitesimal translation δx on the wave function,

$$\psi(x) \rightarrow \psi'(x) = \psi(x - \delta x).$$

To first order in δx , the transformed wave function can be expressed as

$$\psi'(x) = \psi(x) - \frac{\partial \psi}{\partial x} \delta x + \mathcal{O}(\delta x^2) \simeq \psi(x) - i \delta x \left(-i \frac{\partial}{\partial x} \right) \psi.$$

We recognize the operator $-i\partial_x$ as the momentum operator p_x in the coordinate basis. Then $\psi'(x)$ is

$$\psi'(x) \simeq (1 - ip_x \delta x) \psi(x).$$

Here p_x is called the **generator** of the group. That is, the momentum operator is the generator of the group of spatial translations. This is for infinitesimal translations. What about a finite translation of distance a to the wave function along the x -axis? Here we use a very important property of translation: a translation by distance a followed by another translation by distance b is equivalent to a single translation of distance $a + b$. Let $U(\Delta x)$ be the translation operator that translate a wave function by Δx . From the calculations above, we know that $U(\delta x)\psi(x) = (1 - ip_x \delta x)\psi(x)$. From the property of translation, we also know that $U(\Delta x + \delta x)\psi(x) = U(\Delta x)U(\delta x)\psi(x)$. Therefore,

$$U(\Delta x + \delta x)\psi(x) = U(\Delta x)(1 - ip_x \delta x)\psi(x) \implies \frac{U(\Delta x + \delta x) - U(\Delta x)}{\delta x} = -ip_x U(\Delta x).$$

Taking the limit $\delta x \rightarrow 0$, we can easily solve this differential equation to get the finite translation operator

$$U(\Delta x) = e^{-ip_x \Delta x}.$$

Going back to general Lie groups, let the elements in the Lie group can be represented by $U(\boldsymbol{\theta})$, where $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_k)$ are the continuous, real parameters. We define the **generator** of a Lie group to be

$$G_k = i \left. \frac{\partial U(\boldsymbol{\theta})}{\partial \theta_k} \right|_{\boldsymbol{\theta}=0}. \quad (5.3)$$

The factor of i is conventional. This essentially follows from the fact that $U(\boldsymbol{\theta}) = e^{-i\mathbf{G}\cdot\boldsymbol{\theta}} = e^{-iG_k\theta_k}$, where we may just read off the generators $\mathbf{G} = (G_1, G_2, \dots, G_k)$. However, since these unitary operators may not be expressed in terms of an exponential, the definition (5.3) is more useful and general.

Example 5.1. Generators of SO(3)

The group of rotation matrices acting on ordinary vectors $\mathbf{x} = (x, y, z)$, performing a rotation about the origin, is called the **special orthogonal** $\text{SO}(3)$ group. The matrices R in the $\text{SO}(3)$ group are special, meaning $\det R = 1$, and orthogonal, meaning $R^T R = 1$. Rotation matrices are special orthogonal: for two vectors $\mathbf{a}$ and $\mathbf{b}$, a rotation on them does not change their magnitude or their relative angles. In other words, a rotation preserves their inner product,

$$\mathbf{a}'^T \mathbf{b}' = (R\mathbf{a})^T (R\mathbf{b}) = \mathbf{a}^T R^T R \mathbf{b} = \mathbf{a}^T \mathbf{b}.$$

This requires $R^T R = 1$, hence “orthogonal”. Next, let’s look at the determinant of a rotation matrix. The fact that $R^T R = 1$ implies that $\det R^T R = 1$. Meanwhile, $\det(R^T R) = \det R^T \det R = (\det R)^2$, or $\det R = \pm 1$. Rotations are continuous, i.e. an infinitesimal rotation of angle $\delta\phi$ must reduce to the identity when $\delta\phi \rightarrow 0$. Since the identity has determinant 1, infinitesimal rotation matrices must have determinant 1 for continuity. Finite rotation matrices can be constructed by successive infinitesimal matrices, so they must also have determinant 1, hence “special”.

To find the generators of $\text{SO}(3)$, consider rotation matrices around the x -, y -, and z -axis about an angle ϕ :

$$R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{bmatrix}, \quad R_y(\phi) = \begin{bmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{bmatrix}, \quad R_z(\phi) = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

These matrices are not in the form of exponentials (at least not explicitly). To find the generators, we use the formula (5.3). For example, the generator for rotations about the z -axis is

$$G_z = i \left. \frac{dR_z}{d\phi} \right|_{\phi=0} = i \begin{bmatrix} -\sin \phi & -\cos \phi & 0 \\ \cos \phi & -\sin \phi & 0 \\ 0 & 0 & 0 \end{bmatrix}_{\phi=0} = -i \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

The matrix element of G_z can be written simply as $(G_z)_{ij} = -i\epsilon_{ijz}$, where ϵ_{ijk} is the Levi-Civita symbol. Similarly, we should find that $(G_x)_{ij} = -i\epsilon_{ijx}$ and $(G_y)_{ij} = -i\epsilon_{ijy}$. Combining the three directions, the matrix elements of the three generators G_k , where $k = x, y, z$, are

$$(G_k)_{ij} = -i\epsilon_{ijk}.$$

There are many properties of the generators of Lie groups. For our purposes, we concern about unitary groups $\text{U}(n)$ and special unitary groups $\text{SU}(n)$. **Unitary groups** $\text{U}(n)$ have group elements U as unitary transformations, i.e. $U^\dagger U = 1$. **Special unitary groups** have an additional requirement that $\det U = 1$. Here are two important properties of the generators of unitary groups and special unitary groups.

Theorem 5.1.

The generators G_k of a unitary group $\text{U}(n)$ are Hermitian, $G_k^\dagger = G_k$.

Proof. Consider an infinitesimal unitary transformation $U(\delta\theta)$, where $U^\dagger(\delta\theta)U(\delta\theta) = 1$. By the definition of generators (5.3),

$$\begin{aligned} 1 &= U^\dagger(\delta\theta)U(\delta\theta) \\ &= [1 - iG_j\delta\theta_j + \mathcal{O}(\delta\theta^2)]^\dagger [1 - iG_k\delta\theta_k + \mathcal{O}(\delta\theta^2)] \\ &= 1 + iG_j^\dagger\delta\theta_j - iG_k\delta\theta_k + \mathcal{O}(\delta\theta^2) \\ &= 1 + i(G_k^\dagger - G_k)\delta\theta_k + \mathcal{O}(\delta\theta^2). \end{aligned}$$

Taking the limit $\delta\theta \rightarrow 0$, we have $0 = i(G_k^\dagger - G_k)\delta\theta_k$ for any infinitesimal parameter θ . In other words, $G_k^\dagger = G_k$, the generators of a unitary group are Hermitian. $\square$

Theorem 5.2.

The generators G_k of a special unitary group $SU(n)$ are traceless, $\text{Tr } G_k = 0$.

Proof. To be written. □

Theorem 5.3.

The number of independent generators of a special unitary group $SU(n)$ is $n^2 - 1$.

Proof. From the two theorems above, the generators of $SU(n)$ are $n \times n$ Hermitian, traceless matrices. We want to find the number of independent matrices with these properties.

First, consider diagonal matrices. For diagonal matrices to be Hermitian, each diagonal element must be real. There are n independent such matrices. They can be constructed by setting the n th diagonal element to be 1 and others to be 0. The traceless constraint removes one degree of freedom, so there are $n - 1$ independent Hermitian, traceless diagonal matrices.

Second, consider off-diagonal matrices, which are automatically traceless. If the elements are real, a Hermitian matrix have to be symmetric. There are $n(n - 1)/2$ independent real, off-diagonal symmetric matrices. They can be constructed by setting the ij th and ji th element ($i \neq j$) to be 1 and others 0. If the elements are imaginary, a Hermitian matrix have to be antisymmetric. There are again $n(n - 1)/2$ independent imaginary, off-diagonal antisymmetric matrices. They can be constructed by setting the ij th and ji th element ($i \neq j$) to be i and $-i$, respectively, and others 0.

In total, the number of independent $n \times n$, Hermitian, traceless matrices is

$$N = (n - 1) + \frac{n(n - 1)}{2} + \frac{n(n - 1)}{2} = n^2 - 1.$$

□

5.3 Flavor Symmetry

5.3.1 The Origin of Isospin

It was discovered in the early nuclear physics (before the quark model) that the proton and neutron have nearly the same mass and the strong nuclear force is roughly charge independent. Heisenberg suggested that if we ignore the charge of the proton, a proton and a neutron are indistinguishable. He proposed that the proton p and neutron n can be treated as the same particle called the **nucleon** N , but in two different states. We define

$$p = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad n = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \text{and} \quad N = \begin{bmatrix} \alpha \\ \beta \end{bmatrix},$$

where α and β are complex numbers with normalization $|\alpha|^2 + |\beta|^2 = 1$. This introduces the concept of **isospin**. The proton and neutron is an isospin doublet with total isospin $I = 1/2$, and the third component of isospin $I_3 = \pm 1/2$. This is very similar to the spin states of a spin-1/2 particle. In fact, we will show that they indeed obey the same $SU(2)$ algebra (the algebra for angular momentum in quantum mechanics). Then all the tools we developed in the theory of angular momentum can be used exactly the same way for isospin.

Just like the strong nuclear force does not see electric charge of the proton, the strong interaction between quarks does not see electric charge of quarks and the quark flavors. They only see the color charge. In other words, quarks are treated equally by the strong interaction, regardless of their flavors. This symmetry is called **flavor symmetry**.

Mathematically, we define the states of up quark and down quark as

$$u = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad d = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Due to flavor symmetry, the up quark and down quark are treated as indistinguishable to the strong interaction. Then the strong interaction is invariant under a unitary transformation (unitary to preserve normalization) of the *isospin space*,

$$\begin{bmatrix} u' \\ d' \end{bmatrix} = U \begin{bmatrix} u \\ d \end{bmatrix}.$$

Suppose the Hamiltonian of a system of quarks is

$$H = H_0 + H_{\text{strong}} + H_{\text{EM}},$$

where H_0 is the free Hamiltonian containing the kinetic and rest mass energy of the quarks, and H_{strong} and H_{EM} are the strong interaction piece and electromagnetic piece, respectively. Though the strong Hamiltonian cannot tell the difference of the quarks, the electromagnetic Hamiltonian can because the quarks have different charges. Nevertheless, as long as we are considering processes in which the strong interaction dominates over others, flavor symmetry works well.

We say that the symmetry “works well” because it is not an exact symmetry. In QCD, the coupling between gluons and quarks depend only on their color charge, but independent of the quark flavor. Then does that mean all quarks with the same color behave the same way in strong-interaction processes, regardless of their flavor? The answer is no, because quarks have different masses. This is just analogous to Newton’s second law, $F = ma$. Even though the force experienced by quarks are the same, their reaction are different because of the mass difference. Thus, only isospin symmetry regarding the up and down quark works pretty well, because they have approximately the same mass. The isospin symmetry leads to many observational consequences in decays. Historically, this symmetry existed long before the quark model. Until now, we do not have a fundamental understanding of why quarks have those particular masses, so flavor symmetries are said to be *accidental*.

Let’s now look at the quark masses:

$$\begin{pmatrix} u & 2.16 \text{ MeV} \\ d & 4.70 \text{ MeV} \end{pmatrix} \quad \begin{pmatrix} c & 1.27 \text{ GeV} \\ s & 93.5 \text{ MeV} \end{pmatrix} \quad \begin{pmatrix} t & 173 \text{ GeV} \\ b & 4.18 \text{ GeV} \end{pmatrix}.$$

The down quark is over twice as heavy as the up quark. Why do we say they have approximately the same mass? This is because the quarks are confined inside a hadron, where the interaction energy is much larger than the masses of u and d . For example, the mass of a proton ($p = uud$) is $938 \text{ MeV} \gg 2m_u + m_d$. The *effective mass* carried by the quarks inside a proton are $\sim 300 \text{ MeV}$. In this way the up and down quarks can be treated as having the same mass. In fact, the flavor symmetry can be extended to the strange quark s as well. But strange quarks have mass $m_s \approx 93.5 \text{ MeV}$, the same order of magnitude as the strong interaction energy inside a hadron. The flavor symmetry including the strange quark is not as good as isospin symmetry.

5.3.2 Isospin Algebra

Like other continuous unitary transformations, the unitary transformation of the isospin space can be written as

$$U = e^{-i\alpha_i G_i},$$

where G_i are the independent generators of the transformation. The generators must be Hermitian for U to be unitary. For 2×2 matrices representing the generators, there are only four of them that are linearly independent. The generator cannot be proportional to the identity operator because we want transformations between different flavors, so only three independent 2×2 Hermitian matrices are left. They are conventionally chosen to be the Pauli spin matrices

$$\sigma_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \text{and} \quad \sigma_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

The algebra of the ud flavor symmetry (or isospin symmetry) is therefore exactly the same as that of a spin-1/2 particle. We define the isospin operator

$$\mathbf{T} = \frac{1}{2}\boldsymbol{\sigma} \quad \Longrightarrow \quad U = e^{i\boldsymbol{\alpha} \cdot \mathbf{T}}.$$

The general isospin transformation is a rotation in flavor space. Their generators obey the **special unitary** SU(2) group algebra. The three generators of the SU(2) group obey the following commutation relations,

$$[T_1, T_2] = iT_3, \quad [T_2, T_3] = iT_1, \quad [T_3, T_1] = iT_2,$$

exactly the same algebra for angular momentum. Thus, we can define the total isospin operator squared:

$$T^2 = T_1^2 + T_2^2 + T_3^2.$$

For an isospin state $|I, I_3\rangle$ with total isospin I and the third component of isospin I_3 ,

$$T^2 |I, I_3\rangle = I(I+1) |I, I_3\rangle \quad \text{and} \quad T_3 |I, I_3\rangle = I_3 |I, I_3\rangle.$$

There are in total $2I+1$ possible values of I_3 for an isospin state with total isospin I . The ladder operators $T_{\pm}$ are defined as

$$T_+ \equiv T_1 + iT_2 \quad \text{and} \quad T_- \equiv T_1 - iT_2.$$

The action of the ladder operators on an isospin state $|I, I_3\rangle$ are

$$\begin{aligned} T_+ |I, I_3\rangle &= \sqrt{I(I+1) - I_3(I_3+1)} |I, I_3+1\rangle, \\ T_- |I, I_3\rangle &= \sqrt{I(I+1) - I_3(I_3-1)} |I, I_3-1\rangle. \end{aligned} \tag{5.4}$$

As usual, the ladder operators annihilate the extreme states,

$$T_+ |I, I\rangle = 0 \quad \text{and} \quad T_- |I, -I\rangle = 0.$$

Let's take the up quark and down quark for example. In terms of isospin, the up quark and down quark are represented by states

$$u = \left| \frac{1}{2}, \frac{1}{2} \right\rangle \quad \text{and} \quad d = \left| \frac{1}{2}, -\frac{1}{2} \right\rangle.$$

The effects of the isospin ladder operators on them are

$$T_+ u = 0, \quad T_+ d = u, \quad T_- u = d, \quad T_- d = 0.$$

5.3.3 Addition of Isospin

The strong interaction Hamiltonian does not distinguish between up and down quarks. In the limit where their masses are the same, processes involving only the strong interaction are symmetric under unitary transformations in the isospin space. In particular, we can use addition of isospin to predict decay rates for hadronic decays.

The addition of isospin follows the same rule as the addition of angular momentum. Consider two particles with isospin states $|I^a, I_3^a\rangle$ and $|I^b, I_3^b\rangle$. The combined states $|I, I_3\rangle$ of these two particles should have

$$|I^a - I^b| \leq I \leq I^a + I^b \quad \text{and} \quad I_3 = I_3^a + I_3^b.$$

The isospin ladder operators for the combined states are the sum of the ladder operators for individual states,

$$T_{\pm} = T_{\pm}^a + T_{\pm}^b.$$

Example 5.2. Direct product states vs. total isospin states

Consider the up quark and down quark isospin states,

$$u = \left| \frac{1}{2}, +\frac{1}{2} \right\rangle, \quad d = \left| \frac{1}{2}, -\frac{1}{2} \right\rangle.$$

There are two sets of basis vectors for the combined states of two of these light quarks. The first one is the basis of direct product states in the form $|I^a, I_3^a\rangle |I^b, I_3^b\rangle$:

$$\begin{aligned} uu &= \left|\frac{1}{2}, +\frac{1}{2}\right\rangle \left|\frac{1}{2}, +\frac{1}{2}\right\rangle, \\ ud &= \left|\frac{1}{2}, +\frac{1}{2}\right\rangle \left|\frac{1}{2}, -\frac{1}{2}\right\rangle, \\ du &= \left|\frac{1}{2}, -\frac{1}{2}\right\rangle \left|\frac{1}{2}, +\frac{1}{2}\right\rangle, \\ dd &= \left|\frac{1}{2}, -\frac{1}{2}\right\rangle \left|\frac{1}{2}, -\frac{1}{2}\right\rangle. \end{aligned}$$

One can also write $uu = \left|\frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, +\frac{1}{2}\right\rangle$ and similar for others. This notation can be useful when computing inner products. On the other hand, there is the basis of total isospin states in the form $|I, I_3\rangle$:

$$|1, +1\rangle, \quad |1, 0\rangle, \quad |1, -1\rangle, \quad |0, 0\rangle.$$

Of course, the number of basis vectors must match in the two different basis. Then how can we relate these two basis with each other? That is, we want to find the basis transformation law so that one set of basis vectors can be expressed in terms of the other set. There are two ways to do it. (They are in fact the same way.) The first way is to look up the Clebsch-Gordan coefficients and transfer the knowledge of angular momentum to isospin. Again, we can do this because they both obey the $SU(2)$ algebra.

The second way is to use the ladder operators—we are deriving the Clebsch-Gordan coefficients. Following the rules $|I^a - I^b| \leq I \leq I^a + I^b$ and $I_3 = I_3^a + I_3^b$, we identify that

$$uu = \left|\frac{1}{2}, +\frac{1}{2}\right\rangle \left|\frac{1}{2}, +\frac{1}{2}\right\rangle = |1, +1\rangle \quad \text{and} \quad dd = \left|\frac{1}{2}, -\frac{1}{2}\right\rangle \left|\frac{1}{2}, -\frac{1}{2}\right\rangle = |1, -1\rangle.$$

The reason is simple. For the total isospin state to have the third component $I_3 = 1$, it can only have two quarks with third components of isospin $I_3^a = \frac{1}{2}$ and $I_3^b = \frac{1}{2}$, which is uu . The dd case is similar. Now apply a lowering operator $T_- = T_-^a + T_-^b$ to both sides of $uu = |1, +1\rangle$. The LHS is

$$T_- uu = T_- \left|\frac{1}{2}, +\frac{1}{2}\right\rangle \left|\frac{1}{2}, +\frac{1}{2}\right\rangle = T_-^a \left|\frac{1}{2}, +\frac{1}{2}\right\rangle \left|\frac{1}{2}, +\frac{1}{2}\right\rangle + T_-^b \left|\frac{1}{2}, +\frac{1}{2}\right\rangle \left|\frac{1}{2}, +\frac{1}{2}\right\rangle = \left|\frac{1}{2}, -\frac{1}{2}\right\rangle \left|\frac{1}{2}, +\frac{1}{2}\right\rangle + \left|\frac{1}{2}, +\frac{1}{2}\right\rangle \left|\frac{1}{2}, -\frac{1}{2}\right\rangle = ud + du.$$

Referring to (5.4), the RHS is

$$T_- |1, +1\rangle = \sqrt{2} |1, 0\rangle.$$

Thus,

$$|1, 0\rangle = \frac{1}{\sqrt{2}}(ud + du).$$

The last total isospin state $|0, 0\rangle$ can be found by orthogonality. The only normalized, orthogonal state to $|0, 0\rangle$ must be

$$|0, 0\rangle = \frac{1}{\sqrt{2}}(ud - du)$$

off by a phase factor. One can check explicitly that $T_+ |0, 0\rangle = T_- |0, 0\rangle = 0$. The total isospin states with $I = 1$ is a triplet of isospin 1. They are symmetric:

$$|1, +1\rangle = uu, \quad |1, 0\rangle = \frac{1}{\sqrt{2}}(ud + du), \quad |1, -1\rangle = dd.$$

The state with $I = 0$ is a singlet of isospin 0. It is antisymmetric:

$$|0, 0\rangle = \frac{1}{\sqrt{2}}(ud - du).$$

The direct product states are often denoted as $2 \otimes 2$, while the total isospin states are denoted as $3 \oplus 1$ (triplet + singlet).

Hadrons include baryons that is composed of three quarks. We expect the number of independent isospin states with three quarks to be $2 \times 2 \times 2 = 8$. These states can be constructed from states with two quarks and states with one quarks. Suppose the triplet is chosen to be the two-quark state, $I = 1$. Combining it with another single-quark state gives either $I = 1/2$ or $I = 3/2$. The $I = \frac{3}{2}$

states are

$$\begin{aligned} \left|\frac{3}{2}, +\frac{3}{2}\right\rangle &= uuu, \\ \left|\frac{3}{2}, +\frac{1}{2}\right\rangle &= \frac{1}{\sqrt{3}}(uud + udu + duu), \\ \left|\frac{3}{2}, -\frac{1}{2}\right\rangle &= \frac{1}{\sqrt{3}}(udd + dud + ddu), \\ \left|\frac{3}{2}, -\frac{3}{2}\right\rangle &= ddd. \end{aligned}$$

The $I = \frac{1}{2}$ states with a triplet and a single-quark state are

$$\begin{aligned} \left|\frac{1}{2}, +\frac{1}{2}\right\rangle_S &= \frac{1}{\sqrt{6}}(2uud - udu - duu), \\ \left|\frac{1}{2}, -\frac{1}{2}\right\rangle_S &= \frac{1}{\sqrt{6}}(udd + dud - 2ddu). \end{aligned}$$

The subscript S indicates that these states are symmetric under the change of first two quarks (from the triplet state). If the singlet is chosen to be the two-quark state, $I = 0$. Then combining it with a single-quark state gives only $I = 1/2$. These states are antisymmetric in the first two quarks,

$$\begin{aligned} \left|\frac{1}{2}, +\frac{1}{2}\right\rangle_A &= \frac{1}{\sqrt{2}}(udu - duu), \\ \left|\frac{1}{2}, -\frac{1}{2}\right\rangle_A &= \frac{1}{\sqrt{2}}(udd - dud). \end{aligned}$$

In terms of the $SU(2)$ group structure, the eight combinations of three quarks can be expressed as

$$2 \otimes 2 \otimes 2 = 2 \otimes (3 \oplus 1) = (2 \otimes 3) \oplus (2 \otimes 1) = 4 \oplus 2 \oplus 2.$$

Just for reference, we list the spin states of three quarks here. Since $SU(2)$ algebra works for both isospin and angular momentum, the spin states have exactly the same form:

$$\begin{aligned} \left|\frac{3}{2}, +\frac{3}{2}\right\rangle &= \uparrow\uparrow\uparrow, & \left|\frac{1}{2}, +\frac{1}{2}\right\rangle_S &= \frac{1}{\sqrt{6}}(2\uparrow\uparrow\downarrow - \uparrow\downarrow\uparrow - \downarrow\uparrow\uparrow), \\ \left|\frac{3}{2}, +\frac{1}{2}\right\rangle &= \frac{1}{\sqrt{3}}(\uparrow\uparrow\downarrow + \uparrow\downarrow\uparrow + \downarrow\uparrow\uparrow), & \left|\frac{1}{2}, -\frac{1}{2}\right\rangle_S &= \frac{1}{\sqrt{6}}(\uparrow\downarrow\downarrow + \downarrow\uparrow\downarrow - 2\downarrow\downarrow\uparrow), \\ \left|\frac{3}{2}, -\frac{1}{2}\right\rangle &= \frac{1}{\sqrt{3}}(\downarrow\downarrow\uparrow + \downarrow\uparrow\downarrow + \uparrow\downarrow\downarrow), & \left|\frac{1}{2}, +\frac{1}{2}\right\rangle_A &= \frac{1}{\sqrt{2}}(\uparrow\downarrow\uparrow - \downarrow\uparrow\uparrow), \\ \left|\frac{3}{2}, -\frac{3}{2}\right\rangle &= \downarrow\downarrow\downarrow, & \left|\frac{1}{2}, -\frac{1}{2}\right\rangle_A &= \frac{1}{\sqrt{2}}(\uparrow\downarrow\downarrow - \downarrow\uparrow\downarrow). \end{aligned}$$

5.3.4 Baryons

Baryons is made of three quarks. There are 8 possible isospin states and 8 possible spin states, giving 64 possible combined flavor and spin states. They are not necessarily physical. Quarks are fermions, and fermions obey the Pauli exclusion principle. Hence the overall wave function of a three-quark state must be totally antisymmetric. For a bound qqq state (q for quark), the overall has the form

$$\psi = \phi_{\text{flavor}} \chi_{\text{spin}} \xi_{\text{color}} \eta_{\text{space}},$$

where ϕ is the isospin state, χ is the spin state, ξ is the color state, and η is the coordinate space wave function. Later in QCD we will show that ξ must be totally antisymmetric.

Consider $L = 0$ ground state baryons (no orbital angular momentum) so η is symmetric. In order for ψ to be totally antisymmetric, the combined state $\phi\chi$ must symmetric for $L = 0$ baryons. There are two ways to construct a symmetric $\phi\chi$, and they give rise to different baryons. The first way is simple: combine symmetric isospin states and symmetric spin states give 4 possible isospin-3/2, spin-3/2 baryons. These are called **Δ -baryons**,

$$\Delta^{++} = uuu, \quad \Delta^+ = \frac{1}{\sqrt{3}}(uud + udu + duu), \quad \Delta^0 = \frac{1}{\sqrt{3}}(udd + dud + ddu), \quad \Delta^- = ddd.$$

The number of possible baryons is not 4×4 because spin does not distinguish the particle types; only isospin does.

The second way is to somehow use ϕ_S , χ_S , ϕ_A , and χ_A . Notice that $\phi_S\chi_S$ and $\phi_A\chi_A$ are symmetric under quarks $1 \leftrightarrow 2$, but not

$1 \leftrightarrow 3$ or $2 \leftrightarrow 3$. We want to find some linear combination of $\phi_S \chi_S$ and $\phi_A \chi_A$ such that this combination is totally symmetric. A promising linear combination is

$$\psi = \frac{1}{\sqrt{2}}(\phi_S \chi_S + \phi_A \chi_A),$$

where ϕ_S and ϕ_A should have the same quantum number, and so should χ_S and χ_A . In this case, the baryon is isospin-1/2 and spin-1/2. For $\phi(\frac{1}{2}, +\frac{1}{2})$, the particle is called the **proton**. A spin-up proton has the wave function

$$\begin{aligned} p(\uparrow) &= \frac{1}{\sqrt{2}} \left[\phi_S \left(\frac{1}{2}, +\frac{1}{2} \right) \chi_S \left(\frac{1}{2}, +\frac{1}{2} \right) + \phi_A \left(\frac{1}{2}, +\frac{1}{2} \right) \chi_A \left(\frac{1}{2}, +\frac{1}{2} \right) \right] \\ &= \frac{1}{6\sqrt{2}} (2uud - udu - duu)(2\uparrow\uparrow\downarrow - \uparrow\downarrow\uparrow - \downarrow\uparrow\uparrow) + \frac{1}{2\sqrt{2}} (udu - duu)(\uparrow\downarrow\uparrow - \downarrow\uparrow\uparrow). \end{aligned}$$

We can also make the proton to be spin-down,

$$p(\downarrow) = \frac{1}{\sqrt{2}} \left[\phi_S \left(\frac{1}{2}, +\frac{1}{2} \right) \chi_S \left(\frac{1}{2}, -\frac{1}{2} \right) + \phi_A \left(\frac{1}{2}, +\frac{1}{2} \right) \chi_A \left(\frac{1}{2}, -\frac{1}{2} \right) \right].$$

The other particle with isospin-1/2 and spin-1/2, but with $\phi(\frac{1}{2}, -\frac{1}{2})$, is called the **neutron**:

$$\begin{aligned} n(\uparrow) &= \frac{1}{\sqrt{2}} \left[\phi_S \left(\frac{1}{2}, -\frac{1}{2} \right) \chi_S \left(\frac{1}{2}, +\frac{1}{2} \right) + \phi_A \left(\frac{1}{2}, -\frac{1}{2} \right) \chi_A \left(\frac{1}{2}, +\frac{1}{2} \right) \right], \\ n(\downarrow) &= \frac{1}{\sqrt{2}} \left[\phi_S \left(\frac{1}{2}, -\frac{1}{2} \right) \chi_S \left(\frac{1}{2}, -\frac{1}{2} \right) + \phi_A \left(\frac{1}{2}, -\frac{1}{2} \right) \chi_A \left(\frac{1}{2}, -\frac{1}{2} \right) \right]. \end{aligned}$$

Therefore, there are in total 6 baryons with $L = 0$ made up of up quarks and down quarks: Δ^{++} , Δ^+ , Δ^0 , Δ^- , p , and n .

5.3.5 Isospin of Antiquarks

So far we have only discussed quarks. The up quark and down quark are placed in an isospin doublet, and a general SU(2) transformation on the doublet $q \rightarrow q' = Uq$ is

$$\begin{bmatrix} u \\ d \end{bmatrix} \rightarrow \begin{bmatrix} u' \\ d' \end{bmatrix} = \begin{bmatrix} a & b \\ -b^* & a^* \end{bmatrix} \begin{bmatrix} u \\ d \end{bmatrix}, \quad \text{where } q = \begin{bmatrix} u \\ d \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} a & b \\ -b^* & a^* \end{bmatrix}.$$

Here a and b are complex numbers with $aa^* + bb^* = 1$ so that U is special unitary. To get the antiquarks, we take the complex conjugate to this equation,

$$\begin{bmatrix} \bar{u}' \\ \bar{d}' \end{bmatrix} = \begin{bmatrix} a^* & b^* \\ -b & a \end{bmatrix} \begin{bmatrix} \bar{u} \\ \bar{d} \end{bmatrix}. \quad (5.5)$$

The complex conjugate of a quark is interpreted as its corresponding antiquark. A special property of SU(2) is that we can place antiquarks into a doublet that transforms in the same way as quarks. In other words, we want $\bar{q}' = U\bar{q}$ so that we can use all the math in previous sections. The idea is to write the antiquark doublet as

$$\bar{q} \equiv \begin{bmatrix} -\bar{d} \\ \bar{u} \end{bmatrix} \equiv S \begin{bmatrix} \bar{u} \\ \bar{d} \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \bar{u} \\ \bar{d} \end{bmatrix} \iff \begin{bmatrix} \bar{u} \\ \bar{d} \end{bmatrix} = S^{-1} \bar{q} \quad \text{and} \quad \begin{bmatrix} \bar{u}' \\ \bar{d}' \end{bmatrix} = S^{-1} \bar{q}'.$$

Substituting these equations into (5.5), we obtain

$$S^{-1} \bar{q}' = U^* S^{-1} \bar{q} \implies \bar{q}' = SU^* S^{-1} \bar{q}.$$

Explicitly calculating $SU^* S^{-1}$ shows that

$$SU^* S^{-1} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a^* & b^* \\ -b & a \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} a & b \\ -b^* & a^* \end{bmatrix} = U.$$

This gives $\bar{q}' = U\bar{q}$, where the doublet is given by

$$\bar{q} \equiv \begin{bmatrix} -\bar{d} \\ \bar{u} \end{bmatrix}. \quad (5.6)$$

With this antiquark doublet, the $I_3 = 1/2$ antiquark is assigned to $-\bar{d}$, and the $I_3 = -1/2$ antiquark is $\bar{u}$. The isospin ladder operators acting on the antiquark follows

$$T_+ \bar{u} = -\bar{d}, \quad T_+ \bar{d} = 0, \quad T_- \bar{u} = 0, \quad T_- \bar{d} = -\bar{u}.$$

5.3.6 Mesons

A meson is a bound state of a quark and an antiquark (not necessarily the same type). For quark and antiquark isospin states given by

$$u = \left| \frac{1}{2}, +\frac{1}{2} \right\rangle, \quad d = \left| \frac{1}{2}, -\frac{1}{2} \right\rangle, \quad \bar{u} = \left| \frac{1}{2}, -\frac{1}{2} \right\rangle, \quad -\bar{d} = \left| \frac{1}{2}, +\frac{1}{2} \right\rangle,$$

we can immediately construct four mesons. The meson isospin triplet is

$$|1, +1\rangle = -u\bar{d}, \quad |1, 0\rangle = \frac{1}{\sqrt{2}}(u\bar{u} - d\bar{d}), \quad |1, -1\rangle = \bar{u}d,$$

and the isospin singlet is

$$|0, 0\rangle = \frac{1}{\sqrt{2}}(u\bar{u} + d\bar{d}).$$

These results are analogous to those in Example 5.2. Just replace some u with $-\bar{d}$ and some d with $\bar{u}$. Physically, the triplet states are called **pions**:

$$\pi^+ = u\bar{d}, \quad \pi^0 = \frac{1}{\sqrt{2}}(u\bar{u} - d\bar{d}), \quad \pi^- = d\bar{u}.$$

The singlet state is a good basis for the isospin states, but it does not correspond to any physical meson. The $SU(3)$ group and strange mesons s are needed, e.g. the η -meson looks like

$$\eta = \frac{1}{\sqrt{3}}(u\bar{u} + d\bar{d} + s\bar{s}).$$

5.3.7 Amplitudes Related to Isospin

We can use the fact that the strong interaction conserves isospin to make some predictions on decays of hadronic decays. In general, the decay width Γ of a process is proportional to the amplitude squared of that process. We would like to predict the relative decay widths between hadronic decays,

$$\frac{\Gamma_1}{\Gamma_2} = \left| \frac{\mathcal{A}_1}{\mathcal{A}_2} \right|^2.$$

Here are the steps to make such predictions:

1. Identify the isospin quantum numbers of the initial-state and final-state particles. Care only about the up quark and down quark; other quarks do not have isospin.
2. List the relevant transition amplitudes from the initial-state particle to the final-state particles.
3. The final-state particles are usually expressed in direct product states. Express these direct product states into total isospin states using the Clebsch-Gordan coefficients.
4. Use orthonormality conditions to compare the relative amplitudes.

These steps may look a bit abstract, so let's work through some examples for concreteness.

Example 5.3. K^{*+} decays

Consider the initial-state particle $K^{*+} = u\bar{s}$. (A K^{*+} is not the same as a K^+ , though they have the same quark content. The spin state of a K^{*+} is symmetric while that of a K^+ is antisymmetric.) We are interested in the decays $K^{*+} \rightarrow K^+\pi^0$ and $K^{*+} \rightarrow K^0\pi^+$.

Step 1: identify the isospin quantum numbers for the particles involved. The total isospin states of the kaons are

$$K^+ = u\bar{s} = \left|\frac{1}{2}, +\frac{1}{2}\right\rangle, \quad K^0 = d\bar{s} = \left|\frac{1}{2}, -\frac{1}{2}\right\rangle.$$

(The strange quark has no isospin, so K^+ has the same I, I_3 as u and K^0 has the same I, I_3 as d .) The total isospin states of the pions are

$$\pi^+ = u\bar{d} = |1, +1\rangle, \quad \pi^0 = \frac{1}{\sqrt{2}}(u\bar{u} - d\bar{d}) = |1, 0\rangle, \quad \pi^- = d\bar{u} = |1, -1\rangle.$$

Step 2: list the relevant transition amplitudes,

$$\mathcal{A}_1 \equiv \langle K^+\pi^0 | S | K^{*+} \rangle = \langle \frac{1}{2}, +\frac{1}{2}; 1, 0 | S | \frac{1}{2}, +\frac{1}{2} \rangle,$$

$$\mathcal{A}_2 \equiv \langle K^0\pi^+ | S | K^{*+} \rangle = \langle \frac{1}{2}, -\frac{1}{2}; 1, +1 | S | \frac{1}{2}, +\frac{1}{2} \rangle.$$

Step 3: the initial state is in a total isospin state, but the final states are in direct product states (indicated by semicolons). The S operator that contains the strong Hamiltonian conserves total isospin. We therefore need to expand the direct product states in terms of total isospin states. A formal way to do it is to insert an identity operator in $\mathcal{A}$. For example,

$$\mathcal{A}_1 = \langle \frac{1}{2}, +\frac{1}{2}; 1, 0 | S | \frac{1}{2}, +\frac{1}{2} \rangle = \sum_{I', I'_3} \langle \frac{1}{2}, +\frac{1}{2}; 1, 0 | I', I'_3 \rangle \langle I', I'_3 | S | \frac{1}{2}, +\frac{1}{2} \rangle. \quad (5.7)$$

The quantities $\langle \frac{1}{2}, +\frac{1}{2}; 1, 0 | I', I'_3 \rangle$ are the Clebsch-Gordan coefficients (listed in Appendix C), and $\langle I', I'_3 |$ are the possible total isospin states. The results are

$$\begin{aligned} \langle \frac{1}{2}, +\frac{1}{2}; 1, 0 | &= \sqrt{\frac{2}{3}} \langle \frac{3}{2}, +\frac{1}{2} | - \sqrt{\frac{1}{3}} \langle \frac{1}{2}, +\frac{1}{2} |, \\ \langle \frac{1}{2}, -\frac{1}{2}; 1, +1 | &= \sqrt{\frac{1}{3}} \langle \frac{3}{2}, +\frac{1}{2} | + \sqrt{\frac{2}{3}} \langle \frac{1}{2}, +\frac{1}{2} |. \end{aligned}$$

The S -matrix element is given by

$$\langle I', I'_3 | S | I, I_3 \rangle = \delta_{I', I} \delta_{I'_3, I_3} S_I.$$

The Kronecker deltas indicates the conservation of isospin I, I_3 . There is an extra complex factor S_I that depends only on I but not I_3 . We do not care about this factor for now because eventually it will be canceled by the same S_I when we take the ratio of amplitudes. In our case,

$$\langle I', I'_3 | S | \frac{1}{2}, +\frac{1}{2} \rangle = \delta_{I', \frac{1}{2}} \delta_{I'_3, \frac{1}{2}} S_{\frac{1}{2}}.$$

Step 4: perform the summation in (5.7) for $K^{*+} \rightarrow K^+\pi^0$ and use orthonormality conditions,

$$\mathcal{A}_1 = \sum_{I', I'_3} \langle \frac{1}{2}, +\frac{1}{2}; 1, 0 | I', I'_3 \rangle \delta_{I', \frac{1}{2}} \delta_{I'_3, \frac{1}{2}} S_{\frac{1}{2}} = \langle \frac{1}{2}, +\frac{1}{2}; 1, 0 | \frac{1}{2}, +\frac{1}{2} \rangle S_{\frac{1}{2}} = -\sqrt{\frac{1}{3}} S_{\frac{1}{2}}.$$

Similarly, for $K^{*+} \rightarrow K^0\pi^+$,

$$\mathcal{A}_2 = \sum_{I', I'_3} \langle \frac{1}{2}, -\frac{1}{2}; 1, +1 | I', I'_3 \rangle \delta_{I', \frac{1}{2}} \delta_{I'_3, \frac{1}{2}} S_{\frac{1}{2}} = \langle \frac{1}{2}, -\frac{1}{2}; 1, +1 | \frac{1}{2}, +\frac{1}{2} \rangle S_{\frac{1}{2}} = \sqrt{\frac{2}{3}} S_{\frac{1}{2}}.$$

Finally, compute the ratio of amplitudes squared to get the relative decay widths,

$$\frac{\Gamma(K^{*+} \rightarrow K^+\pi^0)}{\Gamma(K^{*+} \rightarrow K^0\pi^+)} = \left| \frac{\mathcal{A}_1}{\mathcal{A}_2} \right|^2 = \frac{1}{2}.$$

Example 5.4. Pion-proton scattering

To be written.

5.4 Discrete Symmetries

5.4.1 Parity

Before 1956, it was widely believed that the mirror image of any physical process is also a perfectly possible physical process. This is called mirror symmetry, or **parity** invariance. In 1956, Lee and Yang found that there are evidence of parity invariance in strong and electromagnetic processes, but none for weak interactions. They asked Wu for an experiment to test the parity invariance in weak processes. In the experiment, cobalt 60 nuclei were aligned so that their spins point at the z -direction, or upward. These nuclei undergoes beta decay ($^{60}\text{Co} \rightarrow ^{60}\text{Ni} + e^- + \bar{\nu}_e$). Most of the emitted electrons (emitted downward) are opposite to the nuclear spin. This concluded the experiment.

Consider the mirror image of this experiment in Figure 5.1:

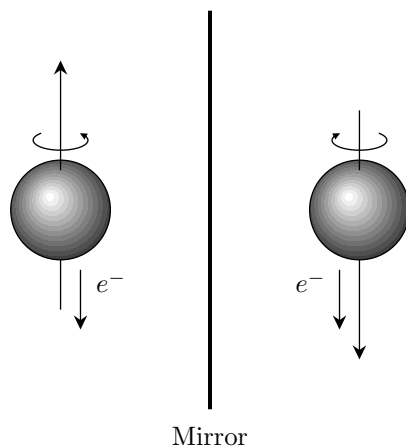


Figure 5.1: Decay of Cobalt 60 (left) and its mirror image (right). In the beta decay of Cobalt 60, most electrons are emitted in the direction opposite to the nuclear spin.

The spins of the nuclei are now downward, while the electrons after mirroring are also downward. Now most of the electrons are emitted in the same direction as the nuclear spin. The mirror image does not occur in nature, and parity is not invariant in weak processes. If parity were invariant, the electrons should come out in equal numbers in both upward and downward direction.

Parity violation is a trace of weak force. It is mostly connected to the behavior of neutrinos. Suppose a particle is traveling with velocity $\mathbf{v}$. We usually pick its direction of motion as the z -axis. The value of m_s/s for this axis is called the **helicity** of the particle. A particle of spin $\frac{1}{2}$ can have a helicity of $+1$ ($m_s = \frac{1}{2}$, right-handed), or -1 ($m_s = -\frac{1}{2}$, left-handed). For electrons, the helicity is not Lorentz-invariant. If you are traveling in the direction of $\mathbf{v}$ in the lab frame, but with a greater speed than $|\mathbf{v}|$, then in your frame, the electron will appear to be traveling opposite to you. The helicity will then flip.

Now we assume that neutrinos are massless so they travel at the speed of light. Then it is impossible to flip its helicity because you cannot travel faster. Thus, for massless particles, the helicity is Lorentz-invariant. It is discovered by experiments that neutrinos are left-handed, while antineutrinos are right handed. There are indirect ways (as neutrinos are hard to detect directly) to determine the helicity of a neutrino using the decay of the pion: $\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$ (see Figure 5.2). Using conservation of momentum and angular momentum with the fact that the pion has spin 0, we will have the muon and the antineutrino traveling in opposite direction, with opposite spin. Hence they should have the same helicity in the pion's rest frame, and measuring the muon's helicity will give the antineutrino's.

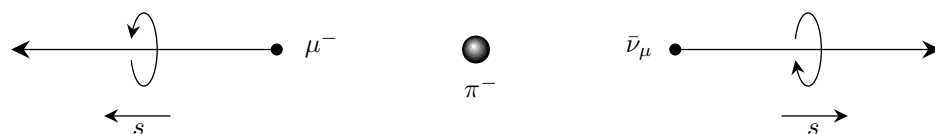


Figure 5.2: Decay of π^- at rest.

On the contrary, the decay of $\pi^0 \rightarrow \gamma + \gamma$ is an electromagnetic process in which parity is invariant, so left-handed photon pairs and right-handed photon pairs are in equal amount. This does not work for neutrinos because they only appear in weak interaction. Therefore, every neutrino is left handed. The mirror image of a neutrino does not exist.

Now, we want to develop some formalism instead of using “mirrors”, or reflection about a plane, e.g. $(x, y, z) \rightarrow (x, -y, z)$. We define the **inversion**, in which every point is reflected about the origin: $(x, y, z) \rightarrow (-x, -y, -z)$. We also define the **parity operator** P to denote inversion. When it is applied to a vector $\mathbf{a}$, it produces the opposite vector, $P(\mathbf{a}) = -\mathbf{a}$. However, for the cross product of two vectors, $\mathbf{c} = \mathbf{a} \times \mathbf{b}$, the parity operator does not change its sign $P(\mathbf{c}) = \mathbf{c}$ because it changes both signs of $\mathbf{a}$ and $\mathbf{b}$. Hence obviously there are two types of vectors: the **(polar) vectors** with sign changed, and the **pseudovectors** with sign unchanged under parity operator. The cross product of a polar vector with a pseudovector is a polar vector. There are pseudovectors that we are familiar with: angular momentum, magnetic field, etc. The dot product of two polar vectors does not change sign under P , but the dot product of a polar vector and a pseudovector (such as the triple scalar product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$) does change sign. The former is called the scalar, and the latter is the **pseudoscalar**.

Evidently,

$$P^2 = I,$$

so the parity group has only two elements: I and P . The eigenvalues of P are 1, for scalars and pseudovectors, and -1 , for pseudoscalars and vectors. According to QFT, the parity of a fermion is opposite to that of its antiparticle. The parity of a boson is the same as its antiparticle. By convention, we let quarks have positive intrinsic parity. The parity of a composite system in its ground state is the product of the parities of its constituents, so parity is a “multiplicative” quantum number. Unlike parity, the charge is “additive”.

With the formalism of parity, we can see how Lee and Yang doubted the parity invariance in weak processes. The question arose from the “tau-theta puzzle” in the early 1950s. The puzzle is the following: the two mesons τ and θ have the same mass same spin, same charge, and so on, but their decay process are different:

$$\begin{aligned} \theta^+ &\rightarrow \pi^+ + \pi^0 & (P = (-1)^2 = +1), \\ \tau^+ &\rightarrow \begin{cases} \pi^+ + \pi^0 + \pi^0 \\ \pi^+ + \pi^+ + \pi^- \end{cases} & (P = (-1)^3 = -1). \end{aligned}$$

Now we know that the τ and the θ are the same particle, the K^+ . The parity is just not conserved in the first process.

5.4.2 Charge Conjugation

A change in the sign of all electric charges reverses all fields potentials in classical electrodynamics, but the Lorentz force law $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ does not change. We say that classical electrodynamics is invariant under the sign change of electric charges. In elementary particle physics, the **charge conjugation**, C , changes all the internal quantum numbers: charge, baryon number, lepton number, etc. In other words, it converts particles into their antiparticles:

$$C |p\rangle = |\bar{p}\rangle.$$

Apparently, $C^2 = I$, so the eigenvalues of C are ± 1 . Now, since $|p\rangle$ and $|\bar{p}\rangle$ differs at most by a sign,

$$C |p\rangle = \pm |p\rangle = |\bar{p}\rangle,$$

only the particles that are their own antiparticles can be eigenstates of C .

Photon has a charge conjugation number -1 ; a system of a spin- $\frac{1}{2}$ particle and its antiparticle, together with orbital angular momentum ℓ and total spin s , is an eigenstate of C with eigenvalue $(-1)^{\ell+s}$. Like parity, charge conjugation is a multiplicative quantum number and is conserved in strong and electromagnetic interactions. For example, for both sides $C = 1$,

$$\pi^0 \rightarrow \gamma + \gamma$$

but $\pi^0 \rightarrow \gamma + \gamma + \gamma$ cannot occur. Charge conjugation is not a symmetry in weak interactions. When C applies to neutrinos, which are always left-handed, they become impossible left-handed (because C does not change spin) antineutrinos.

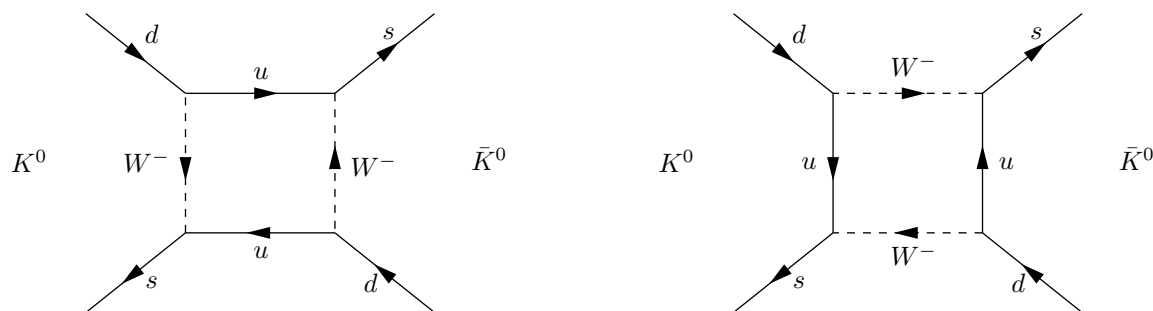
5.4.3 CP

Now we know that the weak interactions are not invariant under the parity operator P and the charge conjugation operator C . Consider the π meson decay and its charge conjugated version:

$$\pi^+ \rightarrow \mu^+ + \nu_\mu \quad \text{and} \quad \pi^- \rightarrow \mu^- + \bar{\nu}_\mu.$$

Recall that the muon is always right-handed and the antimuon is always left-handed. Hence if the π^+ decay is true (with a left-handed muon), its charge conjugated version will never happen because the muon coming out will be left-handed. However, if we apply the parity operator to the charge conjugated version, we will produce a right-handed muon. It seems like the weak interactions are invariant under the CP operator.

Neutral Kaons Gell-Mann and Pais noticed that the K^0 and $\bar{K}^0$ can turn into each other through a second-order weak interaction:



The K 's satisfy

$$P|K^0\rangle = -|K^0\rangle, \quad P|\bar{K}^0\rangle = -|\bar{K}^0\rangle, \quad C|K^0\rangle = |\bar{K}^0\rangle, \quad C|\bar{K}^0\rangle = |K^0\rangle.$$

Thus, $CP|K^0\rangle = -|\bar{K}^0\rangle$ and $CP|\bar{K}^0\rangle = -|K^0\rangle$. The actual particles which are eigenstates of CP that we observe in labs are linear combination of the two particles,

$$|K_1\rangle = \frac{1}{\sqrt{2}}(|K^0\rangle - |\bar{K}^0\rangle) \quad \text{and} \quad |K_2\rangle = \frac{1}{\sqrt{2}}(|K^0\rangle + |\bar{K}^0\rangle)$$

with

$$CP|K_1\rangle = |K_1\rangle \quad \text{and} \quad CP|K_2\rangle = -|K_2\rangle.$$

Note that K_1 and K_2 are not antiparticles of one another, but each is its own antiparticle with $C = -1$ for K_1 and $C = +1$ for K_2 .

If we assume CP is conserved in the weak interactions, then K_1 can only decay into a state with $CP = +1$, and K_2 with $CP = -1$. Neutral kaons decay into two or three pions typically. From the tau-theta puzzle, we know that the two-pion system has $P = +1$, three-pion system $P = -1$, and both $C = +1$. Therefore, K_1 decays into two pions while K_2 decays into three pions. The 2π decay is faster than the 3π decay. If the incident beam is

$$|K^0\rangle = \frac{1}{\sqrt{2}}(|K_1\rangle + |K_2\rangle),$$

we should see most 2π events near the source, but 3π events far away.

CP Violation In 1964, Cronin and Fitch found that at the end of a beam 57 feet long, there were 45 two pion events in 22,700 decays. This indicates that the neutral kaon is not a perfect eigenstate of CP , but an admixture of K_1 :

$$|K_L\rangle = \frac{1}{\sqrt{1+|\epsilon|^2}}(|K_2\rangle + \epsilon|K_1\rangle),$$

where $\epsilon \approx 2.24 \times 10^{-3}$ is a measure of nature's departure from perfect CP invariance. Unlike parity, which is "maximally" violated in weak interactions (all neutrinos are left-handed, not 50.01% of them), CP violation is a small effect.

5.4.4 Time-Reversal and the TCP Theorem

Imagine a movie of an elastic collision. It will depict a possible physical process when the movie is ran backward. The elementary particle interactions was believed to share this time-reversal invariance. However, parity violation made people wonder whether the time-reversal was invariant.

It is hard to test the invariance of time-reversal. First, no particle is an eigenstate of the time-reversal operator T . Second, time-reversal is invariant in strong and electromagnetic interactions, but it is difficult to do inverse reactions in weak interactions. For example, the inverse reaction of $\Lambda \rightarrow p^+ + \pi^-$ would be $p^+ + \pi^- \rightarrow \Lambda$, but the strong interaction of the proton and the pion will dominant over the weak interaction. Meanwhile, measurements of neutrinos (which only exist in weak process) are notoriously difficult. For now, no experiments has shown direct evidence of T violation.

However, in quantum field theory, the TCP theorem states that the combined operation of time reversal , charge conjugation, and parity in any order is an exact symmetry of any interaction. In other words, it is impossible to construct a quantum field theory in which TCP is not conserved. We know that CP is violated, so there must be a violation of T .

Part II

Quantum Field Theory

6 QUANTUM FIELD THEORY: SCALAR FIELDS

WELCOME TO THE WORLD OF SPIN-0.

6.1 The Klein-Gordon Field

6.1.1 The Real Klein-Gordon Field

The equation that describes a relativistic free particle is known as the **Klein-Gordon equation** (or Klein-Gordon equation). In relativistic covariant form, the Klein-Gordon equation is

$$(\partial_\mu \partial^\mu \phi + m^2)\phi = 0. \quad (6.1)$$

In fact, the Klein-Gordon equation is more than just covariant—it is a Lorentz *invariant* equation. The Lagrangian that produces the Klein-Gordon equation is

$$\mathcal{L}_{\text{KG}} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2. \quad (6.2)$$

The solutions to the Klein-Gordon equation are free fields $e^{\pm i p \cdot x}$, where p is the 4-momentum and x is the spacetime coordinate. These solutions give the relation $p_\mu p^\mu = E^2 - \mathbf{p}^2 = m^2$ if you plug them into the Klein-Gordon equation. The most general solution is a linear combination of free fields, integrated over all momentum,

$$\phi(x) = \int \frac{d^3 p}{(2\pi)^3 \sqrt{2E}} [a(p) e^{-i p \cdot x} + a^*(p) e^{i p \cdot x}],$$

for some coefficients $a(p)$ and a particular normalization factor $1/\sqrt{2E}$. The field $\phi(x)$ is the **real Klein-Gordon field**. The way to write out this kind of linear combination of plane waves is called mode expansion.

The Hamiltonian of the Klein-Gordon field is straightforward to compute. First, separate the Lagrangian (6.2) into time-derivative and spatial derivatives, and compute the conjugate momentum $\pi(x)$:

$$\mathcal{L} = \frac{1}{2} [\dot{\phi}^2 - (\nabla \phi)^2 - m^2 \phi^2] \implies \pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi}.$$

Then by the definition of Hamiltonian density, $\mathcal{H} = \pi \dot{\phi} - \mathcal{L}$, the Hamiltonian density of the Klein-Gordon field is

$$\mathcal{H}_{\text{KG}} = \frac{1}{2} \pi^2 + \frac{1}{2} (\nabla \phi)^2 + \frac{1}{2} m^2 \phi^2. \quad (6.3)$$

The first term is the energy density associated with the change of field with respect to time; the second term is the energy density associated with the change of field in space; the third term can be seen as the potential energy of the field.

6.1.2 The Complex Klein-Gordon Field

In the previous section we constructed ϕ to be real. A more general and important case is the complex scalar field. Suppose there are two independent Klein-Gordon fields $\phi_1(x)$ and $\phi_2(x)$ with the *same mass* m . The Lagrangian for this system is just the sum of two independent Lagrangians of ϕ_1 and ϕ_2 :

$$\mathcal{L}(\phi_1, \phi_2, \partial_\mu \phi_1, \partial_\mu \phi_2) = \frac{1}{2} \partial_\mu \phi_1 \partial^\mu \phi_1 + \frac{1}{2} \partial_\mu \phi_2 \partial^\mu \phi_2 - \frac{1}{2} m^2 (\phi_1^2 + \phi_2^2).$$

Varying ϕ_1 and ϕ_2 will give two Klein-Gordon equations,

$$(\partial_\mu \partial^\mu + m^2)\phi_1 = 0, \quad (\partial_\mu \partial^\mu + m^2)\phi_2 = 0.$$

We can write $\mathcal{L}$ in terms of a complex scalar field ϕ , defined by

$$\phi(x) = \frac{1}{\sqrt{2}}(\phi_1 - i\phi_2), \quad \phi^*(x) = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2).$$

It is easy to show that

$$\begin{aligned} \phi\phi^* &= \frac{1}{2}(\phi_1^2 + \phi_2^2) \quad \text{and} \quad \partial_\mu \phi^* \partial^\mu \phi = \left[\frac{1}{\sqrt{2}} \partial_\mu (\phi_1 + i\phi_2) \right] \left[\frac{1}{\sqrt{2}} \partial^\mu (\phi_1 - i\phi_2) \right] \\ &= \frac{1}{2} [\partial_\mu \phi_1 \partial^\mu \phi_1 + i \partial_\mu \phi_1 \partial^\mu \phi_2 - i \partial_\mu \phi_2 \partial^\mu \phi_1 + \partial_\mu \phi_2 \partial^\mu \phi_2] \\ &= \frac{1}{2} [\partial_\mu \phi_1 \partial^\mu \phi_1 + \partial_\mu \phi_2 \partial^\mu \phi_2]. \end{aligned}$$

Hence the Lagrangian of two real fields can be written simply as a single complex field:

$$\mathcal{L}(\phi, \phi^*, \partial_\mu \phi, \partial_\mu \phi^*) = \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi. \quad (6.4)$$

Later we will see that complex fields give rise to the concept of “charge”, while real fields are neutral.

6.2 Quantum Field Theory

6.2.1 Canonical Quantization

In quantum mechanics, the term “quantization” or “canonical quantization” does not mean discrete energy levels or something similar. Canonical quantization means the canonical commutation relation:

$$[x_i, p_j] = i\delta_{ij}, \quad [x_i, x_j] = 0, \quad [p_i, p_j] = 0.$$

where the dynamical variables, position x_i and momentum p_j , are now operators. This is the basis of everything in quantum mechanics, essentially a postulate in quantum mechanics. In quantum field theory, x is *not* a dynamical variable. It is merely a label to indicate where you are in the field. The dynamical variables in quantum field theory are the field $\phi(x)$ itself, its conjugate momentum $\pi(x)$, and other associated derivatives. These dynamical variables are turned to operators. (Of course, they have to operate on something, but we shall not mention it for now.) The canonical commutation relation of quantum field theory is

$$[\phi(\mathbf{x}, t), \pi(\mathbf{x}', t)] = i\delta^3(\mathbf{x} - \mathbf{x}'), \quad [\phi(\mathbf{x}, t), \phi(\mathbf{x}', t)] = 0, \quad [\pi(\mathbf{x}, t), \pi(\mathbf{x}', t)] = 0. \quad (6.5)$$

Since the field and its conjugate momentum are continuous, it is natural to replace the Kronecker delta δ_{ij} with Dirac delta $\delta^3(\mathbf{x} - \mathbf{x}')$. Note that the argument of ϕ and π are at the same time t .

6.2.2 The Real Klein-Gordon Field

First consider the real Klein-Gordon Field. Classically, the field is

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3 \sqrt{2E}} [a(p)e^{-ip \cdot x} + a^*(p)e^{+ip \cdot x}].$$

Remember that in quantum field theory, x and p are just labels. The operators are the fields. Because ϕ is an operator, the only thing in the expansion of ϕ that can be turned into an operator is $a(p)$. For real fields, we demand it to be Hermitian, so $a^*(p)$ turns into $a^\dagger(p)$. The field operator is then

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3 \sqrt{2E}} [a(p)e^{-ip \cdot x} + a^\dagger(p)e^{+ip \cdot x}]. \quad (6.6)$$

With this, we can compute the conjugate momentum field operator of the Klein-Gordon field,

$$\pi(x) = \dot{\phi}(x) = \int \frac{d^3p}{(2\pi)^3\sqrt{2E}} (-iE) [a(p)e^{-ip\cdot x} - a^\dagger(p)e^{ip\cdot x}]. \quad (6.7)$$

There is another set of extremely useful commutator relations of $a(p)$ and $a^\dagger(p)$:

$$[a(p), a^\dagger(p')] = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}'), \quad [a(p), a(p')] = [a^\dagger(p), a^\dagger(p')] = 0. \quad (6.8)$$

Appendix A.3.1 shows that (6.8) are completely equivalent to the commutation relations (6.5). These relations are analogous to the quantum harmonic oscillator commutation relations $[a, a^\dagger] = 1$ in ordinary quantum mechanics. We can put the Klein-Gordon field (6.6) and its conjugate momentum (6.7) into the Hamiltonian density (6.3) and integrate over all space to get the Hamiltonian. The final result (see Appendix A.3.2) can be written in terms of $a(p)$ and $a^\dagger(p)$:

$$\begin{aligned} H_{\text{KG}} &= \frac{1}{2} \int \frac{d^3p}{(2\pi)^3} [a^\dagger(p)a(p) + a(p)a^\dagger(p)] E \\ &= \int \frac{d^3p}{(2\pi)^3} \left[a^\dagger(p)a(p) + \frac{1}{2}(2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}) \right] E \\ &\stackrel{!}{=} \int \frac{d^3p}{(2\pi)^3} \left[a^\dagger(p)a(p) + \frac{1}{2}(2\pi)^3 \delta^3(0) \right] E. \end{aligned}$$

Note that $\delta^3(\mathbf{p} - \mathbf{p})$ does not have a prime in the second $\mathbf{p}$! This means the integral is over $\delta^3(0) = \infty$, which is divergent. A method to deal with it is called **normal ordering**, where we just ignore the infinity $\delta^3(0)$ because we care about only energy difference in quantum field theory. (This will be a problem in the theory of gravity, because the stress-energy tensor cares about all energies, including those of ground states. The problem is not solved yet.) After normal ordering, the Hamiltonian has the form.

$$H_{\text{KG}} = \int \frac{d^3p}{(2\pi)^3} a^\dagger(p)a(p)E. \quad (6.9)$$

This looks very similar to the Hamiltonian for the quantum harmonic oscillator, $H = (a^\dagger a + \frac{1}{2})E$, or $H = a^\dagger a E$ if we remove the ground state energy. We often denote $n = a^\dagger(p)a(p)$ as the number operator. The field operators act on states that live in the **Fock space**. Like the quantum harmonic oscillator in ordinary quantum mechanics, we argue that there is a **vacuum** state $|0\rangle$ such that

$$H_{\text{KG}} |0\rangle = 0.$$

In the vacuum state, there are no excitations of fields that represent particles at any momentum. Excited states are given by

$$|n_1, n_2, \dots, n_j, \dots\rangle = \prod_j \frac{1}{\sqrt{n_j!}} [a^\dagger(p_j)]^{n_j} |0\rangle,$$

where n_j indicates that there are n_j number of particles with 4-momentum p_j . These particles are created by the creation operator $a^\dagger(p_j)$ for each 4-momentum p_j . Note that it does not matter which particle is excited first,

$$a^\dagger(p_1)a^\dagger(p_2)|0\rangle = a^\dagger(p_2)a^\dagger(p_1)|0\rangle.$$

6.2.3 The Complex Klein-Gordon Field

Consider two real Klein-Gordon fields which satisfy the Klein-Gordon equation separately.

$$\phi_1(x) = \int \frac{d^3p}{(2\pi)^3\sqrt{2E}} [a_1(p)e^{-ip\cdot x} + a_1^\dagger(p)e^{ip\cdot x}] \quad \text{and} \quad \phi_2(x) = \int \frac{d^3p}{(2\pi)^3\sqrt{2E}} [a_2(p)e^{ip\cdot x} + a_2^\dagger(p)e^{-ip\cdot x}].$$

We construct a complex Klein-Gordon field by taking a particular linear combination of these two real Klein-Gordon fields,

$$\phi(x) = \frac{1}{\sqrt{2}} [\phi_1(x) - i\phi_2(x)] = \int \frac{d^3p}{(2\pi)^3\sqrt{2E}} \left\{ \frac{1}{\sqrt{2}} [a_1(p) - ia_2(p)] e^{-ip\cdot x} + \frac{1}{\sqrt{2}} [a_1^\dagger(p) - ia_2^\dagger(p)] e^{ip\cdot x} \right\}$$

It is conventional to define an annihilation operator $a(p)$ and a creation operator $b^\dagger(p)$ as the following:

$$a(p) \equiv \frac{1}{\sqrt{2}}[a_1(p) - ia_2(p)], \quad b^\dagger(p) \equiv \frac{1}{\sqrt{2}}[a_1^\dagger(p) - ia_2^\dagger(p)].$$

Note that $a^\dagger(p) \neq b^\dagger(p)$. This means the field $\phi(x)$ is not Hermitian. The mode expansion of $\phi(x)$ can then be expressed in terms of $a(p)$ and $b^\dagger(p)$,

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3 \sqrt{2E}} [a(p)e^{-ip \cdot x} + b^\dagger(p)e^{+ip \cdot x}]. \quad (6.10)$$

The commutation relations for $a(p)$ and $b^\dagger(p)$ follows from those for $a_1(p)$ and $a_2(p)$ (and $a_1(p)$ and $a_2(p)$ are independent, so they commute). The result is

$$[a(p), a^\dagger(p')] = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}'), \quad [b(p), b^\dagger(p')] = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}') \quad (6.11)$$

with all other commutators being zero. This means the complex Klein-Gordon field has two types of quanta, one associated with $a^\dagger(p)$ and the other with $b^\dagger(p)$. This is expected because the construction of this complex scalar field is from two independent real fields which has their own corresponding quanta.

The Lagrangian for a complex scalar (quantum) field, by changing (6.4) into field operators, is

$$\mathcal{L} = \partial_\mu \phi^\dagger \partial^\mu \phi - m^2 \phi^\dagger \phi. \quad (6.12)$$

The Lagrangian is invariant under the following transformation of ϕ , called the **global phase transformation**:

$$\phi \rightarrow \phi' = e^{-i\alpha} \phi, \quad \phi^\dagger \rightarrow \phi'^\dagger = e^{+i\alpha} \phi^\dagger.$$

It is called “global” because we assume that the parameter α is independent of spacetime coordinates x^μ . Evidently, if α depends on x^μ , then the phase factor $e^{i\alpha}$ cannot be moved around derivatives freely. Later we will modify many things in *gauge theory* such that the Lagrangian is invariant under *local* phase transformations.

The global phase invariance of the Lagrangian has a significant consequence. Consider infinitesimal transformations

$$\phi'(x) = (1 - i\delta\alpha)\phi(x) \implies \delta\phi = -i\phi(x)(\delta\alpha).$$

Basically, we just Taylor expand the exponential $e^{-i\delta\alpha}$, assuming that $\delta\alpha$ is infinitesimally small. We know that the Lagrangian is invariant under such transformations (whether $\delta\alpha$ is infinitesimal or not), so

$$0 = \delta\mathcal{L} = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta(\partial_\mu\phi) + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^\dagger)}\delta(\partial_\mu\phi^\dagger) + \frac{\partial\mathcal{L}}{\partial\phi}\delta\phi + \frac{\partial\mathcal{L}}{\partial\phi^\dagger}\delta\phi^\dagger.$$

So far these may not be physical fields. Physical fields need to satisfy $\delta S = 0$, or the Euler-Lagrange equation. Incorporate the Euler-Lagrange equation (5.1) for fields,

$$\partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \right) = \frac{\partial\mathcal{L}}{\partial\phi}, \quad \partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^\dagger)} \right) = \frac{\partial\mathcal{L}}{\partial\phi^\dagger}.$$

Substitute these Euler-Lagrange equations into $\delta\mathcal{L} = 0$ derived above, we can get a total derivative

$$\begin{aligned} 0 = \delta\mathcal{L} &= \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu(\delta\phi) + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^\dagger)}\partial_\mu(\delta\phi^\dagger) + \partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \right) \delta\phi + \partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^\dagger)} \right) \delta\phi^\dagger \\ &= \partial_\mu \left[\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^\dagger)}\delta\phi^\dagger \right]. \end{aligned}$$

This total derivative is in fact a divergence term because the μ are contracted. Whenever a divergence is equal to zero, there should be a conserved current. We call the terms in the square bracket is the conserved **4-current** J^μ so that the above result can be expressed as a continuity equation:

$$\partial_\mu J^\mu = 0 \iff \partial_t J^t + \nabla \cdot \mathbf{J} = 0.$$

We can explicitly compute the derivatives from the Lagrangian and replace $\delta\phi = -i\phi(x)(\delta\alpha)$.

$$\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^\dagger)}\delta\phi^\dagger = -i(\delta\alpha)[(\partial^\mu\phi^\dagger)\phi - (\partial^\mu\phi)\phi^\dagger].$$

Since $\delta\alpha$ is arbitrary, they can be thrown away. The conserved 4-current is now

$$J_\phi^\mu(x) = i(\phi^\dagger \partial^\mu \phi - \phi \partial^\mu \phi^\dagger). \quad (6.13)$$

Here the subscript ϕ denotes the scalar field. We can integrate the continuity equation over all space,

$$\frac{d}{dt} \int_V J_\phi^t d^3x + \int_{\partial V} \mathbf{J}_\phi \cdot d\mathbf{S} = 0.$$

In general, the current decays sufficiently fast at infinity so that the surface integral vanishes. Therefore, the quantity (call it **symmetry operator**)

$$N_\phi = \int J_\phi^t d^3x$$

is a constant in time. This also means that N_ϕ , as a conserved quantity, commutes with the Hamiltonian (after normal ordering) of the complex Klein-Gordon field,

$$H = \int \frac{d^3p}{(2\pi)^3} [a^\dagger(p)a(p) + b^\dagger(p)b(p)]E. \quad (6.14)$$

One can derive this Hamiltonian in essentially two ways. The first way is to go through all the steps: find the conjugate momentum of ϕ and $\phi^\dagger$; express all ϕ , $\phi^\dagger$, π , and $\pi^\dagger$ in mode expansions of $a(p)$ and $b(p)$; and finally do the normal ordering. The second way is to say that the Hamiltonian of two independent fields ϕ and $\phi^\dagger$, which arise from ϕ_1 and ϕ_2 , should be the sum of their individual Hamiltonians. This is the reason why we expect to see both $a^\dagger a$ and $b^\dagger b$ terms. The symmetry operator N_ϕ in terms of the creation and annihilation operators are derived in Appendix A.3.3. The result (after normal ordering) is

$$N_\phi = \int \frac{d^3p}{(2\pi)^3} [a^\dagger(p)a(p) - b^\dagger(p)b(p)]. \quad (6.15)$$

Now with the Hamiltonian (6.14) and the symmetry operator (6.15), we can discuss the interpretation of the a quanta and b quanta. Note that in (6.14), both types of quanta contributes the same amount of energy. In particular, there is *no negative energies* in the Hamiltonian. On the contrary, N_ϕ involves a difference of number operators $a^\dagger a$ and $b^\dagger b$. It counts +1 for each particle of type a and -1 for each particle of type b . This strongly suggests us to view the b 's as antiparticles of the a 's. Then the current J^μ can really be seen as a charge current, where the b 's carry opposite charges to those carried by the a 's, and meanwhile both have the same energy.

7 QUANTUM FIELD THEORY: DIRAC FIELDS

WELCOME TO THE WORLD OF SPIN-1/2.

7.1 The Dirac Equation

The Klein-Gordon equation is a second-order equation which shows that free-particle solutions lead to $E^2 - \mathbf{p}^2 = m^2$. It is also possible to find a first-order equation that also enforces this identity on free particles. The first-order equation we want to find had better automatically satisfy the Klein-Gordon equation at the same time. Suppose the equation of interest is first-order in both ∂_t and ∇ :

$$[\boldsymbol{\alpha} \cdot (-i\nabla) + \beta m]\psi(x) = i\frac{\partial}{\partial t}\psi(x), \quad (7.1)$$

where the coefficients $\boldsymbol{\alpha} = \{\alpha_i\}$ and β are to be determined. The RHS is the same as in the Schrödinger equation, but the LHS (a Hamiltonian term) is first order in ∇ . Because this equation is linear in momentum, it is natural to involve only one power of mass m . As mentioned above, we want (7.1) to have free-particle solutions that automatically satisfy the Klein-Gordon equation (6.1). This means the following operator equation should hold:

$$(i\partial_t)^2 = [\boldsymbol{\alpha} \cdot (-i\nabla) + \beta m][\boldsymbol{\alpha} \cdot (-i\nabla) + \beta m] = (-i\nabla)^2 + m^2.$$

This will impose some conditions on α_i and β . Compute the LHS explicitly,

$$\begin{aligned} (i\partial_t)^2 &= [\boldsymbol{\alpha} \cdot (-i\nabla) + \beta m][\boldsymbol{\alpha} \cdot (-i\nabla) + \beta m] \\ &= [\alpha_i(-i\partial_i) + \beta m][\alpha_j(-i\partial_j) + \beta m] \\ &= [\alpha_i(-i\partial_i)]^2 + (\alpha_i\alpha_j + \alpha_j\alpha_i)(-i)^2\partial_i\partial_j + (\alpha_i\beta + \beta\alpha_i)(-i\partial_i)m + \beta^2m^2, \end{aligned}$$

where repeated indices infer implicit summations, though α_i and ∂_i are both subscripts. (Note that α_i is not the spatial part of a Lorentz 4-vector. It is just a coefficient with a subscript corresponding to partial derivatives ∂_i .) We want the above expression to end up being $(-i\nabla)^2 + m^2$. The requirements are

$$\alpha_i\alpha_j + \alpha_j\alpha_i = 0, \quad \alpha_i\beta + \beta\alpha_i = 0, \quad \text{and} \quad \alpha_1^2 = \alpha_2^2 = \alpha_3^2 = \beta^2 = 1.$$

These conditions can be written compactly as [anticommutators](#),

$$\{\alpha_i, \alpha_j\} = 2\delta_{ij} \quad \text{and} \quad \{\alpha_i, \beta\} = 0.$$

The anticommutators tells us that α_i and β cannot simply be numbers.

7.1.1 The γ -Matrices

Equation (7.1) can be written in a more relativistic covariant way. First, we want to get rid of the β before m . Multiplying (7.1) by β on both sides,

$$[\beta\boldsymbol{\alpha} \cdot (-i\nabla) + \beta^2m]\psi = i\beta\partial_t\psi.$$

Now β^2m is just m because $\beta^2 = 1$. Define

$$\gamma^i \equiv \beta\alpha_i, \quad \gamma^0 = \beta. \quad (7.2)$$

It looks like $\gamma^i \equiv \beta\alpha_i$ violates index notation rules, but remember that the index i in α_i is not a Lorentz index but just a label. Then

$$[\boldsymbol{\gamma} \cdot (-i\nabla) + m]\psi = \gamma^0(i\partial_t)\psi \implies [\gamma^0(i\partial_t) + \boldsymbol{\gamma} \cdot (i\nabla)]\psi = m\psi.$$

Define $\gamma^\mu = (\gamma^0, \boldsymbol{\gamma})$. Then we can write the above equation as

$$\boxed{i\gamma^\mu \partial_\mu \psi = m\psi.} \quad (7.3)$$

This is the **Dirac equation**. It turns out that there will be a lot of terms involving $\gamma^\mu A_\mu$. For any 4-vector A^μ , there is a notation like $\not{A}$ called **Feynman slash notation**. It means $\not{A} = \gamma_\mu A^\mu = \gamma^\mu A_\mu$, to avoid writing $\gamma_\mu A^\mu$ everywhere. Hence the Dirac equation can also be written as

$$i\not{\partial}\psi = m\psi. \quad (7.4)$$

The object γ^μ is independent of x^μ , so it commutes with partial derivatives ∂_μ . Also, because γ^μ are just constant matrices, they do not transform— γ^μ is not a 4-vector. However, we can still raise or lower indices of them: $\eta_{\mu\nu}\gamma^\mu = \gamma_\nu$, $\eta^{\mu\nu}\gamma_\mu = \gamma^\nu$.

7.1.2 Properties of α_i , β , and γ^μ Matrices

Here we list some basic properties in α_i , β , and γ^μ matrices. The proofs and some further results are available in Appendix A.2.

- α_i and β square to the identity: $\alpha_i^2 = \beta^2 = 1$.
- α_i and β anticommute with each other: $\{\alpha_i, \alpha_j\} = 0$ for $i \neq j$, and $\{\alpha_i, \beta\} = 0$.
- α_i and β are Hermitian because they constitutes the Hamiltonian term in (7.1): $\alpha_i^\dagger = \alpha_i$, $\beta^\dagger = \beta$.
- α_i and β are traceless: $\text{Tr}(\alpha_i) = \text{Tr}(\beta) = 0$.
- α_i and β have *even* dimensions. (This is very important.)
- The Hermitian conjugate of γ^μ can be written as $(\gamma^\mu)^\dagger = \gamma^0 \gamma^\mu \gamma^0$. In other words, $(\gamma^0)^\dagger = \gamma^0$, and $(\gamma^i)^\dagger = -\gamma^i$.
- Square of γ^μ : $(\gamma^i)^2 = -1$, $\gamma^0 = 1$. This also implies that $\det \gamma^\mu \neq 0$.
- Anticommutator relation of γ^μ :

$$\boxed{\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}1.} \quad (7.5)$$

Note that γ^μ itself is a matrix, while $\eta^{\mu\nu}$ are elements of the Minkowski metric tensor. This is why it is better to write an identity on the RHS.

7.1.3 Representations of α_i , β , and γ^μ

We are looking for α_i and β with the following properties: Hermitian, anticommute with each other, squaring to the identity, traceless, and even dimensionality. We can start with dimensionality and let $N = 2$. However, there does not exist four 2×2 Hermitian matrices that anticommute with each other.

Proof. By the definition of a Hermitian matrix, its matrix elements satisfy $H_{mn} = H_{nm}^*$. A 2×2 Hermitian matrix H must have real diagonal entries, and its off-diagonal entries are complex conjugate of each other. We can find that the most general form of any 2×2 Hermitian matrix can be written as

$$H = \begin{bmatrix} a & c - id \\ c + id & b \end{bmatrix}$$

for real numbers a, b, c, d . It is useful to express H in terms of the Pauli spin matrices σ_i and the identity:

$$H = \frac{a+b}{2}1 + \frac{a-b}{2}\sigma_z + c\sigma_x + d\sigma_y, \quad \text{where} \quad \sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

The Pauli spin matrices already have good properties: they anticommute with each other, $\{\sigma_i, \sigma_j\} = 0$ for $i \neq j$, they square to the identity, and they are traceless. So now we already get three anticommuting matrices, we want to find the fourth one.

Suppose a Hermitian matrix A anti-commutes with all of $\sigma_x, \sigma_y, \sigma_z$. Then it cannot be a linear combination of them. (Unless $A = 0$, but that is trivial.) Compute $\{A, \sigma_z\}$:

$$\{A, \sigma_z\} = \frac{a+b}{2}\{1, \sigma_z\} + \frac{a-b}{2}\{\sigma_z, \sigma_z\} + c\{\sigma_x, \sigma_z\} + d\{\sigma_y, \sigma_z\}.$$

The last two anticommutator vanish because Pauli spin matrices anticommute with each other. We are left with

$$\{A, \sigma_z\} = (a+b)\sigma_z + (a-b)1 = \begin{bmatrix} 2a & 0 \\ 0 & -2b \end{bmatrix}.$$

We want $\{A, \sigma_z\} = 0$, so $a = b = 0$. However, if $a = b = 0$, then A is a linear combination of σ_y and σ_z , which contradicts with our assumption. Hence there does not exist a nontrivial Hermitian matrix that anticommutes with the three Pauli spin matrices. $\square$

$N = 2$ does not work, so we shall turn to $N = 4$. The good properties of the Pauli spin matrices may give us some hints. It is possible to find four 4×4 Matrices that satisfy all the properties. (In fact, there are infinite possible ways.) There are two common representations: the Weyl representation and the Dirac-Pauli representation. The **Weyl representation** has β off-diagonal and α_i diagonal:

$$\beta = \begin{bmatrix} 0 & 1_{2 \times 2} \\ 1_{2 \times 2} & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad \alpha_x = \begin{bmatrix} \sigma_x & 0 \\ 0 & -\sigma_x \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{bmatrix}.$$

$$\alpha_y = \begin{bmatrix} \sigma_y & 0 \\ 0 & -\sigma_y \end{bmatrix} = \begin{bmatrix} 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & i \\ 0 & 0 & -i & 0 \end{bmatrix}, \quad \alpha_z = \begin{bmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The **Dirac-Pauli representation** has β diagonal and α_i off-diagonal:

$$\beta = \begin{bmatrix} 1_{2 \times 2} & 0 \\ 0 & -1_{2 \times 2} \end{bmatrix}, \quad \alpha_x = \begin{bmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{bmatrix}, \quad \alpha_y = \begin{bmatrix} 0 & \sigma_y \\ \sigma_y & 0 \end{bmatrix}, \quad \alpha_z = \begin{bmatrix} 0 & \sigma_z \\ \sigma_z & 0 \end{bmatrix}.$$

From the Dirac equation (7.1), we can see that the Weyl representation is more convenient in the relativistic limit where $p \gg m$. In this way, the off-diagonal matrix β is negligible when multiplied by m . On the other hand, the Dirac-Pauli representation is more convenient in the non-relativistic limit where $m \gg p$, where the α_i terms are negligible.

Now with α_i and β , we can compute γ^μ in both representations. In the Weyl representation:

$$\gamma^0 = \begin{bmatrix} 0 & 1_{2 \times 2} \\ 1_{2 \times 2} & 0 \end{bmatrix}, \quad \gamma^i = \begin{bmatrix} 0 & -\sigma_i \\ \sigma_i & 0 \end{bmatrix}.$$

In the Dirac-Pauli representation:

$$\gamma^0 = \begin{bmatrix} 1_{2 \times 2} & 0 \\ 0 & -1_{2 \times 2} \end{bmatrix}, \quad \gamma^i = \begin{bmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{bmatrix}.$$

Even though we know the matrix representations of β , α_i and γ^μ , in most calculations we only need their basic properties such as $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}1$.

7.2 Free-Particle Solutions

If the γ -matrices in the Dirac equation (7.3) are represented by 4×4 matrices. It must act on an object ψ that has 4 components,

$$\psi(x) = \begin{bmatrix} \psi_1(x) \\ \psi_2(x) \\ \psi_3(x) \\ \psi_4(x) \end{bmatrix}.$$

This object is called a **Dirac spinor**, which is not a 4-vector because it does not undergo a Lorentz transformation from one inertial frame to another. Hence we will label its components using Latin indices $i = 1, 2, 3, 4$ though it has 4 components. The goal now is to solve the Dirac equation.

7.2.1 The Dirac-Pauli Representation

We will work in the Dirac-Pauli representation for now. Suppose we want to find the free-particle solution. First, write the spinor $\psi(x)$ as two separate 2-component spinors ϕ and χ :

$$\psi(x) = N \begin{bmatrix} \phi \\ \chi \end{bmatrix} e^{\pm i p \cdot x} = N \begin{bmatrix} \phi \\ \chi \end{bmatrix} e^{\pm i(Et - \mathbf{p} \cdot \mathbf{x})},$$

where $N = \sqrt{E + m}$ is the conventional normalization factor. Because we assume that $\psi(x)$ is a plane wave solution, ϕ and χ should not have any dependence on x^μ . All of the spacetime dependence is in the exponential factor $e^{\pm i p \cdot x}$. Plug the free-particle solution $\psi(x)$ into the Dirac equation

$$[\gamma^0(i\partial_t) + \boldsymbol{\gamma} \cdot (i\nabla)] \begin{bmatrix} \phi \\ \chi \end{bmatrix} e^{\pm i p \cdot x} = m \begin{bmatrix} \phi \\ \chi \end{bmatrix} e^{\pm i p \cdot x}.$$

The time-derivative extract one factor of $\pm iE$ and the spatial-derivative extract one factor of $i \mp \mathbf{p}$ from the exponential factor, so

$$[\gamma^0(\mp E) + \boldsymbol{\gamma} \cdot (\pm \mathbf{p})] \begin{bmatrix} \phi \\ \chi \end{bmatrix} e^{\pm i p \cdot x} = m \begin{bmatrix} \phi \\ \chi \end{bmatrix} e^{\pm i p \cdot x}.$$

The exponential factor cancels. Writing out the γ -matrices (in the Dirac-Pauli representation),

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} (\mp E) + \begin{bmatrix} 0 & \boldsymbol{\sigma} \\ -\boldsymbol{\sigma} & 0 \end{bmatrix} \cdot (\pm \mathbf{p}) \right\} \begin{bmatrix} \phi \\ \chi \end{bmatrix} = m \begin{bmatrix} \phi \\ \chi \end{bmatrix}.$$

Gathering the matrices together,

$$\mp \begin{bmatrix} E & -\boldsymbol{\sigma} \cdot \mathbf{p} \\ \boldsymbol{\sigma} \cdot \mathbf{p} & -E \end{bmatrix} \begin{bmatrix} \phi \\ \chi \end{bmatrix} = m \begin{bmatrix} \phi \\ \chi \end{bmatrix}, \quad (7.6)$$

where

$$\boldsymbol{\sigma} \cdot \mathbf{p} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} p_x + \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} p_y + \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} p_z = \begin{bmatrix} p_z & p_x - ip_y \\ p_x + ip_y & -p_z \end{bmatrix}. \quad (7.7)$$

In fact, one can find that

$$\not{p} = \gamma^\mu \partial_\mu = \begin{bmatrix} E & -\boldsymbol{\sigma} \cdot \mathbf{p} \\ \boldsymbol{\sigma} \cdot \mathbf{p} & -E \end{bmatrix},$$

which is just the matrix in (7.6). Equation (7.6) is essentially two equations:

$$\begin{bmatrix} \phi \\ \chi \end{bmatrix}_- = \frac{1}{m} \begin{bmatrix} E & -\boldsymbol{\sigma} \cdot \mathbf{p} \\ \boldsymbol{\sigma} \cdot \mathbf{p} & -E \end{bmatrix} \begin{bmatrix} \phi \\ \chi \end{bmatrix}_-, \quad \begin{bmatrix} \phi \\ \chi \end{bmatrix}_+ = -\frac{1}{m} \begin{bmatrix} E & -\boldsymbol{\sigma} \cdot \mathbf{p} \\ \boldsymbol{\sigma} \cdot \mathbf{p} & -E \end{bmatrix} \begin{bmatrix} \phi \\ \chi \end{bmatrix}_+,$$

corresponding to $e^{-ip \cdot x}$ solution and $e^{ip \cdot x}$ solution, respectively. Let's solve the $e^{-ip \cdot x}$ solution by directly multiplying the matrices out,

$$\begin{aligned} \phi &= \frac{1}{m}(E\phi - \boldsymbol{\sigma} \cdot \mathbf{p}\chi), \\ \chi &= \frac{1}{m}(\boldsymbol{\sigma} \cdot \mathbf{p}\phi - E\chi). \end{aligned}$$

Solving for χ in terms of ϕ using the second equation,

$$\chi = \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \phi.$$

We may want to plug this χ into the first equation, but eventually we should just get $E^2 - \mathbf{p}^2 = m^2$, which is already known for free particles. Therefore, the free-particle solution with $e^{-ip \cdot x}$ is

$$\psi(x) = \begin{bmatrix} \phi \\ \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \phi \end{bmatrix} e^{-ip \cdot x} \equiv u(p, s) e^{-ip \cdot x}, \quad (7.8)$$

where $u(p, s)$ is a 4-component object for the $e^{-ip \cdot x}$ solution that is specified by the 4-momentum p and a quantum number s , which we will later interpret as spin. There seems to be missing some information as we cannot determine ϕ , but it also means that we

have the freedom to choose any ϕ that span the 2-state space for spinors. The simplest basis is

$$\phi_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \phi_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

In this basis, the full Dirac-spinor for free particles:

$$\psi(x) = \begin{bmatrix} 1 \\ 0 \\ \frac{p_z}{E+m} \\ \frac{p_x+ip_y}{E+m} \end{bmatrix} e^{-ip \cdot x}, \quad \phi_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \psi(x) = \begin{bmatrix} 0 \\ 1 \\ \frac{p_x-ip_y}{E+m} \\ -\frac{p_z}{E+m} \end{bmatrix} e^{-ip \cdot x}, \quad \phi_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \quad (7.9)$$

The solution (7.9) is what we called the “positive-energy” solution. The energy is positive because the energy operator $i\partial_t$ brings down $+E$ from the exponential. It has two spin states ϕ_1 and ϕ_2 . For example, (7.9) can represent a free electron with either spin up or spin down. In the rest frame of the electron, $p = 0$ and $E = m$, the solution becomes very simple (in the Dirac-Pauli representation):

$$\psi(x) = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} e^{-imt}, \quad \phi_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \psi(x) = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} e^{-imt}, \quad \phi_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Going through the whole process again for $e^{+ip \cdot x}$, expressing ϕ in terms of χ in this case, we can find the “negative-energy” solution with $e^{+ip \cdot x}$,

$$\psi(x) = \begin{bmatrix} \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E+m} \chi \\ \chi \end{bmatrix} e^{+ip \cdot x} \equiv v(p, s) e^{+ip \cdot x}, \quad (7.10)$$

where $v(p, s)$ is similar to $u(p, s)$, but for the $e^{+ip \cdot x}$ solution. In the simplest basis,

$$\psi(x) = \begin{bmatrix} \frac{p_z}{E+m} \\ \frac{p_x+ip_y}{E+m} \\ 1 \\ 0 \end{bmatrix} e^{ip \cdot x}, \quad \chi_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \psi(x) = \begin{bmatrix} \frac{p_x-ip_y}{E+m} \\ -\frac{p_z}{E+m} \\ 0 \\ 1 \end{bmatrix} e^{ip \cdot x}, \quad \chi_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \quad (7.11)$$

It is called “negative-energy” just for historical reasons; the energy is not truly negative. It is interpreted as the free-antiparticle (e.g. positron) solution of the corresponding free-particle solution. We will find that both particle and antiparticle contribute positive energies in the Dirac Hamiltonian, just like in the complex Klein-Gordon Hamiltonian. The antiparticle solution also has two spin states, just like the particle solution. In the rest frame of the antiparticle,

$$\psi(x) = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} e^{+imt}, \quad \chi_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \psi(x) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} e^{+imt}, \quad \chi_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Normalization The spinors $u(p, s)$ and $v(p, s)$ are normalized such that

$$u^\dagger(p, s_1)u(p, s_2) = 2E \delta_{s_1, s_2}, \quad v^\dagger(p, s_1)v(p, s_2) = 2E \delta_{s_1, s_2}, \quad \bar{u}(p, s_1)v(p, s_2) = 0, \quad (7.12)$$

where $\bar{u}(p, s_1) = u(p, s_1)\gamma^0$, and s_1 and s_2 can take values 1 and 2 for spin-up and spin-down. (The “bars” will be useful when we study the Lorentz properties of Dirac spinors.)

Proof. For positive-energy solutions $u(p, s)$,

$$u(p, s) = \sqrt{E+m} \begin{bmatrix} \phi_s \\ \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E+m} \phi_s \end{bmatrix}, \quad u^\dagger(p, s) = \sqrt{E+m} \left[\phi_s^\dagger \quad \phi_s^\dagger \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E+m} \right].$$

Their inner product is

$$u^\dagger(p, s_1)u(p, s_2) = (E + m) \begin{bmatrix} \phi_{s_1}^\dagger & \phi_{s_1}^\dagger \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \end{bmatrix} \begin{bmatrix} \phi_{s_2} \\ \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \phi_{s_2} \end{bmatrix} = (E + m) \left[\phi_{s_1}^\dagger \phi_{s_2} + \left(\frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \right)^2 \phi_{s_1}^\dagger \phi_{s_2} \right].$$

It is easy to verify explicitly that $(\boldsymbol{\sigma} \cdot \mathbf{p})^2 = \mathbf{p}^2 \mathbf{1} = E^2 - m^2$. The spin basis ϕ_s is already normalized, so we have $\phi_{s_1}^\dagger \phi_{s_2} = \delta_{s_1, s_2}$. Then

$$u^\dagger(p, s_1)u(p, s_2) = (E + m) \left(\delta_{s_1, s_2} + \frac{E - m}{E + m} \delta_{s_1, s_2} \right) = 2E \delta_{s_1, s_2}.$$

Similarly, for negative-energy solutions, we can find that $v(p, s_1)v(p, s_2) = 2E \delta_{s_1, s_2}$. Lastly, the “inner product” of the positive- and negative-energy solutions is zero,

$$\bar{u}(p, s_1)v(p, s_2) = (E + m) \begin{bmatrix} \phi_{s_1}^\dagger & -\phi_{s_1}^\dagger \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \end{bmatrix} \begin{bmatrix} \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \chi_{s_2} \\ \chi_{s_2} \end{bmatrix} = (E + m) \left(\phi_{s_1}^\dagger \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \chi_{s_2} - \phi_{s_1}^\dagger \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \chi_{s_2} \right) = 0.$$

□

7.2.2 The Weyl Representation

In the Weyl representation, the Dirac equation is

$$i\gamma^\mu \partial_\mu \psi = m\psi, \quad \text{where} \quad \gamma^0 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \gamma^i = \begin{bmatrix} 0 & -\sigma_i \\ \sigma_i & 0 \end{bmatrix}.$$

We still assume that the free-particle solution has the form (neglecting normalization)

$$\psi(x) = \begin{bmatrix} \phi \\ \chi \end{bmatrix} e^{\pm i p \cdot x} = \begin{bmatrix} \phi \\ \chi \end{bmatrix} e^{\pm i(Et - \mathbf{p} \cdot \mathbf{x})}.$$

The Dirac equation in the Weyl representation leads to

$$\mp \begin{bmatrix} 0 & E + \boldsymbol{\sigma} \cdot \mathbf{p} \\ E - \boldsymbol{\sigma} \cdot \mathbf{p} & 0 \end{bmatrix} \begin{bmatrix} \phi \\ \chi \end{bmatrix}_\pm,$$

which represent two equations for $e^{-ip \cdot x}$ and $e^{ip \cdot x}$ solutions, respectively,

$$\begin{bmatrix} 0 & E + \boldsymbol{\sigma} \cdot \mathbf{p} \\ E - \boldsymbol{\sigma} \cdot \mathbf{p} & 0 \end{bmatrix} \begin{bmatrix} \phi \\ \chi \end{bmatrix}_- = m \begin{bmatrix} \phi \\ \chi \end{bmatrix}_-, \quad \begin{bmatrix} 0 & -E - \boldsymbol{\sigma} \cdot \mathbf{p} \\ -E + \boldsymbol{\sigma} \cdot \mathbf{p} & 0 \end{bmatrix} \begin{bmatrix} \phi \\ \chi \end{bmatrix}_+ = m \begin{bmatrix} \phi \\ \chi \end{bmatrix}_+.$$

First solve for the positive energy solution with $e^{-ip \cdot x}$,

$$\begin{aligned} \phi &= \frac{1}{m}(E + \boldsymbol{\sigma} \cdot \mathbf{p})\chi, \\ \chi &= \frac{1}{m}(E - \boldsymbol{\sigma} \cdot \mathbf{p})\phi. \end{aligned}$$

This is nice because χ is already expressed in terms of ϕ , so the free-particle solution for $e^{-ip \cdot x}$ is

$$\psi(x) = \begin{bmatrix} \phi \\ \frac{E - \boldsymbol{\sigma} \cdot \mathbf{p}}{m} \phi \end{bmatrix} e^{-ip \cdot x} \equiv u(p, s) e^{-ip \cdot x}, \quad \text{where} \quad \phi = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \quad (7.13)$$

Similarly, the negative energy solution is

$$\psi(x) = \begin{bmatrix} \phi \\ -\frac{E - \boldsymbol{\sigma} \cdot \mathbf{p}}{m} \phi \end{bmatrix} e^{+ip \cdot x} \equiv v(p, s) e^{+ip \cdot x}, \quad \text{where} \quad \phi = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \quad (7.14)$$

We know that the free-particle solution in the rest frame of the particle looks simple in the Dirac-Pauli representation. Here we can show that the free-particle solution looks simple in the relativistic limit in the Weyl representation. Choose the z -axis such that it

aligns with the momentum of the particle, so

$$\boldsymbol{\sigma} \cdot \mathbf{p} = \sigma_z p_z = p_z \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

In the relativistic limit ($|\mathbf{p}| \gg m$), the energy of the particle is

$$E = \sqrt{\mathbf{p}^2 + m^2} \simeq p_z \left(1 + \frac{m^2}{2p_z^2} \right).$$

The quantity $E - \boldsymbol{\sigma} \cdot \mathbf{p}$ simplifies to

$$E - \boldsymbol{\sigma} \cdot \mathbf{p} = p_z \left(1 + \frac{m^2}{2p_z^2} \right) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - p_z \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = p_z \begin{bmatrix} m^2/2p_z^2 & 0 \\ 0 & m^2/2p_z^2 + 2 \end{bmatrix} \simeq \begin{bmatrix} m^2/2p_z & 0 \\ 0 & 2p_z \end{bmatrix}$$

to lowest order in m , and

$$\frac{E - \boldsymbol{\sigma} \cdot \mathbf{p}}{m} = \begin{bmatrix} m/2p_z & 0 \\ 0 & 2p_z/m \end{bmatrix}.$$

For positive energy solutions, the spinors are then

$$\begin{aligned} \psi(x) &= \begin{bmatrix} \phi \\ \frac{E - \boldsymbol{\sigma} \cdot \mathbf{p}}{m} \phi \end{bmatrix} e^{-ip \cdot x} = \begin{bmatrix} 1 \\ 0 \\ m/2p_z \\ 0 \end{bmatrix} e^{-ip \cdot x} \simeq \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} e^{-ip \cdot x}, \quad \phi_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \\ \psi(x) &= \begin{bmatrix} \phi \\ \frac{E - \boldsymbol{\sigma} \cdot \mathbf{p}}{m} \phi \end{bmatrix} e^{-ip \cdot x} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 2p_z/m \end{bmatrix} e^{-ip \cdot x} \rightarrow \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} e^{-ip \cdot x}, \quad \phi_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \end{aligned}$$

In the second line, we normalized the spinor for the spin-down particle because p_z/m blows up. Therefore, while the Dirac-Pauli representation is convenient in the non-relativistic limit, the Weyl representation is more useful in the relativistic limit.

7.3 Lorentz Covariance

As mentioned before, though the spinor $\psi(x)$ is a 4-component object, it does not transform like a 4-vector. Consider an inertial frame Σ , the Dirac equation is

$$i\gamma^\mu \partial_\mu \psi(x) = m\psi(x).$$

Due to Lorentz covariance, we expect the Dirac equation in another inertial frame Σ' to have the same form:

$$i\gamma^\mu \partial'_\mu \psi'(x') = m\psi'(x'),$$

where ∂'_μ , x' , and ψ' are the transformed version of ∂_μ , x , and ψ . The transformation laws of ∂_μ and x^μ are already known:

$$\partial'_\mu = \Lambda^\nu{}_{\mu'} \partial_\nu = \frac{\partial x^\nu}{\partial x'^{\mu'}} \partial_\nu, \quad x'^{\mu'} = \Lambda^{\mu'}{}_\nu x^\nu = \frac{\partial x'^{\mu'}}{\partial x^\nu} x^\nu,$$

where $\Lambda^\mu{}_\nu$ are elements of the usual Lorentz transformation matrix. We need to find the transformation law of the spinor $\psi \rightarrow \psi'$ to achieve Lorentz covariance of the Dirac equation. Since both ∂_μ and x^μ transform linearly in the Dirac equation, we expect that ψ also transform linearly. Let the transformation law be

$$\psi'(x') = S(\Lambda)\psi(x). \tag{7.15}$$

Here the object $S(\Lambda)$ is a 4×4 matrix. It is also a function of the Lorentz transformation matrix Λ because $\psi(x)$ is a function of x , which transforms according to Λ . With this transformation law, we can write $\psi(x)$ in terms of $\psi'(x')$,

$$\psi(x) = S^{-1}(\Lambda)\psi'(x') = S^{-1}(\Lambda)\psi'(\Lambda x).$$

Meanwhile, if we perform a transformation from Σ' to Σ , the Lorentz transformation is now Λ^{-1} . Thus, we also have

$$\psi(x) = S(\Lambda^{-1})\psi'(x') = S(\Lambda^{-1})\psi'(\Lambda x).$$

This implies a basic property of $S(\Lambda)$,

$$S^{-1}(\Lambda) = S(\Lambda^{-1}).$$

To determine $S(\Lambda)$, multiply both sides of the Dirac equation by $S(\Lambda)$ and insert an identity $1 = S^{-1}(\Lambda)S(\Lambda)$ between ∂_μ and $\psi(x)$,

$$S(\Lambda)i\gamma^\mu\partial_\mu S^{-1}(\Lambda)S(\Lambda)\psi(x) = mS(\Lambda)\psi(x).$$

To express everything in primed objects, we write $\partial_\mu = \Lambda^\nu{}_\mu\partial'_\nu$ and $S(\Lambda)\psi(x) = \psi'(x')$. The Dirac equation becomes

$$S(\Lambda)i\gamma^\mu\Lambda^\nu{}_\mu S^{-1}(\Lambda)\partial'_\nu\psi'(x') = m\psi'(x').$$

(Note that $\Lambda^\nu{}_\mu$ commutes with everything in the equation because each $\Lambda^\nu{}_\mu$ is a *number*, an element of the Lorentz transformation matrix.) Comparing this with the desired covariant form $i\gamma^\nu\partial'_\nu\psi'(x') = m\psi'(x')$, we find that

$$\Lambda^\nu{}_\mu S(\Lambda)\gamma^\mu S^{-1}(\Lambda) = \gamma^\nu,$$

or, the object $S(\Lambda)$ must satisfy

$$\boxed{\Lambda^\nu{}_\mu\gamma^\mu = S^{-1}(\Lambda)\gamma^\nu S(\Lambda).} \quad (7.16)$$

If we can solve this equation for $S(\Lambda)$, we know how to transform Dirac spinors using (7.15).

7.3.1 Infinitesimal Transformations

Consider infinitesimal Lorentz transformations of the coordinates

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu = (\delta^\mu{}_\nu + \omega^\mu{}_\nu)x^\nu, \quad \text{where } |\omega^\mu{}_\nu| \ll 1.$$

The metric tensor is preserved under a Lorentz transformation, $\eta_{\mu\nu} = \Lambda^\alpha{}_\mu\Lambda^\beta{}_\nu\eta_{\alpha\beta}$. In terms of $\omega^\mu{}_\nu$,

$$\begin{aligned} \eta_{\mu\nu} &= (\delta^\alpha{}_\mu + \omega^\alpha{}_\mu)(\delta^\beta{}_\nu + \omega^\beta{}_\nu)\eta_{\alpha\beta} \\ &= (\delta^\alpha{}_\mu\delta^\beta{}_\nu + \delta^\alpha{}_\mu\omega^\beta{}_\nu + \omega^\alpha{}_\mu\delta^\beta{}_\nu + \omega^\alpha{}_\mu\omega^\beta{}_\nu)\eta_{\alpha\beta} \\ &= \eta_{\mu\nu} + \eta_{\mu\beta}\omega^\beta{}_\nu + \omega^\alpha{}_\mu\eta_{\alpha\nu} + \mathcal{O}(\omega^2). \end{aligned}$$

Neglecting second order terms for $|\omega^\mu{}_\nu| \ll 1$, an infinitesimal Lorentz transformation requires

$$\omega_{\mu\nu} + \omega_{\nu\mu} = 0. \quad (7.17)$$

Suppose $S(\Lambda)$ for infinitesimal Lorentz transformations is given by

$$S(\Lambda) = 1 - \frac{i}{4}\sigma_{\mu\nu}\omega^{\mu\nu},$$

where the second term is small, and $\sigma_{\mu\nu}$ represent 16 different matrices labeled by μ, ν . We would like to determine these matrices. The first step is to plug $\Lambda^\mu{}_\nu$ and $S(\Lambda)$ into (7.16),

$$(\delta^\nu{}_\mu + \omega^\nu{}_\mu)\gamma^\mu = \left(1 + \frac{i}{4}\sigma_{\alpha\beta}\omega^{\alpha\beta}\right)\gamma^\nu\left(1 - \frac{i}{4}\sigma_{\gamma\delta}\omega^{\gamma\delta}\right),$$

from which we can obtain

$$\begin{aligned} \omega^\nu{}_\mu\gamma^\mu &= \frac{i}{4}(\sigma_{\alpha\beta}\omega^{\alpha\beta}\gamma^\nu - \gamma^\nu\sigma_{\gamma\delta}\omega^{\gamma\delta}) + \mathcal{O}(\omega^2) \\ &= \frac{i}{4}(\sigma_{\alpha\beta}\omega^{\alpha\beta}\gamma^\nu - \gamma^\nu\sigma_{\alpha\beta}\omega^{\alpha\beta}) \\ \omega^\nu{}_\mu\gamma^\mu &= -\frac{i}{4}[\gamma^\nu, \sigma^{\alpha\beta}]\omega_{\alpha\beta}. \end{aligned}$$

We claim that $\sigma^{\alpha\beta} = \frac{i}{2}[\gamma^\alpha, \gamma^\beta]$ solves this equation.

Proof. With $\sigma^{\alpha\beta} = \frac{i}{2}[\gamma^\alpha, \gamma^\beta]$, the commutator $[\gamma^\nu, \sigma^{\alpha\beta}]$ becomes

$$[\gamma^\nu, \sigma^{\alpha\beta}] = \frac{i}{2}(\gamma^\nu \gamma^\alpha \gamma^\beta - \gamma^\nu \gamma^\beta \gamma^\alpha - \gamma^\alpha \gamma^\beta \gamma^\nu + \gamma^\beta \gamma^\alpha \gamma^\nu).$$

Using the anticommutation relation $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$, we can write

$$\gamma^\alpha \gamma^\beta \gamma^\nu = \gamma^\alpha (-\gamma^\nu \gamma^\beta + 2\eta^{\beta\nu}) = -\gamma^\alpha \gamma^\nu \gamma^\beta + 2\eta^{\beta\nu} \gamma^\alpha = \gamma^\nu \gamma^\alpha \gamma^\beta - 2\eta^{\alpha\nu} \gamma^\beta + 2\eta^{\beta\nu} \gamma^\alpha,$$

and similarly for $\gamma^\beta \gamma^\alpha \gamma^\nu$,

$$\gamma^\beta \gamma^\alpha \gamma^\nu = \gamma^\nu \gamma^\beta \gamma^\alpha - 2\eta^{\beta\nu} \gamma^\alpha + 2\eta^{\alpha\nu} \gamma^\beta.$$

Substituting these terms into $[\gamma^\nu, \sigma^{\alpha\beta}]$,

$$[\gamma^\nu, \sigma^{\alpha\beta}] = \frac{i}{2}(\gamma^\nu \gamma^\alpha \gamma^\beta - \gamma^\nu \gamma^\beta \gamma^\alpha - \gamma^\nu \gamma^\alpha \gamma^\beta + 2\eta^{\alpha\nu} \gamma^\beta - 2\eta^{\beta\nu} \gamma^\alpha - 2\eta^{\beta\nu} \gamma^\alpha + 2\eta^{\alpha\nu} \gamma^\beta) = 2i(\eta^{\alpha\nu} \gamma^\beta - \eta^{\beta\nu} \gamma^\alpha).$$

Then

$$-\frac{i}{4}[\gamma^\nu, \sigma^{\alpha\beta}]\omega_{\alpha\beta} = \frac{1}{2}(\eta^{\alpha\nu} \gamma^\beta - \eta^{\beta\nu} \gamma^\alpha)\omega_{\alpha\beta} = \frac{1}{2}\eta^{\alpha\nu} \gamma^\beta (\omega_{\alpha\beta} - \omega_{\beta\alpha}) = \eta^{\alpha\nu} \gamma^\beta \omega_{\alpha\beta} = \omega^\nu{}_\beta \gamma^\beta,$$

where we used $\omega_{\alpha\beta} + \omega_{\beta\alpha} = 0$ in the third equality. $\square$

In conclusion, $S(\Lambda)$ for infinitesimal Lorentz transformation is given by

$$S(\Lambda) = 1 - \frac{i}{4}\sigma_{\mu\nu}\omega^{\mu\nu}, \quad \text{where} \quad \sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]. \quad (7.18)$$

Finally, we can use the transformation law (7.15) to obtain $\psi'(x')$:

$$\psi'(x') = S(\Lambda)\psi(x) = \left(1 + \frac{1}{8}[\gamma_\mu, \gamma_\nu]\omega^{\mu\nu}\right)\psi(x).$$

A finite transformation of the spinor can be constructed by successively multiplying infinitesimal transformation $S(\Lambda)$ in the same direction,

$$\psi'(x') = \lim_{N \rightarrow \infty} \left(1 - \frac{i}{4N}\sigma_{\mu\nu}\omega^{\mu\nu}\right)^N \psi(x) = e^{-i\sigma_{\mu\nu}\omega^{\mu\nu}/4}\psi(x), \quad (7.19)$$

where now $\omega^\mu{}_\nu$ is finite, while $\omega^\mu{}_\nu/N$ is infinitesimal as $N \rightarrow \infty$. Raising $S(\Lambda)$ to the N th power as $N \rightarrow \infty$ gives the exponential by the definition of e .

7.3.2 Bilinear Covariants

Though spinors do not transform like a 4-vector under Lorentz transformations, certain quantities involving combinations of spinors do transform like a 4-vector. First, it is convenient to define a quantity

$$\bar{\psi} \equiv \psi^\dagger \gamma^0.$$

This is different from the Hermitian conjugate of ψ because of the γ^0 multiplied. Quantities that look like $\bar{\psi}[\cdot]\psi$ are called **bilinear covariants**. Before we study those bilinear covariants, it is useful to investigate the quantity $\gamma^0 S^\dagger(\Lambda) \gamma^0$. We claim that

$$\gamma^0 S^\dagger(\Lambda) \gamma^0 = S^{-1}(\Lambda), \quad (7.20)$$

which is proved in Appendix A.4.1. This gives the transformation law of $\bar{\psi}(x)$:

$$\bar{\psi}'(x') = [S\psi(x)]^\dagger \gamma^0 = \psi^\dagger(x) S^\dagger \gamma^0 = \psi^\dagger(x) \gamma^0 \gamma^0 S^\dagger \gamma^0 = \bar{\psi}(x) S^{-1}.$$

Now consider the simplest bilinear covariant is $\bar{\psi}\psi$. This is a Lorentz scalar, because it is invariant under a Lorentz transformation,

$$\bar{\psi}'(x')\psi'(x') = \bar{\psi}(x)S^{-1}S\psi(x) = \bar{\psi}(x)\psi(x).$$

Another important bilinear covariant is the probability current $J^\mu(x)$, defined as

$$J^\mu(x) \equiv \psi^\dagger(x) \gamma^0 \gamma^\mu \psi(x). \quad (7.21)$$

Let's see how $J^\mu(x)$ transform under a Lorentz transformation,

$$J'^\mu(x') = \bar{\psi}'(x') \gamma^\mu \psi'(x') = \bar{\psi}(x) S^{-1} \gamma^\mu S \psi(x).$$

But we know that $S^{-1} \gamma^\mu S = \Lambda^\mu{}_\nu \gamma^\nu$ from (7.16), so

$$J'^\mu(x') = \psi^\dagger(x) \gamma^0 \Lambda^\mu{}_\nu \gamma^\nu \psi(x) = \Lambda^\mu{}_\nu J^\nu(x).$$

This means J^μ is a valid 4-vector as it transforms by one factor of $\Lambda^\mu{}_\nu$.

7.3.3 Lorentz Properties of Spinors: the Dirac-Pauli Representation

To be written.

7.3.4 Lorentz Properties of Spinors: the Weyl Representation

To be written.

7.4 Quantization

So far everything we have done with the Dirac equation has treated $\psi(x)$ essentially as a wave function. We will see that quantization of the Dirac spinor to Dirac fields gets rid of the so-called “negative energies”. Following the usual step for quantization just like when we quantized the Klein-Gordon field, we need to

- Find the Lorentz-invariant action and Lagrangian for the Dirac spinor.
- Obtain the canonical conjugate momentum of the spinor to get the Hamiltonian.
- Impose quantization conditions on the spinor and its corresponding conjugate momentum.
- Expand the spinor in plane-wave modes and find the quantization conditions on the creation and annihilation operators.

7.4.1 The Dirac Lagrangian and Hamiltonian

The Lorentz-invariant Lagrangian (density) $\mathcal{L}$ for the Dirac spinor is

$$\mathcal{L}_D = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi. \quad (7.22)$$

One can check that $\mathcal{L}_D$ is Lorentz-invariant as it is composed of bilinear covariants. From now on we will write $\mathcal{L}_D = \mathcal{L}$ for simplicity. It is easy to check that the Euler-Lagrange equation for this Lagrangian by varying $\bar{\psi}$ gives the Dirac equation:

$$\frac{\partial \mathcal{L}}{\partial \bar{\psi}} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \bar{\psi})} \right) = 0 \implies i(\gamma^\mu \partial_\mu - m)\psi = 0.$$

Alternatively, if we vary ψ , we will get an equivalent Dirac equation for $\bar{\psi}$,

$$\frac{\partial \mathcal{L}}{\partial \psi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \psi)} \right) = 0 \implies i(\gamma^\mu \partial_\mu - m)\bar{\psi} = 0.$$

From the Lagrangian (7.22), we can obtain the conjugate momentum of ψ :

$$\pi(x) = \frac{\partial \mathcal{L}}{\partial \dot{\psi}} = \frac{\partial}{\partial \dot{\psi}} [i\bar{\psi} \gamma^0 \dot{\psi}] = i\bar{\psi} \gamma^0,$$

or, writing out $\bar{\psi} = \psi^\dagger \gamma^0$, we have

$$\pi(x) = i\psi^\dagger(x). \quad (7.23)$$

Of course, the spinor ψ and $\psi^\dagger$ are treated as independent fields, so there should be a conjugate momentum for $\psi^\dagger$ as well, but that is zero because there is no $\dot{\psi}^\dagger$ in the Lagrangian. The Hamiltonian density is given by

$$\mathcal{H} = \sum_i \pi_i \dot{\psi}_i - \mathcal{L} = i\psi^\dagger \dot{\psi} - \bar{\psi}(i\gamma^0 \partial_t + i\boldsymbol{\gamma} \cdot \nabla - m)\psi.$$

Note that $i\psi^\dagger \dot{\psi}$ cancels with $-\bar{\psi}(i\gamma^0 \partial_t)\psi$, so the Hamiltonian density is

$$\mathcal{H} = \bar{\psi}[\boldsymbol{\gamma} \cdot (-i\nabla) + m]\psi. \quad (7.24)$$

The terms in the bracket should look familiar—they are parts of the Dirac equation:

$$i\gamma^\mu \partial_\mu \psi = m\psi \quad \implies \quad i\gamma^0 \partial_t \psi = (-i\boldsymbol{\gamma} \cdot \nabla + m)\psi.$$

This means the Hamiltonian density can also be written simply as

$$\mathcal{H} = i\psi^\dagger \dot{\psi}. \quad (7.25)$$

(This form is useful for practical purposes, though the Hamiltonian density should not contain time-derivatives of the field. Equation (7.24) is a better way to write the Hamiltonian density.)

7.4.2 Mode Expansion

The Dirac equation admits free particle solutions (7.8) and (7.10),

$$\psi(x) = u(p, s)e^{-ip \cdot x} \quad \text{and} \quad \bar{\psi}(x) = v(p, s)e^{+ip \cdot x},$$

where p can be any 4-momentum and $s = 1, 2$ (representing spin up and spin down). The Dirac field is not a real field as it predicts particles and antiparticles. We expect its mode expansion to be similar to that of a complex Klein-Gordon field (6.10). Also, since the particle can take different spins, we should sum over all spin projections to obtain a complete set of plane waves. Hence the mode expansion for the Dirac field should look like

$$\psi(x) = \int \frac{d^3p}{(2\pi)^3 \sqrt{2E}} \sum_s [c(p, s)u(p, s)e^{-ip \cdot x} + d^\dagger(p, s)v(p, s)e^{+ip \cdot x}], \quad (7.26)$$

where $c(p, s)$ and $d^\dagger(p, s)$ are operators. In particular, $c^\dagger(p, s)$ and $c(p, s)$ are creation and annihilation operators of a particle; $d^\dagger(p, s)$ and $d(p, s)$ are creation and annihilation operators of an antiparticle. The creation operators can act on the vacuum $|0\rangle$, creating single-particle states. For example, the state $|p, s\rangle = c^\dagger(p, s)|0\rangle$ represent a particle with 4-momentum p and spin s . The annihilation operators satisfy $c(p, s)|0\rangle = d(p, s)|0\rangle = 0$.

7.4.3 Anticommutation Relations

The problem arises when we want to create two-particle states:

$$|p_1, s_1; p_2, s_2\rangle = c^\dagger(p_1, s_1)c^\dagger(p_2, s_2)|0\rangle.$$

The Dirac equation does not say that p_1 and p_2 must be different, nor that s_1 and s_2 must be different. In our formalism, we can choose the two-particle state to be

$$|p, s; p, s\rangle = c^\dagger(p, s)c^\dagger(p, s)|0\rangle.$$

However, it is well-known (long before the existence of quantum field theory) that spin-1/2 particles obey the **Pauli exclusion principle**: two identical fermions cannot occupy the same state. This means we must build in a condition that

$$c^\dagger(p, s)c^\dagger(p, s) = 0.$$

A more general statement about the Pauli exclusion principle is that multi-particle states of identical fermions must be totally antisymmetric under an interchange of any two particles. In other words,

$$|p_1, s_1; p_2, s_2\rangle = -|p_2, s_2; p_1, s_1\rangle \implies c^\dagger(p_1, s_1)c^\dagger(p_2, s_2)|0\rangle = -c^\dagger(p_2, s_2)c^\dagger(p_1, s_1)|0\rangle.$$

This can be summarized to anticommutation relations

$$\{c^\dagger(p_1, s_1), c^\dagger(p_2, s_2)\} = 0, \quad \{c(p_1, s_1), c(p_2, s_2)\} = 0, \quad \{c(p_1, s_1), c^\dagger(p_2, s_2)\} = (2\pi)^3 \delta^3(\mathbf{p}_1 - \mathbf{p}_2) \delta_{s_1, s_2}, \quad (7.27)$$

and the same set of relations for $d, d^\dagger$. In (7.27), the second relation is the Hermitian conjugate of the first. The third relation is a postulate, which is analogous to $[a(p), a^\dagger(p')] = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}')$ for the Klein-Gordon field, except that the commutator is replaced by an anticommutator, and there is a δ_{s_1, s_2} for spins. The particle and antiparticle operators c and d obey anticommutators as well:

$$\{c(p_1, s_1), d(p_2, s_2)\} = \{c^\dagger(p_1, s_1), d(p_2, s_2)\} = \{c(p_1, s_1), d^\dagger(p_2, s_2)\} = \{c^\dagger(p_1, s_1), d^\dagger(p_2, s_2)\}.$$

We will later see that why we must use anticommutation relations for fermions. These relations for the creation and annihilation operators above are equivalent to the equal-time anticommutation relations for Dirac fields

$$\{\psi_\alpha(t, \mathbf{x}), \pi_\beta(t, \mathbf{x}')\} = i\delta^3(\mathbf{x} - \mathbf{x}')\delta_{\alpha\beta}, \quad \{\psi(t, \mathbf{x}), \psi(t, \mathbf{x}')\} = 0, \quad \{\pi(t, \mathbf{x}), \pi(t, \mathbf{x}')\} = 0, \quad (7.28)$$

where $\alpha, \beta = 1, 2, 3, 4$ indicate the components of the Dirac fields. It is sometimes more useful to use $\pi = i\psi^\dagger$ and write

$$\{\psi_\alpha(t, \mathbf{x}), \psi^\dagger_\beta(t, \mathbf{x}')\} = \delta^3(\mathbf{x} - \mathbf{x}')\delta_{\alpha\beta}, \quad \{\psi(t, \mathbf{x}), \psi(t, \mathbf{x}')\} = 0, \quad \{\psi^\dagger(t, \mathbf{x}), \psi^\dagger(t, \mathbf{x}')\} = 0.$$

Just like the complex Klein-Gordon field, the Dirac field also has a conserved current. First, consider a global phase transformation:

$$\psi \rightarrow \psi' = e^{-i\alpha}\psi, \quad \bar{\psi} \rightarrow \bar{\psi}' = e^{+i\alpha}\bar{\psi},$$

where α is independent of x^μ . The Lagrangian (7.22) is obviously invariant under such a transformation. An infinitesimal global phase transformation has $e^{-i\delta\alpha} \simeq 1 - i\delta\alpha$. The Dirac fields changes by $\delta\psi = -i\psi\delta\alpha$ and $\delta\bar{\psi} = +i\bar{\psi}\delta\alpha$. The Lagrangian is unchanged,

$$0 = \delta\mathcal{L} = \delta\bar{\psi}\frac{\partial\mathcal{L}}{\partial\bar{\psi}} + \frac{\partial\mathcal{L}}{\partial\psi}\delta\psi + \delta(\partial_\mu\bar{\psi})\frac{\partial\mathcal{L}}{\partial(\partial_\mu\bar{\psi})} + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}\delta(\partial_\mu\psi).$$

For physical fields, they satisfy the Euler-Lagrange equations,

$$\frac{\partial\mathcal{L}}{\partial\psi} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}\right) = 0 \quad \text{and} \quad \frac{\partial\mathcal{L}}{\partial\bar{\psi}} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\bar{\psi})}\right) = 0.$$

Then the variation in the Lagrangian is

$$\begin{aligned} 0 = \delta\mathcal{L} &= \delta\bar{\psi}\partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\bar{\psi})}\right) + \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}\right)\delta\psi + \partial_\mu(\delta\bar{\psi})\frac{\partial\mathcal{L}}{\partial(\partial_\mu\bar{\psi})} + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}\partial_\mu(\delta\psi) \\ &= \partial_\mu\left[\delta\bar{\psi}\frac{\partial\mathcal{L}}{\partial(\partial_\mu\bar{\psi})} + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}\delta\psi\right]. \end{aligned}$$

The terms in the brackets is conserved. Computing these derivatives from the Lagrangian (7.22) gives

$$0 = \partial_\mu [i\bar{\psi}\gamma^\mu(-i\psi\delta\alpha)] = \partial_\mu(\bar{\psi}\gamma^\mu\psi)\delta\alpha.$$

Since $\delta\alpha$ is arbitrary, we define the conserved 4-current for the Dirac field to be

$$J^\mu_\psi \equiv \bar{\psi}\gamma^\mu\psi. \quad (7.29)$$

The symmetry operator (or number operator) associated with the conserved current is defined as

$$N_\psi \equiv \int d^3x J^t_\psi = \int d^3x \psi^\dagger\psi.$$

Using the mode expansion form of the Dirac field (7.26), the symmetry operator is given by

$$N_\psi = \int \frac{d^3p}{(2\pi)^3} \sum_s [c^\dagger(p, s)c(p, s) + d(p, s)d^\dagger(p, s)].$$

After normal ordering (using the anticommutation relations instead of commutation relations), it becomes

$$N_\psi = \int \frac{d^3p}{(2\pi)^3} \sum_s [c^\dagger(p, s)c(p, s) - d^\dagger(p, s)d(p, s)]. \quad (7.30)$$

We can also compute the Hamiltonian in mode expansion. The result is

$$H = \int \frac{d^3p}{(2\pi)^3} \sum_s [c^\dagger(p, s)c(p, s) - d(p, s)d^\dagger(p, s)] E,$$

and after normal ordering,

$$H = \int \frac{d^3p}{(2\pi)^3} \sum_s [c^\dagger(p, s)c(p, s) + d^\dagger(p, s)d(p, s)] E. \quad (7.31)$$

The symmetry operator (7.30) and the Hamiltonian (7.31) are derived in Appendix A.4.2 (to be written). Again, both the particle and antiparticle gives positive energies. There is no weird negative energy in the Hamiltonian. On the other hand, the minus sign in N_ψ infers that the antiparticle has opposite charge to the particle. The anticommutation relations are important for spin-1/2 particles. If we were to use commutation relations instead of anticommutation ones, the signs in N_ψ and H would be reversed after normal ordering, i.e. $N_\psi \sim c^\dagger c + d^\dagger d$ and $H \sim c^\dagger c - d^\dagger d$. In this case, the particle and antiparticle have the same charge, but opposite energy, which is not observed in experiments.

7.4.4 Relationship between Wave Functions and Dirac Fields

Consider a state with one quantum of Dirac field with 4-momentum p' and spin s' . For instance, the state can be a free electron,

$$|p', s'\rangle_{e^-} = c^\dagger(p', s') |0\rangle.$$

Here is a quantity to calculate:

$$\langle 0 | \psi(x) | p', s' \rangle_{e^-} = \langle 0 | \int \frac{d^3p}{(2\pi)^3 \sqrt{2E}} \sum_s [c(p, s)u(p, s)e^{-ip \cdot x} + d^\dagger(p, s)v(p, s)e^{+ip \cdot x}] c^\dagger(p', s') | 0 \rangle.$$

We can immediately see that the second term does not contribute. There can be two explanations. Mathematically, $\langle 0 | d^\dagger(p, s) = 0$ because it is the Hermitian conjugate of $d(p, s) | 0 \rangle$, which vanishes as the lowering operator annihilates the vacuum. Physically, if $d^\dagger c^\dagger$ do contribute, then it will create an antiparticle (a positron) when acting on $|0\rangle$. When the final state is projected on the vacuum $\langle 0 |$, the amplitude is zero because the vacuum does not contain any particle or antiparticle. The only nonzero amplitude is the $cc^\dagger$ term:

$$\langle 0 | c(p, s) c^\dagger(p', s') | 0 \rangle = -\langle 0 | c^\dagger(p', s') c(p, s) | 0 \rangle + \langle 0 | (2\pi)^3 \delta^3(\mathbf{p}' - \mathbf{p}) \delta_{s', s} | 0 \rangle = (2\pi)^3 \delta^3(\mathbf{p}' - \mathbf{p}) \delta_{s', s}.$$

In the first equality, we used the anticommutation relation (7.28). Then,

$$\langle 0 | \psi(x) | p', s' \rangle_{e^-} = \int \frac{d^3p}{(2\pi)^3 \sqrt{2E}} (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}') \delta_{s, s'} u(p, s) e^{-ip \cdot x} = \frac{1}{\sqrt{2E}} u(p', s') e^{-ip' \cdot x}.$$

The result is the usual wave function for the free particle solution. We shall conclude that the wave function corresponds to the vacuum-to-one-particle-state matrix elements of the field operators.

7.4.5 The Spin-Statistics Theorem

To be written.

8 ELEMENTARY PARTICLE INTERACTIONS

In the previous two chapters on quantum field theory, we investigated only plane-wave solutions, or free particles. There are not many interesting things happening because plane waves just oscillates forever when there are no interactions. This chapter starts to discuss elementary particle interactions quantitatively. We shall make connections to Feynman diagrams discussed in Chapter 4.

8.1 Perturbation Theory in the Interaction Picture

8.1.1 The Schrödinger and the Heisenberg Picture

In ordinary quantum mechanics, especially wave mechanics, we typically work in the **Schrödinger picture** where the governing equation is the Schrödinger equation,

$$H |\psi(t)\rangle = i \frac{d}{dt} |\psi(t)\rangle.$$

The time dependence is in the state $|\psi(t)\rangle$ but not in operators. If H is independent of time, the solution to the Schrödinger equation is simple:

$$|\psi(t)\rangle = e^{-iHt} |\psi(0)\rangle,$$

where $|\psi(0)\rangle$ is some initial state at $t = 0$. The operator $U(t) = e^{-iHt}$ is the time-evolution operator. The amplitude of some operator A acting on $|\psi(t)\rangle$, projecting on another state $|\phi(t)\rangle$, is given by

$$\langle \phi(t) | A_S | \psi(t) \rangle_S,$$

where we use the subscript S to denote the Schrödinger picture. In terms of the time-evolution operator, we can write the amplitude as

$$\langle \phi(0) | e^{iHt} A_S e^{-iHt} | \psi(0) \rangle_S = \langle \phi | A_H | \psi \rangle_H,$$

where $A_H(t) = e^{iHt} A_S e^{-iHt}$ is the operator A in the **Heisenberg picture**. The states in the Heisenberg picture does not change with time: $|\phi\rangle_H = |\phi(0)\rangle_S$ and $|\psi\rangle_H = |\psi(0)\rangle_S$. Instead, the time dependence is in the operators. The Heisenberg picture is more closely related to classical mechanics because we can obtain equations of motion of operators. Taking the time-derivative of $A_H(t)$,

$$\frac{dA_H}{dt} = iH e^{iHt} A_S e^{-iHt} + e^{iHt} A_S e^{-iHt} (-iH) = i[H, e^{iHt} A_S e^{-iHt}].$$

Turning $e^{iHt} A_S e^{-iHt}$ back to the Heisenberg picture, we get the equation of motion:

$$\frac{dA_H}{dt} = -i[A_H, H].$$

This is analogous to the classical equation of motion, where the commutator is replaced by the Poisson bracket (with some factor of $i\hbar$).

8.1.2 The Interaction Picture

To study particle interactions, we use a third picture called the **interaction picture**. The interaction picture is between the Schrödinger and the Heisenberg picture. We divide the Hamiltonian (in the Schrödinger picture) into two parts, one associated with the free propagation of the system and the other with interactions:

$$H = H_0 + H'(t).$$

In the interaction picture, the operators carry the time dependence generated by the free Hamiltonian H_0 ; the states carry the time dependence associated with the interaction Hamiltonian $H'(t)$. An operator A_I and a state $|\psi(t)\rangle_I$ in the interaction picture are defined as

$$A_I(t) \equiv e^{iH_0t} A_S e^{-iH_0t} \quad \text{and} \quad |\psi(t)\rangle_I \equiv e^{iH_0t} |\psi(t)\rangle_S.$$

To study how $|\psi(t)\rangle_I$ changes with time, we compute

$$i \frac{d}{dt} |\psi(t)\rangle_I = i \frac{d}{dt} [e^{iH_0t} |\psi(t)\rangle_S] = e^{iH_0t} \left[-H_0 |\psi(t)\rangle_S + i \frac{d}{dt} |\psi(t)\rangle_S \right].$$

By the Schrödinger equation, $i d|\psi(t)\rangle_S/dt = H |\psi(t)\rangle = (H_0 + H') |\psi(t)\rangle_S$, so

$$i \frac{d}{dt} |\psi(t)\rangle_I = e^{iH_0t} - H_0 |\psi(t)\rangle_S + (H_0 + H') |\psi(t)\rangle_S = e^{iH_0t} H' |\psi(t)\rangle_S.$$

Inserting the identity operator $e^{-iH't} e^{iH_0t}$,

$$i \frac{d}{dt} |\psi(t)\rangle_I = (e^{iH_0t} H' e^{-iH_0t}) [e^{iH_0t} |\psi(t)\rangle_S].$$

We identify the terms in the parenthesis to be the interaction Hamiltonian in the interaction picture, $e^{iH_0t} H' e^{-iH_0t} = H'_I$, and the terms in the bracket to be the state in the interaction picture, $e^{iH_0t} |\psi(t)\rangle_S = |\psi(t)\rangle_I$. This gives the evolution of the state in the interaction picture,

$$\boxed{i \frac{d}{dt} |\psi(t)\rangle_I = H'_I |\psi(t)\rangle_I, \quad \frac{dH'_I}{dt} = -i[H'_I, H_0].} \quad (8.1)$$

The first equation looks just like the Schrödinger equation, but the Hamiltonian H'_I is only the interaction piece in the interaction picture. The second equation is the equation of motion of H'_I , which can be derived from the definition $H'_I(t) = e^{iH_0t} H_I e^{-iH_0t}$. Note that the prime ' and the subscript I do not mean the same thing. The prime means that H' is the interaction piece of the total Hamiltonian, while the subscript I indicates the picture we are working in—the interaction picture.

In fact, our treatment of quantum field theory in the previous two chapters is consistent with the interaction picture. For example, the free Dirac field is,

$$\psi(t) = \int \frac{d^3p}{(2\pi)^3 \sqrt{2E}} \sum_s [c(p, s) u(p, s) e^{-ip \cdot x} + d^\dagger(p, s) v(p, s) e^{+ip \cdot x}].$$

Since fields are operators, they carry the time dependence (in the exponentials) as expected. The occupation number states, such as $|p, s\rangle$, does not carry any time dependence because there is no interaction Hamiltonian introduced. In other words, $H'_I = 0$, so the state does not evolve, $d|\psi(t)\rangle_I/dt = 0$, according to (8.1).

8.1.3 Perturbation Theory

To study particle interactions, we assume that they occur in some finite region of spacetime. The initial and final states of interest, denoted as $|i\rangle$ and $|f\rangle$, are taken to be free-particle states, which in general contain multiple free particles. This treatment will be an excellent approximation in detectors. To be concrete, let $|i\rangle \equiv |\psi(t \rightarrow -\infty)\rangle$ be the initial state and $|\psi(t \rightarrow +\infty)\rangle$ be the final state. Define the S operator such that

$$|\psi(t \rightarrow +\infty)\rangle = S |\psi(t \rightarrow -\infty)\rangle = S |i\rangle.$$

In general, the final state $|\psi(t \rightarrow +\infty)\rangle$ is a superposition of free-particle states $|f\rangle$. We therefore want to compute the amplitude

$$\langle f | \psi(t \rightarrow +\infty) \rangle = \langle f | S | i \rangle \equiv S_{fi}.$$

The quantities S_{fi} are called S -matrix elements. They can be obtained using perturbation theory in the interaction picture. To begin with, we turn the Schrödinger equation (8.1) into an integral equation,

$$i \frac{d}{dt} |\psi(t)\rangle_I = H'_I |\psi(t)\rangle \implies |\psi(t)\rangle_I = |i\rangle - i \int_{-\infty}^t dt' H'_I(t') |\psi(t')\rangle_I.$$

In order to know $|\psi(t)\rangle_I$, we need to know $|\psi(t)\rangle_I$ because it appears on both sides. The general method is to do a series expansion for $|\psi(t)\rangle_I$ and solve the integral equation iteratively order by order.

The zeroth order is just the initial state, $|\psi^{(0)}(t)\rangle_I = |i\rangle$. Inserting this to the RHS of the integral equation gives the first-order state:

$$|\psi^{(1)}(t)\rangle_I = |i\rangle + (-i) \int_{-\infty}^t dt_1 H'_I(t_1) |i\rangle.$$

Inserting the first-order state into the integral equation again gives the second-order state:

$$\begin{aligned} |\psi^{(2)}(t)\rangle_I &= |i\rangle + (-i) \int_{-\infty}^t H'_I(t_1) \left[|i\rangle + (-i) \int_{-\infty}^{t_1} dt_2 H'_I(t_2) |i\rangle \right] \\ &= |i\rangle + (-i) \int_{-\infty}^t dt_1 H'_I(t_1) |i\rangle + (-i)^2 \int_{-\infty}^t dt_1 \int_{-\infty}^{t_1} dt_2 H'_I(t_1) H'_I(t_2) |i\rangle. \end{aligned}$$

We can do this iteratively to get better and better approximation using

$$|\psi^{(n+1)}(t)\rangle = |i\rangle + (-i) \int_{-\infty}^t dt_1 H'_I(t_1) |\psi^{(n)}(t_1)\rangle.$$

Note that in the second-order state (and higher-order state), we express everything in terms of operators acting on $|i\rangle$. Then the S operator, defined as $|\psi(t \rightarrow \infty)\rangle = S |i\rangle$, will look like

$$S = 1 + (-i) \int_{-\infty}^{\infty} dt_1 H'_I(t_1) + (-i)^2 \int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{t_1} dt_2 H'_I(t_1) H'_I(t_2) + \cdots.$$

In a summation form, the S operator is

$$S = \sum_{n=0}^{\infty} (-i)^n \int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{t_1} dt_2 \cdots \int_{-\infty}^{t_{n-1}} dt_n H'_I(t_1) H'_I(t_2) \cdots H'_I(t_n).$$

In terms of the Hamiltonian density $\mathcal{H}'_I(t)$, we can write the S operator in a more covariant way,

$$\begin{aligned} S &= 1 + (-i) \int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{\infty} d^3x_1 \mathcal{H}'_I(t_1, \mathbf{x}_1) + (-i)^2 \int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{\infty} d^3x_1 \int_{-\infty}^{t_1} dt_2 \int_{-\infty}^{\infty} d^3x_2 \mathcal{H}'_I(t_1, \mathbf{x}_1) \mathcal{H}'_I(t_2, \mathbf{x}_2) + \cdots \\ &= 1 + (-i) \int d^4x_1 \mathcal{H}'_I(x_1) + (-i)^2 \int_{t_1 > t_2} d^4x_1 d^4x_2 \mathcal{H}'_I(x_1) \mathcal{H}'_I(x_2). \end{aligned}$$

The last problem in this perturbation theory is that the integration over t_n always has the upper limit being t_{n-1} . That is, we need to track the integration limits such that $t_1 > t_2$, $t_2 > t_3$, and so on, in the integral. It would be nice if those limits are $-\infty$ and $+\infty$. To do this, we introduce the **time-ordering operator** T , defined as

$$T\{f(t_1)f(t_2)\} = \begin{cases} f(t_1)f(t_2), & t_1 > t_2, \\ f(t_2)f(t_1), & t_2 > t_1. \end{cases} \quad (8.2)$$

Mathematically, it is more useful to express the time-ordering operator in terms of the θ -function,

$$T\{f(t_1)f(t_2)\} = \theta(t_1 - t_2)f(t_1)f(t_2) + \theta(t_2 - t_1)f(t_2)f(t_1), \quad \text{where } \theta(x) = \begin{cases} 0, & x < 0, \\ 1, & x > 0. \end{cases} \quad (8.3)$$

There is an identity about the time-ordering operator saying that

$$\int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{t_1} dt_2 \cdots \int_{-\infty}^{t_{n-1}} dt_n f(t_1)f(t_2) \cdots f(t_n) = \frac{1}{n!} \int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{\infty} dt_2 \cdots \int_{-\infty}^{\infty} dt_n T\{f(t_1)f(t_2) \cdots f(t_n)\} \quad (8.4)$$

See Appendix A.5.1 for proof (to be written). Using this result, the S operator has its final form

$$\begin{aligned} S &= 1 + (-i) \int dt_1 H'_I(t_1) + \frac{(-i)^2}{2} \int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{\infty} dt_2 T\{H'_I(t_1)H'_I(t_2)\} + \cdots \\ &= \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} \int d^4x_1 d^4x_2 \cdots d^4x_n T\{\mathcal{H}'_I(x_1)\mathcal{H}'_I(x_2) \cdots \mathcal{H}'_I(x_n)\} \end{aligned} \quad (8.5)$$

To be written.

8.2 The “ABC” Theory

8.2.1 Setting up a Toy Theory

In this section, we will apply our perturbation theory to a decay process. We shall call this toy theory the “ABC” theory. Let particle A , B , and C be excitations of three independent, real, Klein-Gordon field $\phi_A(x)$, $\phi_B(x)$, $\phi_C(x)$ with corresponding masses m_A , m_B , and m_C . The mode expansion of each Klein-Gordon field is given by

$$\phi_i(x) = \int \frac{d^3p}{(2\pi)^3 \sqrt{2E}} \left[a_i(p) e^{-ip \cdot x} + a_i^\dagger e^{ip \cdot x} \right],$$

for $i = A, B, C$ with commutation relations

$$[a_i(p), a_j^\dagger(p')] = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}') \delta_{ij}, \quad [a_i(p), a_j(p')] = [a_i^\dagger(p), a_j^\dagger(p')] = 0.$$

The free Hamiltonian is the sum of the three separated free Hamiltonians for ϕ_A , ϕ_B , and ϕ_C . Referring to (6.3), the free Hamiltonian should be

$$H_0 = \frac{1}{2} \sum_i \int d^3x \left[\pi_i^2 + (\nabla \phi_i)^2 + m_i^2 \phi_i^2 \right] \quad (8.6)$$

In our toy theory, we assume the interaction Hamiltonian to be

$$H_I' = \int d^3x \mathcal{H}_I'(\phi_1, \phi_2, \phi_3) \equiv g \int d^3x \phi_A(x) \phi_B(x) \phi_C(x),$$

where g is a coupling constant. We see that ϕ_A , ϕ_B , and ϕ_C are multiplied together in the interaction Hamiltonian so that they are coupled. This toy model is pedagogically useful because almost all interaction Hamiltonians are in this form or similar.

Let's compute the S -matrix element for the decay process $C \rightarrow A + B$. The initial state contains one particle C , so the state is $|p_C\rangle = \sqrt{2E_C} a_C^\dagger(p_C) |0\rangle$, where $\sqrt{2E_C}$ is just a normalization factor. We choose the final state to have one particle A and one particle B , so the state is $|p_A, p_B\rangle = \sqrt{4E_A E_B} a_A^\dagger(p_A) a_B^\dagger(p_B) |0\rangle$. The S -matrix element is

$$S_{fi} = \langle p_A, p_B | S | p_C \rangle.$$

Recall that the S operator is given by (8.5),

$$S = 1 + (-i) \int dt_1 H_I'(t_1) + \frac{(-i)^2}{2} \int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{\infty} dt_2 T\{H_I'(t_1) H_I'(t_2)\} + \cdots.$$

The zeroth-order approximation gives nothing unless the initial state and final state are the same,

$$S_{fi}^{(0)} = \langle p_A, p_B | 1 | p_C \rangle = 0.$$

A more general statement would be $S_{fi}^{(0)} = \delta_{fi}$.

8.2.2 First-Order Perturbation: The Fundamental Vertex

The first-order term, here denoted as $\mathcal{A}_{fi} = S_{fi}^{(1)}$, is

$$\mathcal{A}_{fi} = \langle p_A, p_B | \int d^4x [-ig \phi_A(x) \phi_B(x) \phi_C(x)] | p_C \rangle.$$

Since each $\phi_i(x)$ can be expanded to something like $a_i + a_i^\dagger$, we expect the product of three Klein-Gordon fields to have 8 terms:

$$a_A a_B a_C, \quad a_A a_B a_C^\dagger, \quad a_A a_B^\dagger a_C, \quad a_A a_B^\dagger a_C^\dagger, \quad a_A^\dagger a_B a_C, \quad a_A^\dagger a_B a_C^\dagger, \quad a_A^\dagger a_B^\dagger a_C, \quad a_A^\dagger a_B^\dagger a_C^\dagger.$$

Without computing all the terms, we can first analyze the problem physically. The initial state has a particle C but the final state has no particle C . This means we need a piece in the interaction Hamiltonian that annihilates the particle C , so there should be

an a_C in this piece. On the other hand, there is no particle A or B in the initial state but they are in the final state, so there should be $a_A^\dagger$ and $a_B^\dagger$ that creates them. Hence the only term in the interaction Hamiltonian that contributes involves $a_A^\dagger a_B^\dagger a_C$.

Indeed, we can check mathematically that all other terms vanishes in $\mathcal{A}_{fi}$. In particular, $\mathcal{A}_{fi}$ involves

$$\phi_C(x) |p_C\rangle = \int \frac{d^3p}{(2\pi)^3 \sqrt{2E}} \left[a_C(p) e^{-ip \cdot x} + a_C^\dagger(p) e^{ip \cdot x} \right] \sqrt{2E_C} a_C^\dagger(p_C) |0\rangle.$$

and

$$\langle p_A, p_B | \phi_A(x) \phi_B(x) = \langle 0 | \sqrt{4E_A E_B} a_A(p_A) a_B(p_B)$$

so everything is sandwiched between $\langle 0 |$ and $|0\rangle$. The creation and annihilation operators commute if they belong to different fields. Then in $\langle 0 | \phi_C(x) | p_C \rangle$, the $a_C^\dagger$ term does not contribute because $\langle 0 | a_C^\dagger = 0$. Thus, the effective term in $\phi_C(x) | p_C \rangle$ is

$$\begin{aligned} \phi_C(x) | p_C \rangle &= \int \frac{d^3p}{(2\pi)^3 \sqrt{2E}} a_C(p) e^{-ip \cdot x} \sqrt{2E_C} a_C^\dagger(p_C) |0\rangle \\ &= \int \frac{d^3p}{(2\pi)^3 \sqrt{2E}} e^{-ip \cdot x} \left\{ a_C^\dagger(p_C) a_C(p) + [a_C(p), a_C^\dagger(p_C)] \right\} \sqrt{2E_C} |0\rangle \\ &= \int \frac{d^3p}{(2\pi)^3 \sqrt{2E}} e^{-ip \cdot x} \sqrt{2E_C} (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}_C) |0\rangle \\ &= e^{-ip_C \cdot x} |0\rangle. \end{aligned}$$

Similarly, simply replacing C with A or B in this result and take the Hermitian conjugate, we can get

$$\langle p_A, p_B | \phi_A(x) \phi_B(x) = \langle 0 | e^{+i(p_A + p_B) \cdot x}$$

when it acts on $|0\rangle$. Combining these results,

$$\langle p_A, p_B | \phi_A(x) \phi_B(x) \phi_C(x) | p_C \rangle = \langle 0 | e^{i(p_A + p_B - p_C) \cdot x} |0\rangle = e^{i(p_A + p_B - p_C) \cdot x}.$$

Substituting this into the amplitude $\mathcal{A}_{fi}$,

$$\mathcal{A}_{fi} = -ig \int d^4x e^{i(p_A + p_B - p_C) \cdot x} = -ig (2\pi)^4 \delta^4(p_A + p_B - p_C). \quad (8.7)$$

Notice that the conservation of 4-momentum is automatically built in the amplitude. There is a vertex factor $-ig$ in the first-order S -matrix element. We can imagine that the n -order term in the S -matrix element should carry higher powers of vertex factors $(-ig)^n$. Of course, (8.7) is just one of the nonzero S -matrix element to first order. We can choose a different initial state and final state, such as $|i\rangle = |p_A, p_C\rangle$ and $|f\rangle = |p_B\rangle$. Then we can almost immediately read out the amplitude $\mathcal{A}_{fi} = -ig (2\pi)^4 \delta^4(p_B - p_A - p_C)$.

8.2.3 Second-Order Perturbation: The Propagator

$$\mathcal{A}_{fi}^{(2)} = \frac{(-ig)^2}{2} \int d^4x_1 d^4x_2 \langle f | T \{ \mathcal{H}'_I(x_1) \mathcal{H}'_I(x_2) \} | i \rangle.$$

In our ABC theory.

$$\mathcal{A}_{fi}^{(2)} = \frac{(-ig)^2}{2} \sqrt{16E_A E'_A E_B E'_B} \int d^4x_1 d^4x_2 \langle 0 | a_A(p'_A) a_B(p'_B) T \{ \phi_A(x_1) \phi_B(x_1) \phi_C(x_1) \phi_A(x_2) \phi_B(x_2) \phi_C(x_2) a_A^\dagger(p_A) a_B^\dagger(p_B) \} | 0 \rangle.$$

To be written.

8.3 Decay Rates

8.3.1 Non-Relativistic Phase Space

Consider a decay process $A \rightarrow 1 + 2 + \dots$. In non-relativistic quantum mechanics, the transition rate or decay rate Γ_{fi} from an initial state $|i\rangle$ to a final state $|f\rangle$ is given by

$$\Gamma_{fi} = 2\pi |T_{fi}|^2 \rho(E_i).$$

This is known as **Fermi's golden rule**. Here T_{fi} is the transition matrix element and $\rho(E_i)$ is the density of states,

$$\rho(E_i) = \left. \frac{dn}{dE_f} \right|_{E_i},$$

where dn is the number of possible states in the energy range $E \rightarrow E_f + dE$. The density of states can also be written as an integral over all final state energy E ,

$$\rho(E_i) = \int \frac{dn}{dE} \delta(E_i - E) dE$$

so that Fermi's golden rule becomes

$$\Gamma_{fi} = 2\pi \int |T_{fi}|^2 \delta(E_i - E) dn. \quad (8.8)$$

Each particle in the final state is a plane wave, $\psi(\mathbf{x}, t) = A e^{-i(Et - \mathbf{p} \cdot \mathbf{x})}$. We usually adopt the *box normalization* for the free-particle states. Imagine that the state ψ is normalized such that there is only one particle in a box of length L and volume $V = L^3$,

$$\int_V |\psi|^2 d^3x = 1 \quad \Rightarrow \quad \psi(\mathbf{x}, t) = \frac{1}{\sqrt{V}} e^{i(\mathbf{p} \cdot \mathbf{x} - Et)}.$$

We can impose the periodic boundary condition $\psi(x + a, y, z) = \psi(x, y, z)$ and similar for the y - and z -directions. The wave function is restricted to be

$$\psi(\mathbf{x}, t) = \frac{1}{\sqrt{V}} e^{-i(Et - p_x x + p_y y + p_z z)},$$

where the momentum is quantized to

$$(p_x, p_y, p_z) = \frac{2\pi}{L} (n_x, n_y, n_z), \quad n \in \mathbb{Z}.$$

(Alternatively, the box normalization is a natural model for the 3-dimensional infinite square well with $\psi(\mathbf{r}, t) = 0$ at the boundary. The wave functions would be a standing wave $\psi(\mathbf{x}, t) = A \sin(p_x x) \sin(p_y y) \sin(p_z z) e^{-iEt}$. This result is equivalent to the one from the periodic boundary condition.) Each state in momentum space has a volume of

$$d^3p = \left(\frac{2\pi}{L} \right)^3 = \frac{(2\pi)^3}{V}.$$

If we want to find the number of states dn within momentum $p \rightarrow p + dp$, it is more convenient to work in spherical coordinates in momentum space. Then dn would be the total volume of a shell of radius p and thickness dp divided by the volume occupied by one state,

$$dn = \frac{4\pi p^2 dp}{(2\pi)^3/V} \quad \Rightarrow \quad \frac{dn}{dp} = \frac{4\pi p^2}{(2\pi)^3} V.$$

This gives the density of states

$$\rho(E) = \frac{dn}{dE} = \frac{dn}{dp} \left| \frac{dp}{dE} \right|.$$

The final result of the decay rate does not depend on the volume of the normalization box: recall that the transition matrix element is

$$T_{fi} \sim \langle \psi_f | H | \psi_i \rangle = \int_V \psi_f^* H \psi_i d^3x \propto \frac{1}{V}.$$

The V in the density of states will cancel with the $1/V$ in $|T_{fi}|^2$. Therefore, it is convenient to just set $V = 1$ and not worry about factors of V anymore. If there are multiple particles in the final state, then the number of possible states dn_j for particle j in a momentum space volume d^3p_j is

$$dn_j = \frac{d^3p_j}{(2\pi)^3}.$$

In a final state of N particles, there are $N - 1$ independent momenta because the final one can be deduced from the conservation of momentum. The total number of independent states is then

$$dn = \prod_{j=1}^{N-1} dn_j = \prod_{j=1}^{N-1} \frac{d^3 p_j}{(2\pi)^3}.$$

In particle physics, it is more useful to multiply a delta function as a constraint of momentum conservation and write

$$dn = (2\pi)^3 \prod_{j=1}^N \frac{d^3 p_j}{(2\pi)^3} \delta^3 \left[\mathbf{p}_i - \sum_{j=1}^N \mathbf{p}_j \right]. \quad (8.9)$$

This is the non-relativistic **phase space factor** for a decay process.

8.3.2 Lorentz-Invariant Phase Space

Obviously the above discussion is not Lorentz invariant. The volume of the box moving with speed v is Lorentz contracted by a factor of $1/\gamma$ in its direction of motion, $L^3 \rightarrow L^3/\gamma$. We can write the Lorentz factor $\gamma = E/m$ for a particle of mass m . The normalization of “one particle per volume” then becomes “ E/m particles per volume”. This is saying that the wave function normalization should be proportional to E , since m is Lorentz invariant. The conventional relativistic normalization is $2E$ particles per unit volume,

$$\int_V |\psi'|^2 d^3x = 2E,$$

where $\psi' = \sqrt{2E}\psi$ and ψ is the usual wave function normalizing to 1. This is why there is always a factor of $\sqrt{2E}$ in front of an excitation state, $|p\rangle = \sqrt{2E} a^\dagger(p) |0\rangle$. For a general process $A + B + \dots \rightarrow 1 + 2 + \dots$, the S -matrix elements are related to the transition matrix elements by

$$S_{fi} = \sqrt{2E_1 2E_2 \dots 2E_A 2E_B \dots} T_{fi}.$$

Then Fermi’s golden rule (8.8) for a decay process $a \rightarrow 1 + 2 + \dots + N$ becomes

$$\Gamma_{fi} = \frac{(2\pi)^4}{2E_i} \int |S_{fi}|^2 \delta \left[E_i - \sum_j E_j \right] \delta^3 \left[\mathbf{p}_i - \sum_j \mathbf{p}_j \right] \prod_j^N \frac{d^3 p_j}{(2\pi)^3 2E_j}. \quad (8.10)$$

The phase space factor

$$d\text{Lips} = \delta^4 \left[p_i - \sum_j p_j \right] \prod_j^N \frac{d^3 p_j}{(2\pi)^3 2E_j} \quad (8.11)$$

is called the **Lorentz-invariant phase space factor**. The decay rate given by (8.10) is a Lorentz-invariant quantity. Hence it can be evaluated in any frame.

8.3.3 Two-Body Decays

The simplest decay process is a two-body decay $A \rightarrow 1 + 2$. A natural choice of reference frame is the center-of-momentum (CM) frame where the initial-state particle is at rest. The 4-momenta of the three particles are

$$p_A^\mu = (m_A, 0, 0, 0), \quad p_1^\mu = (E_1, \mathbf{p}_1), \quad p_2^\mu = (E_2, \mathbf{p}_2).$$

With this setup, we are ready to apply the decay rate formula (8.10) to this two-body decay in the CM frame:

$$\begin{aligned} \Gamma &= \frac{(2\pi)^4}{2E_i} \int |S_{fi}|^2 \delta(E_A - E_1 - E_2) \delta^3(\mathbf{p}_A - \mathbf{p}_1 + \mathbf{p}_2) \frac{d^3 p_1}{(2\pi)^3 2E_1} \frac{d^3 p_2}{(2\pi)^3 2E_2} \\ &= \frac{1}{8\pi^2 m_A} \int |S_{fi}|^2 \delta(m_A - E_1 - E_2) \delta^3(\mathbf{p}_1 + \mathbf{p}_2) \frac{1}{4E_1 E_2} d^3 p_1 d^3 p_2. \end{aligned}$$

The $\delta^3(\mathbf{p}_1 + \mathbf{p}_2)$ means that the two daughter particles have opposite 3-momenta. When integrated over d^3p_2 ,

$$\Gamma = \frac{1}{8\pi^2 m_A} \int |S_{fi}|^2 \delta(m_A - E_1 - E_2) \frac{1}{4E_1 E_2} d^3p_1, \quad \text{where } E_2^2 = (m_2^2 + \mathbf{p}_1^2).$$

Writing out E_1 and E_2 in terms of $\mathbf{p}_1$ in the delta-function, and expressing the integral in spherical coordinates,

$$\Gamma = \frac{1}{8\pi^2 m_A} \int |S_{fi}|^2 \delta\left(m_A - \sqrt{m_1^2 + \mathbf{p}_1^2} - \sqrt{m_2^2 + \mathbf{p}_1^2}\right) \frac{\mathbf{p}_1^2}{4E_1 E_2} d|\mathbf{p}_1| d\Omega \equiv \frac{1}{8\pi^2 m_A} \int |S_{fi}|^2 \delta(f(\mathbf{p}_1)) g(\mathbf{p}_1) d|\mathbf{p}_1| d\Omega,$$

where we define

$$f(\mathbf{p}_1) = m_A - \sqrt{m_1^2 + \mathbf{p}_1^2} - \sqrt{m_2^2 + \mathbf{p}_1^2} \quad \text{and} \quad g(\mathbf{p}_1) = \frac{\mathbf{p}_1^2}{4E_1 E_2}.$$

Let $\mathbf{p}^*$ be the solution to the equation $f(\mathbf{p}^*) = 0$. Then we can evaluate the integral by an identity of the delta-function,

$$\int |S_{fi}|^2 \delta(f(\mathbf{p}_1)) g(\mathbf{p}_1) d|\mathbf{p}_1| = |S_{fi}|^2 \left| \frac{df}{d|\mathbf{p}_1|} \right|_{|\mathbf{p}_1|=|\mathbf{p}^*|}^{-1} g(\mathbf{p}^*).$$

Evaluating the derivatives,

$$\begin{aligned} \int |S_{fi}|^2 \delta(f(\mathbf{p}_1)) g(\mathbf{p}_1) d|\mathbf{p}_1| &= |S_{fi}|^2 \frac{\mathbf{p}^{*2}}{4E_1 E_2} \left[\frac{|\mathbf{p}_1|}{(m_1^2 + \mathbf{p}_1^2)^{1/2}} + \frac{|\mathbf{p}_1|}{(m_2^2 + \mathbf{p}_1^2)^{1/2}} \right]_{|\mathbf{p}_1|=|\mathbf{p}^*|} \\ &= |S_{fi}|^2 \frac{\mathbf{p}^{*2}}{4E_1 E_2} \frac{E_1 E_2}{|\mathbf{p}^*|(E_1 + E_2)} = \frac{|\mathbf{p}^*|}{4m_A} |S_{fi}|^2. \end{aligned}$$

Note that in the above calculation, E_1 and E_2 are initially functions of $\mathbf{p}_1$. When we are doing the delta-function integral, they are then evaluated at $\mathbf{p}^*$, so $E_1 + E_2 = 4m_A$. Also, the $|S_{fi}|^2$ will in general depend on $\mathbf{p}_1$, so it should also be evaluated at $\mathbf{p}^*$ after the integral. Finally, substituting this delta-function integral into the decay rate, we find the general expression

$$\Gamma = \frac{|\mathbf{p}^*|}{32\pi^2 m_A^2} \int |S_{fi}|^2 d\Omega. \quad (8.12)$$

We can apply this formula to *any* two-body decay as long as we know the S -matrix elements. This is because a two-body decay has a special property that the daughter particles have definite energies and magnitudes of 3-momenta fixed entirely by the masses m_A , m_1 , and m_2 . For details, see Section 2.4.4. The results are

$$E_1 = \frac{m_A^2 + m_1^2 - m_2^2}{2m_A}, \quad E_2 = \frac{m_A^2 + m_2^2 - m_1^2}{2m_A}, \quad |\mathbf{p}^*| = \frac{\sqrt{[m_A^2 - (m_1 + m_2)^2][m_A^2 - (m_1 - m_2)^2]}}{2m_A}.$$

8.3.4 Three-Body Decays

There is no definite formulas for three-body decays. This is because, unlike two-body decays, three-body decays can have infinitely many decay configurations with different energies and momenta of the final particles. We need to start from the most general decay rate formula (8.10). Usually there will be many justified approximations (such as taking the relativistic limit) made to make the problem simpler. Examples of three-body decays are presented in Chapter 11.

8.4 Scattering Cross Sections

For particle scattering, we are interested in the scattering rate and the cross section for a particular process. Consider an incident beam of particles a and target particles b in a volume. The cross-sectional area of the beam is A and the velocity of the beam is v_a . The number density of particles b in the volume is n_b , and particles b are moving in a velocity v_b in the opposite direction to v_a . The relative velocity of the beam particles and target particles are then $v = v_a + v_b$. In time δt , one particle a travels through a region with $\delta N = n_b A v \delta t$ particles b . Suppose the interaction probability is δP . Then the scattering cross section σ is defined as

$$\delta P = \frac{\sigma \delta N}{A} = \frac{\sigma n_b A v \delta t}{A} = \sigma n_b v \delta t.$$

The scattering rate for each particle a is

$$r_a = \frac{dP}{dt} = \sigma n_b v.$$

The total scattering rate for a beam of particles a with number density n_a in a volume V is

$$R = r_a n_a V = (n_a v)(n_b V) \sigma = \phi N_b \sigma.$$

Here ϕ is the flux of the beam, and N_b is the total number of target particles. Hence

$$\sigma = \frac{R}{\phi N_b} = \frac{\text{number of interactions per unit time per target particle}}{\text{incident flux}}.$$

8.4.1 Lorentz-Invariant Flux Factor

For the rate $R = (v_a + v_b) n_a n_b \sigma V$, we can normalize the wave function of particles to one particle in a volume V , so $n_a = n_b = 1/V$. The interaction rate then becomes

$$\Gamma_{fi} = \frac{v_a + v_b}{V} \sigma.$$

Additionally, since the final result will not depend on the normalization volume V , we can just set $V = 1$. Then the cross section is related to the interaction rate by

$$\sigma = \frac{\Gamma_{fi}}{v_a + v_b}.$$

For a scattering process $a + b \rightarrow 1 + 2$, Fermi's golden rule says

$$\sigma = \frac{(2\pi)^4}{v_a + v_b} \int |T_{fi}|^2 \delta(E_a + E_b - E_1 - E_2) \delta^3(\mathbf{p}_a + \mathbf{p}_b - \mathbf{p}_1 - \mathbf{p}_2) \frac{d^3 p_1}{(2\pi)^3} \frac{d^3 p_2}{(2\pi)^3}.$$

In terms of the Lorentz-invariant S -matrix element, $S_{fi} = \sqrt{2E_a 2E_b 2E_1 2E_2} T_{fi}$, the cross section is

$$\sigma = \frac{1}{(2\pi)^2 4E_a E_b (v_a + v_b)} \int |S_{fi}|^2 \delta(E_a + E_b - E_1 - E_2) \delta^3(\mathbf{p}_a + \mathbf{p}_b - \mathbf{p}_1 - \mathbf{p}_2) \frac{d^3 p_1}{2E_1} \frac{d^3 p_2}{2E_2}. \quad (8.13)$$

The quantity $F = 4E_a E_b (v_a + v_b)$ is known as the **Lorentz-invariant flux factor**. To see this is Lorentz invariant, write

$$F = 4E_a E_b (v_a + v_b) = 4E_a E_b \left(\frac{|\mathbf{p}_a|}{E_a} + \frac{|\mathbf{p}_b|}{E_b} \right) = 4(E_a |\mathbf{p}_b| + E_b |\mathbf{p}_a|).$$

Squaring the flux factor,

$$F^2 = 16(E_a^2 \mathbf{p}_b^2 + E_b^2 \mathbf{p}_a^2 + 2E_a E_b |\mathbf{p}_a| |\mathbf{p}_b|).$$

In a frame where p_a and p_b are antiparallel, $\mathbf{p}_a = -\mathbf{p}_b$, we have

$$(p_a \cdot p_b)^2 = (E_a E_b + |\mathbf{p}_a| |\mathbf{p}_b|)^2 = E_a^2 E_b^2 + \mathbf{p}_a^2 \mathbf{p}_b^2 + 2E_a E_b |\mathbf{p}_a| |\mathbf{p}_b|.$$

Then substituting $2E_a E_b |\mathbf{p}_a| |\mathbf{p}_b|$ in terms of $(p_a \cdot p_b)$,

$$F^2 = 16[(p_a \cdot p_b)^2 - (E_a^2 - \mathbf{p}_a^2)(E_b^2 - \mathbf{p}_b^2)].$$

This expression is manifestly Lorentz-invariant,

$$F = 4[(p_a \cdot p_b) - m_a^2 m_b^2]^{1/2} \quad (8.14)$$

8.4.2 Scattering in the CM Frame

The cross section is a Lorentz-invariant quantity, so we can evaluate it in any frame. The most convenient frame is the CM frame. Consider the scattering process $a + b \rightarrow 1 + 2$ in the CM frame, where $\mathbf{p}_a = -\mathbf{p}_b = \mathbf{p}_i$ and $\mathbf{p}_1 = -\mathbf{p}_2 = \mathbf{p}_f$. The CM energy is $\sqrt{s} = (E_a + E_b)$. The Lorentz-invariant flux factor is

$$F = 4E_a E_b (v_a + v_b) = 4|\mathbf{p}_i| (E_a + E_b) = 4|\mathbf{p}_i| \sqrt{s}.$$

Using (8.13), the cross section is

$$\sigma = \frac{1}{(2\pi)^2 4|\mathbf{p}_i| \sqrt{s}} \int |S_{fi}|^2 \delta(\sqrt{s} - E_1 - E_2) \delta^3(\mathbf{p}_1 + \mathbf{p}_2) \frac{d^3 p_1}{2E_1} \frac{d^3 p_2}{2E_2}.$$

This integral is exactly the same as the delta-function integral in the previous section (with m_A replaced by $\sqrt{s}$). The result is

$$\sigma = \frac{1}{64\pi^2 s} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|} \int |S_{fi}|^2 d\Omega. \quad (8.15)$$

The differential cross section in the CM frame is given by the integrand

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|} |S_{fi}|^2, \quad (8.16)$$

where the scattering angle θ is defined as $\mathbf{p}_i \cdot \mathbf{p}_f = |\mathbf{p}_i||\mathbf{p}_f| \cos \theta$. The formula (8.16) is particularly useful when the two initial particles have the same mass, such as in electron-positron scattering. Even though the total cross section (8.15) is Lorentz-invariant, the differential cross section does depend on the frame of reference. In other frames, the scattering angle would be different due to a Lorentz transformation.

8.4.3 Scattering in the Fixed Target Frame

There are some occasions where we are interested the differential cross section in a fixed target experiment, such as in electron-proton scattering. The initial proton is assumed to be at rest as it is much heavier than the electron. This frame is usually called the “lab frame”. Consider the scattering process $a + b \rightarrow 1 + 2$ in the lab frame, and particle b is initially at rest. The differential cross section in the lab frame is much more complicated:

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 |\mathbf{p}_a| m_b} \frac{\mathbf{p}_1^2}{|\mathbf{p}_1| E_2 + (|\mathbf{p}_1| - |\mathbf{p}_a| \cos \theta) E_1} |S_{fi}|^2. \quad (8.17)$$

The scattering angle θ is defined as $\mathbf{p}_a \cdot \mathbf{p}_1 = |\mathbf{p}_a||\mathbf{p}_1| \cos \theta$. The energy of particle 2 is related to $\mathbf{p}_a$ and $\mathbf{p}_1$ by

$$E_2 = \sqrt{|\mathbf{p}_a - \mathbf{p}_1|^2 + m_2^2} = \sqrt{\mathbf{p}_a^2 + \mathbf{p}_1^2 - 2|\mathbf{p}_a||\mathbf{p}_1| \cos \theta + m_2^2}.$$

The momentum magnitude $|\mathbf{p}_1|$ can be solved from energy conservation,

$$E_a + m_b = \sqrt{\mathbf{p}_1^2 + m_1^2} + \sqrt{\mathbf{p}_a^2 + \mathbf{p}_1^2 - 2|\mathbf{p}_a||\mathbf{p}_1| \cos \theta + m_2^2}. \quad (8.18)$$

Example 8.1. Electron-proton Elastic Scattering

Consider the process $a + b \rightarrow 1 + 2$ where a and 1 are the same particle and b and 2 are the same particle. Take the relativistic limit $m_a \rightarrow 0$. This, for example, can be the process $e^- p \rightarrow e^- p$ where m_e is negligible. In this case,

$$|\mathbf{p}_a| = E_a, \quad |\mathbf{p}_1| = E_1, \quad m_a = m_1 = 0, \quad m_b = m_2.$$

The differential cross section (8.17) simplifies to

$$\frac{d\sigma}{d\Omega} = \frac{E_1}{64\pi^2 E_a m_b (E_2 + E_1 - E_a \cos \theta)}.$$

Using $E_2 = E_a + m_b - E_1$,

$$\frac{d\sigma}{d\Omega} = \frac{E_1}{64\pi^2 E_a m_b [m_b + E_a (1 - \cos \theta)]}.$$

From the conservation of energy (8.18) explicitly, it is easy to solve for $E_1 = \sqrt{\mathbf{p}_1^2 + m_1^2}$,

$$E_1 = \frac{E_a m_b}{m_b + E_a (1 - \cos \theta)}.$$

Plugging this into $d\sigma/d\Omega$, we find that in the relativistic limit of fixed target scattering with initial and final particles being the same, the differential cross section is

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 [m_b + E_a (1 - \cos \theta)]^2} |S_{fi}|^2. \quad (8.19)$$

Part III

Gauge Theory

WELCOME TO THE WORLD OF SPIN-1.

9 QUANTUM ELECTRODYNAMICS

9.1 U(1) Gauge Invariance

9.1.1 Global U(1) Phase Invariance

In group theory, the U(1) group is the group of 1×1 unitary matrices. It is just a fancier way to say the group of all complex numbers with absolute value 1. Each element in the group can be written as a phase factor $e^{i\chi}$, where χ is any real number. When doing physics, we change the notation $e^{i\chi} \rightarrow e^{iq\chi}$, where q is just a constant. Later we will see that q represents the charge of a particle.

Consider a global U(1) transformation

$$\psi(x) \rightarrow \psi'(x) = e^{iq\chi}\psi(x),$$

where χ is independent of the spacetime coordinate x , i.e. global. The Dirac equation is invariant under such a transformation. This can be proved by multiplying an $e^{iq\chi}$ on both sides of the Dirac equation,

$$e^{iq\chi}i\gamma^\mu\partial_\mu\psi(x) = e^{iq\chi}m\psi(x).$$

Since χ is independent of x , it can move through the partial derivative and meet with $\psi(x)$,

$$i\gamma^\mu\partial_\mu(e^{iq\chi}\psi) = m(e^{iq\chi}\psi) \implies i\gamma^\mu\partial_\mu\psi' = m\psi'.$$

The Lagrangian (7.22) for the Dirac field is invariant under a global U(1) transformation. We know that whenever the Lagrangian is invariant under some transformation, a conserved current appears, and that will be the global conservation of charge.

However, if χ is a function of x , then the above procedure is no longer valid. The transformation $\psi(x) \rightarrow e^{iq\chi(x)}\psi(x)$ is called a local U(1) transformation. Before we study such a transformation, we need to review some classical electrodynamics.

9.1.2 Review of Electromagnetic Fields and Potentials

In classical electromagnetism before special relativity, there are three kinds of objects: sources, fields, and potentials. Sources (charge density ρ and current density $\mathbf{J}$) and fields (electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$) are observable quantities, while potentials (scalar potential ϕ and vector potential $\mathbf{A}$) are not. (In quantum mechanics, the potential is “more physical”. Particles going through a region with zero magnetic field but nonzero vector potential has different wave functions depending on the path taken. The phase difference between these particles can be observed by interference. This effect is known as the [Aharonov-Bohm effect](#).) The governing equations of electromagnetism that relate sources to fields are Maxwell’s equations,

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \rho, & \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \nabla \cdot \mathbf{B} &= 0, & \nabla \times \mathbf{B} &= \mathbf{J} + \frac{\partial \mathbf{E}}{\partial t}. \end{aligned}$$

The electric and magnetic fields can be written in terms of potentials ϕ and $\mathbf{A}$ by

$$\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}. \quad (9.1)$$

The potentials are not observable quantities because they cannot be uniquely determined by $\mathbf{E}$ and $\mathbf{B}$. There is a type of freedom called [gauge freedom](#) lying in the above equations. We can show that under a [gauge transformation](#)

$$\phi \rightarrow \phi' = \phi - \frac{\partial \chi}{\partial t} \quad \text{and} \quad \mathbf{A} \rightarrow \mathbf{A}' = \mathbf{A} + \nabla\chi \quad (9.2)$$

for any scalar function $\chi(\mathbf{x}, t)$, the electric and magnetic fields are invariant.

Proof. The transformed electric field is

$$\mathbf{E}' = -\nabla\phi' - \frac{\partial\mathbf{A}'}{\partial t} = -\nabla\left(\phi - \frac{\partial\chi}{\partial t}\right) - \frac{\partial}{\partial t}(\mathbf{A} + \nabla\chi).$$

Partial derivatives commute, so

$$\mathbf{E}' = -\nabla\phi + \frac{\partial}{\partial t}(\nabla\chi) - \frac{\partial\mathbf{A}}{\partial t} - \frac{\partial}{\partial t}(\nabla\chi) = -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t} = \mathbf{E}.$$

The transformed magnetic field is

$$\mathbf{B}' = \nabla \times \mathbf{A}' = \nabla \times (\mathbf{A} + \nabla\chi) = \nabla \times \mathbf{A} = \mathbf{B}$$

because the curl of a gradient is always zero. $\square$

Since quantum field theory involves special relativity, we had better write equations in electromagnetism in covariant forms. Luckily, the theory of electromagnetism is automatically compatible with special relativity, so all we need to do is to define some 4-vectors and tensors. The sources are unified into a **4-current**, $J^\mu = (\rho, \mathbf{J})$. The potentials are unified into a **4-potential**, $A^\mu = (\phi, \mathbf{A})$. Their generalization to 4-vectors seems natural, but it is not possible for both electric and magnetic fields to get unified into a 4-vector because there are in total six components in $\mathbf{E}$ and $\mathbf{B}$. In fact, they are stored in an object called the **electromagnetic field tensor** $F_{\mu\nu}$:

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu, \quad F_{\mu\nu} = \begin{bmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & -B_z & B_y \\ -E_y & B_z & 0 & -B_x \\ -E_z & -B_y & B_x & 0 \end{bmatrix} \quad (9.3)$$

It is straightforward to verify the components of $F_{\mu\nu}$ using the original relation between fields and potentials (9.1). Finally, the gauge transformation of potentials (9.2) becomes

$$A^\mu \rightarrow A'^\mu = A^\mu - \partial^\mu \chi. \quad (9.4)$$

9.1.3 Local U(1) Phase Invariance

Recall that the invariance of the Lagrangian in a global U(1) transformation implies the global conservation of charge. However, we know that the conservation of charge is true not only globally, but also locally. This suggests us to modify the Dirac equation such that if $\psi(x)$ is a solution of the Dirac equation, so is $\psi'(x) = e^{iq\chi(x)}\psi(x)$, where $\chi(x)$ is a differentiable real function of x . This is called the *principle of local phase invariance*, or *gauge invariance under U(1)*.

Let's see what happens if we want to obtain an equation for $\psi'(x) = e^{iq\chi(x)}\psi(x)$ with no modification to the Dirac equation. A mature way is to start with the Dirac equation for $\psi(x)$ and multiply both sides by $e^{iq\chi}$,

$$e^{iq\chi} i\gamma^\mu \partial_\mu \psi(x) = m e^{iq\chi} \psi(x).$$

Then insert an identity $1 = e^{-iq\chi} e^{iq\chi}$ between ∂_μ and $\psi(x)$,

$$e^{iq\chi} i\gamma^\mu \partial_\mu (e^{-iq\chi} e^{iq\chi} \psi) = m e^{iq\chi} \psi \implies e^{iq\chi} i\gamma^\mu \partial_\mu (e^{-iq\chi} \psi') = m \psi'.$$

Because now $\chi(x)$ is a function of x , it no longer commutes with the partial derivative. Using the product rule, we should get

$$i\gamma^\mu (\partial_\mu - iq\partial_\mu \chi) \psi' = m \psi'.$$

There is an extra term involving $\partial_\mu \chi$ that differs from the Dirac equation for $\psi'(x)$. We see that if χ is independent of x (a global U(1) transformation), then it reduces to the usual Dirac equation. The terms in the parenthesis should be familiar, they look like the gauge transformation (9.4) of the 4-potential.

Now we take the following steps:

1. Insist on local phase invariance of the Dirac equation under U(1) transformations.
2. Introduce the 4-potential A^μ into the Dirac equation such that the gauge transformation of A^μ cancels the extra term $-i\partial_\mu \chi$.

It turns out that modifying the Dirac equation in the following way solves the problem:

$$\boxed{i\gamma^\mu(\partial_\mu + iqA_\mu)\psi = m\psi}, \quad (9.5)$$

and the gauge transformation is given by

$$\psi \rightarrow \psi' = e^{iq\chi}\psi, \quad A^\mu \rightarrow A'^\mu = A^\mu - \partial^\mu\chi \quad (9.6)$$

Proof. Suppose $\psi(x)$ solves the modified Dirac equation (9.5),

$$i\gamma^\mu(\partial_\mu + iqA_\mu)\psi = m\psi.$$

Multiplying both sides by $e^{iq\chi}$ and inserting an identity $1 = e^{-iq\chi}e^{iq\chi}$,

$$\begin{aligned} ie^{iq\chi}\gamma^\mu(\partial_\mu + iqA_\mu)(e^{-iq\chi}\psi) &= me^{iq\chi}\psi \\ ie^{iq\chi}\gamma^\mu(\partial_\mu + iqA_\mu)(e^{-iq\chi}\psi') &= m\psi' \\ i\gamma^\mu(\partial_\mu - iq\partial_\mu\chi + iqA_\mu)\psi' &= m\psi'. \end{aligned}$$

The gauge transformation (9.6) says $A'_\mu = A_\mu - \partial_\mu\chi$, so

$$i\gamma^\mu(\partial_\mu + iqA'_\mu)\psi' = m\psi'.$$

The modified Dirac equation is true for $\psi'(x)$. □

Conventionally, we define a *gauge-covariant derivative* D_μ for convenience,

$$D_\mu = \partial_\mu + iqA_\mu \quad \text{such that} \quad i\gamma^\mu D_\mu\psi = m\psi. \quad (9.7)$$

The achievement of (9.5) and the associated gauge transformation (9.6) is our first example of **gauge theory**. We have a new equation with both the Dirac field $\psi(x)$ and electromagnetism in terms of A^μ , a **gauge field**. Their coupling means that we bring in electromagnetic interactions to elementary particles. Later, when we quantize gauge fields, their associated quanta are known as **gauge bosons**. Quantizing A^μ gives rise to **photons**, and the corresponding quantized gauge theory is **quantum electrodynamics (QED)**. QED is an abelian gauge theory because the $U(1)$ group is an abelian group, which means that *photons do not couple to each other directly*. We will revisit and explain this in later chapters when we encounter non-abelian gauge theories.

9.2 Perturbation Theory

9.2.1 Interaction Lagrangian and Hamiltonian

We now apply perturbation theory to a spin-1/2 particle with charge q in an electromagnetic field. This means we want to modify the derivative operator in the Dirac Lagrangian, $\mathcal{L}_D = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi$. The correct Dirac Lagrangian in QED is

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi. \quad (9.8)$$

It is more convenient to separate the free Lagrangian $\mathcal{L}_0$ and the interaction piece $\mathcal{L}_{\text{int}}$. All we need to do is to write out $D_\mu = \partial_\mu + iqA_\mu$:

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi - q\bar{\psi}\gamma^\mu A_\mu\psi. \quad (9.9)$$

The first part is the free Dirac Lagrangian, and the second part with A_μ is the interaction piece,

$$\mathcal{L}_0 = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi \quad \text{and} \quad \mathcal{L}_{\text{int}} = -q\bar{\psi}\gamma^\mu\psi A_\mu.$$

Note that in $\mathcal{L}_{\text{int}}$, we are allowed to bring out the 4-potential—its components are different fields from the Dirac fields, so A_μ and ψ commute. Using the conserved 4-current (7.29) derived from the Dirac Lagrangian, the interaction piece can be written compactly as

$$\mathcal{L}_{\text{int}} = -J^\mu A_\mu, \quad \text{where} \quad J^\mu = q\bar{\psi}\gamma^\mu\psi. \quad (9.10)$$

Note that this interaction Lagrangian is correct in classical electromagnetism: $\mathcal{L}_{\text{classical}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - A_\mu J^\mu$, where the first term is the Lagrangian for the electromagnetic fields and the second term is for the interaction between charged matter with the field. This is why interpreting J^μ as the conserved *electric charge* current is appropriate.

The interaction Lagrangian does not contain $\dot{\psi}$, so the corresponding conjugate momentum of the Dirac field is still $\pi = i\psi^\dagger$. Then we can compute the Hamiltonian density,

$$\begin{aligned}\mathcal{H} &= \pi\dot{\psi} - \mathcal{L} = i\psi^\dagger\dot{\psi} - [\bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi + \mathcal{L}_{\text{int}}] \\ &= i\bar{\psi}\gamma^0\partial_t\psi - \psi(i\gamma^0\partial_t + i\boldsymbol{\gamma}\cdot\nabla - m)\psi - \mathcal{L}_{\text{int}} \\ &= \bar{\psi}[\boldsymbol{\gamma}\cdot(-i\nabla) + m]\psi - \mathcal{L}_{\text{int}}.\end{aligned}$$

Comparing to the free Dirac Hamiltonian density (7.24), we find that

$$\mathcal{H}_0 = \bar{\psi}[\boldsymbol{\gamma}\cdot(-i\nabla) + m]\psi \quad \text{and} \quad \mathcal{H}_{\text{int}} = -\mathcal{L}_{\text{int}} = J^\mu A_\mu. \quad (9.11)$$

The interaction Hamiltonian here for QED is the product of three fields: $\bar{\psi}$, ψ , and A^μ . This means we can use similar techniques from the “ABC theory” in section 8.2 to obtain scattering amplitudes.

9.2.2 Scattering Amplitudes

We are interested in the scattering amplitudes from the initial state $|i\rangle$ to the final state $|f\rangle$. First consider the case where the 4-potential is still a classical field (no quantization). Suppose both the initial and final states are field states for the electron-positron field. We regard the electron as the “particle” and positron as the “antiparticle”, so the charge quanta for the conserved current is $q = -e$. The S -matrix is

$$S_{fi} = \delta_{fi} + (-i) \langle f | \int d^4x_1 \mathcal{H}'_I(x_1) | i \rangle + \frac{(-i)^2}{2} \langle f | \int d^4x_1 d^4x_2 T\{\mathcal{H}'_I(x_1)\mathcal{H}'_I(x_2)\} | i \rangle + \cdots.$$

To find the fundamental vertices of the interaction, we focus on the first-order terms in $\mathcal{H}'_I$,

$$\mathcal{A}_{fi} = (-i) \langle f | \int d^4x \mathcal{H}'_I(x) | i \rangle.$$

Inserting the interaction Hamiltonian density $\mathcal{H}_{\text{int}}$ in (9.11) as $\mathcal{H}'_I$ and writing out all the mode expansions:

$$\begin{aligned}\mathcal{A}_{fi} &= -i(-e) \int d^4x \langle f | \int \frac{d^3p' d^3p''}{(2\pi)^6 \sqrt{4E'E''}} \sum_{s',s''} \left[c^\dagger(p'', s'') u^\dagger(p'', s'') e^{+ip''\cdot x} + d(p'', s'') v^\dagger(p'', s'') e^{-ip''\cdot x} \right] \\ &\quad \times \gamma^0 \gamma^\mu A_\mu \left[c(p', s') u(p', s') e^{-ip'\cdot x} + d^\dagger(p', s') v(p', s') e^{ip'\cdot x} \right] | i \rangle.\end{aligned}$$

The structure of this amplitude is quite clear. There are basically four terms:

$$\begin{aligned}\text{(I)} &= c^\dagger(p'', s'') c(p', s') \bar{u}(p'', s'') \gamma^\mu u(p', s') e^{i(p''-p')\cdot x}, \\ \text{(II)} &= c^\dagger(p'', s'') d^\dagger(p', s') \bar{u}(p'', s'') \gamma^\mu v(p', s') e^{i(p''+p')\cdot x}, \\ \text{(III)} &= d(p'', s'') c(p', s') \bar{v}(p'', s'') \gamma^\mu u(p', s') e^{-i(p''+p')\cdot x}, \\ \text{(IV)} &= d(p'', s'') d^\dagger(p', s') \bar{v}(p'', s'') \gamma^\mu v(p', s') e^{i(p'-p'')\cdot x}.\end{aligned}$$

Electron Scattering. Let’s do a full calculation in the scenario with both the initial and final states being single-electron states,

$$|i\rangle = |e^-, p_1, s_1\rangle = \sqrt{2E_1} c^\dagger(p_1, s_1) |0\rangle, \quad |f\rangle = |e^-, p_2, s_2\rangle = \sqrt{2E_2} c^\dagger(p_2, s_2) |0\rangle.$$

The first-order amplitude is

$$\mathcal{A}_{fi} = -i(-e) \int d^4x \langle 0 | c(p_2, s_2) \sqrt{2E_2} (\bar{\psi} \gamma^\mu \psi A_\mu) \sqrt{2E_1} c^\dagger(p_1, s_1) | 0 \rangle.$$

In this amplitude, $c^\dagger(p_1, s_1)|0\rangle$ represent the state of an ingoing electron with 4-momentum p_1 and spin s_1 . One part of ψ annihilates the incoming electron. One part of $\bar{\psi}$ creates an outgoing electron. Then the whole thing is projected into the final state $\langle 0|c(p_2, s_2)$, an outgoing electron with 4-momentum p_2 and spin s_2 . Here the 4-potential A_μ is treated classically. For example, it can represent the $1/r$ potential in a Coulomb scattering.

Inside $\bar{\psi}\gamma^\mu\psi$, there are the four terms, (I), (II), (III), (IV), mentioned above. We want to find the term that annihilates an ingoing electron and creates an outgoing one, which will be term (I) with $c^\dagger c$. (This term is already normal ordered.) All other terms do not contribute. Mathematically, the $d^\dagger(p', s')$ in (II) will lead to $\langle 0|d^\dagger = 0$, while the $d(p'', s'')$ in (III) will lead to $d|0\rangle = 0$. For (IV), it is more subtle, and we shall see that why normal ordering is necessary. (IV) contributes to the amplitude as

$$\begin{aligned}\mathcal{A}_{fi}^{(IV)} &\propto \int \frac{d^3p' d^3p'' d^4x}{(2\pi)^6 \sqrt{4E'E''}} \langle 0|c(p_2, s_2)d(p'', s'')d^\dagger(p', s')e^{i(p'-p'')\cdot x}c^\dagger(p_1, s_1)|0\rangle \\ &\propto \int \frac{d^3p' d^3p'' d^4x}{(2\pi)^6 \sqrt{4E'E''}} \langle 0|c(p_2, s_2)[-d^\dagger(p', s')d(p'', s'') + (2\pi)^3\delta^3(\mathbf{p}' - \mathbf{p}'')]e^{i(p'-p'')\cdot x}c^\dagger(p_1, s_1)|0\rangle \\ &= \int \frac{d^3p' d^3p''}{(2\pi)^3 \sqrt{4E'E''}} \langle 0|c(p_2, s_2)\delta^3(\mathbf{p}' - \mathbf{p}'')(2\pi)^4\delta^4(p' - p'')c^\dagger(p_1, s_1)|0\rangle \\ &= \int \frac{d^3p'}{(2\pi)^3(2E')} \langle 0|c(p_2, s_2)\delta^3(0)c^\dagger(p_1, s_1)|0\rangle.\end{aligned}$$

Without normal ordering, when integrating over all spacetime d^4x , the familiar infinity $\delta^3(0)$ appears. Hence normal ordering is necessary to avoid this. We only keep the term with $d^\dagger d$, which then gives no contribution as $d|0\rangle = 0$. For fermionic fields obeying anticommutation relations, normal ordering simply means $dd^\dagger \stackrel{N}{=} -d^\dagger d$; for bosonic field obeying commutation relations, it means $aa^\dagger \stackrel{N}{=} a^\dagger a$. Only the creation and annihilation operators *from the Hamiltonian* can be normal ordered. Normal ordering is not allowed for operators that create particles in the initial or final states, e.g. $\langle 0|c(p', s')c^\dagger(p_1, s_1)|0\rangle \stackrel{N}{=} \langle 0|c^\dagger(p_1, s_1)c(p', s')|0\rangle$ is not allowed.

Back to the amplitude. Now we know that only the term (I) contributes. Note that

$$c(p', s')c^\dagger(p_1, s_1)|0\rangle = -c^\dagger(p_1, s_1)c(p', s')|0\rangle + (2\pi)^3\delta^3(\mathbf{p}_1 - \mathbf{p}')\delta_{s_1, s'}|0\rangle = (2\pi)^3\delta^3(\mathbf{p}_1 - \mathbf{p}')\delta_{s_1, s'}|0\rangle.$$

Similarly,

$$\langle 0|c(p_2, s_2)c^\dagger(p'', s'') = -\langle 0|c^\dagger(p'', s'')c(p_2, s_2) + \langle 0|(2\pi)^3\delta^3(\mathbf{p}_2 - \mathbf{p}'')\delta_{s_2, s''} = \langle 0|(2\pi)^3\delta^3(\mathbf{p}_2 - \mathbf{p}'')\delta_{s_2, s''}.$$

The amplitude becomes

$$\begin{aligned}\mathcal{A}_{fi} &= -i(-e)\sqrt{4E_1E_2} \int \frac{d^3p' d^3p'' d^4x}{(2\pi)^6 \sqrt{4E'E''}} \sum_{s, s''} \langle 0|c(p_2, s_2)c^\dagger(p'', s'')c(p', s')c^\dagger(p_1, s_1)|0\rangle \bar{u}(p'', s'')\gamma^\dagger u(p', s')e^{i(p'-p'')\cdot x}A_\mu \\ &= -i(-e)\sqrt{4E_1E_2} \int \frac{d^3p' d^3p'' d^4x}{\sqrt{4E'E''}} \sum_{s, s''} \langle 0|\delta_{s_1, s'}\delta_{s_2, s''}\delta^3(\mathbf{p}_1, \mathbf{p}')\delta^3(\mathbf{p}_2, \mathbf{p}'')|0\rangle \bar{u}(p'', s'')\gamma^\mu u(p', s')e^{i(p'-p'')\cdot x}A_\mu \\ &= -i(-e) \int d^4x \bar{u}(p_2, s_2)\gamma^\mu u(p_1, s_1)e^{-i(p_1-p_2)\cdot x}A_\mu.\end{aligned}$$

There is no x -dependence in the spinors $\bar{u}(p_2, s_2)$ and $u(p_1, s_1)$, so they can be brought out of the integral. The amplitude for electron scattering from A_μ to first order is

$$\boxed{\mathcal{A}_{fi} = -i(-e)\bar{u}(p_2, s_2)\gamma^\mu u(p_1, s_1) \int d^4x e^{-i(p_1-p_2)\cdot x}A_\mu(x).} \quad (9.12)$$

This amplitude contains three things:

1. It is proportional to the vertex factor $-iq\gamma^\mu$ for QED.
2. The spinors $u(p_1, s_2)$ and $\bar{u}(p_2, s_2)$ represent the ingoing electron and outgoing electron, respectively.
3. There is an integral over all spacetime, which we recognize as the Fourier transform of the 4-potential with respect to the 4-momentum transfer $q^\mu = p_1^\mu - p_2^\mu$!

Positron Scattering. Consider positron scattering with both the initial and final states being single-positron states,

$$|i\rangle = |e^+, p_1, s_1\rangle = \sqrt{2E_1}d^\dagger(p_1, s_1)|0\rangle, \quad |f\rangle = |e^+, p_2, s_2\rangle = \sqrt{2E_2}d^\dagger(p_2, s_2)|0\rangle.$$

The first-order amplitude is

$$\mathcal{A}_{fi} = -i(-e) \int d^4x \langle 0 | d(p_2, s_2) \sqrt{2E_2} (\bar{\psi} \gamma^\mu \psi A_\mu) \sqrt{2E_1} d^\dagger(p_1, s_1) | 0 \rangle.$$

Notice that the charge in the vertex factor is still $-e$, not $+e$. The charge is determined by the electron-positron field, not the particular particles involved. Since both ingoing and outgoing particles are positrons, the only term in $\bar{\psi} \gamma^\mu \psi$ that contributes is $d(p'', s'') d^\dagger(p', s')$ in (IV). This time, normal ordering is required because $d^\dagger$ is on the right of d . After normal ordering, the amplitude reads

$$\mathcal{A}_{fi} = -i(-e) \sqrt{4E_1 E_2} \int \frac{d^3 p' d^3 p'' d^4 x}{(2\pi)^6 \sqrt{4E' E''}} \sum_{s', s''} \langle 0 | d(p_2, s_2) [-d^\dagger(p', s') d(p'', s'')] d^\dagger(p_1, s_1) | 0 \rangle \bar{v}(p'', s'') \gamma^\mu v(p', s') e^{-i(p'' - p') \cdot x} A_\mu.$$

This time, because of the normal ordering, p' is paired with p_2 , while p'' is paired with p_1 . In particular,

$$\langle 0 | d(p_2, s_2) d^\dagger(p', s') = -\langle 0 | d^\dagger(p', s') d(p_2, s_2) + \langle 0 | (2\pi)^3 \delta^3(\mathbf{p}_2 - \mathbf{p}') \delta_{s_2, s'} = \langle 0 | (2\pi)^3 \delta^3(\mathbf{p}_2 - \mathbf{p}') \delta_{s_2, s'},$$

$$d(p'', s'') d^\dagger(p_1, s_1) | 0 \rangle = -d^\dagger(p_1, s_1) d(p'', s'') | 0 \rangle + (2\pi)^3 \delta^3(\mathbf{p}_1 - \mathbf{p}'') \delta_{s_1, s''} | 0 \rangle = (2\pi)^3 \delta^3(\mathbf{p}_1 - \mathbf{p}'') \delta_{s_1, s''} | 0 \rangle.$$

Substituting these expressions into the amplitude,

$$\begin{aligned} \mathcal{A}_{fi} &= -(-i)(-e) \sqrt{4E_1 E_2} \int \frac{d^3 p' d^3 p'' d^4 x}{\sqrt{4E' E''}} \sum_{s, s''} \langle 0 | \delta_{s_2, s'} \delta_{s_1, s''} \delta^3(\mathbf{p}_2, \mathbf{p}') \delta^3(\mathbf{p}_1, \mathbf{p}'') | 0 \rangle \bar{v}(p'', s'') \gamma^\mu v(p', s') e^{-i(p'' - p') \cdot x} A_\mu \\ &= i(-e) \int d^4x \bar{v}(p_1, s_1) \gamma^\mu v(p_2, s_2) e^{-i(p_1 - p_2) \cdot x} A_\mu. \end{aligned}$$

Pulling out the spinors from the integral as they do not depend on x , the amplitude for positron scattering from A_μ to first order is

$$\boxed{\mathcal{A}_{fi} = i(-e) \bar{v}(p_1, s_1) \gamma^\mu v(p_2, s_2) \int d^4x e^{-i(p_1 - p_2) \cdot x} A_\mu(x).}$$

Compared to electron scattering, we find that there is an overall minus sign in the amplitude. Though practically we often square the amplitude and the minus sign goes away, it does have some effect when there is interference between different amplitudes. Another difference is that positron scattering has spinors entering the amplitude as $\bar{v}(p_1, s_1) \gamma^\mu v(p_2, s_2)$, while electron scattering as $\bar{u}(p_2, s_2) \gamma^\mu u(p_1, s_1)$. The order of p_1 and p_2 is reversed. This is indicated by the convention that antiparticles have reversed arrows in Feynman diagrams as particles.

Electron-Positron Annihilation. The other two terms (II) and (III) contribute when an $e^+ e^-$ pair is created or annihilated. We will analyze the case of $e^+ e^-$ annihilation here. The initial and final states are given by

$$|i\rangle = |e^-, p_1, s_1; e^+, p_2, s_2\rangle = \sqrt{4E_1 E_2} d^\dagger(p_2, s_2) c^\dagger(p_1, s_1) | 0 \rangle, \quad |f\rangle = | 0 \rangle.$$

The final state will just be vacuum, since we treat A_μ as classical; the quanta of electromagnetic field is not introduced yet. Only term (III) contributes. The first-order amplitude is

$$\begin{aligned} \mathcal{A}_{fi} &= -i(-e) \int d^4x \langle 0 | (\bar{\psi} \gamma^\mu \psi A_\mu) \sqrt{4E_1 E_2} d^\dagger(p_2, s_2) c^\dagger(p_1, s_1) | 0 \rangle \\ &= -i(-e) \sqrt{4E_1 E_2} \int \frac{d^3 p' d^3 p'' d^4 x}{(2\pi)^6 \sqrt{4E' E''}} \sum_{s', s''} \langle 0 | d(p'', s'') c(p', s') d^\dagger(p_2, s_2) c^\dagger(p_1, s_1) | 0 \rangle \bar{v}(p'', s'') \gamma^\mu u(p', s') e^{-i(p' + p'') \cdot x} A_\mu \\ &= -i(-e) \sqrt{4E_1 E_2} \int \frac{d^3 p' d^3 p'' d^4 x}{(2\pi)^6 \sqrt{4E' E''}} \sum_{s', s''} \langle 0 | d(p'', s'') [-d^\dagger(p_2, s_2) c(p', s')] c^\dagger(p_1, s_1) | 0 \rangle \bar{v}(p'', s'') \gamma^\mu u(p', s') e^{-i(p' + p'') \cdot x} A_\mu \\ &= i(-e) \sqrt{4E_1 E_2} \int \frac{d^3 p' d^3 p'' d^4 x}{\sqrt{4E' E''}} \sum_{s', s''} \delta_{s_2, s''} \delta_{s_1, s'} \delta^3(\mathbf{p}_2 - \mathbf{p}'') \delta^3(\mathbf{p}_1 - \mathbf{p}') \bar{v}(p'', s'') \gamma^\mu u(p', s') e^{-i(p' + p'') \cdot x} A_\mu. \end{aligned}$$

The result is

$$\boxed{\mathcal{A}_{fi} = i(-e) \bar{v}(p_2, s_2) \gamma^\mu u(p_1, s_1) \int d^4x e^{-i(p_1 + p_2) \cdot x} A_\mu(x).}$$

9.3 Scattering from a Static Charge Distribution

In this section, we apply the results of the perturbation theory on an electron scattering from a static charge distribution. In particular, we are interested in probably the simplest scattering we can think of—the **Coulomb scattering** over a $1/r$ potential.

9.3.1 The Coulomb Scattering

An ideal source for A_μ of a Coulomb scattering is an infinitely heavy point particle of charge Ze . Since the source is infinitely heavy, we don't have to worry about its recoil when the electron approaches. Then it will be a truly static $1/r$ potential:

$$A^\mu = \left(\frac{Ze}{4\pi|\mathbf{x}|}, 0, 0, 0 \right).$$

In reality, it is a good approximation for an electron scattering from, for example, a gold nucleus. Referring to (9.12), the amplitude of electron scattering is

$$\mathcal{A}_{fi} = -i(-e)\bar{u}(p_f, s_f)\gamma^\mu u(p_i, s_i) \int d^4x e^{-iq\cdot x} A_\mu(x), \quad \text{where } q = p_i - p_f.$$

Using our assumption of static potential, we should find that $\gamma^\mu A_\mu$ becomes just $\gamma^0 A_t$. This γ^0 can then be absorbed into $\bar{u}$ to give

$$\mathcal{A}_{fi} = -i(-e)u^\dagger(p_f, s_f)u(p_i, s_i) \int d^4x e^{-iq\cdot x} \frac{Ze}{4\pi|\mathbf{x}|},$$

but we shall keep the γ^0 for now. We need to compute the 4-dimensional Fourier transform of $Ze/4\pi|\mathbf{x}|$ with respect to the 4-momentum transfer q^μ . Call this Fourier transform $A^t(q)$. Observing that there is no time-dependence in the potential, so we should first compute the integral over t ,

$$A^t(q) \equiv \int d^4x e^{-iq\cdot x} A^t(x) = \int d^3x dt e^{-i(E_i - E_f)t} e^{i(\mathbf{p}_i - \mathbf{p}_f)\cdot \mathbf{x}} \frac{Ze}{4\pi|x|} = 2\pi\delta(E_i - E_f) \int d^3x e^{i\mathbf{q}\cdot \mathbf{x}} \frac{Ze}{4\pi|\mathbf{x}|}.$$

This delta function makes sense. Because the source is infinitely heavy, no energy is lost by the electron during scattering ($E_i = E_f$). In other words, this process is an elastic scattering. Now it is basically a math problem: what is the 3-dimensional Fourier transform of $1/|\mathbf{x}|$? This is a classic problem, so we should derive it here carefully.

Fourier Transform of $1/|\mathbf{x}|$

The first thing to do is to change to spherical coordinates, and choose $\mathbf{q}$ to lie along the z -axis. The Fourier transform integral becomes

$$\begin{aligned} \int d^3x e^{i\mathbf{q}\cdot \mathbf{x}} \frac{1}{|\mathbf{x}|} &= \int_0^\infty dr \int_0^{2\pi} d\phi \int_0^\pi d\theta r^2 \sin\theta \frac{e^{i|\mathbf{q}|r\cos\theta}}{r} \\ &= 2\pi \int_0^\infty dr \int_0^\pi d(\cos\theta) r e^{i|\mathbf{q}|r\cos\theta} \\ &= 2\pi \int_0^\infty dr \frac{1}{i|\mathbf{q}|r} r e^{i|\mathbf{q}|r\cos\theta} \Big|_0^\pi \\ &= 2\pi \int_0^\infty dr \frac{1}{i|\mathbf{q}|} \left[e^{i|\mathbf{q}|r} - e^{-i|\mathbf{q}|r} \right]. \end{aligned}$$

The integrand is a pure oscillatory function. To perform this integral, we need to insert a convergence factor $e^{-\mu r}$ with $\mu > 0$ and take the limit as $\mu \rightarrow 0$,

$$\begin{aligned} \int d^3x e^{i\mathbf{q}\cdot \mathbf{x}} \frac{1}{|\mathbf{x}|} &= \lim_{\mu \rightarrow 0} \frac{2\pi}{i|\mathbf{q}|} \int_0^\infty dr \left[e^{(i|\mathbf{q}| - \mu)r} - e^{(-i|\mathbf{q}| - \mu)r} \right] \\ &= \lim_{\mu \rightarrow 0} \frac{2\pi}{i|\mathbf{q}|} \left[\frac{1}{i|\mathbf{q}| - \mu} e^{(i|\mathbf{q}| - \mu)r} + \frac{1}{i|\mathbf{q}| + \mu} e^{(-i|\mathbf{q}| - \mu)r} \right]_0^\infty \\ &= \lim_{\mu \rightarrow 0} \frac{2\pi}{i|\mathbf{q}|} \left[-\frac{2i|\mathbf{q}|}{-\mathbf{q}^2 - \mu^2} \right] \\ &= \lim_{\mu \rightarrow 0} \frac{4\pi}{\mathbf{q}^2 + \mu^2}. \end{aligned}$$

In the limit $\mu \rightarrow 0$, we see that the Fourier transform of $1/r$ is convergent. The result is

$$\int d^3x e^{i\mathbf{p}\cdot\mathbf{x}} \frac{1}{|\mathbf{x}|} = \frac{4\pi}{\mathbf{q}^2}.$$

Inserting this result into the scattering amplitude, we obtain

$$\mathcal{A}_{fi} = -2\pi i (Ze^2) \bar{u}\gamma^0(p_f, s_f) u(p_i, s_i) \delta(E_f - E_i) \frac{1}{\mathbf{q}^2}. \quad (9.13)$$

Let's comment on this result. Eventually, we are going to square the amplitude to get a probability density. It is easy to see that the probability density $\mathcal{P}_{fi} = |\mathcal{A}_{fi}|^2 \propto 1/|\mathbf{q}|^4$. Because the initial and final energies of the electron are the same, its initial and final 3-momenta differ only by direction, not magnitude. Thus, the momentum transfer squared is

$$\mathbf{q}^2 = \mathbf{p}_i^2 + \mathbf{p}_f^2 - 2\mathbf{p}_i \cdot \mathbf{p}_f = 2\mathbf{p}_i^2(1 - \cos\theta),$$

where θ is the scattering angle. Using some trig-formula, we can write $1 - \cos\theta = 2\sin^2\frac{\theta}{2}$, which gives the angular dependence of the probability density,

$$\mathcal{P}_{fi} \propto \frac{1}{|\mathbf{q}|^4} \propto \frac{1}{\sin^4\frac{\theta}{2}}.$$

This is the famous angular dependence in the **Rutherford scattering** cross section,

$$\frac{d\sigma}{d\Omega} \propto \frac{1}{\sin^4\frac{\theta}{2}}, \quad (9.14)$$

which is the spinless version of a Coulomb scattering calculation above. (A detailed derivation of the Rutherford scattering cross section is given in Appendix A.1.2. It requires only classical mechanics and electromagnetism.) The Coulomb scattering that takes spins into account is known as the **Mott scattering**. The Mott cross section is what we are interested in. In the end, we will see that it is different from (but contains) the Rutherford scattering cross section. The difference arises from the fact that an electron is a spin-1/2 particle, whereas an α -particle in the Rutherford scattering is spin-0.

9.3.2 Lips, Transition Rates, and Cross Sections

The Mott scattering is a particle scattering from a potential. It is not as “canonical” as a scattering process that has two particles scattering into many particles. We can use this opportunity of review some concepts related to scattering.

The Lorentz-Invariant Phase Space Factor. The probability density $\mathcal{P}_{fi} = |\mathcal{A}_{fi}|^2$ is a function of 4-momentum. To convert it into a probability, some integral over momentum is expected. For a fixed initial state, the probability for it to transition to some range of final states (since 4-momentum is a continuous variable) should be the integral of $\mathcal{P}_{fi}$ over **phase space**, a set of all possible final states. To find the phase space factor, we need to see how spinors are normalized.

By convention, the spinors are normalized such that the corresponding probability current is given by

$$\rho \equiv \bar{\psi}\gamma^0\psi = \frac{2E}{V}.$$

where V is the spatial volume. This normalization is saying that there are $2E$ particles in a volume V , or the volume occupied by one particle is $V_1 = V/2E$. Usually V are omitted (as in previous chapters) because they do not affect any physical results, but we keep them here to see how they cancel each other. With this normalization factor, the spinors should look like

$$u(p, s) \rightarrow \frac{1}{\sqrt{V}} u(p, s) = \frac{1}{\sqrt{V}} \sqrt{E+m} \begin{bmatrix} \phi \\ \frac{\boldsymbol{\sigma}\cdot\mathbf{p}}{E+m}\phi \end{bmatrix}, \quad v(p, s) \rightarrow \frac{1}{\sqrt{V}} v(p, s) = \frac{1}{\sqrt{V}} \sqrt{E+m} \begin{bmatrix} \frac{\boldsymbol{\sigma}\cdot\mathbf{p}}{E+m}\chi \\ \chi \end{bmatrix},$$

and the scattering amplitude (9.13) becomes

$$\mathcal{A}_{fi} = -2\pi i (Ze^2) \frac{1}{V} \bar{u}(p_f, s_f) \gamma^0 u(p_i, s_i) \delta(E_f - E_i) \frac{1}{\mathbf{q}^2}. \quad (9.15)$$

From ordinary quantum mechanics, the number of states for a particle with momentum $\mathbf{p} \rightarrow \mathbf{p} + d\mathbf{p}$ in a box of volume L^3 is given

by

$$d^3n = \left(\frac{L}{2\pi}\right)^3 d^3p.$$

(This works for both non-relativistic and relativistic cases.) Since one particle occupies a volume of $V/2E$, we set $L^3 = V/2E$, and the number of states is occupied by one particle in d^3p is

$$d^3n = \frac{V d^3p}{(2\pi)^3 2E}. \quad (9.16)$$

For n -particle scattering, (9.16) is generalized to

$$\text{dLips} = \prod_{f=1}^n \frac{V d^3p_f}{(2\pi)^3 2E_f}. \quad (9.17)$$

This factor is called the **Lorentz-invariant phase space factor (Lips)**. It converts the probability density $\mathcal{P}_{fi} = |\mathcal{A}_{fi}|^2$ into an actual probability. It is more common to see the definition of dLips as

$$\text{dLips} = \delta^4 \left[p_A + p_B - \sum_{f=1}^n p_f \right] \prod_{f=1}^n \frac{V d^3p_f}{(2\pi)^3 2E_f}.$$

In our case, this delta-function is from $|\mathcal{A}_{fi}|^2$ so it is not contained in the dLips.

Observables. All above calculations are theoretical. It is time to connect them to measurable quantities. In our calculations, we use plane waves because they are easy to work with. It is well-known that plane waves extend over all spacetime, making them not normalizable. Therefore, it is more appropriate to consider transition rates rather than the actual number of transitions, so we don't have to worry about. the rate of transitions for an initial state into a certain set of final states in d^3p_f around $\mathbf{p}_f$ is

$$dR = \frac{|\mathcal{A}_{fi}|^2}{T} \frac{V d^3p_f}{(2\pi)^3 2E_f}. \quad (9.18)$$

In this expression, V is a volume containing a large number of small target particles for a beam interact with; T is the time elapse of an experiment. Physically, the target particle is uniformly distributed over an cross-sectional area A seen by an incident particle. Each of them has an effective area $d\sigma$ (the cross section). Hence the cross section is a *mutual* property of both the incident particle and target particle. We also assume that target particles do not shadow each other. That is, target particles do not block each other so that each of them has the chance to interact with an incident particle. If the beam is also distributed uniformly over this area, the probability of an incident particle that interact with any target particle is

$$P = \frac{\text{total effective area}}{\text{total area of detector}} = \frac{N_T d\sigma}{A}.$$

Then the number of incident particles that interact with target particles is

$$N_{\text{int}} = \frac{N_I N_T d\sigma}{A}.$$

Converting it into a rate of interaction by dividing N_{int} by T , an experimentalist would measure the rate according to

$$dR = \frac{N_I N_T d\sigma}{AT}.$$

Then we equate the theoretical expression (9.18) and this experimental expression:

$$\frac{|\mathcal{A}_{fi}|^2}{T} \frac{V d^3p_f}{(2\pi)^3 2E_f} = \frac{N_I N_T d\sigma}{AT}.$$

To proceed, we need to figure out how to write the number of incident particles and targets consistent with the LHS. In general, the number of incident particles is $N_I = J_I AT$, where $J_I = |\mathbf{J}_I| = |\bar{\psi}_I \boldsymbol{\gamma} \psi_I|$ is the probability current of incident particles. For target particles there are two cases:

1. If the target particles are seen as plane waves, just like the incident particles, then the number of target particles is $N_T = \rho_T V$, where ρ_T is the probability density of target particles. For a scattering of a 2-particle state (one incident and one target) into

an n -particle final state, the transition rate is

$$dR = \frac{|\mathcal{A}_{fi}|^2}{T} \prod_{f=1}^n \frac{V d^3 p_f}{(2\pi)^3 2E_f} = \frac{(J_I AT)(\rho_T V) d\sigma}{AT},$$

which gives a differential cross section

$$d\sigma = \frac{|\mathcal{A}_{fi}|^2}{(\rho_T V) J_I T} \prod_{f=1}^n \frac{V d^3 p_f}{(2\pi)^3 2E_f}.$$

This works for almost all scattering processes.

2. Our case is a special case. There is a 1-particle state and one target potential ($N_T = 1$), so the target is not a plane wave. In this case, the transition rate is

$$dR = \frac{|\mathcal{A}_{fi}|^2}{T} \frac{V d^3 p_f}{(2\pi)^3 2E_f} = \frac{N_I d\sigma}{AT}.$$

With the number of incident particles being $N_I = J_I AT$, the differential cross section is

$$d\sigma = \frac{|\mathcal{A}_{fi}|^2}{J_I T} \frac{V d^3 p_f}{(2\pi)^3 2E_f}. \quad (9.19)$$

9.3.3 The Mott Scattering

Now we shall square the scattering amplitude (9.15) and find the scattering cross sections. Let's analyze what we need to compute in $|\mathcal{A}_{fi}|^2$,

$$|\mathcal{A}_{fi}|^2 = \left(\frac{Ze^2}{V} \right)^2 [(2\pi)\delta(E_f - E_i)]^2 |\bar{u}(p_f, s_f) \gamma^0 u(p_i, s_i)|^2 \frac{1}{|\mathbf{q}|^4}.$$

There are basically three things to deal with: a delta-function squared, a square of some spinors, and the evaluation of J_I in (9.19).

Delta-Function Squared. The square of the delta-function can be written as

$$[(2\pi)\delta(E_i - E_f)]^2 = \left[\lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} dt e^{i(E_i - E_f)t} \right] 2\pi\delta(E_i - E_f) = \lim_{T \rightarrow \infty} 2\pi T \delta(E_i - E_f).$$

The parameter T is the time over which the process occurs, and it runs from $-\infty$ to ∞ . This is essentially an artifact of plane waves, because they extend over all spacetime and keep bringing particle probability whatever detectors we are using. As mentioned above, we care only about the transition rate, so this factor of T will be divided out somewhere.

Evaluation of J_I . In the Dirac-Pauli representation, the spinor for a particle and the γ -matrices are given by

$$\psi(p, s) = \frac{1}{\sqrt{V}} u(p, s) e^{-ip \cdot x} = \frac{1}{\sqrt{V}} \sqrt{E+m} \begin{bmatrix} \phi_s \\ \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E+m} \phi_s \end{bmatrix} e^{-ip \cdot x}, \quad \gamma^0 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \gamma^i = \begin{bmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{bmatrix}.$$

The probability current J^i is

$$\begin{aligned} J^i &= \bar{\psi} \gamma^i \psi = \psi^\dagger \gamma^0 \gamma^i \psi = \frac{E+m}{V} \begin{bmatrix} \phi_s^\dagger & \phi_s^\dagger \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E+m} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{bmatrix} \begin{bmatrix} \phi_s \\ \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E+m} \phi_s \end{bmatrix} \\ &= \frac{E+m}{V} \begin{bmatrix} \phi_s^\dagger & -\phi_s^\dagger \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E+m} \end{bmatrix} \begin{bmatrix} \sigma_i \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E+m} \phi_s \\ -\sigma_i \phi_s \end{bmatrix} \\ &= \frac{E+m}{V} \left(\phi_s^\dagger \sigma_i \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E+m} \phi_s + \phi_s^\dagger \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E+m} \sigma_i \phi_s \right) \\ &= \frac{1}{V} \phi_s^\dagger (\sigma_i \sigma_j p^j + \sigma_j p^j \sigma_i) \phi_s. \end{aligned}$$

Using the fact that the Pauli matrices obey $\sigma_i \sigma_j + \sigma_j \sigma_i = 2\delta_{ij}$, and that $\phi_s^\dagger \phi_s = 1$,

$$J^i = \frac{1}{V} \phi_s^\dagger (2\delta_{ij} p^j) \phi_s = \frac{2p^i}{V}.$$

It is easy to verify that this result also works for antiparticles with $\psi(p, s) = \frac{1}{\sqrt{V}} v(p, s) e^{+ip \cdot x}$.

Now substituting the probability current $J_I = 2|\mathbf{p}_i|/V$ and delta-function squared into the differential cross section (9.19), we obtain

$$d\sigma = \frac{V}{2|\mathbf{p}_i|T} \left(\frac{Ze^2}{V} \right)^2 |\bar{u}(p_f, s_f) \gamma^0 u(p_i, s_i)|^2 [2\pi T \delta(E_i - E_f)] \frac{1}{|\mathbf{q}|^4} \frac{V d^3 p_f}{(2\pi)^3 2E_f}.$$

The delta-function is here for us to evaluate some integral in future times, but the integration now is over $d^3 p_f$ rather than dE_f . Thus, a change of variable is needed: by the famous relation $m^2 = E_f^2 - |\mathbf{p}_f|^2$, the differentials are related by

$$E_f dE_f = |\mathbf{p}_f| d|\mathbf{p}_f|.$$

In spherical coordinates (in momentum space), the dLips simplifies to

$$\frac{V d^3 p_f}{(2\pi)^3 2E_f} = \frac{|\mathbf{p}_f| d|\mathbf{p}_f| d\Omega_f}{(2\pi)^3 2E_f} = \frac{|\mathbf{p}_f| E_f dE_f d\Omega_f}{(2\pi)^3 2E_f} = \frac{V}{2(2\pi)^3} |\mathbf{p}_f| dE_f d\Omega_f. \quad (9.20)$$

Plugging this into the differential cross section,

$$d\sigma = \frac{|\mathbf{p}_f|}{4(2\pi)^2 |\mathbf{p}_i|} \left(\frac{Ze^2}{\mathbf{q}^2} \right)^2 |\bar{u}(p_f, s_f) \gamma^0 u(p_i, s_i)|^2 \delta(E_f - E_i) dE_f d\Omega_f.$$

Notice that both T and V goes away, as promised. The delta-function sets $E_f = E_i$ and hence $|\mathbf{p}_f| = |\mathbf{p}_i|$ when we integrate over E_f . The differential cross section further simplifies,

$$\frac{d\sigma}{d\Omega_f} = \frac{1}{4(2\pi)^2} \left(\frac{Ze^2}{\mathbf{q}^2} \right)^2 |\bar{u}(p_f, s_f) \gamma^0 u(p_i, s_i)|^2. \quad (9.21)$$

We are left only with the (hardest) task: the evaluation of $|\bar{u}(p_f, s_f) \gamma^0 u(p_i, s_i)|^2$. This requires a lot of Diracology and some advanced techniques called **trace techniques** for the γ -matrices.

Trace Techniques and Theorems. In regular experiments, the spin of final-state particles are not measured. This means in the cross-section, we need to sum over all possible spin quantum number for the final-state particles. (If the spins are measured then this sum is removed.) Also, the spin of initial-state particles cannot be controlled, which means the average polarization of the beam particles is zero. In this case, we average over all the initial spin quantum number with equal probabilities. Thus, the actual cross section that is more practical to experiments is

$$\frac{d\bar{\sigma}}{d\Omega} = \frac{1}{2} \sum_{s_i = \pm 1/2} \left[\sum_{s_f = \pm 1/2} \frac{d\sigma(s_i, s_f)}{d\Omega_f} \right],$$

where $\bar{\sigma}$ denotes the spin-averaged cross section. The relevant quantity associated with spin in it is

$$L^{\mu\nu} \equiv \frac{1}{2} \sum_{s_i, s_f} (\bar{u}_f \gamma^\mu u_i)^\dagger (\bar{u}_f \gamma^\nu u_i), \quad (9.22)$$

where u_f and u_i are abbreviated versions of $u(p_f, s_f)$ and $u(p_i, s_i)$, respectively. The object $L^{\mu\nu}$ is known as the **lepton tensor**. It is a tensor because $\bar{u} \gamma^\mu u$ and $\bar{u} \gamma^\nu u$ are Lorentz 4-vectors, so their product is a rank-2 tensor. In our case in (9.21), $\mu = \nu = 0$, but it is better to generalize to arbitrary μ and ν .

This is where trace techniques enter. Removing the Hermitian adjoint,

$$\sum_{s_i, s_f} (\bar{u}_f \gamma^\mu u_i)^\dagger (\bar{u}_f \gamma^\nu u_i) = \sum_{s_i, s_f} [u_i^\dagger (\gamma^\mu)^\dagger \gamma^0 u_f^\dagger] (\bar{u}_f \gamma^\nu u_i) = \sum_{s_i, s_f} (\bar{u}_i \gamma^\mu u_f) (\bar{u}_f \gamma^\nu u_i),$$

where $(\gamma^\mu)^\dagger = \gamma^0 \gamma^\mu \gamma^0$ is used. In order to perform the sum, it would be nice to have $\bar{u}_i$ and u_i near each other and $\bar{u}_f$ and u_f near each other, but all these quantities are spinors and matrices that do not commute. To deal with this, we rewrite matrix multiplication in component form:

$$\sum_{s_i, s_f} (\bar{u}_i \gamma^\mu u_f) (\bar{u}_f \gamma^\nu u_i) = \sum_{s_i, s_f} [(\bar{u}_i)_\alpha \gamma^\mu_{\alpha\beta} (u_f)_\beta] [(\bar{u}_f)_\rho \gamma^\nu_{\rho\sigma} (u_i)_\sigma],$$

where Greek subscripts can be 1, 2, 3, 4, and repeated indices are summed over. (These indices are not Lorentz indices because spinors do not transform like a tensor.) Now in components, we are allowed to bring $(u_i)_\sigma$ to the front and then perform the sum,

$$\sum_{s_i, s_f} (\bar{u}_i \gamma^\mu u_f) (\bar{u}_f \gamma^\nu u_i) = \sum_{s_i, s_f} (u_i)_\sigma (\bar{u}_i)_\alpha \gamma^\mu_{\alpha\beta} (u_f)_\beta (\bar{u}_f)_\rho \gamma^\nu_{\rho\sigma} = \sum_{s_i} (u_i)_\sigma (\bar{u}_i)_\alpha \gamma^\mu_{\alpha\beta} \sum_{s_f} (u_f)_\beta (\bar{u}_f)_\rho \gamma^\nu_{\rho\sigma}.$$

Note that $u_i \bar{u}_i$ is a 4×4 matrix, and should not be confused with $\bar{u}_i u_i$, which is a number. Similarly, γ^μ , $u_f \bar{u}_f$, and γ^ν are all 4×4 matrices. In the above expression, if the last index were not σ but some other Greek letter (e.g. δ), then the resulting object would be a 4×4 matrix

$$M_{\sigma\delta} = \sum_{s_i} (u_i)_\sigma (\bar{u}_i)_\alpha \gamma^\mu_{\alpha\beta} \sum_{s_f} (u_f)_\beta (\bar{u}_f)_\rho \gamma^\nu_{\rho\delta}.$$

But the last index is σ . This means we are taking the *trace* of the matrix M . In other words,

$$\sum_{s_i, s_f} (\bar{u}_f \gamma^\mu u_i)^\dagger (\bar{u}_f \gamma^\nu u_i) = \text{Tr} \left[\sum_{s_i} u_i \bar{u}_i \gamma^\mu \sum_{s_f} u_f \bar{u}_f \gamma^\nu \right].$$

The summations over spin is crucial, because they give very useful identities:

$$\sum_s u(p, s) \bar{u}(p, s) = \not{p} + m, \quad \sum_s v(p, s) \bar{v}(p, s) = \not{p} - m. \quad (9.23)$$

These are proved in the Diracology Appendix A.2.7. With the identities in (9.23), the lepton tensor reads

$$L^{\mu\nu} = \frac{1}{2} \sum_{s_i, s_f} (\bar{u}_f \gamma^\mu u_i)^\dagger (\bar{u}_f \gamma^\nu u_i) = \frac{1}{2} \text{Tr} \left[(\not{p}_i + m) \gamma^\mu (\not{p}_f + m) \gamma^\nu \right] \quad (9.24)$$

To fully evaluate it by eliminating all the γ -matrices, we need some **trace theorems**. Here are some of them:

Theorem 9.1. Trace theorems I

1. $\text{Tr } 1_{4 \times 4} = 4$.
2. $\text{Tr}(\text{odd number of } \gamma\text{'s}) = 0$.
3. For any 4-component objects a and b , $\text{Tr}(\not{a} \not{b}) = 4(a \cdot b)$.
4. For any 4-component objects a , b , c , and d , $\text{Tr}(\not{a} \not{b} \not{c} \not{d}) = 4[(a \cdot b)(c \cdot d) + (a \cdot d)(b \cdot c) - (a \cdot c)(b \cdot d)]$.

A useful trick: one can write $\gamma^\mu = \delta^\mu_\nu \bar{\gamma}^\nu \equiv \not{\delta}^\mu$.

Using these trace theorems, the lepton tensor can be written solely in terms of initial and final momenta p_i and p_f or the momentum transfer $q = p_i - p_f$,

$$L^{\mu\nu} \equiv 2[p_i^\mu p_f^\nu + p_i^\nu p_f^\mu - (p_i \cdot p_f) \eta^{\mu\nu}] + 2m^2 \eta^{\mu\nu} = 2[p_i^\mu p_f^\nu + p_i^\nu p_f^\mu - (q^2/2) \eta^{\mu\nu}]. \quad (9.25)$$

Proof. Starting with (9.24) and expanding the products,

$$2L^{\mu\nu} = \text{Tr}(\not{p}_i \gamma^\mu \not{p}_f \gamma^\nu) + m \text{Tr}(\not{p}_i \gamma^\mu \gamma^\nu) + m \text{Tr}(\gamma^\mu \not{p}_f \gamma^\nu) + m^2 \text{Tr}(\gamma^\mu \gamma^\nu).$$

Because the trace of odd number of γ -matrices always vanish, the terms linear in m go away,

$$\begin{aligned} 2L^{\mu\nu} &= \text{Tr}(\not{p}_i \gamma^\mu \not{p}_f \gamma^\nu) + m^2 \text{Tr}(\gamma^\mu \gamma^\nu) \\ &= \text{Tr}(\not{p}_i \delta^\mu_\alpha \gamma^\alpha \not{p}_f \delta^\nu_\beta \gamma^\beta) + \frac{1}{2} m^2 \text{Tr}(\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu) \\ &= \text{Tr}(\not{p}_i \not{\delta}^\mu \not{p}_f \not{\delta}^\nu) + 4m^2 \eta^{\mu\nu}. \end{aligned}$$

In the second equality, we used the cyclic property of trace $\text{Tr}(AB) = \text{Tr}(BA)$, and in the third equality, we used the identity

$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}1$. The first term is in the form of $\text{Tr}(\not{p}_i \not{p}_f \not{\epsilon})$, so we can use trace theorem 3,

$$\begin{aligned}\text{Tr}(\not{p}_i \not{\epsilon}^\mu \not{p}_f \not{\epsilon}^\nu) &= 4[(p_i \cdot \delta^\mu)(p_f \cdot \delta^\nu) + (p_i \cdot \delta^\nu)(p_f \cdot \delta^\mu) - (p_i \cdot p_f)(\delta^\mu \cdot \delta^\nu)] \\ &= 4[(p_f^\alpha \delta^\mu_\alpha)(p_f^\beta \delta^\nu_\beta) + (p_i^\beta \delta^\nu_\beta)(p_f^\alpha \delta^\mu_\alpha) - (p_i \cdot p_f)\eta^{\alpha\beta}\delta^\mu_\alpha \delta^\nu_\beta] \\ &= 4[p_i^\mu p_f^\nu + p_i^\nu p_f^\mu - (p_i \cdot p_f)\eta^{\mu\nu}].\end{aligned}$$

Substituting these quantities into $2L^{\mu\nu}$ gives the first equality of (9.25). The second equality is simple of (9.25). By the definition of momentum transfer, $q = p_i - p_f$, its square is $q^2 = (p_i - p_f)^2 = p_i^2 + p_f^2 - 2p_i \cdot p_f = 2(m^2 - p_i \cdot p_f)$. Therefore, the lepton tensor can also be expressed as

$$L^{\mu\nu} = 2[p_i^\mu p_f^\nu + p_i^\nu p_f^\mu + (q^2/2)\eta^{\mu\nu}].$$

□

The Mott Cross Section. We have obtained the general lepton tensor $L^{\mu\nu}$. What we need in the spin-averaged cross section for the Mott scattering is just L^{00} ,

$$L^{00} = 2[E_i E_f + E_f E_i - p_i \cdot p_f] + 2m^2 = 2[2E_i E_f - p_i \cdot p_f + m^2].$$

We know that $E_i = E_f$ and $|\mathbf{p}_i| = |\mathbf{p}_f|$ for the Mott scattering. Then in terms of the scattering angle θ , defined as $\mathbf{p}_i \cdot \mathbf{p}_f = \mathbf{p}^2 \cos \theta$,

$$L^{00} = 2[2E^2 - (E^2 - \mathbf{p}^2 \cos \theta) + E^2 - \mathbf{p}^2] = 2E^2 \left[2 - \frac{\mathbf{p}^2}{E^2}(1 - \cos \theta) \right] = 4E^2 \left(1 - v^2 \sin^2 \frac{\theta}{2} \right),$$

where $v = |\mathbf{p}|/E$ is the speed of the particle (in natural units). According to (9.21), the spin-averaged cross section is

$$\frac{d\bar{\sigma}}{d\Omega_f} = \frac{1}{4(2\pi)^2} \left(\frac{Ze^2}{\mathbf{q}^2} \right)^2 L^{00}.$$

The 3-momentum transfer can also be expressed in θ :

$$\mathbf{q}^2 = (\mathbf{p}_i - \mathbf{p}_f)^2 = 2\mathbf{p}^2(1 - \cos \theta) = 4\mathbf{p}^2 \sin^2 \frac{\theta}{2}.$$

Substituting $\mathbf{q}^2$ and L^{00} into the spin-averaged cross section,

$$\frac{d\bar{\sigma}}{d\Omega} = \frac{1}{(2\pi)^2} \frac{(Ze^2)^2}{16|\mathbf{p}|^4 \sin^2 \frac{\theta}{2}} E^2 \left(1 - v^2 \sin^2 \frac{\theta}{2} \right) = \frac{1}{4} \left(\frac{Ze^2}{4\pi} \right)^2 \frac{E^2}{|\mathbf{p}|^4} \frac{(1 - v^2 \sin^2 \frac{\theta}{2})}{\sin^4 \frac{\theta}{2}}.$$

Finally, replacing $e^2/4\pi$ with the fine-structure constant α , we obtain the famous **Mott cross section**:

$$\boxed{\frac{d\bar{\sigma}}{d\Omega_f} = (Z\alpha)^2 \frac{E^2}{4|\mathbf{p}|^4} \frac{1 - v^2 \sin^2 \frac{\theta}{2}}{\sin^4 \frac{\theta}{2}}.} \quad (9.26)$$

Let's comment on this cross section:

- Compared to the Rutherford cross section (9.14), there is an extra angular dependence $1 - v^2 \sin^2 \frac{\theta}{2}$. This term arises from the spin-1/2 nature of the electron.
- In the non-relativistic limit $v \rightarrow 0$, this extra factor also goes away. In Rutherford's original scattering experiment, the α -particle is non-relativistic and is a spin-0 particle, so this extra factor is not present.
- In the relativistic limit $v \rightarrow 1$, the Mott cross section becomes

$$\frac{d\bar{\sigma}}{d\Omega_f} = (Z\alpha)^2 \frac{E^2}{4|\mathbf{p}|^2} \frac{\cos^2 \frac{\theta}{2}}{\sin^4 \frac{\theta}{2}}.$$

Now if we take $\theta = \pi$, the cross section goes to zero. In other words, the back scattering of an electron off a Coulomb potential is forbidden in the relativistic limit.

9.3.4 Helicity Conservation

To be written.

9.4 Quantization

9.4.1 Quantizing A^μ

To be written.

9.4.2 The Photon Propagator

To be written.

9.4.3 The Feynman Rules

The Feynman rules for QED are summarized in Table 9.1.

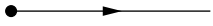
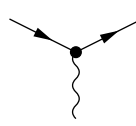
Particle	Amplitude element	Feynman diagram element
Initial-state fermion	$u(p, s)$	
Final-state fermion	$\bar{u}(p, s)$	
Initial-state antifermion	$\bar{v}(p, s)$	
Final-state antifermion	$v(p, s)$	
Initial-state photon	$\epsilon_\mu(p, \lambda)$	
Final-state photon	$\epsilon_\mu^*(p, \lambda)$	
Fermion propagator	$-\frac{i(\gamma^\mu q_\mu + m)}{q^2 - m^2}$	
Photon propagator	$-\frac{i\eta_{\mu\nu}}{q^2}$	
QED vertex	$-iQ\gamma^\mu$	e.g. 

Table 9.1: QED Feynman rules for lowest-order amplitudes. For clarity, the vertex is labeled by a dot. Q is the charge of the fermion. E.g. for charged leptons, $Q = -e$.

Note that there is no gauge dependence ξ in the photon propagator. This will be explained later in the electron-muon scattering example.

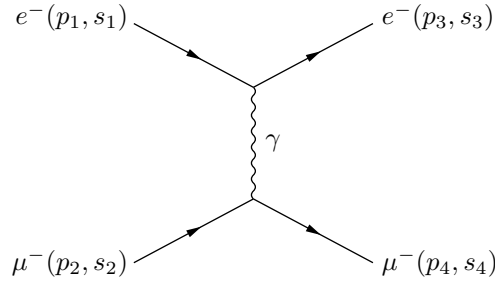
9.5 Electron-Muon Scattering

9.5.1 The Scattering Amplitude

In this section, we introduce our first example of scattering of two spin-1/2 particles in QED and our first application of the propagator. Consider the process of electron-muon scattering

$$e^- \mu^- \rightarrow e^- \mu^-.$$

Denote the electron mass as m and the muon mass as M . The lowest-order (second-order in this case) QED Feynman diagram is given by



The 4-momentum of the virtual photon is $q = p_1 - p_3 = p_4 - p_2$. Let us count the elements in this diagram and apply the Feynman rules listed in Table 9.1:

1. There are two initial-state fermions and two final-state fermions. According to Feynman rules, initial particles correspond to spinors $u(p_1, s_1)$ and $u(p_2, s_2)$, while final particles correspond to $\bar{u}(p_3, s_3)$ and $\bar{u}(p_4, s_4)$. We should also pair up the corresponding momenta from the Feynman diagram: p_1 to p_3 for the electron, and p_2 to p_4 for the muon.
2. There are two vertices, each with a vertex factor of $(-i)(-e)\gamma^\mu = ie\gamma^\mu$.
3. Finally, there is a photon propagator, given by $-i\eta_{\mu\nu}/q^2$.

Putting them altogether, the lowest-order scattering amplitude is

$$\mathcal{A} = [ie\bar{u}(p_3, s_3)\gamma^\mu u(p_1, s_1)] \frac{-i\eta_{\mu\nu}}{q^2} [ie\bar{u}(p_4, s_4)\gamma^\nu u(p_2, s_2)].$$

For simplicity, abbreviate $u(p_1, s_1) \rightarrow u_1$ and similarly for other spinors. The square of the amplitude is

$$|\mathcal{A}|^2 = \left(\frac{e^2}{q^2}\right)^2 |\bar{u}_3\gamma_\mu u_1 \bar{u}_4\gamma^\mu u_2|^2 = \left(\frac{4\pi\alpha}{q^2}\right)^2 (\bar{u}_3\gamma_\mu u_1 \bar{u}_4\gamma^\mu u_2)^\dagger (u_3\gamma_\nu u_1 \bar{u}_4\gamma^\nu u_2).$$

Using $(\bar{u}_f\gamma^\mu \bar{u}_i)^\dagger = \bar{u}_i\gamma^\mu u_f$,

$$|\mathcal{A}|^2 = \left(\frac{4\pi\alpha}{q^2}\right)^2 (\bar{u}_2\gamma^\mu u_4)(\bar{u}_1\gamma_\mu u_3)(\bar{u}_3\gamma_\nu u_1)(\bar{u}_4\gamma^\nu u_2).$$

In fact, the quantity like $\bar{u}_f\gamma^\mu u_i$ is just a number (or 1×1 matrix), so it commutes with everything. In particular, we want to move $\bar{u}_2\gamma^\mu u_4$ to the end,

$$|\mathcal{A}|^2 = \left(\frac{4\pi\alpha}{q^2}\right)^2 (\bar{u}_1\gamma_\mu u_3)(\bar{u}_3\gamma_\nu u_1)(\bar{u}_4\gamma^\nu u_2)(\bar{u}_2\gamma^\mu u_4).$$

To relate the amplitude to observables (the scattering cross section), we compute the spin-averaged amplitude squared:

$$\overline{|\mathcal{A}|^2} \equiv \frac{1}{2} \sum_{s_1, s_3} \frac{1}{2} \sum_{s_2, s_4} |\mathcal{A}|^2 = \frac{1}{4} \sum_{s_1, \dots, s_4} \left(\frac{4\pi\alpha}{q^2}\right)^2 (\bar{u}_1\gamma_\mu u_3 \bar{u}_3\gamma_\nu u_1)(u_2\gamma^\mu u_4 \bar{u}_4\gamma^\nu u_2) = \left(\frac{4\pi\alpha}{q^2}\right)^2 L(e)_{\mu\nu} L(\mu)^{\mu\nu}.$$

In the last step, we identify the sums as the lepton tensors (9.22) for the electron and muon, and then by (9.25),

$$L(e)^{\mu\nu} \equiv \frac{1}{2} \sum_{s_1, s_3} \bar{u}_1\gamma^\mu u_3 \bar{u}_3\gamma^\nu u_1 = 2[p_1^\mu p_3^\nu + p_1^\nu p_3^\mu + (q^2/2)\eta^{\mu\nu}],$$

$$L(\mu)^{\mu\nu} \equiv \frac{1}{2} \sum_{s_2, s_4} \bar{u}_2\gamma^\mu u_4 \bar{u}_4\gamma^\nu u_2 = 2[p_2^\mu p_4^\nu + p_2^\nu p_4^\mu + (q^2/2)\eta^{\mu\nu}].$$

9.5.2 Lepton Tensor Contraction; Conserved Currents

Before we evaluate the contraction $L(e)_{\mu\nu}L(\mu)^{\mu\nu}$, it is important to know an identity:

$$q^\mu L_{\mu\nu} = q^\nu L_{\mu\nu} = 0. \quad (9.27)$$

This is basically the statement of conserved currents, also shown in the following proof.

Proof. We present two approaches prove (9.27). The first approach is by a direct calculation of $q_\mu L^{\mu\nu}$:

$$\begin{aligned} q^\mu L_{\mu\nu} &= 2q^\mu \left(p_{1\mu}p_{3\nu} + p_{1\nu}p_{3\mu} + \frac{1}{2}q^2\eta_{\mu\nu} \right) \\ &= 2(p_1^\mu - p_3^\mu) \left[p_{1\mu}p_{3\nu} + p_{1\nu}p_{3\mu} + \frac{1}{2}(p_1^2 + p_3^2 - 2p_1 \cdot p_3)\eta_{\mu\nu} \right] \\ &= 2[m^2p_{3\nu} + p_{1\nu}(p_1 \cdot p_3) - (p_1 \cdot p_3)p_{3\nu} - p_1m^2 + (p_{1\nu} - p_{3\nu})(m^2 - p_1 \cdot p_3)] \\ &= 0. \end{aligned}$$

The second approach has more physical significance as it is related to the conserved current. This time we use the Lepton tensor in the form (9.22)

$$L^{\mu\nu} \propto (\bar{u}_3\gamma^\mu u_1)^\dagger (\bar{u}_3\gamma^\nu u_1) = (\bar{u}_1\gamma^\mu u_3)(\bar{u}_3\gamma^\nu u_1).$$

Now consider the Dirac field of an electron, $\psi_1 = u_1 e^{-ip_1 \cdot x}$. The Dirac equation implies that

$$i\gamma^\mu \partial_\mu \psi_1 = m\psi_1 \implies p_{1\mu}\gamma^\mu u_1 = m_1 u_1.$$

Similarly, for $\psi_3 = u_3 e^{-ip_3 \cdot x}$, the Dirac equation also implies $p_{3\mu}\gamma^\mu u_3 = m_3 u_3$. Taking the complex conjugate of the ψ_1 equation and then multiply by γ^0 ,

$$\begin{aligned} u_1^\dagger p_{1\mu}(\gamma^\mu)^\dagger \gamma^0 &= m_1 u_1^\dagger \gamma^0 \\ u_1^\dagger p_{1\mu}(\gamma^0 \gamma^\mu \gamma^0) \gamma^0 &= m_1 u_1^\dagger \gamma^0 \\ \bar{u}_1 p_{1\mu} \gamma^\mu &= m_1 \bar{u}_1. \end{aligned}$$

Now calculate $q^\mu L_{\mu\nu}$,

$$q^\mu L_{\mu\nu} \propto (p_1^\mu - p_3^\mu)(\bar{u}_1\gamma^\mu u_3)(\bar{u}_3\gamma^\nu u_1) = (m_1 - m_3)(\bar{u}_1\gamma^\mu u_3)(\bar{u}_3\gamma^\nu u_1).$$

But both the initial and final particles are electrons, $m_1 = m_3 = m$, the result is $q^\mu L_{\mu\nu} = 0$. Recall that $\bar{u}_1\gamma^\mu u_3 e^{i(p_1 - p_3) \cdot x} = \bar{\psi}_1 \gamma^\mu \psi_3$ has the form of the conserved current J_ψ . Taking the divergence $\partial_\mu J_\psi^\mu$ pulls down a momentum transfer $q = p_1 - p_3$. Thus, the current conservation $\partial_\mu J_\psi^\mu = 0$ is equivalent to $q^\mu L_{\mu\nu} = 0$. $\square$

As an aside, recall that we used $-i\eta_{\mu\nu}/q^2$ as the photon propagator instead of $i[-\eta_{\mu\nu} + (1 - \xi)q_\mu q_\nu / q^2]/q^2$. Now it is clear why, as all terms like $q_\mu(\bar{u}_i\gamma^\mu u_f)$ gives zero. Hence the gauge-dependent part $(1 - \xi)q_\mu q_\nu$, when contracted with $(\bar{u}_1\gamma^\mu u_3)$ and $(\bar{u}_3\gamma^\nu u_1)$, all do not contribute. This makes sense: the amplitude squared is a physically measurable quantity, so it must be gauge invariant.

With (9.27), all $q^\mu L_{\mu\nu}$ terms can be neglected. we can simplify $|\overline{\mathcal{A}}|^2$ by writing $p_4 = p_2 + q$,

$$\begin{aligned} |\overline{\mathcal{A}}|^2 &= 2 \left(\frac{4\pi\alpha}{q^2} \right)^2 L_{\mu\nu} [p_2^\mu(p_2^\nu + q^\nu) + p_2^\nu(p_2^\mu + q^\mu) + (q^2/2)\eta^{\mu\nu}] \\ &= 4 \left(\frac{4\pi\alpha}{q^2} \right)^2 [p_{1\mu}p_{3\nu} + p_{1\nu}p_{3\mu} + (q^2/2)\eta_{\mu\nu}][2p_2^\mu p_2^\nu + (q^2/2)\eta^{\mu\nu}] \\ &= 4 \left(\frac{4\pi\alpha}{q^2} \right)^2 [4(p_1 \cdot p_2)(p_3 \cdot p_2) + M^2 q^2 + (p_1 \cdot p_3)q^2 + q^4]. \end{aligned}$$

In principle, this is all we can do before specifying any information about the muon's initial momentum p_2 and the reference frame we are working in. In general, for situations where one light particle (such as an electron with $m = 0.511$ MeV) scattering from a much heavier particle (such as a muon with $M = 105.6$ MeV), we often work in the so-called "lab frame", or the rest frame of the heavier particle. On the other hand, if the two initial particles have roughly (or exactly) the same mass, such as an electron-positron scattering, we may want to work in the center-of-momentum (CM) frame.

9.5.3 The Cross Section in the Lab Frame

For the electron-muon scattering, we shall work in the lab frame in which the muon is at rest, $p_2^\mu = (M, 0, 0, 0)$. Let $p_1^\mu = (E_1, \mathbf{p}_1)$ and $p_3^\mu = (E_1, \mathbf{p}_3)$, and θ be the scattering angle defined as $\mathbf{p}_1 \cdot \mathbf{p}_3 = |\mathbf{p}_1||\mathbf{p}_3| \cos \theta$. In this case,

$$p_1 \cdot p_2 = ME_1, \quad p_3 \cdot p_2 = ME_3, \quad p_1 \cdot p_3 = E_1 E_3 - |\mathbf{p}_1||\mathbf{p}_3| \cos \theta, \quad q^2 = (p_1 - p_3)^2 = 2m^2 - 2p_1 \cdot p_3,$$

and the spin-averaged amplitude squared is

$$\overline{|\mathcal{A}|^2} = 4 \left(\frac{4\pi\alpha}{q^2} \right)^2 [4M^2 E_1 E_3 + M^2 q^2 + (m^2 - q^2/2)q^2 + q^4].$$

Taking the relativistic limit of the electron, $m = 0$, $E_1 = |\mathbf{p}_1|$, $E_3 = |\mathbf{p}_3|$, and

$$q^2 = -2p_1 \cdot p_3 = 2E_1 E_3 (1 - \cos \theta) = -4E_1 E_3 \sin^2 \frac{\theta}{2}.$$

$$\overline{|\mathcal{A}|^2} = 4 \left(\frac{4\pi\alpha}{q^2} \right)^2 = 4 \left(\frac{4\pi\alpha}{q^2} \right)^2 \left(-\frac{M^2 q^2}{\sin^2 \frac{\theta}{2}} + M^2 q^2 + \frac{1}{2} q^4 \right) = 2(4\pi\alpha)^2 \left(1 - \frac{2M^2}{q^2} \cot^2 \frac{\theta}{2} \right).$$

Using the formula (8.17), we obtain the electron-muon scattering differential cross section in the lab frame,

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{2} \left(M - \frac{q^2}{4E_3} \right)^{-2} \left(1 - \frac{2M^2}{q^2} \cot^2 \frac{\theta}{2} \right).$$

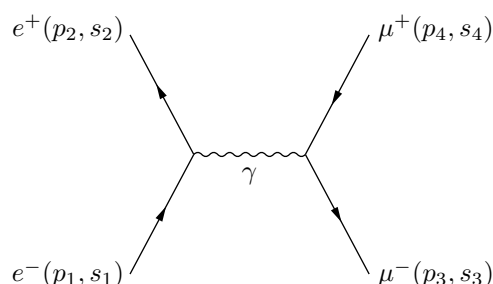
9.6 Electron-Positron Annihilation

9.6.1 The Scattering Amplitude

Consider the process of electron-positron annihilation, creating a muon-antimuon pair:

$$e^+ e^- \rightarrow \mu^+ \mu^-.$$

Let the electron mass be m and the muon mass be M . The lowest-order QED Feynman diagram for this process is



There are antiparticles involved. The 4-momentum of the virtual photon is now $q = p_1 + p_2 = p_3 + p_4$, as seen from the Feynman diagram and 4-momentum conservation. According to the Feynman rules in Table 9.1:

1. The spinors for the initial-state electron is $u(p_1, s_1)$; for the initial-state positron is $\bar{v}(p_2, s_2)$; for the final-state muon is $\bar{u}(p_3, s_3)$; and for the final-state antimuon is $v(p_4, s_4)$.
2. There are two vertex factors $ie\gamma^\mu$ and $ie\gamma^\nu$.
3. There is a photon propagator $-i\eta_{\mu\nu}/q^2$.

The resulting scattering amplitude is

$$\mathcal{A} = [ie\bar{v}(p_2, s_2)\gamma^\mu u(p_1, s_1)] \frac{-i\eta_{\mu\nu}}{q^2} [ie\bar{u}(p_3, s_3)\gamma^\nu v(p_4, s_4)].$$

Again, we adopt the abbreviation $u(p_1, s_1) \rightarrow u_1$ and so on. The spin-averaged amplitude squared is

$$\begin{aligned} \overline{|\mathcal{A}|^2} &= \frac{1}{4} \left(\frac{e^2}{q^2} \right)^2 \sum_{s_1, \dots, s_4} |\bar{v}_2 \gamma_\mu u_1 \bar{u}_3 \gamma^\mu v_4|^2 \\ &= \frac{1}{4} \left(\frac{4\pi\alpha}{q^2} \right)^2 \sum_{s_1, \dots, s_4} (\bar{v}_4 \gamma^\nu u_3) (\bar{u}_1 \gamma_\mu v_2) (\bar{v}_2 \gamma_\nu u_1) (\bar{u}_3 \gamma^\mu v_4) \\ &= \left(\frac{4\pi\alpha}{q^2} \right)^2 \left(\frac{1}{2} \sum_{s_1, s_2} \bar{u}_1 \gamma_\mu v_2 \bar{v}_2 \gamma_\nu u_1 \right) \left(\frac{1}{2} \sum_{s_3, s_4} \bar{u}_3 \gamma^\mu v_4 \bar{v}_4 \gamma^\nu u_3 \right). \end{aligned}$$

9.6.2 Trace Techniques Revisited

The terms in the brackets has the form of lepton tensors. We will later identify them as lepton tensors, but since they involve antiparticle spinors v , it is better to handle them with care for the first time. Applying trace techniques,

$$\sum_{s_1, s_2} \bar{u}_1 \gamma_\mu v_2 \bar{v}_2 \gamma_\nu u_1 = \text{Tr} \left(\sum_{s_1} u_1 \bar{u}_1 \gamma_\mu \sum_{s_2} v_2 \bar{v}_2 \gamma_\nu \right) = \text{Tr}[(\not{p}_1 + m) \gamma_\mu (\not{p}_2 - m) \gamma_\nu].$$

Notice the minus sign in $\not{p}_2 - m$. It comes from $\sum_{s_2} v_2 \bar{v}_2 = \not{p}_2 - m$, which is different from what we had before. (Compare it to Equation 9.24.) Evaluated the trace using trace theorems:

$$\begin{aligned} \text{Tr}[(\not{p} + m) \gamma_\mu (\not{p} - m) \gamma_\nu] &= \text{Tr}(\not{p}_1 \gamma_\mu \not{p}_2 \gamma_\nu) - m^2 \text{Tr}(\gamma_\mu \gamma_\nu) + \text{Tr}(\text{odd number of } \gamma\text{'s}) \\ &= 4[p_{1\mu} p_{2\nu} + p_{1\nu} p_{2\mu} - (p_1 \cdot p_2) \eta_{\mu\nu}] - 4m^2 \eta_{\mu\nu} \\ &= 4[p_{1\mu} p_{2\nu} + p_{1\nu} p_{2\mu} - (q^2/2) \eta_{\mu\nu}]. \end{aligned}$$

(Recall that the momentum transfer is now $q = q_1 + q_2 = q_3 + q_4$, so $q^2 = 2(m^2 + q_1 \cdot q_2)$, and also note the minus sign before $q^2/2$.) By a similar procedure,

$$\sum_{s_3, s_4} \bar{u}_3 \gamma^\mu v_4 \bar{v}_4 \gamma^\nu u_3 = \text{Tr}[(\not{p}_3 + M) \gamma^\mu (\not{p}_4 - M) \gamma^\nu] = 4[p_3^\mu p_4^\nu + p_3^\nu p_4^\mu - (q^2/2) \eta^{\mu\nu}].$$

Inserting these results into $\overline{|\mathcal{A}|^2}$,

$$\overline{|\mathcal{A}|^2} = 4 \left(\frac{4\pi\alpha}{q^2} \right)^2 [p_{1\mu} p_{2\nu} + p_{1\nu} p_{2\mu} - (q^2/2) \eta_{\mu\nu}] [p_3^\mu p_4^\nu + p_3^\nu p_4^\mu - (q^2/2) \eta^{\mu\nu}] \equiv \left(\frac{4\pi\alpha}{q^2} \right)^2 L(e)_{\mu\nu} L(\mu)^{\mu\nu}.$$

Here we identify the lepton tensors $L(e)^{\mu\nu}$ for the electron-positron field, and $L(\mu)^{\mu\nu}$ for the muon-antimuon field:

$$\begin{aligned} L(e)^{\mu\nu} &\equiv 2[p_1^\mu p_2^\nu + p_1^\nu p_2^\mu - (q^2/2) \eta^{\mu\nu}], \\ L(\mu)^{\mu\nu} &\equiv 2[p_3^\mu p_4^\nu + p_3^\nu p_4^\mu - (q^2/2) \eta^{\mu\nu}]. \end{aligned}$$

Even though the lepton tensors look different from before, they are well-defined here because they still obey $q^\mu L_{\mu\nu} = 0$, given that $q = p_1 + p_2 = p_3 + p_4$. The proof is similar to that of (9.27). Then rewrite $p_4 = q - p_3$ in the muon lepton tensor to find $L(e)_{\mu\nu} L(\mu)^{\mu\nu}$,

$$\begin{aligned} L(e)_{\mu\nu} L(\mu)^{\mu\nu} &= 2L(e)_{\mu\nu} [p_3^\mu (q^\nu - p_3^\nu) + p_3^\nu (q^\mu - p_3^\mu) - (q^2/2) \eta^{\mu\nu}] \\ &= 4[p_{1\mu} p_{2\nu} + p_{1\nu} p_{2\mu} - (q^2/2) \eta_{\mu\nu}] [-2p_3^\mu p_3^\nu - (q^2/2) \eta^{\mu\nu}] \\ &= 4[-4(p_1 \cdot p_3)(p_2 \cdot p_3) + M^2 q^2 - (p_1 \cdot p_2) q^2 + q^4]. \end{aligned}$$

This gives the spin-averaged amplitude squared as

$$\overline{|\mathcal{A}|^2} = 4 \left(\frac{4\pi\alpha}{q^2} \right)^2 [-4(p_1 \cdot p_3)(p_2 \cdot p_3) + M^2 q^2 - (p_1 \cdot p_2) q^2 + q^4].$$

9.6.3 The Cross Section in the CM Frame

Consider the center-of-momentum frame of the process. The 4-momentum conservation tells that $\mathbf{p}_1 = -\mathbf{p}_2$, $\mathbf{p}_3 = -\mathbf{p}_4$, and $E_1 = E_2 = E_3 = E_4 = E$. Define the scattering angle θ by $\mathbf{p}_1 \cdot \mathbf{p}_3 = |\mathbf{p}_1||\mathbf{p}_3|\cos\theta$. Then

$$\begin{aligned}\overline{|\mathcal{A}|^2} &= 4 \left(\frac{4\pi\alpha}{q^2} \right)^2 [-4(E^2 - |\mathbf{p}_1||\mathbf{p}_3|\cos\theta)(E^2 + |\mathbf{p}_1||\mathbf{p}_3|\cos\theta) + M^2 q^2 - (E^2 + \mathbf{p}_1^2)q^2 + q^4] \\ &= 4 \left(\frac{4\pi\alpha}{q^2} \right)^2 [-4(E^4 - \mathbf{p}_1^2 \mathbf{p}_3^2 \cos^2\theta) + M^2 q^2 - (E^2 + \mathbf{p}_1^2)q^2 + q^4].\end{aligned}$$

Taking the relativistic limit where both $m, M \rightarrow 0$. This gives $|\mathbf{p}_1| = |\mathbf{p}_2| = |\mathbf{p}_3| = |\mathbf{p}_4| = E$, and

$$\overline{|\mathcal{A}|^2} = 4 \left(\frac{4\pi\alpha}{q^2} \right)^2 [-4E^4(1 - \cos^2\theta) - 2E^2 q^2 + q^4].$$

In the CM frame, using the relation $q^2 = (2E)^2 = 4E^2$ gives

$$\overline{|\mathcal{A}|^2} = (4\pi\alpha)^2(1 + \cos^2\theta). \quad (9.28)$$

The differential cross section in the CM frame is given by (8.16),

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|} \overline{|\mathcal{A}|^2}.$$

In the CM frame, $q^2 = s$ is the CM energy, and $|\mathbf{p}_i| = |\mathbf{p}_f|$ for electron-positron annihilation, so

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4s}(1 + \cos^2\theta). \quad (9.29)$$

Integrating over all angles, we obtain the cross section,

$$\boxed{\sigma = \frac{4\pi\alpha^2}{3s}}. \quad (9.30)$$

Note that the cross section is inversely proportional to the CM energy. That is, for electron-positron annihilation, higher beam energies lead to low interaction rates. This is a common feature of the electromagnetic interaction.

9.7 Scattering from a Charged, Scalar Particle

To be written.

10 QUANTUM CHROMODYNAMICS

In the last chapter, we impose a local $U(1)$ symmetry to the Lagrangian and the results are electromagnetic interactions. Quantizing the electromagnetic 4-potential A^μ gives a gauge boson known as the photon. Throughout the process, we see that the form of the interaction is set once the $U(1)$ group is given. In this chapter, we develop a gauge theory describing the **strong interaction** between quarks and gluons. This gauge theory is called **quantum chromodynamics (QCD)**.

10.1 The Color Charge

10.1.1 Internal Degrees of Freedom

The theory of strong interaction is associated with the gauge group called $SU(3)$. The group elements are unitary 3×3 matrices with determinant 1. The first question to ask could be: why 3? You may try to recall where have you seen the number 3 in particle physics.

Experimental evidence suggests that there are three unknown degrees of freedom in quarks (or hadrons), which we now call the **color**. A typical one is the quantity R , defined in Example 4.1:

$$R = \frac{\sigma(e^+e^- \rightarrow q_i \bar{q}_i)}{\sigma(e^+e^- \rightarrow \mu^+ \mu^-)}.$$

This is a QED process, which seemingly gives no clue to strong interactions, but recall that we multiply the numerator by 3 in the calculation to account for the 3 possible colors of a quark. The factor of 3 is not present in the denominator because muons do not have color. If the 3 degrees of freedom were not there, the theoretical calculation cannot match the experimental data.

Historically, before the proposal of the quark model, many mesons and baryons were discovered. Physicists started to think that hadrons might not be the most fundamentals in nature. In 1964, Murray Gell-Mann and George Zweig proposed that hadrons are composed of some spin-1/2 elementary particles (which we now know as quarks). In 1965, the first evidence of the color charge comes from observations of the Δ -baryon, in particular $\Delta^{++} = uuu$. The full wave function of the Δ^{++} is the product of the spatial wave function ψ , spin χ , flavor ϕ , and maybe something else denoted as C . Let's suppose the full wave function is

$$\Psi = \psi(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) \chi(m_1, m_2, m_3) \phi(f_1, f_2, f_3) C.$$

The flavor wave function is obviously symmetric because of the quark content $\Delta^{++} = uuu$. Also, the Δ^{++} is measured to be $s = 3/2$ and $L = 0$, so its spatial wave function and spin states are both symmetric. With everything symmetric, the Δ^{++} violates the Pauli exclusion principle because quarks are fermions. Therefore, there has to be an antisymmetric piece $C(c_1, c_2, c_3)$ attached to Ψ so that the total wave function is antisymmetric. There should be (at least) three quantum numbers in C , the *color state*, so that all the three quarks are exchangeable, and the three quantum numbers are called red, green, and blue (R,G,B). For baryons, the simplest color state that we can construct is

$$|C\rangle = \frac{1}{\sqrt{6}} \sum_{a,b,c=R,G,B} |a, b, c\rangle = \frac{1}{\sqrt{6}} (|R, G, B\rangle - |R, B, G\rangle + |G, B, R\rangle - |G, R, B\rangle + |B, R, G\rangle - |B, G, R\rangle).$$

One can check that it is totally antisymmetric.

Now the question of 3 is answered. Other questions might be: why $SU(3)$ but not other groups like $SO(3)$ or $U(3)$. The short answer is that $SU(3)$ is the only one that matches the experiments. For example, the elements in the $SO(3)$ group (the group of rotation matrices) is *real*. This is saying that quarks and antiquarks transform the same way in the color space, but in fact it does not. On the other hand, the $U(3)$ group is not a gauge group for the strong interaction because it would give rise to a colorless gluon in addition to 8 known colorful gluons. The colorless gluon is not observed experimentally.

10.1.2 The SU(3) Algebra

Quarks are spin-1/2 particles, so they obey the Dirac equation and can be represented by spinors. Specifically, we write the spinor of a quark as

$$\psi(x) \equiv \begin{bmatrix} \psi_R(x) \\ \psi_G(x) \\ \psi_B(x) \end{bmatrix}.$$

Here ψ_R , ψ_G , and ψ_B are each 4-component spinors, so the whole $\psi(x)$ is some 12-component object. An SU(3) transformation of the color space does not care about the individual components in the spinors, so we can treat $\psi(x)$ as a 3-component object in color space. The generators of an SU(n) group must be Hermitian and traceless. For SU(3), there are 8 independent generators. Naturally, one can construct 9 generators with 2×2 block of Pauli spin matrices inside the 3×3 matrix:

$$\begin{aligned} \lambda_1 &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & \lambda_2 &= \begin{bmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & \lambda_3 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \\ \lambda_4 &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, & \lambda_5 &= \begin{bmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{bmatrix}, & \lambda_X &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \\ \lambda_6 &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, & \lambda_7 &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{bmatrix}, & \lambda_Y &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}. \end{aligned}$$

Only 8 of the 9 λ -matrices above are independent. One of the three diagonal matrices can be expressed in terms of the other two. We will let

$$\lambda_8 = \frac{1}{\sqrt{3}}(\lambda_X + \lambda_Y) = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}.$$

The matrices λ_a for $a = 1, 2, \dots, 8$ are known as the **Gell-Mann matrices**:

$$\begin{aligned} \lambda_1 &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & \lambda_2 &= \begin{bmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & \lambda_3 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \\ \lambda_4 &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, & \lambda_5 &= \begin{bmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{bmatrix}, & \lambda_6 &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \\ \lambda_7 &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{bmatrix}, & \lambda_8 &= \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}. \end{aligned} \tag{10.1}$$

Conventionally, the generators of the SU(3) group, denoted as T_a , are related to the Gell-Mann matrices by

$$T_a = \frac{1}{2}\lambda_a.$$

The algebra of a group is given by the commutation relations:

$$[T_a, T_b] = if_{abc}T_c.$$

The coefficients f_{abc} are called the **structure constants** of the group. For SU(n), the structure constants are totally antisymmetric.

10.2 SU(3) Gauge Invariance

Consider an SU(3) gauge transformation

$$U = e^{ig\alpha_a T_a},$$

where g is a coupling constant, T_a are the generators of the SU(3) group, α_a are 8 parameters for $a = 1, 2, \dots, 8$, and repeated indices are summed over. The color state of a quark transforms as

$$\psi \rightarrow \psi' = U\psi.$$

(Since the matrix U acts on the color space, it should be a 3×3 matrix. However, each component of the color space is a spinor ψ_a for $a = R, G, B$. Hence if we write out U , it is in fact a 12×12 block diagonal matrix. But the key is that it commutes with γ^μ , which changes only the spinor components inside each ψ_a .) The free Dirac Lagrangian for quarks is then

$$\mathcal{L}_0 = \bar{\psi}(i\not{\partial} - m)\psi \iff \mathcal{L}_0 = \begin{bmatrix} \bar{\psi}_R & \bar{\psi}_G & \bar{\psi}_B \end{bmatrix} (i\not{\partial} - m) \begin{bmatrix} \psi_R \\ \psi_G \\ \psi_B \end{bmatrix}.$$

If U is independent of x^μ , or in other words α_a are independent of x^μ , the free Dirac Lagrangian has a global SU(3) gauge symmetry:

$$\mathcal{L}'_0 = \bar{\psi}'(i\not{\partial} - m)\psi' = \bar{\psi}(i\not{\partial} - m)\psi = \mathcal{L}_0.$$

This is not very interesting. Recall that in QED, we modify the 4-derivative ∂^μ into a gauge-covariant derivative $D^\mu = \partial^\mu + iqA^\mu$ so that the Lagrangian is invariant (or the Dirac equation is *covariant*) under a local gauge transformation, and the term iqA^μ gives interaction with the photon field. In QCD, we are going to do the same: to obtain interactions, we need to impose *local* SU(3) gauge invariance. In other words, we insist that the Dirac equation remains the same form under an SU(3) gauge transformation with U , or α_a , depending on x^μ .

Introduce a gauge-covariant derivative D^μ with some gauge transformation rule $D^\mu \rightarrow D'^\mu$. The covariant derivative D^μ and its transformed version D'^μ should satisfy

$$i\not{D}'\psi' = m\psi' \implies i\not{D}'(U\psi) = mU\psi.$$

Multiplying both sides by U^{-1} on the left,

$$iU^{-1}\not{D}'U\psi = m\psi.$$

If ψ satisfy the modified Dirac equation $i\not{D}\psi = m\psi$, this means

$$D'^\mu U = UD^\mu. \quad (10.2)$$

Consider the ordinary derivative on ψ' first,

$$i\partial^\mu\psi' = i\partial^\mu(e^{ig\alpha_a T_a}\psi) = -g(\partial^\mu\alpha_a)T_a e^{ig\alpha_a T_a}\psi + ie^{ig\alpha_a T_a}\partial^\mu\psi.$$

Next, we need to introduce gauge fields whose transformations include terms that cancel the first term. In fact, there are 8 terms to cancel because there is an implicit sum over a , so there are in total 8 gauge fields $G_a^\mu(x)$. Let the gauge-covariant derivative be

$$D^\mu = \partial^\mu + igG_a^\mu T_a. \quad (10.3)$$

Consider infinitesimal gauge transformation

$$U = 1 + ig\alpha_a T_a \quad \text{and} \quad G_a'^\mu = G_a^\mu + \delta G_a^\mu. \quad (10.4)$$

Then the LHS of the transformed Dirac equation contains

$$iD'^\mu(U\psi) = [i\partial^\mu - g(G_a^\mu + \delta G_a^\mu)T_a](1 + ig\alpha_b T_b)\psi.$$

Before we proceed, note that the SU(3) group is a non-abelian group. The generators T_a and T_b do not commute so they need to be handled with care. Keeping the leading order of infinitesimal transformation ($\alpha_a \ll 1$ and $\delta G_a^\mu \ll G_a^\mu$),

$$\begin{aligned} iD'^\mu(U\psi) &= -g(\partial^\mu\alpha_b)T_b\psi + (1 + ig\alpha_b T_b)(i\partial^\mu)\psi - gG_a^\mu T_a(1 + ig\alpha_b T_b)\psi - g\delta G_a^\mu T_a\psi + \mathcal{O}(\alpha_a\delta G_a^\mu) \\ &\simeq -g(\partial^\mu\alpha_a + \delta G_a^\mu)T_a\psi + (1 + ig\alpha_b T_b)(i\partial^\mu)\psi - gG_a^\mu T_a(1 + ig\alpha_b T_b)\psi. \end{aligned}$$

If $[T_a, T_b] = 0$, the second and third terms can be combined into $iUD^\mu\psi$. However, SU(3) is non-abelian, so switching T_a and T_b

needs to be compensated by a commutator,

$$\begin{aligned}
iD'^\mu(U\psi) &\simeq -g(\partial^\mu\alpha_a + \delta G_a^\mu)T_a\psi + (1 + ig\alpha_b T_b)(i\partial^\mu)\psi - g(1 + ig\alpha_b T_b)G_a^\mu T_a\psi - ig^2\alpha_b G_a^\mu [T_a, T_b]\psi \\
&= -g(\partial^\mu\alpha_a + \delta G_a^\mu)T_a\psi - ig^2\alpha_b G_a^\mu [T_a, T_b]\psi + iUD^\mu\psi \\
&= -g(\partial^\mu\alpha_c + \delta G_c^\mu)T_c\psi - ig^2\alpha_b G_a^\mu (if_{abc}T_c)\psi + iUD^\mu\psi \\
&= -g(\partial^\mu\alpha_c + \delta G_c^\mu - gf_{abc}\alpha_b G_a^\mu)T_c\psi + iUD^\mu\psi.
\end{aligned}$$

According to (10.2), in order for the Dirac equation to be gauge covariant, we need to have $D'^\mu U = UD^\mu$ for any gauge transformation. In other words,

$$\partial^\mu\alpha_c + \delta G_c^\mu - gf_{abc}\alpha_b G_a^\mu = 0.$$

This gives the infinitesimal gauge transformation of the field G_a^μ ,

$$\delta G_c^\mu = -\partial^\mu\alpha_c + gf_{abc}\alpha_b G_a^\mu.$$

In fact, this result holds for any infinitesimal gauge transformation, not only for SU(3), as long as the structure constants f_{abc} are known. For SU(3), its structure constant is totally antisymmetric, so $f_{abc} = -f_{cba}$, and

$$\delta G_c^\mu = -\partial^\mu\alpha_c - gf_{cba}\alpha_b G_a^\mu.$$

Relabeling the indices $a \leftrightarrow c$, and combining with (10.4) we obtain the gauge transformation rule

$$\psi \rightarrow \psi' = e^{ig\alpha_a T_a}\psi, \quad G_a^\mu \rightarrow G_a'^\mu = G_a^\mu - \partial^\mu\alpha_a - gf_{abc}\alpha_b G_c^\mu. \quad (10.5)$$

The gauge fields G_a^μ are known as the **gluon fields**. There are 8 different gluons, corresponding to 8 generators of SU(3). In the gauge transformation of the gluon field (10.5), the first two terms $G_a^\mu - \partial^\mu\alpha_a$ look familiar: they are just like the gauge transformation $A^\mu - \partial^\mu\chi$ of the photon field. The third term $gf_{abc}\alpha_b G_c^\mu$ arises from the nonabelian nature of SU(3), in contrast to QED, which is an abelian gauge theory. Later in this chapter, we will see that this term leads to self-coupling of the gluon fields, i.e. *gluons do couple to each other directly*.

10.3 Perturbation Theory

10.3.1 Interaction Lagrangian and Hamiltonian

Incorporating gauge invariance, the full Dirac Lagrangian in QCD is

$$\mathcal{L} = \bar{\psi}(i\not{D} - m)\psi = \bar{\psi}(i\not{\partial} - m)\psi - g_s \bar{\psi} \not{G}_a T_a \psi,$$

where we changed the coupling constant $g \rightarrow g_s$ to indicate that it is for the strong interaction. The first part is the free Dirac Lagrangian, and the second part is the interaction piece for QCD,

$$\mathcal{L}_0 = \bar{\psi}(i\not{\partial} - m)\psi \quad \text{and} \quad \mathcal{L}_{\text{int}} = -g_s \bar{\psi} \not{G}_a T_a \psi. \quad (10.6)$$

The Hamiltonian density $\mathcal{H}$ derived from $\mathcal{L}$ is

$$\mathcal{H} = \pi\dot{\psi} - \mathcal{L} = \bar{\psi}[\boldsymbol{\gamma} \cdot (-i\nabla) + m]\psi - \mathcal{L}_{\text{int}},$$

where $\pi = i\psi^\dagger$ is the conjugate momentum of ψ . The interaction Hamiltonian is then

$$\mathcal{H}_{\text{int}} = -\mathcal{L}_{\text{int}} = J_a^\mu G_a^\mu, \quad \text{where} \quad J_a^\mu = g_s \bar{\psi} \boldsymbol{\gamma}^\mu T_a \psi. \quad (10.7)$$

10.3.2 Scattering Amplitudes

Let the initial and final states be $|i\rangle$ and $|f\rangle$, respectively. The S -matrix is given by

$$S_{fi} = \delta_{fi} + (-i) \langle f | \int d^4x_1 \mathcal{H}'_I(x_1) | i \rangle + \frac{(-i)^2}{2} \langle f | \int d^4x_1 d^4x_2 T \{ \mathcal{H}'_I(x_1) \mathcal{H}'_I(x_2) \} | i \rangle + \dots$$

Substituting $\mathcal{H}'_I = \mathcal{H}_{\text{int}}$ in (10.7), we find the first-order amplitude for QCD scattering to be

$$\mathcal{A}_{fi} = (-i) \langle f | \int d^4x \mathcal{H}'_I(x) | i \rangle = -ig_s \gamma^\mu T_a \langle f | \int d^4x \bar{\psi} G_\mu^a \psi | i \rangle.$$

Most of the steps are just like what we did in QED, except that there are eight generators T_a and eight gluon fields G_μ^a , and that the spinors have color degrees of freedom. Let the initial quark have 4-momentum p_1 and spin s_1 , and the final quark have p_2 and s_2 . One can go through all the steps exactly the same as in QED (manipulating creation and annihilation operators) and find that

$$\mathcal{A}_{fi} = -ig_s \bar{u}(p_2, s_2) \gamma^\mu T_a u(p_1, s_1) \int d^4x e^{-i(p_1 - p_2) \cdot x} G_\mu^a.$$

Since the spinors have color degrees of freedom, we often let

$$u(p, s) \rightarrow c_i u(p, s),$$

where $u(p, s)$ is a 4-component Dirac spinor and $c_i = R, G, B$ represents one of the color states

$$c_1 = R = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad c_2 = G = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad c_3 = B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \quad (10.8)$$

Thus, the first-order amplitude can be written as

$$\mathcal{A}_{fi} = \bar{u}(p_2, s_2) c_j^\dagger (-ig_s \gamma^\mu T_a) c_i u(p_1, s_1) \int d^4x e^{-i(p_1 - p_2) \cdot x} G_\mu^a. \quad (10.9)$$

Again, the amplitude contains three elements:

1. The vertex factor is $-ig_s \gamma^\mu T_a$.
2. The spinors $u(p_1, s_1)$ and $\bar{u}(p_2, s_2)$ represent the ingoing and outgoing quarks.
3. The integral over the gluon field will turn into a delta function that conserves 4-momentum at each vertex once the gluon field is written in mode expansion.

We can do even better. Note that the γ -matrices does not care about the color states c_i , and meanwhile, the generators T_a does not affect the Dirac spinors $u(p, s)$. In other words, the product of the spinors and γ 's does not mix with the product of color states and T_a 's. Hence we can write

$$\bar{u}(p_2, s_2) c_j^\dagger (-ig_s \gamma^\mu T_a) c_i u(p_1, s_1) = -ig_s (c_j^\dagger T_a c_i) [\bar{u}(p_2, s_2) \gamma^\mu u(p_1, s_1)].$$

The product $c_j^\dagger T_a c_i$ is just a number. By direct matrix multiplication using (10.8), it is easy to verify that $c_j^\dagger T_a c_i$ picks out the j th matrix element of T_a . To avoid mixing the indices, we raise the index of $T_a \rightarrow T^a$. Then

$$c_j^\dagger T^a c_i = T_{ji}^a \implies \bar{u}(p_2, s_2) c_j^\dagger (-ig_s \gamma^\mu T_a) c_i u(p_1, s_1) = -\frac{1}{2} ig_s \lambda_{ij}^a [\bar{u}(p_2, s_2) \gamma^\mu u(p_1, s_1)],$$

where $\lambda^a = 2T_a$ are the Gell-Mann matrices. We can name our new vertex factor to be

$$\text{QCD vertex factor} = -\frac{1}{2} ig_s \lambda_{ij}^a \gamma^\mu. \quad (10.10)$$

11 THE WEAK INTERACTION

The weak interaction is mediated by three gauge bosons: $W^\pm$ and Z . Unlike photons or gluons, W and Z -bosons are not massless particles. In fact, they are quite heavy in the Standard Model:

$$m_W = 80.4 \text{ GeV} \quad \text{and} \quad m_Z = 91.2 \text{ GeV}.$$

They are nearly 100 times as heavy as a proton. Hence it was not possible to produce the on-shell W and Z until 1983 when the energy of the collider was high enough. The decay widths of the W and Z are

$$\Gamma(W) = 2.1 \text{ GeV} \quad \text{and} \quad \Gamma(Z) = 2.5 \text{ GeV},$$

which means both mean lifetimes are of the order 10^{-25} s.

Despite the large mass of the W , there are many processes mediated by the (off-shell) W observed throughout the past century. These processes have energy scales $E \ll m_W$, but are easy to detect. The discovery of the first W -mediated process was the β -decay in 1890s, almost a century before the discovery of the on-shell W . Many of the decays involve hadrons, such as mesons with quark content $q_i \bar{q}_j$ with $i \neq j$ in their ground state, or the ground state of a baryon. Here, a ground state hadron means that its spatial and spin wave functions are such that the hadron has the lowest mass. For example, the π^- and ρ^- have the same quark content $d\bar{u}$, but very different masses, $m_\pi = 140 \text{ MeV}$ and $m_\rho = 770 \text{ MeV}$. This is because the ρ^- has spin 1 while the π^- has spin 0; π^- is the ground state. If the hadron is not in its ground state, then its decay is dominated by the strong and electromagnetic interactions, e.g. $\rho^- \rightarrow \pi^- \pi^0$ with branching fraction ~ 1 : Only when a hadron cannot simply decay to lower mass particles with the same quarks, a W -mediated decay that changes the quark content can happen, e.g. $\pi^- \rightarrow \mu^- \bar{\nu}_\mu$ with branching fraction ~ 1 .

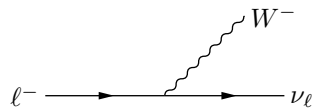
On the contrary, the Z -mediated processes were notoriously difficult to study. This is because any Z -mediated process involving charged particles can also be mediated by the photon. At low energies $E \ll m_Z$, the electromagnetic process is the dominant one. This suggests that we should look for processes involving neutral particles that couple to the Z . The only ones remaining are the neutrinos. Since the neutrino scattering have extremely small cross sections, such processes were not discovered until the 1970s.

11.1 W -Mediated Processes; Vertex Factors and the Propagator

In this section, we discuss the W -mediated processes with Feynman diagrams. Then we quote the vertex factors in leptonic and hadronic processes. These vertex factors are to be derived later once we obtain the Lagrangian for the weak interaction.

11.1.1 Leptonic Processes

A **leptonic process** is a process in which there are no quarks in the final state, e.g. $\pi^- \rightarrow \mu^- \bar{\nu}_\mu$. A *pure* leptonic process has no quarks in both its initial and final states, e.g. $\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$. Pure leptonic processes are the simplest W -mediated processes because there are no strong interaction effects. Here is a leptonic vertex involving a lepton ℓ^- , a neutrino ν_ℓ , and a W -boson:



The vertex factor is given by

$$W\text{-boson leptonic vertex factor} = \frac{-ig}{\sqrt{2}} \gamma^\mu \frac{1}{2} (1 - \gamma^5). \quad (11.1)$$

Here g is the weak coupling constant, and $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ is a γ -matrix associated with chirality:

$$P_L = \frac{1}{2}(1 - \gamma^5), \quad P_R = \frac{1}{2}(1 + \gamma^5).$$

The operators P_L and P_R are called left-handed (L) and right-handed (R) **chiral projection operators**. Since they are projection operators, they satisfy $P_L^2 = P_L$, $P_R^2 = P_R$, and $P_L + P_R = 1$.

Note that (11.1) is the same no matter which lepton is involved. This is called the **lepton universality** of the weak interaction. However, due to the conservation of lepton number within each generator, the ℓ on both sides should be consistent. For example, $\mu^- \rightarrow \nu_e$ is not allowed. Recall that the QED vertex factor is $-iq\gamma^\mu$. This leads to a current described by the spinors $\bar{u}\gamma^\mu u$ and we know that it is a Lorentz 4-vector,

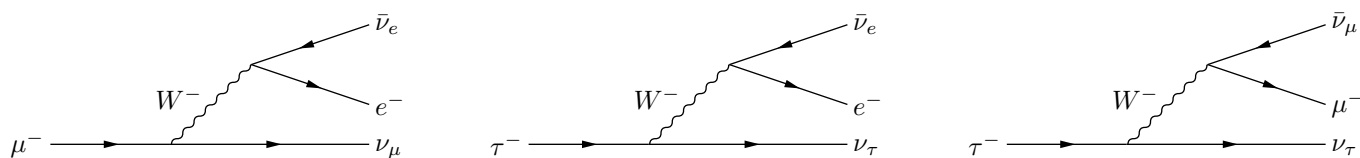
$$\bar{u}(p', s')\gamma^\mu u(p, s) \sim V^\mu.$$

On the other hand, the vertex factor (11.1) gives an extra term

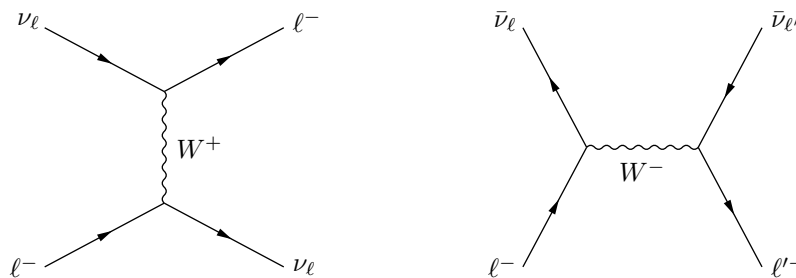
$$\bar{u}(p', s')\gamma^\mu(1 - \gamma^5)u(p, s) \sim V^\mu - A^\mu,$$

where we will find that A^μ is an axial 4-vector. This is the term that results in symmetry violations in weak interactions.

Here are more examples of Feynman diagrams involving purely leptons and a W -boson. There are decays of leptons:



Also, the W is massive, with $m_W \gg m_e, m_\mu, m_\tau$, so there is no problem for an on-shell W -boson to decay into leptons. Beside decays, there are also purely leptonic scattering processes in the t-channels and s-channels,



As an aside, if the leptons in the initial and final states are the same, there can be an s-channel mediated by a W -boson and meanwhile a t-channel mediated by a Z -boson. The total amplitude should be the sum of the two.

11.1.2 Semileptonic and Hadronic Processes

The final state of a **semileptonic process** includes both leptons and hadrons. A classic example is the β -decay: $n \rightarrow pe^- \bar{\nu}_e$ or, at the quark level, $d \rightarrow ue^- \bar{\nu}_e$. The process must involve strong interactions because the quarks must end up in a hadronic system. The vertex factor for quarks is modified to

$$W\text{-boson quark vertex factor} = -\frac{ig}{\sqrt{2}}\gamma^\mu \frac{1}{2}(1 - \gamma^5)V_{ij}, \quad (11.2)$$

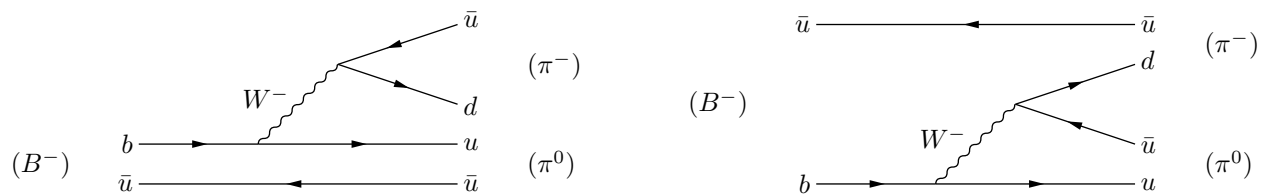
where V_{ij} ($i = u, c, t$ and $j = d, s, b$) are elements in the **Cabibbo–Kobayashi–Maskawa (CKM) matrix**:

$$[|V_{ij}|] = \begin{bmatrix} |V_{ud}| & |V_{us}| & |V_{ub}| \\ |V_{cd}| & |V_{cs}| & |V_{cb}| \\ |V_{td}| & |V_{ts}| & |V_{tb}| \end{bmatrix} = \begin{bmatrix} 0.9744 & 0.2250 & 0.0037 \\ 0.2249 & 0.9735 & 0.0418 \\ 0.0086 & 0.0411 & 0.9991 \end{bmatrix}.$$

The matrix elements V_{ij} are complex, but in most cases we only need their magnitudes. The CKM matrix is a unitary matrix. That is, the rows and columns are orthonormal. Though the off-diagonal elements are small, the CKM matrix is not exactly an identity matrix. This means transitions of quarks between different generations are allowed but suppressed.

The W -mediated hadronic processes have final-state particles entirely made of hadrons. Even though the key mediator is a W -boson,

strong interactions are everywhere in a hadronic process. Hence hadronic processes are the most complicated ones. An example could be $B^- \rightarrow \pi^- \pi^0$:



11.1.3 The Propagator

The propagator factor for the W -boson is

$$W\text{-boson propagator} = \frac{-i(\eta_{\mu\nu} - q_\mu q_\nu)/m_W^2}{q^2 - m_W^2}. \quad (11.3)$$

This propagator is to be derived later. (Historically, the calculation of weak interactions were established before the W -boson was known.) Remember the mass of the W is huge. In many decay processes, the q^2 is the rest mass squared in the CM frame and is much smaller than m_W^2 . The propagator simplifies to an excellent approximation:

$$W\text{-boson propagator} \simeq \frac{i\eta_{\mu\nu}}{m_W^2}.$$

In examples presented in the following sections, this simplified propagator is be used frequently.

11.2 Muon Decay

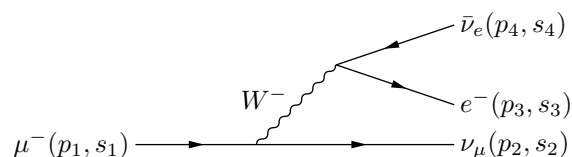
The simplest W -mediated decay is the following muon decay, a pure leptonic decay:

$$\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu.$$

This is the dominant decay mode of the muon. Other decay modes made up only $\sim 10^{-5}$ of the branching fraction. By calculating the decay width $\Gamma(\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu)$ only, we can determine the muon mean lifetime to a good approximation.

11.2.1 The Decay Amplitude

Starting with the Feynman diagram for the muon decay:



Identifying the elements in this Feynman diagram and use Feynman rules:

1. There is one incoming fermion: $u(p_1, s_1)$ for the μ^- . There are two outgoing fermions and one outgoing antifermion: $\bar{u}(p_2, s_2)$ for ν_μ , $\bar{u}(p_3, s_3)$ for e^- , and $v(p_4, s_4)$ for $\bar{\nu}_e$.
2. There are two vertex factors connecting the W^- with leptons. They contribute two factors of $-ig\gamma^\mu(1 - \gamma^5)/2\sqrt{2}$.
3. The W -boson propagator is given by $-i(\eta_{\mu\nu} - q_\mu q_\nu/m_W^2)/(q^2 - m_W^2)$, where q is the 4-momentum of the W . The mass of the muon is only $m_\mu = 106 \text{ MeV}$, which is much smaller than $m_W = 80.4 \text{ GeV}$. The propagator can be simplified to $i\eta_{\mu\nu}/m_W^2$.

The decay amplitude is then

$$\mathcal{A} = \left[-\frac{ig}{\sqrt{2}} \bar{u}(p_2, s_2) \gamma^\mu \frac{1}{2} (1 - \gamma^5) u(p_1, s_1) \right] \frac{i\eta_{\mu\nu}}{m_W^2} \left[-\frac{ig}{\sqrt{2}} \bar{u}(p_3, s_3) \gamma^\nu \frac{1}{2} (1 - \gamma^5) v(p_4, s_4) \right].$$

Abbreviating $u(p_1, s_1) \equiv u_1$ and so on, we can write the amplitude as

$$\mathcal{A} = -i \frac{g^2}{8m_W^2} [\bar{u}_2 \gamma_\mu (1 - \gamma^5) u_1] [\bar{u}_3 \gamma^\mu (1 - \gamma^5) v_4] = -i \frac{G_F}{\sqrt{2}} [\bar{u}_2 \gamma_\mu (1 - \gamma^5) u_1] [\bar{u}_3 \gamma^\mu (1 - \gamma^5) v_4],$$

where we define the **Fermi constant** G_F :

$$G_F = \frac{g^2}{4\sqrt{2}m_W^2} \approx 1.166 \times 10^{-5} \text{ GeV}^{-2}. \quad (11.4)$$

The calculation of spin-averaged amplitude squared requires many new Diracology and trace theorems involving the γ^5 . We should now state them here:

Theorem 11.1. γ^5 and Trace Theorems II

1. $(\gamma^5)^2 = 1$.
2. $(\gamma^5)^\dagger = \gamma^5$.
3. γ^5 anticommutes with all γ -matrices: $\{\gamma^5, \gamma^\mu\} = 0$ for all γ^μ .
4. $\text{Tr}(\not{p}_1 \gamma^5 \not{p}_2) = 0$.
5. $\text{Tr}(\not{p}_1 \gamma^\mu \not{p}_2 \gamma^\nu) = 4[p_1^\mu p_2^\nu + p_1^\nu p_2^\mu - (p_1 \cdot p_2) \eta^{\mu\nu}]$. This is in fact derived in Section 9.3.3 about the Mott scattering and the lepton tensor.
6. $\text{Tr}(\gamma^\alpha \gamma^\mu \gamma^\beta \gamma^\nu \gamma^5) = 4i \epsilon^{\alpha\mu\beta\nu}$, where $\epsilon^{\alpha\mu\beta\nu}$ is the Levi-Civita tensor.
A useful identity would be $\epsilon^{\alpha\mu\beta\nu} \epsilon_{\rho\mu\sigma\nu} = -2\delta^\alpha_\rho \delta^\beta_\sigma + 2\delta^\alpha_\sigma \delta^\beta_\rho$.

Compute the spin-averaged decay amplitude,

$$|\overline{\mathcal{A}}|^2 = \frac{1}{2} \sum_{s_1, \dots, s_4} |\mathcal{A}|^2 = \frac{G_F^4}{4} \sum_{s_1, \dots, s_4} [\bar{u}_3 \gamma^\mu (1 - \gamma^5) v_4]^\dagger [\bar{u}_2 \gamma_\mu (1 - \gamma^5) u_1]^\dagger [\bar{u}_2 \gamma_\nu (1 - \gamma^5) u_1] [\bar{u}_3 \gamma^\nu (1 - \gamma^5) v_4].$$

It is straightforward to show that $[\bar{u}_3 \gamma^\mu (1 - \gamma^5) v_4]^\dagger = \bar{v}_4 \gamma^\mu (1 - \gamma^5) u_3$. With this identity, the spin-averaged amplitude can be turned to

$$\begin{aligned} |\overline{\mathcal{A}}|^2 &= \frac{G_F^4}{4} \sum_{s_1, \dots, s_4} [\bar{v}_4 \gamma^\mu (1 - \gamma^5) u_3] [\bar{u}_1 \gamma_\mu (1 - \gamma^5) u_2] [\bar{u}_2 \gamma_\nu (1 - \gamma^5) u_1] [\bar{u}_3 \gamma^\nu (1 - \gamma^5) v_4] \\ &= \frac{G_F^2}{4} \sum_{s_3, s_4} [\bar{v}_4 \gamma^\mu (1 - \gamma^5) u_3] [\bar{u}_3 \gamma^\nu (1 - \gamma^5) v_4] \sum_{s_1, s_2} [\bar{u}_1 \gamma_\mu (1 - \gamma^5) u_2] [\bar{u}_2 \gamma_\nu (1 - \gamma^5) u_1] \\ &= \frac{G_F^2}{4} \text{Tr} \left[\sum_{s_4} v_4 \bar{v}_4 \gamma^\mu (1 - \gamma^5) \sum_{s_3} u_3 \bar{u}_3 \gamma^\nu (1 - \gamma^5) \right] \text{Tr} \left[\sum_{s_1} u_1 \bar{u}_1 \gamma_\mu (1 - \gamma^5) \sum_{s_2} u_2 \bar{u}_2 \gamma_\nu (1 - \gamma^5) \right] \\ &\equiv G_F^2 E^{\mu\nu} M_{\mu\nu}, \end{aligned}$$

where we define $E^{\mu\nu}$ and $M_{\mu\nu}$ to be

$$\begin{aligned} E^{\mu\nu} &\equiv \frac{1}{2} \text{Tr} \left[\sum_{s_4} v_4 \bar{v}_4 \gamma^\mu (1 - \gamma^5) \sum_{s_3} u_3 \bar{u}_3 \gamma^\nu (1 - \gamma^5) \right], \\ M_{\mu\nu} &\equiv \frac{1}{2} \text{Tr} \left[\sum_{s_1} u_1 \bar{u}_1 \gamma_\mu (1 - \gamma^5) \sum_{s_2} u_2 \bar{u}_2 \gamma_\nu (1 - \gamma^5) \right]. \end{aligned}$$

The goal is to find $E^{\mu\nu}$ and similarly for $M_{\mu\nu}$. We will do $E^{\mu\nu}$ as an example. First, eliminate the sum using $\sum_s u(p, s) \bar{u}(p, s) = \not{p} + m$ and $\sum_s v(p, s) \bar{v}(p, s) = \not{p} - m$,

$$E^{\mu\nu} = \frac{1}{2} \text{Tr}[(\not{p}_4 - m_4) \gamma^\mu \gamma^\mu (1 - \gamma^5) (\not{p}_3 + m_3) \gamma^\nu (1 - \gamma^5)].$$

Next, multiply out the expression and extract the products of even and odd number of γ -matrices. With some work,

$$E^{\mu\nu} = \frac{1}{2} \text{Tr}[\not{p}_4 \gamma^\mu (1 - \gamma^5) \not{p}_3 \gamma^\nu (1 - \gamma^5) - m_3 m_4 \gamma^\mu (1 - \gamma^5) \gamma^\nu (1 - \gamma^5) + (\text{odd number of } \gamma\text{'s})].$$

The trace of odd number of γ -matrices is zero. Furthermore, we can assume that the neutrino is massless, $m_4 = 0$, which is a standard practice in calculating the amplitude. Then we are left with only the first term,

$$E^{\mu\nu} = \frac{1}{2} \text{Tr}[\not{p}_4 \gamma^\mu (1 - \gamma^5) \not{p}_3 \gamma^\nu (1 - \gamma^5)] = \frac{1}{2} \text{Tr}[\not{p}_4 \gamma^\mu (1 - \gamma^5)^2 \not{p}_3 \gamma^\nu].$$

In the second equality, we used $\{\gamma^5, \gamma^\mu\} = 0$ so when $1 - \gamma^5$ commutes with $\not{p}_3 \gamma^\nu$ (there are two γ 's) the sign before γ^5 is unchanged. Then because $P_L = \frac{1}{2}(1 - \gamma^5)$ is the left-handed chiral projection operator, its square is itself, and

$$E^{\mu\nu} = \text{Tr}[\not{p}_4 \gamma^\mu (1 - \gamma^5) \not{p}_3 \gamma^\nu] = \text{Tr}(\not{p}_4 \gamma^\mu \not{p}_3 \gamma^\nu) - \text{Tr}(\not{p}_4 \gamma^\mu \gamma^5 \not{p}_3 \gamma^\nu) = \text{Tr}(\not{p}_4 \gamma^\mu \not{p}_3 \gamma^\nu) - p_{4\alpha} p_{3\beta} \text{Tr}(\gamma^\alpha \gamma^\mu \gamma^\beta \gamma^\nu \gamma^5).$$

Finally, using the trace theorems 4 and 5 listed above,

$$E^{\mu\nu} = 4[p_3^\mu p_4^\nu + p_3^\nu p_4^\mu - (p_3 \cdot p_4) \eta^{\mu\nu}] - 4i p_{4\alpha} p_{3\beta} \epsilon^{\alpha\mu\beta\nu}.$$

Similarly for $M_{\mu\nu}$, assuming that $m_2 = 0$,

$$M_{\mu\nu} = 4[p_{1\mu} p_{2\nu} + p_{1\nu} p_{2\mu} - (p_1 \cdot p_2) \eta_{\mu\nu}] - 4i p_1^\rho p_2^\sigma \epsilon_{\rho\mu\sigma\nu}.$$

Now we want to contract $E^{\mu\nu} M_{\mu\nu}$. Note that $E^{\mu\nu}$ do not obey $q_\mu E^{\mu\nu} = 0$ (or similarly for $M_{\mu\nu}$), unlike the lepton tensor $L_{\mu\nu}$. This is an artifact of the $1 - \gamma^5$. Luckily, it does not take much longer to contract $E^{\mu\nu} M_{\mu\nu}$ via directly using symmetries. Observe that $E^{\mu\nu}$ has its first part $4[p_3^\mu p_4^\nu + p_3^\nu p_4^\mu - (p_3 \cdot p_4) \eta^{\mu\nu}]$ symmetric in μ, ν , while the second part $-4i p_{4\alpha} p_{3\beta} \epsilon^{\alpha\mu\beta\nu}$ is antisymmetric. $M_{\mu\nu}$ also has the same structure. A symmetric tensor contracts with an antisymmetric tensor gives zero, so we only need to find the product of symmetric parts and the product of antisymmetric parts:

$$\begin{aligned} E^{\mu\nu} M_{\mu\nu} &= \{4[p_3^\mu p_4^\nu + p_3^\nu p_4^\mu - (p_3 \cdot p_4) \eta^{\mu\nu}] - 4i p_{4\alpha} p_{3\beta} \epsilon^{\alpha\mu\beta\nu}\} \{4[p_{1\mu} p_{2\nu} + p_{1\nu} p_{2\mu} - (p_1 \cdot p_2) \eta_{\mu\nu}] - 4i p_1^\rho p_2^\sigma \epsilon_{\rho\mu\sigma\nu}\} \\ &= 16[p_3^\mu p_4^\nu + p_3^\nu p_4^\mu - (p_3 \cdot p_4) \eta^{\mu\nu}][p_{1\mu} p_{2\nu} + p_{1\nu} p_{2\mu} - (p_1 \cdot p_2) \eta_{\mu\nu}] - 16 p_1^\rho p_2^\sigma p_{3\beta} p_{4\alpha} \epsilon^{\alpha\mu\beta\nu} \epsilon_{\rho\mu\sigma\nu} \end{aligned}$$

The rest of the calculation is straightforward. One should find that the only term left is

$$E^{\mu\nu} M_{\mu\nu} = 64(p_1 \cdot p_4)(p_2 \cdot p_3),$$

and hence the spin-averaged amplitude is

$$|\overline{\mathcal{A}}|^2 = G_F^2 E^{\mu\nu} M_{\mu\nu} = 64 G_F^2 (p_1 \cdot p_4)(p_2 \cdot p_3). \quad (11.5)$$

11.2.2 Muon Decay Rate

The muon decay is a three-body decay. Using the most general form of the decay rate (8.10),

$$\Gamma = \frac{(2\pi)^4}{2E_1} \int |\overline{\mathcal{A}}|^2 \delta(E_1 - E_2 - E_3 - E_4) \delta^3(\mathbf{p}_1 - \mathbf{p}_2 - \mathbf{p}_3 - \mathbf{p}_4) \frac{d^3 p_2}{(2\pi)^3 2E_2} \frac{d^3 p_3}{(2\pi)^3 2E_3} \frac{d^3 p_4}{(2\pi)^3 2E_4}.$$

Since this is our first example of three-body decay, we shall work it out step-by-step. Here are some general steps to handle this problem:

1. Take the relativistic limit for the final-state particles.
2. Use the momentum delta-function to eliminate the $d^3 p_2$ integral.
3. Identify the observable particle in the final state. This particle in general should not be the neutrino. Then write $d^3 p_4$ (the neutrino integral) in spherical coordinates and evaluate the delta-function integral over the energy E_4 . After eliminating the energy delta-function integral, integrate over $\sin \theta_4 d\theta_4 d\phi_4$.
4. Finally, find the differential decay rate for the process in terms of the observable particle energy, e.g. the electron energy E_3 in our case. Integrating the differential decay rate gives the decay rate for the process.

First plug in the spin-averaged decay amplitude and simplify the expression a bit,

$$\Gamma = \frac{64G_F^2}{(2\pi)^5 2E_1} \int (p_1 \cdot p_4)(p_2 \cdot p_3) \delta(m_1 - E_2 - E_3 - E_4) \delta^3(\mathbf{p}_1 + \mathbf{p}_2 + \mathbf{p}_3) \frac{d^3 p_2 d^3 p_3 d^3 p_4}{E_2 E_3 E_4}.$$

We now assume that all the particle masses except the muon mass are negligible in the relativistic limit, so that

$$E_1 = m_1, \quad E_2 = |\mathbf{p}_2|, \quad E_3 = |\mathbf{p}_3|, \quad E_4 = |\mathbf{p}_4|.$$

In this limit, $p_1 \cdot p_4 = m_1 E_4$. To find $p_2 \cdot p_3$, we examine $(p_1 - p_4)^2$ and $(p_2 + p_3)^2$,

$$\begin{aligned} (p_1 - p_4)^2 &= p_1^2 + p_4^2 - 2p_1 \cdot p_4 = m_1^2 - 2p_1 \cdot p_4, \\ (p_2 + p_3)^2 &= p_2^2 + p_3^2 + 2p_2 \cdot p_3 = 2p_2 \cdot p_3. \end{aligned}$$

By 4-momentum conservation, $p_1 - p_4 = p_2 + p_3$, so

$$p_2 \cdot p_3 = \frac{1}{2}(m_1^2 - p_1 \cdot p_4) = \frac{1}{2}m_1(m_1 - 2E_4).$$

Substituting $p_1 \cdot p_4$ and $p_2 \cdot p_3$ into the decay rate,

$$\Gamma = \frac{G_F^2}{16\pi^5} \int m_1 E_4 (m_1 - 2E_4) \delta(m_1 - E_2 - E_3 - E_4) \delta^3(\mathbf{p}_2 + \mathbf{p}_3 + \mathbf{p}_4) \frac{d^3 p_2 d^3 p_3 d^3 p_4}{E_2 E_3 E_4}.$$

The next step is to evaluate the $d^3 p_2$ integral using the momentum delta-function: just set $\mathbf{p}_2 = \mathbf{p}_3 + \mathbf{p}_4$ and $E_2 = |\mathbf{p}_2| = |\mathbf{p}_3 + \mathbf{p}_4|$,

$$\Gamma = \frac{G_F^2}{16\pi^5} \int m_1 E_4 (m_1 - 2E_4) \delta(m_1 - |\mathbf{p}_3 + \mathbf{p}_4| - E_3 - E_4) \frac{d^3 p_3 d^3 p_4}{|\mathbf{p}_3 + \mathbf{p}_4| E_3 E_4}.$$

Eventually we are looking for the differential decay rate $d\Gamma/dE$, and this E should be observable. Usually neutrinos are not detected, so the only observable energy in the final state is the electron energy E_3 . Thus, we want to perform the $d^3 p_4$ integral first so that in the end we are left with $d\Gamma/dE_3$ as a function of E_3 . In spherical coordinates, $d^3 p_4 = \mathbf{p}_4^2 d|\mathbf{p}_4| d\Omega_4 = E_4^2 \sin \theta_4 dE_4 d\theta_4 d\phi_4$, and

$$\Gamma = \frac{G_F^2}{16\pi^5} \int m_1 E_4 (m_1 - 2E_4) \delta(m_1 - |\mathbf{p}_3 + \mathbf{p}_4| - E_3 - E_4) \frac{E_4 dE_4 d^3 p_3}{|\mathbf{p}_3 + \mathbf{p}_4| E_3} \sin \theta_4 d\theta_4 d\phi_4.$$

The angle θ_4 is arbitrary, but it is convenient to define θ_4 by $\mathbf{p}_3 \cdot \mathbf{p}_4 = |\mathbf{p}_3||\mathbf{p}_4| \cos \theta_4$. In this way, we can write

$$|\mathbf{p}_3 + \mathbf{p}_4| = (\mathbf{p}_3^2 + \mathbf{p}_4^2 + 2|\mathbf{p}_3||\mathbf{p}_4| \cos \theta_4)^{1/2}.$$

The decay rate is now a delta-function integral over E_4 ,

$$\Gamma = \frac{G_F^2}{16\pi^5} \int \frac{m_1}{E_3} \delta(f(E_4)) g(E_4) dE_4 d^3 p_3 \sin \theta_4 d\theta_4 d\phi_4,$$

where

$$\begin{aligned} f(E_4) &= m_1 - (E_3^2 + E_4^2 + 2E_3 E_4 \cos \theta_4)^{1/2} - E_3 - E_4, \\ g(E_4) &= E_4^2 (m_1 - 2E_4) (E_3^2 + E_4^2 + 2E_3 E_4 \cos \theta_4)^{-1/2}. \end{aligned}$$

Then using the delta-function identity,

$$\int \delta(f(E_4)) g(E_4) dE_4 = \left| \frac{df}{dE_4} \right|_{E_4'}^{-1} g(E_4'), \quad \text{where } f(E_4') = 0.$$

To find the correct E_4' , we solve the equation

$$m_1 - (E_3^2 + E_4'^2 + 2E_3 E_4' \cos \theta_4)^{1/2} - E_3 - E_4' = 0.$$

After some algebra, one can find that

$$E_4' = \frac{m_1(m_1 - 2E_3)}{2m_1 - 2E_3(1 - \cos \theta_4)}. \quad (11.6)$$

Now find $df(E'_4)/dE_4$ and $g(E'_4)$ separately,

$$\left. \frac{df}{dE_4} \right|_{E'_4} = -(E_3^2 + E_4'^2 + 2E_3E_4' \cos \theta_4)^{-1/2} (E_3 \cos \theta_4 + E_4) - 1 = -\frac{E_3 \cos \theta_4 + E_4}{m_1 - E_3 - E_4'} - 1 = -\frac{m_1 - E_3(1 - \cos \theta_4)}{m_1 - E_3 - E_4'}.$$

In the second equality, we used the fact that $m_1 - (E_3^2 + E_4'^2 + 2E_3E_4' \cos \theta_4)^{1/2} - E_3 - E_4' = 0$ in the denominator. Leaving E'_4 in df/dE_4 for now, we evaluate $g(E'_4)$,

$$g(E'_4) = \frac{E_4'^2(m_1 - 2E_4')}{m_1 - E_3 - E_4'}.$$

We find that $m_1 - E_3 - E_4'$ cancel:

$$\left| \frac{df}{dE_4} \right|_{E'_4}^{-1} g(E'_4) = \left[\frac{m_1 - E_3(1 - \cos \theta_4)}{m_1 - E_3 - E_4'} \right]^{-1} \frac{E_4'^2(m_1 - 2E_4')}{m_1 - E_3 - E_4'} = \frac{E_4'^2(m_1 - 2E_4')}{m_1 - E_3(1 - \cos \theta_4)}.$$

Note that the absolute value should be handled with care. In our decay process, $m_1 \geq 2E_3$. This is because if the electron has energy E_3 , the two neutrinos must have energy at least E_3 due to momentum conservation. The equality $m_1 = 2E_3$ is reached only when one of the neutrinos has $\mathbf{p} = -\mathbf{p}_3$ and $E = E_3$ and the other neutrino has energy approaching zero. Therefore, $m_1 \geq 2E_3$ suggests that $m_1 - E_3(1 - \cos \theta_4) > 0$, so we should remove the minus sign in $df(E'_4)/dE_4$ using the absolute value. After all these analysis, we plug in E'_4 using (11.6) and get

$$\int \delta(f(E_4))g(E_4) dE_4 = \left| \frac{df}{dE_4} \right|_{E'_4}^{-1} g(E'_4) = \frac{m_1^3 E_3 (m_1 - 2E_3)^2 (1 + \cos \theta_4)}{4[m_1 - E_3(1 - \cos \theta_4)]^4}.$$

The decay rate becomes

$$\Gamma = \frac{G_F^2}{16\pi^5} \int \frac{m_1}{E_3} \delta(f(E_4))g(E_4) dE_4 d^3p_3 \sin \theta_4 d\theta_4 d\phi_4 = \frac{G_F^2}{64\pi^5} \int \frac{m_1^4 (m_1 - 2E_3)^2 (1 + \cos \theta_4)}{[m_1 - E_3(1 - \cos \theta_4)]^4} \sin \theta_4 d\theta_4 d\phi_4 d^3p_3.$$

Performing the angular integral over θ_4 and ϕ_4 ,

$$\Gamma = \frac{G_F^2}{64\pi^5} \int m_1^4 (m_1 - 2E_3)^2 \frac{4\pi(3m_1 - 4E_3)}{3m_1^3(m_1 - 2E_3)} d^3p_3 = \frac{G_F^2}{48\pi^4} \int m_1(3m_1 - 4E_3) d^3p_3.$$

Finally, writing d^3p_3 in spherical coordinates and integrating over $d\Omega_3$ gives the differential decay rate,

$$\Gamma = \frac{G_F^2 m_1}{12\pi^3} \int_0^{m_1/2} E_3^2 (3m_1 - 4E_3) dE_3 \implies \frac{d\Gamma}{dE_3} = \frac{G_F^2 m_1 E_3^2}{12\pi^3} (3m_1 - 4E_3). \quad (11.7)$$

The limits of integration is again because $m_1 \geq 2E_3$. Integrating the differential decay rate from 0 to $m_1/2$, and replacing m_1 by the muon mass m_μ , we obtain the muon decay rate:

$$\boxed{\Gamma(\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu) = \frac{G_F^2}{192\pi^3} m_\mu^5.} \quad (11.8)$$

11.2.3 Predictions and Experiments

There are many implications from the muon decay rate (11.8). First, it depends only on the muon mass based on our analysis in the relativistic limit. The power law $\Gamma \propto m_\mu^5$ is consistent with the dimensional analysis presented in Example 4.2. We expect that the formula can be generalized to the decays $\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau$ and $\tau^- \rightarrow \mu^- \bar{\nu}_\mu \nu_\tau$. In particular, the decay rate should depend only on m_τ^5 but not on m_e or m_μ ,

$$\Gamma(\tau^- \rightarrow \ell^- \bar{\nu}_\ell \nu_\tau) = \frac{G_F^2}{192\pi^3} m_\tau^5 \quad \text{for } \ell = e, \mu. \quad (11.9)$$

There are many ways to test these formula:

- Since the muon decay is dominated by the process $\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$, the muon mean lifetime can be directly calculated from $\Gamma(\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu)$. The predicted muon decay width is

$$\Gamma_\mu = \frac{G_F^2}{192\pi^3} m_\mu^5 = \frac{(1.166 \times 10^{-5} \text{ GeV}^{-2})^2}{192\pi^3} (0.106 \text{ GeV})^5 \approx 3.06 \times 10^{-19} \text{ GeV}.$$

In SI units, the muon mean lifetime would be

$$\tau_\mu = \frac{\hbar}{\Gamma_\mu} = \frac{6.582 \times 10^{-25} \text{ GeV} \cdot \text{s}}{3.06 \times 10^{-19} \text{ GeV}} \approx 2.15 \times 10^{-6} \text{ s}.$$

The measured muon mean lifetime is $2.2 \mu\text{s}$. Our prediction matches the experimental value to a good approximation.

- According to (11.9), the branching fraction $B(\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau)$ and $B(\tau^- \rightarrow \mu^- \bar{\nu}_\mu \nu_\tau)$ should be approximately the same. Indeed, the branching fraction of the two processes are very similar

$$B(\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau) = (17.85 \pm 0.04)\% \quad \text{and} \quad B(\tau^- \rightarrow \mu^- \bar{\nu}_\mu \nu_\tau) = (17.37 \pm 0.04)\%.$$

- The experimental value of τ mean lifetime is $\tau_\tau = 2.903 \times 10^{-13} \text{ s}$, which corresponds to a total decay width of $\Gamma_\tau = 2.27 \times 10^{-12} \text{ GeV}$. The predicted partial decay width by (11.9) is

$$\Gamma(\tau^- \rightarrow \ell^- \bar{\nu}_\ell \nu_\tau) = \frac{(1.166 \times 10^{-5} \text{ GeV}^{-2})^2}{192\pi^3} (1.777 \text{ GeV})^5 \approx 4.05 \times 10^{-13} \text{ GeV}.$$

The predicted branching fraction is

$$B(\tau^- \rightarrow \ell^- \bar{\nu}_\ell \nu_\tau) = \frac{4.05 \times 10^{-13}}{2.27 \times 10^{-12}} \approx 17.84\%,$$

consistent with the experimental branching fraction.

11.3 Pion Decay

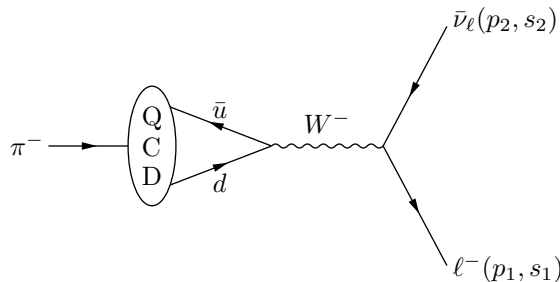
Consider a leptonic two-body decay with the initial particle being a hadron: the pion decay

$$\pi^- \rightarrow \ell^- \bar{\nu}_\ell.$$

Here $\ell = e, \mu$ but not τ because $m_\pi < m_\tau$. The dominant decay mode is $\pi^- \rightarrow \mu^- \bar{\nu}_\mu$, with a branching fraction $B(\pi^- \rightarrow \mu^- \bar{\nu}_\mu) = (99.98770 \pm 0.00004)\%$. Once we understand this decay process, we can use the result in many other decays from a meson to two leptons.

11.3.1 The Charged Pion Decay Constant

Though the pion decay is a two-body decay, it is more subtle than pure leptonic decays because the initial state is a bound state of quarks $\pi^- = d\bar{u}$. In the bound state, there are QCD effects moving around the quarks; they are not momentum eigenstates. This means each quark cannot be assigned with a spinor with a fixed momentum or spin. The Feynman diagram for this process is more appropriately drawn in the following way:



The solution turns out to be the triumph of relativity. First, we see that there are no quarks in the final state. This suggests that the weak leptonic current associated with the $\mu^- \bar{\nu}_\mu$ vertex can be factor out

$$J_\ell^\mu = -\frac{ig}{\sqrt{2}} \bar{u}(p_1, s_1) \frac{1}{2} \gamma^\mu (1 - \gamma^5) v(p_2, s_2),$$

where $u(p_1, s_1)$ and $v(p_2, s_2)$ represent the muon and the neutrino, respectively. In order for the decay amplitude to be a Lorentz-invariant quantity, the current associated with the pion should also be a 4-vector so that when it contracts with the leptonic current,

the result is a scalar. Moreover, the pion is a spin-0 particle, so there is no polarization vector associated with it. Then only 4-vector quantity relevant to the pion is its 4-momentum. We therefore deduce that the pion current should have the form

$$J_\pi^\mu = -\frac{ig}{2\sqrt{2}}f_\pi V_{ud}q^\mu, \quad (11.10)$$

where f_π is called the **charged pion decay constant** (to be determined later), and $q = p_1 + p_2$ due to 4-momentum conservation. There is a V_{ud} present in the pion current because the $\bar{u}$ and d need to annihilate somehow. It is shown explicitly and is conventionally not absorbed into f_π . Taking into account the propagator, the decay amplitude should be

$$\mathcal{A} = \left[-\frac{ig}{2\sqrt{2}}f_\pi V_{ud}q^\mu \right] \frac{i\eta_{\mu\nu}}{m_W^2} \left[-\frac{ig}{\sqrt{2}}\bar{u}(p_1, s_2)\gamma^\nu \frac{1}{2}(1 - \gamma^5)v(p_2, s_2) \right].$$

In terms of the Fermi constant,

$$\mathcal{A} = -i\frac{G_F}{\sqrt{2}}f_\pi V_{ud}(p_1^\mu + p_2^\mu)\bar{u}_1\gamma_\mu(1 - \gamma^5)v_2 \quad (11.11)$$

11.3.2 The Decay Rate

The pion decay is a two body decay, so the rest of the calculation should be simpler than the muon decay calculation. There is no initial spin to averaged over, since π^- is a spin-0 particle. The spin-summed decay amplitude squared (still denoted as $|\mathcal{A}|^2$ for convenience) is

$$|\mathcal{A}|^2 = \frac{1}{2}G_F^2 f_\pi^2 |V_{ud}|^2 \sum_{s_1, s_2} |\bar{u}_1(\not{p}_1 + \not{p}_2)(1 - \gamma^5)v_2|^2 \equiv G_F^2 f_\pi^2 |V_{ud}|^2 \Sigma,$$

where we define

$$\Sigma \equiv \frac{1}{2} \sum_{s_1, s_2} |\bar{u}_1(\not{p}_1 + \not{p}_2)(1 - \gamma^5)v_2|^2.$$

From the Dirac equation, we know that $\bar{u}_1\not{p}_1 = m_1\bar{u}_1$ and $\not{p}_2 v_2 = -m_2 v_2$, so

$$\Sigma = \frac{1}{2} \sum_{s_1, s_2} |\bar{u}_1\not{p}_1(1 - \gamma^5)v_2 + \bar{u}_1(1 + \gamma^5)\not{p}_2 v_2| = \frac{1}{2} \sum_{s_1, s_2} m_1^2 |\bar{u}_1(1 - \gamma^5)v_2|^2.$$

In the second equality, we set the neutrino mass to zero, $m_2 = 0$. It is straightforward to show that $[\bar{u}_1(1 - \gamma^5)v_2]^\dagger = \bar{v}_2(1 + \gamma^5)u_1$ (in contrast to $[\bar{u}_1\gamma^\mu(1 - \gamma^5)v_2]^\dagger = \bar{v}_2\gamma^\mu(1 - \gamma^5)u_1$). Then by some trace techniques,

$$\begin{aligned} \Sigma &= \frac{1}{2} \sum_{s_1, s_2} m_1^2 [\bar{u}_1(1 - \gamma^5)v_2]^\dagger [\bar{u}_1(1 - \gamma^5)v_2] \\ &= \frac{1}{2} \sum_{s_1, s_2} m_1^2 [\bar{v}_2(1 + \gamma^5)u_1] [\bar{u}_1(1 - \gamma^5)v_2] \\ &= \frac{1}{2} m_1^2 \text{Tr} \left[\sum_{s_2} v_2 \bar{v}_2 (1 + \gamma^5) \sum_{s_1} u_1 \bar{u}_1 (1 - \gamma^5) \right] \\ &= \frac{1}{2} m_1^2 \text{Tr} [(\not{p}_2 - m_2)(1 + \gamma^5)(\not{p}_1 + m_1)(1 - \gamma^5)]. \end{aligned}$$

Again, the neutrino mass is zero, $m_2 = 0$,

$$\Sigma = \frac{1}{2} m_1^2 \text{Tr} [\not{p}_2(1 + \gamma^5)(\not{p}_1 + m_1)(1 - \gamma^5)] = \frac{1}{2} m_1^2 \text{Tr} [\not{p}_2(1 + \gamma^5)^2(\not{p}_1 + m_1)].$$

Note that $P_R = \frac{1}{2}(1 + \gamma^5)$ is the right-handed projection operator; its square is itself, and

$$\Sigma = m_1^2 \text{Tr} [\not{p}_2(1 + \gamma^5)(\not{p}_1 + m_1)] = m_1^2 \text{Tr} [\not{p}_2\not{p}_1 + \not{p}_2\gamma^5\not{p}_1 + (\text{odd number of } \gamma\text{'s})].$$

Using the trace theorems, $\text{Tr}(\not{p}_2\not{p}_1) = 4(p_1 \cdot p_2)$, $\text{Tr}(\not{p}_2\gamma^5\not{p}_1) = 0$, and $\text{Tr}(\text{odd number of } \gamma\text{'s}) = 0$, we arrive at

$$\Sigma = 4m_1^2(p_1 \cdot p_2) \quad \text{and} \quad |\mathcal{A}|^2 = 4G_F^2 f_\pi^2 |V_{ud}|^2 m_1^2(p_1 \cdot p_2).$$

In the CM frame of the pion, the 4-momentum of the lepton and the neutrino are

$$p_1 = (E_1, \mathbf{p}_1) \quad \text{and} \quad p_2 = (E_2, -\mathbf{p}_1).$$

Since the neutrino is assumed to be massless, we have $|\mathbf{p}_1| = |\mathbf{p}_2| = E_2$, and

$$p_1 \cdot p_2 = E_1 E_2 + |\mathbf{p}_1| |\mathbf{p}_2| = E_2(E_1 + E_2).$$

For a two-body decay, the energies are definite and the decay rate is given by (8.12),

$$\Gamma = \frac{|\mathbf{p}^*|}{32\pi^2 m_A^2} \int |\mathcal{A}|^2 d\Omega,$$

where

$$E_1 = \frac{m_A^2 + m_1^2 - m_2^2}{2m_A}, \quad E_2 = \frac{m_A^2 + m_2^2 - m_1^2}{2m_A}, \quad |\mathbf{p}^*| = \frac{\sqrt{[m_A^2 - (m_1 + m_2)^2][m_A^2 - (m_1 - m_2)^2]}}{2m_A}.$$

Taking $m_A = m_\pi$, $m_1 = m_\ell$, and $m_2 = 0$, this gives

$$p_1 \cdot p_2 = E_2(E_1 + E_2) = \frac{1}{2}(m_\pi^2 - m_\ell^2) \implies |\mathcal{A}|^2 = 2G_F^2 f_\pi^2 |V_{ud}|^2 m_\ell^2 (m_\pi^2 - m_\ell^2).$$

Finally, the decay rate is

$$\Gamma = \frac{|\mathbf{p}^*|}{32\pi^2 m_\pi^2} \int |\mathcal{A}|^2 d\Omega = \frac{1}{32\pi^2 m_\pi^2} \left(\frac{m_\pi^2 - m_\ell^2}{2m_\pi} \right) [8\pi G_F^2 f_\pi^2 |V_{ud}|^2 m_\ell^2 (m_\pi^2 - m_\ell^2)].$$

Simplifying this expression, we obtain the pion decay rate:

$$\boxed{\Gamma(\pi^- \rightarrow \ell^- \bar{\nu}_\ell) = \frac{1}{8\pi} G_F^2 f_\pi^2 |V_{ud}|^2 m_\pi m_\ell^2 \left(\frac{m_\pi^2 - m_\ell^2}{m_\pi} \right)^2.} \quad (11.12)$$

11.3.3 Predictions and Experiments

Just like for the muon decay, let's look at some predictions vs. experimental results for the pion decay:

- One prediction that can be made without knowing anything about f_π is the ratio

$$\frac{\Gamma(\pi^- \rightarrow e^- \bar{\nu}_e)}{\Gamma(\pi^- \rightarrow \mu^- \bar{\nu}_\mu)} = \frac{m_e^2(m_\pi^2 - m_e^2)^2}{m_\mu^2(m_\pi^2 - m_\mu^2)^2} = \frac{0.511^2(140^2 - 0.511^2)^2}{106^2(140^2 - 106^2)^2} \approx 1.28 \times 10^{-4}.$$

The experimental branching fractions are $B(\pi^- \rightarrow e^- \bar{\nu}_e) = (1.230 \pm 0.004) \times 10^{-4}$ and $B(\pi^- \rightarrow \mu^- \bar{\nu}_\mu) = (99.98770 \pm 0.00004)\%$, consistent with our prediction. This ratio is in fact a very famous result in particle physics. The two-body decay rate formula seems to be proportional to $|\mathbf{p}^*|$: lighter particles appears to have higher decay rate because their momentum is larger. But the experiments indicate that the decay rate for $\pi^- \rightarrow e^- \bar{\nu}_e$ is highly suppressed. The key hides in the amplitude squared which gives a m_ℓ^2 .

Let's analyze the problem the perspective of angular momentum conservation and helicity. In the CM frame, $\bar{\nu}_\ell$ (an antiparticle) has helicity $\lambda_{\bar{\nu}} = +1/2$ and the charged lepton has helicity $\lambda_\ell = -1/2$ in the relativistic limit. The helicity is the spin projection in the direction of momentum. Both the momenta and helicities of $\bar{\nu}_\ell$ and ℓ^- are opposite, meaning that their spins are parallel instead of antiparallel. This violates the conservation of angular momentum because the pion is spin-0, so the process should be forbidden. The pion can decay only when the relativistic limit *does not* hold. In reality, the relativistic limit works for the electron but not for the muon since $m_e \ll m_\pi$ while $m_\mu \sim m_\pi$. The result is that $\pi^- \rightarrow e^- \bar{\nu}_e$ is suppressed in the relativistic limit, but $\pi^- \rightarrow \mu^- \bar{\nu}_\mu$ is not so suppressed as the muon is not so relativistic.

- To infer about f_π , we need to compute the decay rate of the pion. The $\ell = \mu$ mode is the dominant one, so the decay width of the pion is approximately $\Gamma_\pi \simeq \Gamma(\pi^- \rightarrow \mu^- \bar{\nu}_\mu)$. We can write f_π in terms of measurable quantities,

$$\Gamma(\pi^- \rightarrow \mu^- \bar{\nu}_\mu) = \frac{1}{8\pi} G_F^2 f_\pi^2 |V_{ud}|^2 m_\pi m_\mu^2 \left(\frac{m_\pi^2 - m_\mu^2}{m_\pi} \right)^2 \implies f_\pi = \left[\frac{8\pi m_\pi^3 \Gamma_\pi}{G_F^2 |V_{ud}|^2 m_\mu^2 (m_\pi^2 - m_\mu^2)} \right]^{1/2}.$$

The measured π^- mean lifetime is $\tau_\pi = 2.60 \times 10^{-8}$ s, which corresponds to a total decay width

$$\Gamma_\pi = \frac{\hbar}{\tau_\pi} \approx 2.54 \times 10^{-14} \text{ MeV}.$$

Plugging in the numbers,

$$f_\pi = \left[\frac{8\pi \cdot 140^3 \cdot 2.54 \times 10^{-14}}{(1.17 \times 10^{-11})^2 \cdot 0.9744^2 \cdot 106^2 \cdot (140^2 - 106^2)^2} \right]^{1/2} \text{ MeV} \approx 131 \text{ MeV}.$$

The pion decay constant f_π is not an empirical value. It can be calculated theoretically from **lattice QCD**, a theory that is able to handle non-perturbative physics. The theoretical value is $f_\pi = 130$ MeV.

- The pion decay serves as a prototype for a spin-0 meson decaying into a charged lepton and a neutrino. For example, consider the kaon decay $K^- \rightarrow \ell^- \bar{\nu}_\ell$. The kaon has quark content $K^- = \bar{u}s$ and it is a spin-0 meson. The steps are exactly the same. Just replace $m_\pi \rightarrow m_K$, $f_\pi \rightarrow f_K$, and $V_{ud} \rightarrow V_{us}$ and write down

$$\Gamma(K^- \rightarrow \ell^- \bar{\nu}_\ell) = \frac{1}{8\pi} G_F^2 f_K^2 |V_{us}|^2 m_K m_\ell^2 \left(\frac{m_K^2 - m_\ell^2}{m_K^2} \right)^2.$$

We don't need to know the decay constants for these particles to predict ratio of decay rates:

$$\frac{\Gamma(K^- \rightarrow e^- \bar{\nu}_e)}{\Gamma(K^- \rightarrow \mu^- \bar{\nu}_\mu)} = \frac{m_e^2 (m_K^2 - m_e^2)^2}{m_\mu^2 (m_K^2 - m_\mu^2)^2} \approx 2.55 \times 10^{-5},$$

which agrees with the measured value 2.488×10^{-5} . For heavier mesons like D_s^+ or B^+ (though their decay modes are not mainly leptonic), the lepton ℓ can even be the τ -lepton.

11.4 W-Mediated Neutrino Scattering

11.4.1 Neutrinos

Neutrinos are particles with many mysterious properties, many of which are not well understood yet:

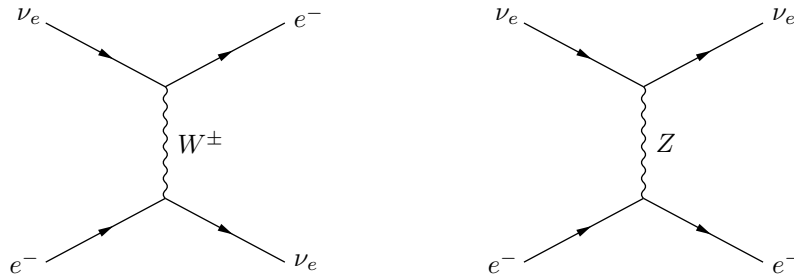
1. Neutrinos are the only fermions that have weak interactions only.
2. Neutrinos are the only particles for which the true field is not known yet. In our discussion, we assume that neutrinos are described by the Dirac field. On the other hand, they can also be described by the **Majorana field**, which correspond to neutral fermions that are their own antiparticles. Many theoretical models suggest that neutrinos are Majorana, not Dirac.
3. Neutrinos have a mass scale much smaller than the scale of the other fermions, and their masses are not determined yet. They may acquire masses not from the Higgs mechanism.
4. Neutrinos are the only fundamental particles that undergo oscillations. That is, they spontaneously change from one neutrino flavor to another, $\nu_e \leftrightarrow \nu_\mu \leftrightarrow \nu_\tau$. In this case, the lepton number for each flavor is not conserved. Neutrino oscillation proves that neutrinos must have nonzero masses. (This is the only place where we cannot assume neutrinos are massless in calculations.)
5. Neutrinos are crucial in many astrophysical processes. For example, most energy in a supernova is carried away by energetic neutrinos.

11.4.2 Neutrino-Electron Scattering

Consider the neutrino-electron scattering at low energy ($s \ll m_W^2$),

$$\nu_e e^- \rightarrow \nu_e e^-.$$

There are two lowest-order Feynman diagrams, one mediated by a W , and the other mediated by a Z :



For now, we are going to neglect the Z -mediated contribution. Let $u(p_1, s_1)$, $u(p_2, s_2)$, $u(p_3, s_3)$, and $u(p_4, s_4)$ be the spinors for the initial electron, initial neutrino, final electron, and final neutrino, respectively. We can easily write down the scattering amplitude:

$$\mathcal{A} = \left[-\frac{ig}{\sqrt{2}} \bar{u}(p_3, s_3) \gamma^\mu \frac{1}{2} (1 - \gamma^5) u(p_2, s_2) \right] \frac{i\eta_{\mu\nu}}{m_W^2} \left[-\frac{ig}{\sqrt{2}} \bar{u}(p_4, s_4) \gamma^\nu \frac{1}{2} (1 - \gamma^5) u(p_1, s_1) \right] = -i \frac{G_F}{\sqrt{2}} [\bar{u}_3 \gamma_\mu (1 - \gamma^5) u_2] [\bar{u}_4 \gamma^\mu (1 - \gamma^5) u_1].$$

The spin-averaged amplitude squared is given by

$$\overline{|\mathcal{A}|^2} = \frac{1}{2} \sum_{s_1, \dots, s_4} |\mathcal{A}|^2.$$

Note that the averaging factor is $\frac{1}{2}$ instead of $\frac{1}{4}$. This is because we only average over the spin of the electron. The neutrino produced is guaranteed to be left-handed if they are assumed to be massless; its spin is always polarized opposite to its momentum. Antineutrinos are always right-handed, or polarized along its momentum. Nevertheless, for calculation simplicity, we still sum over all polarizations for the neutrino. We can do this because the left-handed chiral projection operator $P_L = \frac{1}{2}(1 - \gamma^5)$ ensures that right-handed neutrinos contribute nothing in the amplitude. See Appendix A.6.2 for proof. All neutrinos experimentally detected are left-handed. However, right-handed neutrinos exist theoretically because neutrinos are not exactly massless.

The spin-averaged amplitude has exactly the same form as the one for the muon decay. The result for the neutrino-electron scattering is very similar to (11.5),

$$\overline{|\mathcal{A}|^2} = 64G_F^2 (p_1 \cdot p_2)(p_3 \cdot p_4). \quad (11.13)$$

In actual experiments, the electron is viewed at rest in the detector because the speed of atomic electrons are much less than the speed of light. We define the energy of the neutrino to be E_ν as measured in the rest frame of the electron. Suppose the electron mass is negligible to the CM energy, $m_e^2 \ll s$, then

$$s = (p_1 + p_2)^2 = p_1^2 + p_2^2 + 2p_1 \cdot p_2 \simeq 2p_1 \cdot p_2.$$

Similarly, $s = (p_3 + p_4)^2 \simeq 2p_3 \cdot p_4 = 2p_1 \cdot p_2$. The spin-averaged amplitude squared can then be written as

$$\overline{|\mathcal{A}|^2} = 16G_F^2 s^2.$$

The CM energy is Lorentz-invariant. We can evaluate it in the electron rest frame, $s = 2p_1 \cdot p_2 = 2m_e E_\nu$. The differential cross section in the CM frame is given by (8.16),

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|} \overline{|\mathcal{A}|^2}.$$

If the two initial-state particles are the same as the two final-state particles, then $|\mathbf{p}|_i = |\mathbf{p}|_f$, so

$$\frac{d\sigma}{d\Omega}(\nu_e e^- \rightarrow \nu_e e^-) = \frac{1}{4\pi^2} G_F^2 s. \quad (11.14)$$

The total cross section (W -mediated contribution only) is

$$\sigma(\nu_e e^- \rightarrow \nu_e e^-) = \frac{1}{\pi} G_F^2 s = \frac{2}{\pi} G_F^2 m_e E_\nu. \quad (11.15)$$

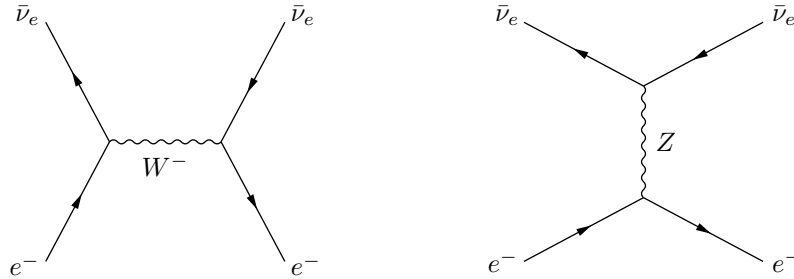
From (11.14), we note that the neutrino-electron scattering differential cross section has no angular dependence. The scattering process is *isotropic*. Also, this weak interaction cross section scales linearly as the CM energy, $\sigma_{\text{weak}} \propto s$. Recall that the electromagnetic interactions often have cross sections that scale like $\sigma_{\text{EM}} \propto s^{-1}$, e.g. electron-positron annihilation. This is a direct result of the different propagators, $i\eta_{\mu\nu}/m_W^2$ vs. $-i\eta_{\mu\nu}/q^2$. For low-energy W -mediated weak interactions, the mass of the W -boson sets the energy scale; q^2 is negligible compared to m_W^2 . On the other hand, for electromagnetic interactions, the photon is massless, so the propagator itself carries $1/q^2 = s^{-1}$. Squaring the propagator gives s^{-2} , which accounts for the ratio $\sigma_{\text{EM}}/\sigma_{\text{weak}} \propto s^{-2}$.

11.4.3 Antineutrino-Electron Scattering

Consider the antineutrino-electron scattering at low energy ($s \ll m_W^2$),

$$\bar{\nu}_e e^- \rightarrow \bar{\nu}_e e^-.$$

It also has two lowest-order Feynman diagrams, one mediated by a W , and the other mediated by a Z , but this time the W -mediated process proceeds in an s -channel:



Again, we are going to neglect the Z -mediated contribution. Let $u(p_1, s_1)$, $v(p_2, s_2)$, $u(p_3, s_3)$, $v(p_4, s_4)$ be the spinors for the initial electron, initial antineutrino, final electron, and final antineutrino, respectively. The scattering amplitude is

$$\mathcal{A} = \left[-\frac{ig}{\sqrt{2}} \bar{v}(p_2, s_2) \gamma^\mu \frac{1}{2} (1 - \gamma^5) u(p_1, s_1) \right] \frac{i\eta_{\mu\nu}}{m_W^2} \left[-\frac{ig}{\sqrt{2}} \bar{u}(p_3, s_3) \gamma^\nu \frac{1}{2} (1 - \gamma^5) v(p_4, s_4) \right] = -i \frac{G_F}{\sqrt{2}} [\bar{v}_2 \gamma_\mu (1 - \gamma^5) u_1] [u_3 \gamma^\mu (1 - \gamma^5) v_4].$$

It also has exactly the same form as the amplitude for the muon decay. We can immediately write down the result for the spin-averaged amplitude squared,

$$|\overline{\mathcal{A}}|^2 = 64 G_F^2 (p_1 \cdot p_4) (p_2 \cdot p_3). \quad (11.16)$$

Note that this is slightly different from (11.13) for neutrino-electron scattering, which proceeds through a t -channel. Assume that the electron mass is small compared to the CM energy, $m_e \ll s$; in other words, the electron is relativistic in the CM frame. Let the 4-momentum configuration of the system be

$$p_1 = (E_i, 0, 0, -E_i), \quad p_2 = (E_i, 0, 0, E_i), \quad p_3 = (E_f, 0, E_f \sin \theta, E_f \cos \theta), \quad p_4 = (E_f, 0, -E_f \sin \theta, -E_f \cos \theta).$$

Then

$$(p_1 - p_4)^2 = p_1^2 + p_4^2 - p_1 \cdot p_4 \simeq -2p_1 \cdot p_4,$$

The 4-momentum conservation gives $p_1 + p_2 = p_3 + p_4$, or $(p_2 - p_3)^2 \simeq -2p_2 \cdot p_3 = (p_1 - p_4)^2 = -p_1 \cdot p_4$. The CM energy is $s \simeq 2p_1 \cdot p_2 = 4E_i E_f$. Then spin-averaged amplitude squared can then be expressed as

$$|\overline{\mathcal{A}}|^2 = 64 G_F^2 (p_2 \cdot p_3)^2 = 64 G_F^2 [E_i E_f (1 - \cos \theta)]^2 = 4 G_F^2 s^2 (1 - \cos \theta)^2.$$

In the CM frame, the differential cross section is

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|} |\overline{\mathcal{A}}|^2.$$

Due to energy conservation, $E_i = E_f = |\mathbf{p}_i| = |\mathbf{p}_f|$ in the CM frame. Plugging in the amplitude,

$$\frac{d\sigma}{d\Omega}(\bar{\nu}_e e^- \rightarrow \bar{\nu}_e e^-) = \frac{1}{16\pi^2} G_F^2 s (1 - \cos \theta)^2. \quad (11.17)$$

The total cross section is

$$\sigma(\bar{\nu}_e e^- \rightarrow \bar{\nu}_e e^-) = \frac{1}{3\pi} G_F^2 s = \frac{1}{3} \sigma(\nu_e e^- \rightarrow \nu_e e^-). \quad (11.18)$$

For antineutrino-electron scattering, the differential cross section *does* have angular dependence. Because of the $(1 - \cos \theta)^2$ angular dependence, the final-state electron is forbidden to have a momentum opposite to the momentum of the initial-state electron. Equivalently, we say that we can never see an antineutrino bouncing back in an antineutrino-electron scattering experiment.

11.4.4 Helicity Configuration

We can analyze the two cases $\nu_e e^- \rightarrow \nu_e e^-$ and $\bar{\nu}_e e^- \rightarrow \bar{\nu}_e e^-$ via angular momentum conservation and helicity. Figure 11.1 shows the helicity configuration of the two process in the CM frame, assuming that all particles are relativistic.

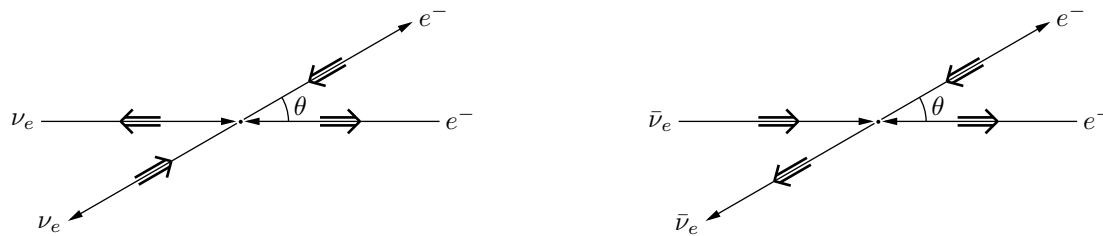


Figure 11.1: Neutrino-electron scattering (left) and antineutrino-electron scattering (right) helicity configuration. The total spin of the neutrino-electron system is zero, leaving the scattering process isotropic. The total spin of the antineutrino-electron system is nonzero, resulting in angular dependence in the differential cross section.

For the neutrino-electron scattering (left), all particles in the process are left-handed. The spin of the particles (indicated by large arrows) sum up to zero. There is no preferred direction in space, so the neutrino-electron scattering is isotropic. For the antineutrino-electron scattering, antineutrinos are right-handed but electrons are left-handed. The initial state has a total spin of 1 pointing in the $+z$ -direction; the final state has a total spin of 1 pointing in θ away from the $+z$ -direction. If $\theta = 0$, the total spin is reversed after interaction. This violates the conservation of angular momentum, leading to zero amplitude for $\theta = 0$. For $\theta \neq 0$, the spin projection on the z -axis is nonzero, giving nonzero amplitude. Evidently, the most favored spin projection is $\theta = \pi$ because it is consistent with the initial total spin. Indeed, the amplitude is the largest when $\theta = \pi$.

11.5 Neutrino Oscillations

11.5.1 History of Neutrino Oscillation

To be written.

11.5.2 Formalism

To be written.

11.6 CP Violation

To be written.

12 ELECTROWEAK THEORY

12.1 Z -Mediated Processes; Vertex Factors and Propagators

To be written.

13 THE HIGGS BOSON

To be written.

Acknowledgements

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Lecture notes by professor Richman.

Part IV

Appendices

A PROOFS AND DERIVATIONS

A.1 Classical Particle Physics and Relativity

A.1.1 Lorentz Transformation

To derive the Lorentz transformation, we need to use both postulates of special relativity:

1. The laws of physics take the same form in all inertial reference frames. All inertial frames are fundamentally indistinguishable based on experiments in that frame only.
2. The speed of light (and other massless particles) c is the maximum speed in the universe and has the same value in all inertial reference frames.

Consider two frames S and S' , moving at a velocity v relative to S , with parallel coordinate axes. The x -axis and x' -axis are aligned. When origins of S and S' coincide (call them $\mathcal{O}$ and $\mathcal{O}'$ respectively), the time in both frames is set to zero, $t = t' = 0$. Since the x -axes are aligned, we will ignore y - and z -components here because they are not interesting. Suppose the Lorentz transformation from inertial frame S to another inertial frame S' takes the form

$$t' = f(x, t), \quad x' = g(x, t),$$

and f and g are linearly independent. Consider a free particle moving with constant velocity $v_x = dx/dt$ in S frame. By postulate 1, this particle observed in S' must also move at a constant velocity $v'_x = dx'/dt'$, which means

$$v'_x = \frac{dx'}{dt'} = \frac{dx'/dt}{dt'/dt} = \frac{\frac{\partial g}{\partial x} \frac{dx}{dt} + \frac{\partial g}{\partial t}}{\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial t}} = \frac{\frac{\partial g}{\partial x} v_x + \frac{\partial g}{\partial t}}{\frac{\partial f}{\partial x} v_x + \frac{\partial f}{\partial t}}.$$

Since v_x and v'_x can be arbitrary constants, all partial derivatives must also be constants. This means the Lorentz transformation must be linear:

$$\begin{aligned} t' &= a_s + a_0 t + a_1 x, \\ x' &= b_s + b_0 t + b_1 x. \end{aligned}$$

The goal is to find the constants a_s , a_0 , a_1 , b_s , b_0 , and b_1 . When $\mathcal{O}$ and $\mathcal{O}'$ coincide, we have $x = x' = 0$ at $t' = t = 0$. This synchronization event sets $a_s = b_s = 0$. Now, the frame S' is moving at a velocity v relative to S , then the position of $\mathcal{O}'$ in S is $x = vt$. In S' , its origin $\mathcal{O}'$ is always at $x' = 0$, so

$$0 = x' = b_0 t + b_1 (vt) \implies b_0 = -b_1 v \implies x' = b_1 (x - vt).$$

Moreover, by symmetry, S is moving at a velocity of $-v$ relative to S' . If the transformation works for all frames (by postulate 1), then we should obtain

$$x = b_1 (x' + vt').$$

The transformation equations now look like

$$\begin{aligned} t' &= a_0 t + a_1 x, \\ x' &= b_1 (x - vt), \\ x &= b_1 (x' + vt'). \end{aligned}$$

It's time to use postulate 2. Suppose a light pulse is emitted at $(t, x) = (0, 0)$ and $(t', x') = (0, 0)$, i.e., at synchronization. By the constancy of the speed of light, the event $(t, x) = (t, ct)$ in S should have coordinates $(t', x') = (t', ct')$ in S' . Thus,

$$\begin{aligned} x' = ct' &= b_1 (x - vt) = b_1 (ct - vt) = b_1 (1 - \beta) ct, \\ x = ct &= b_1 (x' + vt') = b_1 (ct' + vt') = b_1 (1 + \beta) ct', \end{aligned}$$

where $\beta \equiv v/c$. Substituting ct into ct' , we have

$$ct' = b_1^2(1 - \beta^2)ct' \implies b_1 = \pm \frac{1}{\sqrt{1 - \beta^2}}.$$

We will keep the positive solution as the negative solution does not satisfy $x = x'$ and $t = t'$ at $v = 0$. This b_1 is the Lorentz factor,

$$\gamma \equiv \frac{1}{\sqrt{1 - \beta^2}}.$$

Using $x' = \gamma(x - vt)$ and $x = \gamma(x' + vt')$ to solve for t' and we will get

$$t' = \gamma t - \gamma \beta (x/c).$$

The Lorentz transformation is

$$ct' = \gamma(ct - \beta x),$$

$$x' = \gamma(x - \beta ct),$$

$$y' = y,$$

$$z' = z.$$

A.1.2 The Classical Rutherford Scattering

The Rutherford scattering describes a process where an α -particle (charge $Z_\alpha = 2e$) scattering from a gold nucleus (charge $Z_{\text{Au}} = 79e$).

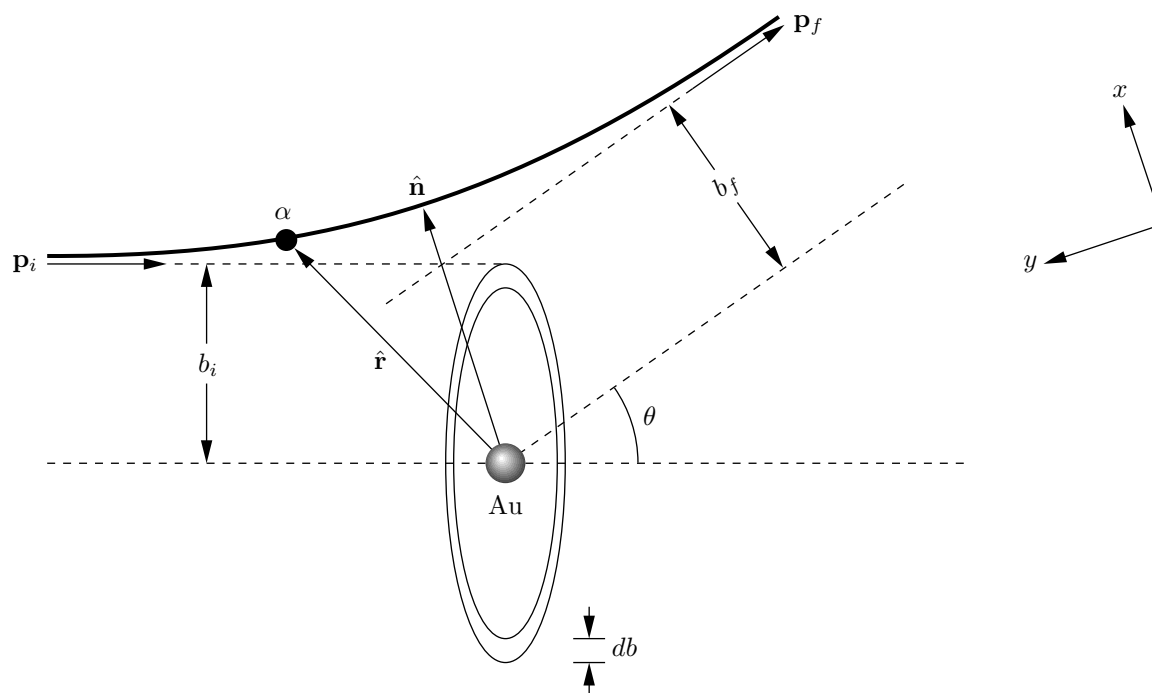


Figure A.1: A diagram for the Rutherford scattering process.

The momentum transfer of the α -particle is defined by

$$\mathbf{q} \equiv \mathbf{p}_i - \mathbf{p}_f,$$

where $\mathbf{p}_i$ and $\mathbf{p}_f$ are initial and final momenta of the particle. We assume that there is no loss in α -particle's energy because $m_{\text{Au}} \gg m_\alpha$. Thus, $\mathbf{p}_i$ and $\mathbf{p}_f$ have the same magnitude and differ only by a scattering angle θ . The square of the momentum transfer is then

$$\mathbf{q}^2 = \mathbf{p}_i^2 + \mathbf{p}_f^2 - 2\mathbf{p}_i \cdot \mathbf{p}_f = 2\mathbf{p}^2(1 - \cos \theta) = 4\mathbf{p}^2 \sin^2 \frac{\theta}{2},$$

where we define $|\mathbf{p}| \equiv |\mathbf{p}_i| = |\mathbf{p}_f|$. The potential generated by the gold nucleus is a Coulomb potential. The potential and force

between the α -particle and the nucleus are given by

$$V(r) = \frac{Z_{\text{Au}}e}{4\pi\epsilon_0 r}, \quad \mathbf{F} = \frac{Z_{\text{Au}}Z_{\alpha}e^2}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}}.$$

The Coulomb force is both radial and conservative. This means both the angular momentum and energy are conserved during the scattering process. Let the initial and final impact parameters of the α -particle be b_i and b_f . The conservation of angular momentum tells us that

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} = b_i |\mathbf{p}_i| \hat{\mathbf{z}} = b_f |\mathbf{p}_f| \hat{\mathbf{z}} = b |\mathbf{p}| \hat{\mathbf{z}} \implies b_i = b_f \equiv b.$$

The initial and final impact parameters are the same. Referring to Figure A.1, let $\hat{\mathbf{n}}$ point in the direction of the point of closest approach. Let the x -axis lie along $\hat{\mathbf{n}}$, and define ϕ to be the angle between $\hat{\mathbf{r}}$ of the α -particle and $\hat{\mathbf{n}}$. This is just a tilted version of polar coordinates. The radial unit vector is still $\hat{\mathbf{r}} = (\cos \phi, \sin \phi, 0)$. In this way, the angular momentum is

$$\mathbf{L} = mr^2 \frac{d\phi}{dt} \hat{\mathbf{z}}.$$

Meanwhile, we also know that $\mathbf{L} = b |\mathbf{p}| \hat{\mathbf{z}}$, so the time-derivative of ϕ is

$$\frac{d\phi}{dt} = \frac{b |\mathbf{p}|}{mr^2}.$$

By the impulse-momentum theorem, the momentum transfer is the integral of force over time,

$$\mathbf{q} = - \int_{-\infty}^{\infty} dt \mathbf{F} = - \int_{-\infty}^{\infty} dt \frac{Z_{\text{Au}}Z_{\alpha}e^2}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}}.$$

Note the minus sign before the integral. This is because we defined the momentum transfer as $\mathbf{q} = \mathbf{p}_i - \mathbf{p}_f$ instead of $\mathbf{p}_f - \mathbf{p}_i$. Making a change of variable to ϕ , the integral becomes

$$\mathbf{q} = - \int_{+\frac{1}{2}(\pi-\theta)}^{-\frac{1}{2}(\pi-\theta)} d\phi \frac{dt}{d\phi} \frac{Z_{\text{Au}}Z_{\alpha}e^2}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}} = \int_{-\frac{1}{2}(\pi-\theta)}^{\frac{1}{2}(\pi-\theta)} d\phi \frac{Z_{\text{Au}}Z_{\alpha}me^2}{4\pi\epsilon_0 b |\mathbf{p}|} \hat{\mathbf{r}}.$$

The ϕ -dependence in the integrand is hidden inside $\hat{\mathbf{r}}$. To avoid doing a vector integral, we take the inner product with $\hat{\mathbf{n}} = \hat{\mathbf{x}}$ on both sides,

$$\mathbf{q} \cdot \hat{\mathbf{n}} = - \frac{Z_{\text{Au}}Z_{\alpha}me^2}{4\pi\epsilon_0 b |\mathbf{p}|} \int_{-\frac{1}{2}(\pi-\theta)}^{\frac{1}{2}(\pi-\theta)} d\phi \cos \phi.$$

The whole process has reflection symmetry about $\hat{\mathbf{n}}$, so $\mathbf{q}$ is actually in the direction of $\hat{\mathbf{n}}$. The LHS is just the magnitude $|\mathbf{q}| = 2|\mathbf{p}| \sin \frac{\theta}{2}$. Performing the integral on the RHS,

$$2|\mathbf{p}| \sin \frac{\theta}{2} = \frac{Z_{\text{Au}}Z_{\alpha}me^2}{2\pi b |\mathbf{p}|} \cos \frac{\theta}{2}.$$

Rearranging some terms, we obtain the impact parameter as a function of scattering angle,

$$b = \frac{Z_{\text{Au}}Z_{\alpha}me^2}{4\pi\epsilon_0 \mathbf{p}^2} \cot \frac{\theta}{2}, \quad db = - \frac{Z_{\text{Au}}Z_{\alpha}me^2}{8\pi \mathbf{p}^2} \csc^2 \frac{\theta}{2} d\theta.$$

The differential cross section in terms of the impact parameter is

$$\begin{aligned} d\sigma &\equiv 2\pi b db = -\pi \left(\frac{Z_{\text{Au}}Z_{\alpha}me^2}{4\pi\epsilon_0 \mathbf{p}^2} \right)^2 \cot \frac{\theta}{2} \csc^2 \frac{\theta}{2} d\theta \\ &= -\pi \left(\frac{Z_{\text{Au}}Z_{\alpha}me^2}{4\pi\epsilon_0 \mathbf{p}^2} \right)^2 \frac{\sin \frac{\theta}{2} \cos \frac{\theta}{2}}{\sin^4 \frac{\theta}{2}} d\theta \\ &= -\frac{\pi}{2} \left(\frac{Z_{\text{Au}}Z_{\alpha}me^2}{4\pi\epsilon_0 \mathbf{p}^2} \right)^2 \frac{\sin \theta}{\sin^4 \frac{\theta}{2}} d\theta \\ &= \frac{1}{16} \left(\frac{Z_{\text{Au}}Z_{\alpha}e^2}{4\pi\epsilon_0} \frac{2m}{\mathbf{p}^2} \right)^2 \frac{1}{\sin^4 \frac{\theta}{2}} 2\pi d(\cos \theta). \end{aligned}$$

This gives the Rutherford scattering angular dependence of the differential cross section:

$$\frac{d\sigma}{2\pi d(\cos\theta)} = \frac{1}{16} \left(Z_{\text{Au}} Z_{\alpha} \alpha \hbar c \frac{2m}{\mathbf{p}^2} \right)^2 \frac{1}{\sin^4 \frac{\theta}{2}},$$

where $\alpha = e^2/4\pi\epsilon_0\hbar c \approx 1/137$ is the fine-structure constant.

A.2 Diracology

A.2.1 Useful Identities

To be written.

A.2.2 Trace of α_i and β

First, we need the cyclic property of the trace from linear algebra:

$$\text{Tr}(AB) = \text{Tr}(BA), \quad \text{Tr}(ABC) = \text{Tr}(BCA) = \text{Tr}(CAB), \quad \text{and so on.}$$

Proof. The matrix element of AB can be calculated as

$$(AB)_{ik} = \sum_j A_{ij} B_{jk},$$

by the definition of matrix multiplication. Using this to calculate the trace:

$$\text{Tr}(AB) \equiv \sum_i (AB)_{ii} = \sum_{i,j} A_{ij} B_{ji}.$$

Since A_{ij} and B_{ji} are just numbers, they commute,

$$\text{Tr}(AB) = \sum_{i,j} B_{ji} A_{ij} = \sum_j (BA)_{jj} = \text{Tr}\{BA\}.$$

By this result (and the associativity of matrix multiplication), it is straightforward to show cyclic property of the trace of more products of matrices:

$$\text{Tr}(ABC) = \text{Tr}[(AB)C] = \text{Tr}[C(AB)] = \text{Tr}(CAB), \quad \text{Tr}(ABC) = \text{Tr}[A(BC)] = \text{Tr}[(BC)A] = \text{Tr}[BCA],$$

and so on. □

The trace of α_i : because $\beta\alpha_i = -\alpha_i\beta$,

$$\text{Tr}(\alpha_i\beta^2) = -\text{Tr}(\beta\alpha_i\beta) = -\text{Tr}(\beta^2\alpha_i).$$

The second equality is from the cyclic property of the trace. But $\text{Tr}(\alpha_i\beta^2) = -\text{Tr}(\beta^2\alpha_i)$ infers that $\text{Tr}(\alpha_i) = -\text{Tr}(\alpha_i)$ because $\beta^2 = I$, so $\text{Tr}(\alpha_i) = 0$. Similarly, by $\alpha_i\beta = -\beta\alpha_i$ and the cyclic property of the trace,

$$\text{Tr}(\alpha_i\beta\alpha_i) = -\text{Tr}(\beta\alpha_i^2) = -\text{Tr}(\alpha_i^2\beta),$$

$$\text{Tr}(\alpha_i\beta\alpha_i) = \text{Tr}(\alpha_i^2\beta),$$

$$\text{Tr}(\beta) = -\text{Tr}(\beta) \implies \text{Tr}(\beta) = 0.$$

Hence both α_i and β are traceless.

A.2.3 Determinants

Taking the determinant of $\alpha_i\beta$:

$$\det(\alpha_i\beta) = \det(-\beta\alpha_i).$$

$$\det \alpha_i \det \beta = \det(-I) \det \beta \det \alpha_i.$$

$$1 = \det(-I) = (-1)^N,$$

where N is the dimension of the identity matrix, α_i and β . In this case N must be even.

A.2.4 Hermitian Conjugate of γ^μ

For $\gamma^0 = \beta$, because β is Hermitian, $(\gamma^0)^\dagger = \gamma^0$. For γ^i ,

$$(\gamma^i)^\dagger = (\beta\alpha_i)^\dagger = \alpha_i^\dagger\beta^\dagger = \alpha_i\beta = -\beta\alpha_i = -\gamma^i,$$

where we used the anticommutation relation $\{\alpha_i, \beta\} = 0$, or $\alpha_i\beta = -\beta\alpha_i$.

A.2.5 Anticommutator $\{\gamma^\mu, \gamma^\nu\}$

We can split the anticommutator $\{\gamma^\mu, \gamma^\nu\}$ into three cases: $\{\gamma^0, \gamma^0\}$, $\{\gamma^0, \gamma^i\}$, and $\{\gamma^i, \gamma^j\}$.

$$\begin{aligned}\{\gamma^0, \gamma^0\} &= 2(\gamma^0)^2 = 2\beta^2 = 2I, \\ \{\gamma^0, \gamma^i\} &= \gamma^0\gamma^i + \gamma^i\gamma^0 = \beta\beta\alpha_i + \beta\alpha_i\beta = \alpha_i - \beta\beta\alpha_i = \alpha_i - \alpha_i = 0, \\ \{\gamma^i, \gamma^j\} &= \beta\alpha_i\beta\alpha_j + \beta\alpha_j\beta\alpha_i = -\beta^2\alpha_i\alpha_j - \beta^2\alpha_j\alpha_i = -(\alpha_i\alpha_j + \alpha_j\alpha_i) = -2\delta_{ij}I.\end{aligned}$$

They can be summarized into

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}I.$$

A.2.6 More Properties of γ^μ

Let $\gamma'^\mu = U^{-1}\gamma^\mu U$, where U is an arbitrary unitary transformation. The anticommutation property of the γ matrices are preserved under these transformations: $\{\gamma'^\mu, \gamma'^\nu\} = 2\eta^{\mu\nu}I$.

Proof.

$$\begin{aligned}\{\gamma'^\mu, \gamma'^\nu\} &= \{U^{-1}\gamma^\mu U, U^{-1}\gamma^\nu U\} \\ &= U^{-1}\gamma^\mu U U^{-1}\gamma^\nu U + U^{-1}\gamma^\nu U U^{-1}\gamma^\mu U \\ &= U^{-1}\{\gamma^\mu, \gamma^\nu\}U \\ &= U^{-1}(2\eta^{\mu\nu}I)U^{-1} \\ &= 2\eta^{\mu\nu}I U^{-1}U \\ &= 2\eta^{\mu\nu}I.\end{aligned}$$

□

The contraction of γ^μ with itself is $\gamma_\mu\gamma^\mu = 4I$,

Proof.

$$\gamma_\mu\gamma^\mu = \eta_{\mu\nu}\gamma^\nu\gamma^\mu = \gamma_0^2 - \gamma_1^2 - \gamma_2^2 - \gamma_3^2 = I - (-I) - (-I) - (-I) = 4I.$$

□

The trace of any two slashed 4-vectors is $\text{Tr}(\not{a}\not{b}) = 4a \cdot b$.

Proof. Write

$$\text{Tr}(\not{a}\not{b}) = \text{Tr}(a_\mu\gamma^\mu b_\nu\gamma^\nu) = a_\mu b_\nu \text{Tr}(\gamma^\mu\gamma^\nu).$$

We can pull out a_μ and b_ν because they are just numbers. Then write

$$\begin{aligned}a_\mu b_\nu \text{Tr}(\gamma^\mu\gamma^\nu) &= a_\mu b_\nu (\gamma^\nu\gamma^\mu), \\ a_\mu b_\nu \text{Tr}(\gamma^\mu\gamma^\nu) &= a_\mu b_\nu [\text{Tr}(2\eta^{\mu\nu}I) - \text{Tr}(\gamma^\nu\gamma^\mu)].\end{aligned}$$

In the first line we use the cyclic property of the trace. In the second line we use $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}I$. Setting the two RHS's equal gives

$$a_\mu b_\nu \text{Tr}(\gamma^\nu\gamma^\mu) = a_\mu b_\nu [\text{Tr}(2\eta^{\mu\nu}I) - \text{Tr}(\gamma^\nu\gamma^\mu)] \implies 2a_\mu b_\nu \text{Tr}(\gamma^\nu\gamma^\mu) = \eta^{\mu\nu}a_\mu b_\nu \text{Tr}(2I).$$

We know that $\text{Tr}(2I) = 8$, so

$$2a_\mu b_\nu \text{Tr}(\gamma^\nu \gamma^\mu) = 8\eta^{\mu\nu} a_\mu b_\nu = 8a \cdot b \implies \text{Tr}(\not{a}\not{b}) = 4a \cdot b.$$

□

A.2.7 The Outer Product of Spinors

In contrast to $\bar{u}(p, s)u(p, s)$, the inner product of spinors, the expression like $u(p, s)\bar{u}(p, s)$ is called the outer product of spinors. When it is summed over all possible spin quantum numbers s , the outer product of spinors is

$$\sum_s u(p, s)\bar{u}(p, s) = \not{p} + m, \quad \sum_s v(p, s)\bar{v}(p, s) = \not{p} - m,$$

where

$$\not{p} = \eta_{\mu\nu} \gamma^\mu p^\nu = \begin{bmatrix} E & -\boldsymbol{\sigma} \cdot \mathbf{p} \\ \boldsymbol{\sigma} \cdot \mathbf{p} & -E \end{bmatrix}.$$

Proof. In the Dirac-Pauli representation,

$$u(p, s) = \sqrt{E + m} \begin{bmatrix} \phi_s \\ \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \phi_s \end{bmatrix}, \quad v(p, s) = \sqrt{E + m} \begin{bmatrix} \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \chi_s \\ \chi_s \end{bmatrix}, \quad \gamma^0 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \gamma^i = \begin{bmatrix} 0 & -\sigma_i \\ \sigma_i & 0 \end{bmatrix}.$$

The matrix $u(p, s)\bar{u}(p, s)$ is

$$u(p, s)\bar{u}(p, s) = u(p, s)u^\dagger(p, s)\gamma^0 = (E + m) \begin{bmatrix} \phi_s \\ \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \phi_s \end{bmatrix} \begin{bmatrix} \phi_s^\dagger & \phi_s^\dagger \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = (E + m) \begin{bmatrix} \phi_s \phi_s^\dagger & -\phi_s \phi_s^\dagger \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \\ \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \phi_s \phi_s^\dagger & -\frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \phi_s \phi_s^\dagger \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \end{bmatrix}.$$

Since $\{\phi_s\}$ form an orthonormal basis, $\sum_s \phi_s \phi_s^\dagger = 1$ is just the identity matrix. Then

$$\sum_s u(p, s)\bar{u}(p, s) = \begin{bmatrix} E + m & -\boldsymbol{\sigma} \cdot \mathbf{p} \\ \boldsymbol{\sigma} \cdot \mathbf{p} & -\frac{(\boldsymbol{\sigma} \cdot \mathbf{p})^2}{E + m} \end{bmatrix} = \begin{bmatrix} E + m & -\boldsymbol{\sigma} \cdot \mathbf{p} \\ \boldsymbol{\sigma} \cdot \mathbf{p} & -E + m \end{bmatrix} = \not{p} + m.$$

In the second equality, we used the identity $(\boldsymbol{\sigma} \cdot \mathbf{p})^2 = \mathbf{p}^2 1 = E^2 - m^2$. Similarly,

$$v(p, s)\bar{v}(p, s) = \sum_s (E + m) \begin{bmatrix} \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \chi_s \\ \chi_s \end{bmatrix} \begin{bmatrix} \chi_s^\dagger \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} & \chi_s^\dagger \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \sum_s (E + m) \begin{bmatrix} \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \chi_s \chi_s^\dagger \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} & -\frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \chi_s \chi_s^\dagger \\ \chi_s \chi_s^\dagger \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} & -\chi_s \chi_s^\dagger \end{bmatrix}.$$

Summing over all spin,

$$\sum_s v(p, s)\bar{v}(p, s) = \begin{bmatrix} \frac{(\boldsymbol{\sigma} \cdot \mathbf{p})^2}{E + m} & -\boldsymbol{\sigma} \cdot \mathbf{p} \\ \boldsymbol{\sigma} \cdot \mathbf{p} & -E - m \end{bmatrix} = \begin{bmatrix} E - m & -\boldsymbol{\sigma} \cdot \mathbf{p} \\ \boldsymbol{\sigma} \cdot \mathbf{p} & -E - m \end{bmatrix} = \not{p} - m.$$

□

A.3 The Klein-Gordon Field

A.3.1 Commutator Relations

The Klein-Gordon field and its conjugate momentum are given by

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3\sqrt{2E}} \left[a(p)e^{-ip \cdot x} + a^\dagger(p)e^{+ip \cdot x} \right] \quad \text{and} \quad \pi(x) = \int \frac{d^3p}{(2\pi)^3\sqrt{2E}} (-iE) \left[a(p)e^{-ip \cdot x} - a^\dagger(p)e^{+ip \cdot x} \right].$$

The goal is to verify the commutation relations

$$[\phi(\mathbf{x}, t), \pi(\mathbf{x}', t)] = i\delta^3(\mathbf{x} - \mathbf{x}'), \quad [\phi(\mathbf{x}, t), \phi(\mathbf{x}', t)] = 0, \quad [\pi(\mathbf{x}, t), \pi(\mathbf{x}, t)] = 0$$

using the commutation relations of the creation and annihilation operators

$$[a(p), a^\dagger(p')] = (2\pi)^3\delta^3(\mathbf{p} - \mathbf{p}'), \quad [a(p), a(p')] = [a^\dagger(p), a^\dagger(p')] = 0.$$

In the derivation, we will use the fact that

$$\int \frac{d^3p}{(2\pi)^3} e^{i\mathbf{p} \cdot \mathbf{x}} = \delta^3(\mathbf{x}).$$

First, for $[\phi(\mathbf{x}, t), \pi(\mathbf{x}', t)]$,

$$\begin{aligned} \phi(\mathbf{x}, t)\pi(\mathbf{x}', t) &= \int \frac{d^3p d^3p'}{(2\pi)^6\sqrt{4EE'}} (-iE') \left[a(p)e^{-i(Et-\mathbf{p} \cdot \mathbf{x})} + a^\dagger(p)e^{+i(Et-\mathbf{p} \cdot \mathbf{x})} \right] \left[a(p')e^{-i(E't-\mathbf{p}' \cdot \mathbf{x}')} - a^\dagger(p')e^{+i(E't-\mathbf{p}' \cdot \mathbf{x}')} \right] \\ &= \frac{1}{2i} \int \frac{d^3p d^3p'}{(2\pi)^6} \sqrt{\frac{E'}{E}} \left\{ a(p)a(p')e^{-i[(E+E')t-\mathbf{p} \cdot \mathbf{x}-\mathbf{p}' \cdot \mathbf{x}']} - a(p)a^\dagger(p')e^{-i[(E-E')t-\mathbf{p} \cdot \mathbf{x}+\mathbf{p}' \cdot \mathbf{x}']} \right. \\ &\quad \left. + a^\dagger(p)a(p')e^{i[(E-E')t-\mathbf{p} \cdot \mathbf{x}+\mathbf{p}' \cdot \mathbf{x}']} - a^\dagger(p)a^\dagger(p')e^{i[(E+E')t-\mathbf{p} \cdot \mathbf{x}-\mathbf{p}' \cdot \mathbf{x}']} \right\}, \\ \pi(\mathbf{x}', t)\phi(\mathbf{x}, t) &= \int \frac{d^3p d^3p'}{(2\pi)^6\sqrt{4EE'}} (-iE') \left[a(p')e^{-i(E't-\mathbf{p}' \cdot \mathbf{x}')} - a^\dagger(p')e^{+i(E't-\mathbf{p}' \cdot \mathbf{x}')} \right] \left[a(p)e^{-i(Et-\mathbf{p} \cdot \mathbf{x})} + a^\dagger(p)e^{+i(Et-\mathbf{p} \cdot \mathbf{x})} \right] \\ &= \frac{1}{2i} \int \frac{d^3p d^3p'}{(2\pi)^6} \sqrt{\frac{E'}{E}} \left\{ a(p')a(p)e^{-i[(E+E')t-\mathbf{p} \cdot \mathbf{x}-\mathbf{p}' \cdot \mathbf{x}']} + a(p')a^\dagger(p)e^{i[(E-E')t-\mathbf{p} \cdot \mathbf{x}+\mathbf{p}' \cdot \mathbf{x}']} \right. \\ &\quad \left. - a^\dagger(p')a(p)e^{-i[(E-E')t-\mathbf{p} \cdot \mathbf{x}+\mathbf{p}' \cdot \mathbf{x}']} - a^\dagger(p')a^\dagger(p)e^{i[(E+E')t-\mathbf{p} \cdot \mathbf{x}-\mathbf{p}' \cdot \mathbf{x}']} \right\}. \end{aligned}$$

Compute their difference to get the commutator. We collect terms according to their exponentials.

$$\begin{aligned} [\phi(\mathbf{x}, t), \pi(\mathbf{x}, t)] &= \frac{1}{2i} \int \frac{d^3p d^3p'}{(2\pi)^6} \left\{ [a(p), a(p')]e^{-i[(E+E')t-\mathbf{p} \cdot \mathbf{x}-\mathbf{p}' \cdot \mathbf{x}']} - [a(p), a^\dagger(p')]e^{-i[(E-E')t-\mathbf{p} \cdot \mathbf{x}+\mathbf{p}' \cdot \mathbf{x}']} \right. \\ &\quad \left. + [a^\dagger(p), a(p')]e^{i[(E-E')t-\mathbf{p} \cdot \mathbf{x}+\mathbf{p}' \cdot \mathbf{x}']} - [a^\dagger(p), a^\dagger(p')]e^{i[(E+E')t-\mathbf{p} \cdot \mathbf{x}-\mathbf{p}' \cdot \mathbf{x}']} \right\} \\ &= \frac{1}{2i} \int \frac{d^3p d^3p'}{(2\pi)^6} \left[-(2\pi)^3\delta^3(\mathbf{p} - \mathbf{p}')e^{-i[(E-E')t-\mathbf{p} \cdot \mathbf{x}+\mathbf{p}' \cdot \mathbf{x}']} - (2\pi)^3\delta^3(\mathbf{p}' - \mathbf{p})e^{i[(E-E')t-\mathbf{p} \cdot \mathbf{x}+\mathbf{p}' \cdot \mathbf{x}']} \right] \\ &= -\frac{1}{2i} \int \frac{d^3p}{(2\pi)^3} \left[e^{i\mathbf{p} \cdot (\mathbf{x}-\mathbf{x}')} + e^{-i\mathbf{p} \cdot (\mathbf{x}-\mathbf{x}')} \right] \\ &= +\frac{i}{2} [\delta^3(\mathbf{x} - \mathbf{x}') + \delta^3(\mathbf{x}' - \mathbf{x})] \\ &= i\delta^3(\mathbf{x} - \mathbf{x}'). \quad \checkmark \end{aligned}$$

Second, for $[\phi(\mathbf{x}, t), \phi(\mathbf{x}', t)]$,

$$\begin{aligned} \phi(\mathbf{x}, t)\phi(\mathbf{x}', t) &= \int \frac{d^3p d^3p'}{(2\pi)^6\sqrt{4EE'}} \left[a(p)e^{-i(Et-\mathbf{p} \cdot \mathbf{x})} + a^\dagger(p)e^{+i(Et-\mathbf{p} \cdot \mathbf{x})} \right] \left[a(p')e^{-i(E't-\mathbf{p}' \cdot \mathbf{x}')} + a^\dagger(p')e^{+i(E't-\mathbf{p}' \cdot \mathbf{x}')} \right] \\ &= \int \frac{d^3p d^3p'}{(2\pi)^6\sqrt{4EE'}} \left\{ a(p)a(p')e^{-i[(E+E')t-\mathbf{p} \cdot \mathbf{x}-\mathbf{p}' \cdot \mathbf{x}']} + a(p)a^\dagger(p')e^{-i[(E-E')t-\mathbf{p} \cdot \mathbf{x}+\mathbf{p}' \cdot \mathbf{x}']} \right. \\ &\quad \left. + a^\dagger(p)a(p')e^{i[(E-E')t-\mathbf{p} \cdot \mathbf{x}+\mathbf{p}' \cdot \mathbf{x}']} + a^\dagger(p)a^\dagger(p')e^{i[(E+E')t-\mathbf{p} \cdot \mathbf{x}-\mathbf{p}' \cdot \mathbf{x}']} \right\}, \\ \phi(\mathbf{x}', t)\phi(\mathbf{x}, t) &= \int \frac{d^3p d^3p'}{(2\pi)^6\sqrt{4EE'}} \left[a(p')e^{-i(E't-\mathbf{p}' \cdot \mathbf{x}')} + a^\dagger(p')e^{+i(E't-\mathbf{p}' \cdot \mathbf{x}')} \right] \left[a(p)e^{-i(Et-\mathbf{p} \cdot \mathbf{x})} + a^\dagger(p)e^{+i(Et-\mathbf{p} \cdot \mathbf{x})} \right] \\ &= \int \frac{d^3p d^3p'}{(2\pi)^6\sqrt{4EE'}} \left\{ a(p')a(p)e^{-i[(E+E')t-\mathbf{p} \cdot \mathbf{x}-\mathbf{p}' \cdot \mathbf{x}']} + a(p')a^\dagger(p)e^{i[(E-E')t-\mathbf{p} \cdot \mathbf{x}+\mathbf{p}' \cdot \mathbf{x}']} \right. \\ &\quad \left. + a^\dagger(p')a(p)e^{-i[(E-E')t-\mathbf{p} \cdot \mathbf{x}+\mathbf{p}' \cdot \mathbf{x}']} + a^\dagger(p')a^\dagger(p)e^{i[(E+E')t-\mathbf{p} \cdot \mathbf{x}-\mathbf{p}' \cdot \mathbf{x}']} \right\}. \end{aligned}$$

Compute their difference,

$$\begin{aligned}
[\phi(\mathbf{x}, t), \phi(\mathbf{x}', t)] &= \int \frac{d^3p d^3p'}{(2\pi)^6 \sqrt{4EE'}} \left\{ [a(p), a(p')] e^{-i[(E+E')t - \mathbf{p} \cdot \mathbf{x} - \mathbf{p}' \cdot \mathbf{x}']} + [a(p), a^\dagger(p')] e^{-i[(E-E')t - \mathbf{p} \cdot \mathbf{x} + \mathbf{p}' \cdot \mathbf{x}']} \right. \\
&\quad \left. + [a^\dagger(p), a(p')] e^{i[(E-E')t - \mathbf{p} \cdot \mathbf{x} + \mathbf{p}' \cdot \mathbf{x}']} + [a^\dagger(p), a^\dagger(p')] e^{i[(E+E')t - \mathbf{p} \cdot \mathbf{x} - \mathbf{p}' \cdot \mathbf{x}']} \right\} \\
&= \frac{d^3p d^3p'}{(2\pi)^6 \sqrt{4EE'}} \left[(2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}') e^{-i[(E-E')t - \mathbf{p} \cdot \mathbf{x} + \mathbf{p}' \cdot \mathbf{x}']} - (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}') e^{i[(E-E')t - \mathbf{p} \cdot \mathbf{x} + \mathbf{p}' \cdot \mathbf{x}']} \right] \\
&= \int \frac{d^3p}{(2\pi)^3 (2E)} \left[e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{x}')} - e^{-i\mathbf{p} \cdot (\mathbf{x} - \mathbf{x}')} \right] \\
&= i \int \frac{d^3p}{(2\pi)^3 \sqrt{\mathbf{p}^2 + m^2}} \sin[\mathbf{p} \cdot (\mathbf{x} - \mathbf{x}')] \quad \leftarrow \quad \text{The integrand is odd.} \\
&= 0. \quad \checkmark
\end{aligned}$$

Third, for $[\pi(\mathbf{x}, t), \pi(\mathbf{x}', t)]$

$$\begin{aligned}
\pi(\mathbf{x}, t) \pi(\mathbf{x}', t) &= -\frac{1}{2} \int \frac{d^3p d^3p'}{(2\pi)^6} \sqrt{EE'} \left[a(p) e^{-i(Et - \mathbf{p} \cdot \mathbf{x})} - a^\dagger(p) e^{+i(Et - \mathbf{p} \cdot \mathbf{x})} \right] \left[a(p') e^{-i(E't - \mathbf{p}' \cdot \mathbf{x}')} - a^\dagger(p') e^{+i(E't - \mathbf{p}' \cdot \mathbf{x}')} \right] \\
&= -\frac{1}{2} \int \frac{d^3p d^3p'}{(2\pi)^6} \sqrt{EE'} \left\{ a(p) a(p') e^{-i[(E+E')t - \mathbf{p} \cdot \mathbf{x} - \mathbf{p}' \cdot \mathbf{x}']} - a(p) a^\dagger(p') e^{-i[(E-E')t - \mathbf{p} \cdot \mathbf{x} + \mathbf{p}' \cdot \mathbf{x}']} \right. \\
&\quad \left. - a^\dagger(p) a(p') e^{i[(E-E')t - \mathbf{p} \cdot \mathbf{x} + \mathbf{p}' \cdot \mathbf{x}']} + a^\dagger(p) a^\dagger(p') e^{i[(E+E')t - \mathbf{p} \cdot \mathbf{x} - \mathbf{p}' \cdot \mathbf{x}']} \right\}, \\
\pi(\mathbf{x}', t) \pi(\mathbf{x}, t) &= -\frac{1}{2} \int \frac{d^3p d^3p'}{(2\pi)^6} \sqrt{EE'} \left[a(p') e^{-i(E't - \mathbf{p}' \cdot \mathbf{x}')} - a^\dagger(p') e^{+i(E't - \mathbf{p}' \cdot \mathbf{x}')} \right] \left[a(p) e^{-i(Et - \mathbf{p} \cdot \mathbf{x})} - a^\dagger(p) e^{+i(Et - \mathbf{p} \cdot \mathbf{x})} \right] \\
&= -\frac{1}{2} \int \frac{d^3p d^3p'}{(2\pi)^6} \sqrt{EE'} \left\{ a(p') a(p) e^{-i[(E+E')t - \mathbf{p} \cdot \mathbf{x} - \mathbf{p}' \cdot \mathbf{x}']} - a(p') a^\dagger(p) e^{i[(E-E')t - \mathbf{p} \cdot \mathbf{x} + \mathbf{p}' \cdot \mathbf{x}']} \right. \\
&\quad \left. - a^\dagger(p') a(p) e^{-i[(E-E')t - \mathbf{p} \cdot \mathbf{x} + \mathbf{p}' \cdot \mathbf{x}']} + a^\dagger(p') a^\dagger(p) e^{i[(E+E')t - \mathbf{p} \cdot \mathbf{x} - \mathbf{p}' \cdot \mathbf{x}']} \right\}.
\end{aligned}$$

Compute their difference,

$$\begin{aligned}
[\pi(\mathbf{x}, t), \pi(\mathbf{x}', t)] &= -\frac{1}{2} \int \frac{d^3p d^3p'}{(2\pi)^6} \sqrt{EE'} \left\{ [a(p), a(p')] e^{-i[(E+E')t - \mathbf{p} \cdot \mathbf{x} - \mathbf{p}' \cdot \mathbf{x}']} - [a(p), a^\dagger(p')] e^{-i[(E-E')t - \mathbf{p} \cdot \mathbf{x} + \mathbf{p}' \cdot \mathbf{x}']} \right. \\
&\quad \left. - [a^\dagger(p), a(p')] e^{i[(E-E')t - \mathbf{p} \cdot \mathbf{x} + \mathbf{p}' \cdot \mathbf{x}']} + [a^\dagger(p), a^\dagger(p')] e^{i[(E+E')t - \mathbf{p} \cdot \mathbf{x} - \mathbf{p}' \cdot \mathbf{x}']} \right\} \\
&= -\frac{1}{2} \int \frac{d^3p d^3p'}{(2\pi)^6} \sqrt{EE'} \left[-(2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}') e^{-i[(E-E')t - \mathbf{p} \cdot \mathbf{x} + \mathbf{p}' \cdot \mathbf{x}']} + (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}') e^{i[(E-E')t - \mathbf{p} \cdot \mathbf{x} + \mathbf{p}' \cdot \mathbf{x}']} \right] \\
&= +\frac{1}{2} \int \frac{d^3p}{(2\pi)^3} E \left[e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{x}')} - e^{-i\mathbf{p} \cdot (\mathbf{x} - \mathbf{x}')} \right] \\
&= i \int \frac{d^3p}{(2\pi)^3} \sqrt{\mathbf{p}^2 + m^2} \sin[\mathbf{p} \cdot (\mathbf{x} - \mathbf{x}')] \quad \leftarrow \quad \text{The integrand is odd.} \\
&= 0. \quad \checkmark
\end{aligned}$$

In fact, if we Fourier transform $\phi(x)$ and $\pi(x)$ to $\phi(p)$ and $\pi(p)$, it is possible show in the other direction: the field commutation relations implies creation and annihilation operator commutation relations.

A.3.2 The Klein-Gordon Hamiltonian

The Hamiltonian of the real Klein-Gordon field is

$$H_{\text{KG}} = \frac{1}{2} \int d^3x \left[\pi^2 + (\nabla \phi)^2 + m^2 \phi^2 \right].$$

We will directly plug in the Klein-Gordon field and its momentum,

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3 \sqrt{2E}} \left[a(p) e^{-ip \cdot x} + a^\dagger(p) e^{+ip \cdot x} \right] \quad \text{and} \quad \pi(x) = \int \frac{d^3p}{(2\pi)^3 \sqrt{2E}} (-iE) \left[a(p) e^{-ip \cdot x} - a^\dagger(p) e^{+ip \cdot x} \right].$$

First, find the quadratic term $\phi^2(\mathbf{x}, t)$, gradient term $(\nabla\phi)^2$, and kinetic term $\pi^2(\mathbf{x}, t)$ separately,

$$\begin{aligned}
\phi^2(\mathbf{x}, t) &= \int \frac{d^3p d^3p'}{(2\pi)^6 \sqrt{4EE'}} \left[a(p) e^{-i(Et - \mathbf{p} \cdot \mathbf{x})} + a^\dagger(p) e^{+i(Et - \mathbf{p} \cdot \mathbf{x})} \right] \left[a(p') e^{-i(E't - \mathbf{p}' \cdot \mathbf{x})} + a^\dagger(p') e^{+i(E't - \mathbf{p}' \cdot \mathbf{x})} \right] \\
&= \int \frac{d^3p d^3p'}{(2\pi)^6 \sqrt{4EE'}} \left\{ a(p) a(p') e^{-i[(E+E')t - (\mathbf{p} + \mathbf{p}') \cdot \mathbf{x}]} + a(p) a^\dagger(p') e^{-i[(E-E')t - (\mathbf{p} - \mathbf{p}') \cdot \mathbf{x}]} \right. \\
&\quad \left. + a^\dagger(p) a(p') e^{i[(E-E')t - (\mathbf{p} - \mathbf{p}') \cdot \mathbf{x}]} + a^\dagger(p) a^\dagger(p') e^{i[(E+E')t - (\mathbf{p} + \mathbf{p}') \cdot \mathbf{x}]} \right\}, \\
(\nabla\phi)^2 &= \int \frac{d^3p d^3p'}{(2\pi)^6 \sqrt{4EE'}} (i\mathbf{p}) \cdot (i\mathbf{p}') \left[a(p) e^{-i(Et - \mathbf{p} \cdot \mathbf{x})} - a^\dagger(p) e^{+i(Et - \mathbf{p} \cdot \mathbf{x})} \right] \left[a(p') e^{-i(E't - \mathbf{p}' \cdot \mathbf{x})} - a^\dagger(p') e^{+i(E't - \mathbf{p}' \cdot \mathbf{x})} \right] \\
&= - \int \frac{d^3p d^3p'}{(2\pi)^6 \sqrt{4EE'}} (\mathbf{p} \cdot \mathbf{p}') \left\{ a(p) a(p') e^{-i[(E+E')t - (\mathbf{p} + \mathbf{p}') \cdot \mathbf{x}]} - a(p) a^\dagger(p') e^{-i[(E-E')t - (\mathbf{p} - \mathbf{p}') \cdot \mathbf{x}]} \right. \\
&\quad \left. - a^\dagger(p) a(p') e^{i[(E-E')t - (\mathbf{p} - \mathbf{p}') \cdot \mathbf{x}]} + a^\dagger(p) a^\dagger(p') e^{i[(E+E')t - (\mathbf{p} + \mathbf{p}') \cdot \mathbf{x}]} \right\}, \\
\pi^2(\mathbf{x}, t) &= -\frac{1}{2} \int \frac{d^3p d^3p'}{(2\pi)^6} \sqrt{EE'} \left[a(p) e^{-i(Et - \mathbf{p} \cdot \mathbf{x})} - a^\dagger(p) e^{+i(Et - \mathbf{p} \cdot \mathbf{x})} \right] \left[a(p') e^{-i(E't - \mathbf{p}' \cdot \mathbf{x})} - a^\dagger(p') e^{+i(E't - \mathbf{p}' \cdot \mathbf{x})} \right] \\
&= -\frac{1}{2} \int \frac{d^3p d^3p'}{(2\pi)^6} \sqrt{EE'} \left\{ a(p) a(p') e^{-i[(E+E')t - (\mathbf{p} + \mathbf{p}') \cdot \mathbf{x}]} - a(p) a^\dagger(p') e^{-i[(E-E')t - (\mathbf{p} - \mathbf{p}') \cdot \mathbf{x}]} \right. \\
&\quad \left. - a^\dagger(p) a(p') e^{i[(E-E')t - (\mathbf{p} - \mathbf{p}') \cdot \mathbf{x}]} + a^\dagger(p) a^\dagger(p') e^{i[(E+E')t - (\mathbf{p} + \mathbf{p}') \cdot \mathbf{x}]} \right\}.
\end{aligned}$$

Then integrate over all space for each term,

$$\begin{aligned}
\int d^3x \phi^2 &= \int \frac{d^3p d^3p' d^3x}{(2\pi)^6 \sqrt{4EE'}} \left\{ a(p) a(p') e^{-i[(E+E')t - (\mathbf{p} + \mathbf{p}') \cdot \mathbf{x}]} + a(p) a^\dagger(p') e^{-i[(E-E')t - (\mathbf{p} - \mathbf{p}') \cdot \mathbf{x}]} \right. \\
&\quad \left. + a^\dagger(p) a(p') e^{i[(E-E')t - (\mathbf{p} - \mathbf{p}') \cdot \mathbf{x}]} + a^\dagger(p) a^\dagger(p') e^{i[(E+E')t - (\mathbf{p} + \mathbf{p}') \cdot \mathbf{x}]} \right\} \\
&= \int \frac{d^3p d^3p'}{(2\pi)^3 \sqrt{4EE'}} \left[a(p) a(p') e^{-i(E+E')t} \delta^3(\mathbf{p} + \mathbf{p}') + a(p) a^\dagger(p') e^{-i(E-E')t} \delta^3(\mathbf{p} - \mathbf{p}') \right. \\
&\quad \left. + a^\dagger(p) a(p') e^{i(E-E')t} \delta^3(\mathbf{p} - \mathbf{p}') + a^\dagger(p) a^\dagger(p') e^{i(E+E')t} \delta^3(\mathbf{p} + \mathbf{p}') \right] \\
&= \int \frac{d^3p}{(2\pi)^3 (2E)} \left[a(\mathbf{p}) a(-\mathbf{p}) e^{-2iEt} + a(\mathbf{p}) a^\dagger(\mathbf{p}) + a^\dagger(\mathbf{p}) a(\mathbf{p}) + a^\dagger(\mathbf{p}) a^\dagger(-\mathbf{p}) e^{2iEt} \right], \\
\int d^3x (\nabla\phi)^2 &= - \int \frac{d^3p d^3p' d^3x}{(2\pi)^6 \sqrt{4EE'}} (\mathbf{p} \cdot \mathbf{p}') \left\{ a(p) a(p') e^{-i[(E+E')t - (\mathbf{p} + \mathbf{p}') \cdot \mathbf{x}]} - a(p) a^\dagger(p') e^{-i[(E-E')t - (\mathbf{p} - \mathbf{p}') \cdot \mathbf{x}]} \right. \\
&\quad \left. - a^\dagger(p) a(p') e^{i[(E-E')t - (\mathbf{p} - \mathbf{p}') \cdot \mathbf{x}]} + a^\dagger(p) a^\dagger(p') e^{i[(E+E')t - (\mathbf{p} + \mathbf{p}') \cdot \mathbf{x}]} \right\} \\
&= - \int \frac{d^3p d^3p'}{(2\pi)^3 \sqrt{4EE'}} (\mathbf{p} \cdot \mathbf{p}') \left[a(p) a(p') e^{-i(E+E')t} \delta^3(\mathbf{p} + \mathbf{p}') - a(p) a^\dagger(p') e^{-i(E-E')t} \delta^3(\mathbf{p} - \mathbf{p}') \right. \\
&\quad \left. - a^\dagger(p) a(p') e^{i(E-E')t} \delta^3(\mathbf{p} - \mathbf{p}') + a^\dagger(p) a^\dagger(p') e^{i(E+E')t} \delta^3(\mathbf{p} + \mathbf{p}') \right] \\
&= + \int \frac{d^3p}{(2\pi)^3 \sqrt{4EE'}} \mathbf{p}^2 \left[a(\mathbf{p}) a(-\mathbf{p}) e^{-2iEt} + a(\mathbf{p}) a^\dagger(\mathbf{p}) + a^\dagger(\mathbf{p}) a(\mathbf{p}) + a^\dagger(\mathbf{p}) a^\dagger(-\mathbf{p}) e^{2iEt} \right], \\
\int d^3x \pi^2 &= -\frac{1}{2} \int \frac{d^3p d^3p' d^3x}{(2\pi)^6} \sqrt{EE'} \left\{ a(p) a(p') e^{-i[(E+E')t - (\mathbf{p} + \mathbf{p}') \cdot \mathbf{x}]} - a(p) a^\dagger(p') e^{-i[(E-E')t - (\mathbf{p} - \mathbf{p}') \cdot \mathbf{x}]} \right. \\
&\quad \left. - a^\dagger(p) a(p') e^{i[(E-E')t - (\mathbf{p} - \mathbf{p}') \cdot \mathbf{x}]} + a^\dagger(p) a^\dagger(p') e^{i[(E+E')t - (\mathbf{p} + \mathbf{p}') \cdot \mathbf{x}]} \right\} \\
&= -\frac{1}{2} \int \frac{d^3p d^3p'}{(2\pi)^3} \sqrt{EE'} \left[a(p) a(p') e^{-i(E+E')t} \delta^3(\mathbf{p} + \mathbf{p}') - a(p) a^\dagger(p') e^{-i(E-E')t} \delta^3(\mathbf{p} - \mathbf{p}') \right. \\
&\quad \left. - a^\dagger(p) a(p') e^{i(E-E')t} \delta^3(\mathbf{p} - \mathbf{p}') + a^\dagger(p) a^\dagger(p') e^{i(E+E')t} \delta^3(\mathbf{p} + \mathbf{p}') \right] \\
&= -\frac{1}{2} \int \frac{d^3p}{(2\pi)^3} E \left[a(\mathbf{p}) a(-\mathbf{p}) e^{-2iEt} - a(\mathbf{p}) a^\dagger(\mathbf{p}) - a^\dagger(\mathbf{p}) a(\mathbf{p}) + a^\dagger(\mathbf{p}) a^\dagger(-\mathbf{p}) e^{2iEt} \right].
\end{aligned}$$

Combining them, we can compute the Hamiltonian,

$$\begin{aligned}
H &= \frac{1}{2} \int \frac{d^3x}{(2\pi)^3} [\pi^2 + (\nabla\phi)^2 + m^2 \phi^2] \\
&= \frac{1}{2} \left\{ - \int \frac{d^3p}{(2\pi)^3} \frac{E}{2} \left[a(\mathbf{p}) a(-\mathbf{p}) e^{-2iEt} - a(\mathbf{p}) a^\dagger(\mathbf{p}) - a^\dagger(\mathbf{p}) a(\mathbf{p}) + a^\dagger(\mathbf{p}) a^\dagger(-\mathbf{p}) e^{2iEt} \right] \right. \\
&\quad + \int \frac{d^3p}{(2\pi)^3} \frac{\mathbf{p}^2}{2E} \left[a(\mathbf{p}) a(-\mathbf{p}) e^{-2iEt} + a(\mathbf{p}) a^\dagger(\mathbf{p}) + a^\dagger(\mathbf{p}) a(\mathbf{p}) + a^\dagger(\mathbf{p}) a^\dagger(-\mathbf{p}) e^{2iEt} \right] \\
&\quad \left. + \int \frac{d^3p}{(2\pi)^3} \frac{m^2}{2E} \left[a(\mathbf{p}) a(-\mathbf{p}) e^{-2iEt} + a(\mathbf{p}) a^\dagger(\mathbf{p}) + a^\dagger(\mathbf{p}) a(\mathbf{p}) + a^\dagger(\mathbf{p}) a^\dagger(-\mathbf{p}) e^{2iEt} \right] \right\} \\
&= \frac{1}{4} \int \frac{d^3p}{(2\pi)^3} \frac{1}{E} \left[a(\mathbf{p}) a(-\mathbf{p}) e^{-2iEt} (-E^2 + \mathbf{p}^2 + m^2) + a(\mathbf{p}) a^\dagger(\mathbf{p}) (E^2 + \mathbf{p}^2 + m^2) + a^\dagger(\mathbf{p}) a(\mathbf{p}) (E^2 + \mathbf{p}^2 + m^2) + a^\dagger(\mathbf{p}) a^\dagger(-\mathbf{p}) (-E^2 + \mathbf{p}^2 + m^2) \right] \\
&= \frac{1}{2} \int \frac{d^3p}{(2\pi)^3} \left[a(p) a^\dagger(p) + a^\dagger(p) a(p) \right].
\end{aligned}$$

A.3.3 The Symmetry Operator

The time component of the conserved current (6.13) is

$$\begin{aligned}
J_\phi^t &= i(\phi^\dagger \partial_t \phi - \phi \partial_t \phi^\dagger) \\
&= i \int \frac{d^3 p d^3 p'}{(2\pi)^6 \sqrt{4EE'}} \left[a^\dagger(p) e^{+i(Et - \mathbf{p} \cdot \mathbf{x})} + b(p) e^{-i(Et - \mathbf{p} \cdot \mathbf{x})} \right] \partial_t \left[a(p') e^{-i(E't - \mathbf{p}' \cdot \mathbf{x})} + b^\dagger(p') e^{+i(E't - \mathbf{p}' \cdot \mathbf{x})} \right] \\
&\quad - i \int \frac{d^3 p d^3 p'}{(2\pi)^6 \sqrt{4EE'}} \left[a(p) e^{-i(Et - \mathbf{p} \cdot \mathbf{x})} + b^\dagger(p) e^{+i(Et - \mathbf{p} \cdot \mathbf{x})} \right] \partial_t \left[a^\dagger(p') e^{+i(E't - \mathbf{p}' \cdot \mathbf{x})} + b(p') e^{-i(E't - \mathbf{p}' \cdot \mathbf{x})} \right] \\
&= \frac{1}{2} \int \frac{d^3 p d^3 p'}{(2\pi)^6} \sqrt{\frac{E'}{E}} \left[a^\dagger(p) e^{+i(Et - \mathbf{p} \cdot \mathbf{x})} + b(p) e^{-i(Et - \mathbf{p} \cdot \mathbf{x})} \right] \left[a(p') e^{-i(E't - \mathbf{p}' \cdot \mathbf{x})} - b^\dagger(p') e^{+i(E't - \mathbf{p}' \cdot \mathbf{x})} \right] \\
&\quad + \frac{1}{2} \int \frac{d^3 p d^3 p'}{(2\pi)^6} \sqrt{\frac{E'}{E}} \left[a(p) e^{-i(Et - \mathbf{p} \cdot \mathbf{x})} + b^\dagger(p) e^{+i(Et - \mathbf{p} \cdot \mathbf{x})} \right] \left[a^\dagger(p') e^{+i(E't - \mathbf{p}' \cdot \mathbf{x})} - b(p') e^{-i(E't - \mathbf{p}' \cdot \mathbf{x})} \right] \\
&= \frac{1}{2} \int \frac{d^3 p d^3 p'}{(2\pi)^6} \sqrt{\frac{E'}{E}} \left\{ a^\dagger(p) a(p') e^{i[(E-E')t - (\mathbf{p}-\mathbf{p}') \cdot \mathbf{x}]} - a^\dagger(p) b^\dagger(p') e^{i[(E+E')t - (\mathbf{p}+\mathbf{p}') \cdot \mathbf{x}]} + b(p) a(p') e^{-i[(E+E')t - (\mathbf{p}+\mathbf{p}') \cdot \mathbf{x}]} - b(p) b^\dagger(p') e^{-i[(E-E')t - (\mathbf{p}-\mathbf{p}') \cdot \mathbf{x}]} \right\} \\
&\quad + \frac{1}{2} \int \frac{d^3 p d^3 p'}{(2\pi)^6} \sqrt{\frac{E'}{E}} \left\{ a(p) a^\dagger(p') e^{-i[(E-E')t - (\mathbf{p}-\mathbf{p}') \cdot \mathbf{x}]} - a(p) b(p') e^{-i[(E+E')t - (\mathbf{p}+\mathbf{p}') \cdot \mathbf{x}]} + b^\dagger(p) a^\dagger(p') e^{i[(E+E')t - (\mathbf{p}+\mathbf{p}') \cdot \mathbf{x}]} - b^\dagger(p) b(p') e^{i[(E-E')t - (\mathbf{p}-\mathbf{p}') \cdot \mathbf{x}]} \right\}.
\end{aligned}$$

The symmetry operator N_ϕ is the spatial integral of J_ϕ^t :

$$\begin{aligned}
N_\phi &= \int d^3 x J_\phi^t \\
&= \frac{1}{2} \int \frac{d^3 p d^3 p'}{(2\pi)^6} \sqrt{\frac{E'}{E}} \left[a^\dagger(p) a(p') e^{i(E-E')t} \delta^3(\mathbf{p} - \mathbf{p}') - a^\dagger(p) b^\dagger(p') e^{i(E+E')t} \delta^3(\mathbf{p} + \mathbf{p}') + b(p) a(p') e^{-i(E+E')t} \delta^3(\mathbf{p} + \mathbf{p}') - b(p) b^\dagger(p') e^{i(E-E')t} \delta^3(\mathbf{p} - \mathbf{p}') \right] \\
&\quad + \frac{1}{2} \int \frac{d^3 p d^3 p'}{(2\pi)^6} \sqrt{\frac{E'}{E}} \left[a(p) a^\dagger(p') e^{i(E-E')t} \delta^3(\mathbf{p} - \mathbf{p}') - a(p) b(p') e^{-i(E+E')t} \delta^3(\mathbf{p} + \mathbf{p}') + b^\dagger(p) a^\dagger(p') e^{i(E+E')t} \delta^3(\mathbf{p} + \mathbf{p}') - b^\dagger(p) b(p') e^{i(E-E')t} \delta^3(\mathbf{p} - \mathbf{p}') \right] \\
&= \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} \left[a^\dagger(\mathbf{p}) a(\mathbf{p}) - a^\dagger(\mathbf{p}) b^\dagger(-\mathbf{p}) e^{2iEt} + b(\mathbf{p}) a(-\mathbf{p}) e^{-2iEt} - b(\mathbf{p}) b^\dagger(\mathbf{p}) + a(\mathbf{p}) a^\dagger(\mathbf{p}) - a(\mathbf{p}) b(-\mathbf{p}) e^{-2iEt} + b^\dagger(\mathbf{p}) a^\dagger(-\mathbf{p}) e^{2iEt} - b^\dagger(\mathbf{p}) b(\mathbf{p}) \right].
\end{aligned}$$

The integral is over all momentum space, so $\int d^3 p a^\dagger(\mathbf{p}) b^\dagger(-\mathbf{p}) = \int d^3 p a^\dagger(-\mathbf{p}) b(\mathbf{p})$, and similar for other terms. In this way, the terms with exponential factors all cancel because a and b commute. We are left with

$$\begin{aligned}
N_\phi &= \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} \left[a^\dagger(p) a(p) + a(p) a^\dagger(p) - b^\dagger(p) b(p) - b(p) b^\dagger(p) \right] \\
&= \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} \left\{ 2a^\dagger(p) a(p) + [a(p), a^\dagger(p)] - 2b^\dagger(p) b(p) - [b(p), b^\dagger(p)] \right\} \\
&= \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} \left[2a^\dagger(p) a(p) + (2\pi)^3 \delta^3(0) - 2b^\dagger(p) b(p) - (2\pi)^3 \delta^3(0) \right].
\end{aligned}$$

After normal ordering, the symmetry operator is

$$N_\phi = \int \frac{d^3 p}{(2\pi)^3} [a^\dagger(p) a(p) - b^\dagger(p) b(p)].$$

A.4 The Dirac Field

A.4.1 Inverse Spinor Transformation Matrix

The finite transformation $S(\Lambda)$ takes the form (7.19),

$$S(\Lambda) = e^{i\sigma^{\mu\nu}\omega_{\mu\nu}/4}.$$

Its Hermitian conjugate involves

$$\left(-\frac{i}{4}\sigma^{\mu\nu}\omega_{\mu\nu}\right)^\dagger = +\frac{i}{4}(\omega_{\mu\nu})^\dagger(\sigma^{\mu\nu})^\dagger = \frac{i}{4}(\sigma^{\mu\nu})^\dagger\omega_{\mu\nu},$$

with

$$\begin{aligned}(\sigma^{0i})^\dagger &= \left(\frac{i}{2}[\gamma^0, \gamma^i]\right)^\dagger = -\frac{i}{2}[(\gamma^i)^\dagger, (\gamma^0)^\dagger] = -\frac{i}{2}[-\gamma^i, \gamma^0] = -\frac{i}{2}[\gamma^0, \gamma^i] = -\sigma^{0i}, \\(\sigma^{ij})^\dagger &= \left(\frac{i}{2}[\gamma^i, \gamma^j]\right)^\dagger = -\frac{i}{2}[(\gamma^j)^\dagger, (\gamma^i)^\dagger] = -\frac{i}{2}[-\gamma^j, -\gamma^i] = \frac{i}{2}[\gamma^i, \gamma^j] = \sigma^{ij},\end{aligned}$$

and of course $\sigma^{00} = \sigma^{ii} = 0$. If $\omega_{\mu\nu}$ is a Lorentz boost, then $\omega_{ij} = 0$ for all $i \neq j$. But $\sigma_{ij} = 0$ for all $i = j$ (recall the Lorentz boost matrix 2.8), so the contraction $(\sigma^{\mu\nu})^\dagger\omega_{\mu\nu}$ leaves only

$$(\sigma^{\mu\nu})^\dagger\omega_{\mu\nu} = (\sigma^{0i})^\dagger\omega_{0i} + (\sigma^{i0})^\dagger\omega_{i0} = -\sigma^{0i}\omega_{0i} - \sigma^{i0}\omega_{i0} = -\sigma^{\mu\nu}\omega_{\mu\nu}.$$

Therefore, for Lorentz boosts,

$$S^\dagger(\Lambda) = e^{(-i\sigma^{\mu\nu}\omega_{\mu\nu}/4)^\dagger} = e^{+i(-\sigma^{\mu\nu})\omega_{\mu\nu}/4} = e^{-i\sigma^{\mu\nu}\omega_{\mu\nu}/4} = S(\Lambda).$$

On the other hand, for rotations, $\omega_{0i} = 0$, so

$$(\sigma^{\mu\nu})^\dagger\omega_{\mu\nu} = (\sigma^{ij})^\dagger\omega_{ij} = \sigma^{ij}\omega_{ij} = \sigma^{\mu\nu}\omega_{\mu\nu}.$$

This means for rotations,

$$S^\dagger(\Lambda) = e^{(-i\sigma^{\mu\nu}\omega_{\mu\nu}/4)^\dagger} = e^{+i\sigma^{\mu\nu}\omega_{\mu\nu}/4} = S^{-1}(\Lambda).$$

We want to find a result that works for both boosts and rotations, and that is

$$\gamma^0 S^\dagger(\Lambda) \gamma^0 = S^{-1}.$$

Proof. For Lorentz boosts, $S^\dagger(\Lambda)$ is basically (up to a power of $-i/4$)

$$S^\dagger(\Lambda) \sim e^{-\sigma^{0i}\omega_{0i} - \sigma^{i0}\omega_{i0}} = e^{-2\sigma^{0i}\omega_{0i}} = 1 - 2\sigma^{0i}\omega_{0i} + \frac{(2\sigma^{0i}\omega_{0i})^2}{2!} - \dots,$$

where in the second equality we used the fact that both $\sigma^{\mu\nu}$ and $\omega_{\mu\nu}$ are antisymmetric. Sandwiching $S^\dagger(\Lambda)$ between γ^0 ,

$$\gamma^0 S^\dagger(\Lambda) \gamma^0 \sim 1 - \gamma^0(2\sigma^{0i}\omega_{0i})\gamma^0 + \gamma^0 \frac{(2\sigma^{0i}\omega_{0i})^2}{2!} \gamma^0 - \dots$$

Notice that we can write

$$\gamma^0 A^2 \gamma^0 = \gamma^0 A \gamma^0 \gamma^0 A \gamma^0 = (\gamma^0 A \gamma^0)^2,$$

and similarly for higher powers, $\gamma^0 A^n \gamma^0 = (\gamma^0 A \gamma^0)^n$ because $(\gamma^0)^2 = 1$. Now determine $\gamma^0 \sigma^{0i} \omega_{0i} \gamma^0$ explicitly,

$$\gamma^0 \sigma^{0i} \omega_{0i} \gamma^0 = \frac{i}{2} \omega_{0i} \gamma^0 (\gamma^0 \gamma^i - \gamma^i \gamma^0) \gamma^0 = \frac{i}{2} \omega_{0i} (\gamma^i \gamma^0 - \gamma^0 \gamma^i) = -\sigma^{0i} \omega_{0i}.$$

Then

$$\gamma^0 S^\dagger(\Lambda) \gamma^0 \sim 1 - (2\gamma^0 \sigma^{0i} \omega_{0i} \gamma^0) + \frac{(2\gamma^0 \sigma^{0i} \omega_{0i} \gamma^0)^2}{2!} - \dots = e^{-2\gamma^0 \sigma^{0i} \omega_{0i} \gamma^0} = e^{+2\sigma^{0i} \omega_{0i}} = (S^\dagger)^{-1} = S^{-1}.$$

Similarly, for rotations,

$$\gamma^0 S^\dagger(\Lambda) \gamma^0 \sim e^{\gamma^0 \sigma^{ij} \omega_{ij} \gamma^0}$$

up to a power of $i/4$. Calculate $\gamma^0 \sigma^{ij} \omega_{ij} \gamma^0$ explicitly,

$$\gamma^0 \sigma^{ij} \omega_{ij} \gamma^0 = \frac{i}{2} \omega_{ij} (\gamma^0 \gamma^i \gamma^j \gamma^0 - \gamma^0 \gamma^j \gamma^i \gamma^0) = -\frac{i}{2} \omega_{ij} (\gamma^0 \gamma^i \gamma^0 \gamma^j - \gamma^0 \gamma^j \gamma^0 \gamma^i) = \frac{i}{2} \omega_{ij} (\gamma^0 \gamma^0 \gamma^i \gamma^j - \gamma^0 \gamma^0 \gamma^j \gamma^i) = \frac{i}{2} \sigma^{ij} \omega_{ij}.$$

Hence

$$\gamma^0 S^\dagger(\Lambda) \gamma^0 \sim e^{\gamma^0 \sigma^{ij} \omega_{ij} \gamma^0} = e^{\sigma^{ij} \omega_{ij}} \sim S^\dagger(\Lambda) = S^{-1}.$$

In conclusion, $\gamma^0 S^\dagger(\Lambda) \gamma^0 = S^{-1}$ works for both Lorentz boosts and rotations. $\square$

A.4.2 The Symmetry Operator

The symmetry operator is defined as

$$N_\psi = \int d^3x J_\psi^t = \int d^3x \psi^\dagger \psi,$$

where the mode expansion of the Dirac field is

$$\psi(x) = \int \frac{d^3p}{(2\pi)^3 \sqrt{2E}} \sum_s \left[c(p, s) u(p, s) e^{-ip \cdot x} + d^\dagger(p, s) v(p, s) e^{+ip \cdot x} \right].$$

First, calculate $\psi^\dagger \psi$:

$$\begin{aligned} \psi^\dagger \psi &= \int \frac{d^3p_1 d^3p_2}{(2\pi)^6 \sqrt{4E_1 E_2}} \sum_{s_1, s_2} \left[c^\dagger(p_1, s_1) u^\dagger(p_1, s_1) e^{+ip_1 \cdot x} + d(p_1, s_1) v^\dagger(p_1, s_1) e^{-ip_1 \cdot x} \right] \left[c(p_2, s_2) u(p_2, s_2) e^{-ip_2 \cdot x} + d^\dagger(p_2, s_2) v(p_2, s_2) e^{+ip_2 \cdot x} \right] \\ &= \int \frac{d^3p_1 d^3p_2}{(2\pi)^6 \sqrt{4E_1 E_2}} \sum_{s_1, s_2} \left[c^\dagger(p_1, s_1) c(p_2, s_2) u^\dagger(p_1, s_1) u(p_2, s_2) e^{i(p_1 - p_2) \cdot x} + c^\dagger(p_1, s_1) d^\dagger(p_2, s_2) u^\dagger(p_1, s_1) v(p_2, s_2) e^{i(p_1 + p_2) \cdot x} \right. \\ &\quad \left. + d(p_1, s_1) c(p_2, s_2) v^\dagger(p_1, s_1) u(p_2, s_2) e^{-i(p_1 + p_2) \cdot x} + d(p_1, s_1) d^\dagger(p_2, s_2) v^\dagger(p_1, s_1) v(p_2, s_2) e^{-i(p_1 - p_2) \cdot x} \right]. \end{aligned}$$

Next, integrate over all space:

$$\begin{aligned} \int d^3x \psi^\dagger \psi &= \int \frac{d^3p_1 d^3p_2}{(2\pi)^3 \sqrt{4E_1 E_2}} \sum_{s_1, s_2} \left[c^\dagger(p_1, s_1) c(p_2, s_2) u^\dagger(p_1, s_1) u(p_2, s_2) \delta^3(\mathbf{p}_1 - \mathbf{p}_2) e^{i(E_1 - E_2)t} + c^\dagger(p_1, s_1) d^\dagger(p_2, s_2) u^\dagger(p_1, s_1) v(p_2, s_2) \delta^3(\mathbf{p}_1 + \mathbf{p}_2) e^{i(E_1 + E_2)t} \right. \\ &\quad \left. + d(p_1, s_1) c(p_2, s_2) v^\dagger(p_1, s_1) u(p_2, s_2) \delta^3(\mathbf{p}_1 + \mathbf{p}_2) e^{-i(E_1 + E_2)t} + d(p_1, s_1) d^\dagger(p_2, s_2) v^\dagger(p_1, s_1) v(p_2, s_2) \delta^3(\mathbf{p}_1 - \mathbf{p}_2) e^{-i(E_1 - E_2)t} \right] \\ &= \int \frac{d^3p}{(2\pi)^3 2E} \sum_{s_1, s_2} \left[c^\dagger(\mathbf{p}, s') c(\mathbf{p}, s) u^\dagger(\mathbf{p}, s') u(\mathbf{p}, s) + c^\dagger(\mathbf{p}, s) d^\dagger(-\mathbf{p}, s) u^\dagger(\mathbf{p}, s') v(-\mathbf{p}, s) e^{2iEt} \right. \\ &\quad \left. + d(\mathbf{p}, s') c(-\mathbf{p}, s) v^\dagger(\mathbf{p}, s') u(-\mathbf{p}, s) e^{-2iEt} + d(\mathbf{p}, s') d^\dagger(\mathbf{p}, s) v(\mathbf{p}, s') v(\mathbf{p}, s) \right]. \end{aligned}$$

Then we have to deal with the spinors $u(p_1, s_1)$ and $v(p_1, s_1)$. The normalization of spinors (7.12) says that

$$u^\dagger(p, s_1) u(p, s_2) = 2E \delta_{s_1, s_2}, \quad v^\dagger(p, s_1) v(p, s_2) = 2E \delta_{s_1, s_2}, \quad \bar{u}(p, s_1) v(p, s_2) = 0.$$

In the Dirac-Pauli representation, it is easy to see that the third normalization condition is equivalent to $u^\dagger(-\mathbf{p}, s_1) v(\mathbf{p}, s_2) = 0$:

$$\bar{u}(p, s_1) v(p, s_2) = u^\dagger(-\mathbf{p}, s_1) v(\mathbf{p}, s_2) = (E + m) \left[\phi_{s_1}^\dagger \quad -\phi_{s_1}^\dagger \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \right] \begin{bmatrix} \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \chi_{s_2} \\ \chi_{s_2} \end{bmatrix} = 0.$$

This result also works for $v^\dagger(\mathbf{p}, s_1) u(\mathbf{p}, s_2) = 0$. By these normalization conditions, the symmetry operator simplifies to

$$\begin{aligned} N_\psi &= \int d^3x \psi^\dagger \psi = \int \frac{d^3p}{(2\pi)^3 2E} \sum_{s_1, s_2} \left[c^\dagger(\mathbf{p}, s_1) c(\mathbf{p}, s_2) (2E \delta_{s_1, s_2}) + d(\mathbf{p}, s_1) d^\dagger(\mathbf{p}, s_2) (2E \delta_{s_1, s_2}) \right] \\ &= \int \frac{d^3p}{(2\pi)^3} \sum_s \left[c^\dagger(p, s) c(p, s) + d(p, s) d^\dagger(p, s) \right]. \end{aligned}$$

Applying normal ordering and using the anticommutation relation $\{d(p_1, s_1), d^\dagger(p_2, s_2)\} = (2\pi)^3 \delta^3(\mathbf{p}_1 - \mathbf{p}_2) \delta_{s_1, s_2}$,

$$N_\psi = \int \frac{d^3p}{(2\pi)^3} \sum_s \left[c^\dagger(p, s) c(p, s) - d^\dagger(p, s) d(p, s) + (2\pi)^3 \delta^3(0) \delta_{s, s} \right] \stackrel{N}{=} \int \frac{d^3p}{(2\pi)^3} \sum_s \left[c^\dagger(p, s) c(p, s) - d^\dagger(p, s) d(p, s) \right].$$

A.4.3 The Dirac Hamiltonian

For convenience, we shall use (7.25) for the Hamiltonian density,

$$\mathcal{H} = i\psi^\dagger \dot{\psi},$$

where

$$\psi^\dagger(x) = \int \frac{d^3p}{(2\pi)^3 \sqrt{2E}} \sum_s \left[c^\dagger(p, s) u^\dagger(p, s) e^{+ip \cdot x} + d(p, s) v^\dagger(p, s) e^{-ip \cdot x} \right] \quad \text{and} \quad \dot{\psi} = \int \frac{d^3p}{(2\pi)^3 \sqrt{2E}} (-iE) \sum_s \left[c(p, s) u(p, s) e^{-ip \cdot x} - d^\dagger(p, s) v(p, s) e^{+ip \cdot x} \right].$$

The Hamiltonian density in mode expansion is

$$\begin{aligned}
\mathcal{H} &= i\psi^\dagger \dot{\psi} \\
&= \frac{1}{2} \int \frac{d^3 p_1 d^3 p_2}{(2\pi)^6} \sqrt{\frac{E_2}{E_1}} \sum_{s_1, s_2} \left[c^\dagger(p_1, s_1) u^\dagger(p_1, s_1) e^{+ip_1 \cdot x} + d(p_1, s_1) v^\dagger(p_1, s_1) e^{-ip_1 \cdot x} \right] \left[c(p_2, s_2) u(p_2, s_2) e^{-ip_2 \cdot x} - d^\dagger(p_2, s_2) v(p_2, s_2) e^{+ip_2 \cdot x} \right] \\
&= \frac{1}{2} \int \frac{d^3 p_1 d^3 p_2}{(2\pi)^6} \sqrt{\frac{E_2}{E_1}} \sum_{s_1, s_2} \left[c^\dagger(p_1, s_1) c(p_2, s_2) u^\dagger(p_1, s_1) u(p_2, s_2) e^{i(p_1 - p_2) \cdot x} - c^\dagger(p_1, s_1) d^\dagger(p_2, s_2) u^\dagger(p_1, s_1) v(p_2, s_2) e^{i(p_1 + p_2) \cdot x} \right. \\
&\quad \left. + d(p_1, s_1) c(p_2, s_2) v^\dagger(p_1, s_1) u(p_2, s_2) e^{-i(p_1 + p_2) \cdot x} - d(p_1, s_1) d^\dagger(p_2, s_2) v^\dagger(p_1, s_1) v(p_2, s_2) e^{-i(p_1 - p_2) \cdot x} \right].
\end{aligned}$$

Integrating over all space to get the Hamiltonian,

$$\begin{aligned}
H &= \int d^3 x \mathcal{H} \\
&= \frac{1}{2} \int \frac{d^3 p_1 d^3 p_2}{(2\pi)^3} \sqrt{\frac{E_2}{E_1}} \sum_{s_1, s_2} \left[c^\dagger(p_1, s_1) c(p_2, s_2) u^\dagger(p_1, s_1) u(p_2, s_2) \delta^3(\mathbf{p}_1 - \mathbf{p}_2) e^{i(E_1 - E_2)t} - c^\dagger(p_1, s_1) d^\dagger(p_2, s_2) u^\dagger(p_1, s_1) v(p_2, s_2) \delta^3(\mathbf{p}_1 + \mathbf{p}_2) e^{i(E_1 + E_2)t} \right. \\
&\quad \left. + d(p_1, s_1) c(p_2, s_2) v^\dagger(p_1, s_1) u(p_2, s_2) \delta^3(\mathbf{p}_1 + \mathbf{p}_2) e^{-i(E_1 + E_2)t} - d(p_1, s_1) d^\dagger(p_2, s_2) v^\dagger(p_1, s_1) v(p_2, s_2) \delta^3(\mathbf{p}_1 - \mathbf{p}_2) e^{-i(E_1 - E_2)t} \right] \\
&= \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} \sum_{s_1, s_2} \left[c^\dagger(\mathbf{p}, s_1) c(\mathbf{p}, s_2) u^\dagger(\mathbf{p}, s_1) u(\mathbf{p}, s_2) - c^\dagger(\mathbf{p}, s_1) d^\dagger(-\mathbf{p}, s_2) u^\dagger(\mathbf{p}, s_1) v(-\mathbf{p}, s_2) e^{2iEt} \right. \\
&\quad \left. + d(\mathbf{p}, s_1) c(-\mathbf{p}, s_2) v^\dagger(\mathbf{p}, s_1) u(-\mathbf{p}, s_2) e^{-2iEt} - d(\mathbf{p}, s_1) d^\dagger(\mathbf{p}, s_2) v^\dagger(\mathbf{p}, s_1) v(\mathbf{p}, s_2) \right].
\end{aligned}$$

Now use the normalization conditions,

$$u^\dagger(p, s_1) u(p, s_2) = 2E \delta_{s_1, s_2}, \quad v^\dagger(p, s_1) v(p, s_2) = 2E \delta_{s_1, s_2}, \quad u^\dagger(\mathbf{p}, s_1) v(-\mathbf{p}, s_2) = v^\dagger(\mathbf{p}, s_1) u(-\mathbf{p}, s_2) = 0.$$

The Hamiltonian becomes

$$\begin{aligned}
H &= \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} \sum_{s_1, s_2} \left[c^\dagger(p, s_1) c(p, s_2) (2E \delta_{s_1, s_2}) - d(p, s_1) d^\dagger(p, s_2) (2E \delta_{s_1, s_2}) \right] \\
&= \int \frac{d^3 p}{(2\pi)^3} \sum_s \left[c^\dagger(p, s) c(p, s) - d(p, s) d^\dagger(p, s) \right] E.
\end{aligned}$$

Using the anticommutation relations $\{d(p_1, s_1), d^\dagger(p_2, s_2)\} = (2\pi)^3 \delta^3(\mathbf{p}_1 - \mathbf{p}_2) \delta_{s_1, s_2}$ and normal ordering,

$$H = \int \frac{d^3 p}{(2\pi)^3} \sum_s \left[c^\dagger(p, s) c(p, s) + d^\dagger(p, s) d(p, s) \right] E.$$

A.5 Interactions

A.5.1 Time-Ordering Operator Identity

To be written.

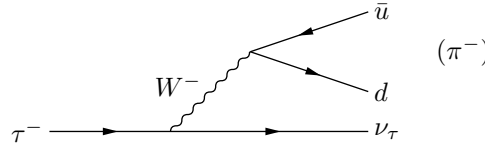
A.6 Decays and Scatterings

A.6.1 Polarized τ -Lepton Decaying to Pions

Consider the τ -lepton decay

$$\tau^- \rightarrow \pi^- \nu_\tau.$$

The Feynman diagram for this process is



Let $u(p_1, s_1)$ and $u(p_2, s_2)$ be the spinors representing the τ -lepton and ν_τ , respectively. The final particles $\bar{u}$ and d are not free particles. They are bound in the pion state, so they cannot be described by free-particle spinors. Hence we use the current for the pion,

$$J_\pi^\mu = -\frac{ig}{2\sqrt{2}} f_\pi V_{ud} q^\mu, \quad \text{where } q = p_1 - p_2.$$

Since $m_\tau \ll m_W$, we can work in the low energy limit and write the W -boson propagator as $i\eta_{\mu\nu}/m_W^2$. The amplitude for this process is

$$\mathcal{A} = \left[-\frac{ig}{\sqrt{2}} \bar{u}_2 \gamma^\mu \frac{1}{2} (1 - \gamma^5) u_1 \right] \frac{i\eta_{\mu\nu}}{m_W^2} \left[-\frac{ig}{2\sqrt{2}} f_\pi V_{ud} q^\mu \right] = -i \frac{G_F}{\sqrt{2}} f_\pi V_{ud} (p_1^\mu - p_2^\mu) \bar{u}_2 \gamma_\mu (1 - \gamma^5) u_1.$$

In this example, we assume that the τ -lepton is polarized in the z -direction. In the rest frame of the τ -lepton, let the 4-momentum and spin configuration be

$$p_1 = (m_1, 0, 0, 0), \quad p_2 = |\mathbf{p}^*|(1, \sin\theta, 0, \cos\theta), \quad s_1 = (0, 0, 0, 1).$$

Here s_1 is the polarization vector saying that the τ is polarized in the $+z$ -direction, and $|\mathbf{p}^*|$ is the energy (or magnitude of 3-momentum) of the neutrino. In general, the polarization vector should satisfy $s_1 \cdot p_1 = 0$ and $s^2 = -1$. Summing over the final spin but not averaging over the initial spin because of the polarization,

$$|\overline{\mathcal{A}}|^2 = \frac{1}{2} G_F^2 f_\pi^2 |V_{ud}|^2 \sum_{s_2} |\bar{u}_2 (\not{p}_1 - \not{p}_2) (1 - \gamma^5) u_1|^2.$$

(There is no sum over the pion spin because it is spin-0.) Then the basic goal is to find $\Sigma = \sum_{s_2} |\bar{u}_2 (\not{p}_1 - \not{p}_2) (1 - \gamma^5) u_1|^2$. Assuming that the neutrino is massless, $m_2 = 0$,

$$\begin{aligned} \Sigma &= \sum_{s_2} |\bar{u}_2 \not{p}_1 (1 - \gamma^5) u_1|^2 \\ &= \sum_{s_2} |\bar{u}_2 (1 + \gamma^5) \not{p}_1 u_1|^2 \\ &= m_1^2 \sum_{s_2} \bar{u}_1 (1 - \gamma^5) u_2 \bar{u}_2 (1 + \gamma^5) u_1 \\ &= m_1^2 \text{Tr} \left[u_1 \bar{u}_1 (1 - \gamma^5) \sum_{s_2} u_2 \bar{u}_2 (1 + \gamma^5) \right]. \end{aligned}$$

For spinors $u(p, s)$ and $v(p, s)$ polarized in a polarization vector s^μ , we have two useful identities

$$u(p, s) \bar{u}(p, s) = (\not{p} + m) \frac{1}{2} (1 + \gamma^5 \not{s}) \quad \text{and} \quad v(p, s) \bar{v}(p, s) = (\not{p} - m) \frac{1}{2} (1 + \gamma^5 \not{s}). \quad (\text{A.1})$$

Using (A.1),

$$\begin{aligned} \Sigma &= \frac{1}{2} m_1^2 \text{Tr} [(\not{p}_1 + m_1) (1 + \gamma^5 \not{s}_1) (1 - \gamma^5) \not{p}_2 (1 + \gamma^5)] \\ &= \frac{1}{2} m_1^2 \text{Tr} [(\not{p}_1 + \not{p}_1 \gamma^5 \not{s}_1 + m_1 + m_1 \gamma^5 \not{s}_1) (1 - \gamma^5)^2 \not{p}_2] \\ &= m_1^2 \text{Tr} [(\not{p}_1 + \not{p}_1 \gamma^5 \not{s}_1 + m_1 + m_1 \gamma^5 \not{s}_1) (1 - \gamma^5) \not{p}_2] \\ &= m_1^2 \text{Tr} [\not{p}_1 \not{p}_2 - \not{p}_1 \gamma^5 \not{p}_2 + m_1 \gamma^5 \not{s}_1 \not{p}_2 - m \gamma^5 \not{s}_1 \gamma^5 \not{p}_2 + (\text{odd number of } \gamma\text{'s})] \\ &= m_1^2 \text{Tr} (\not{p}_1 \not{p}_2) - \text{Tr} (\not{p}_1 \gamma^5 \not{p}_2) - m_1^3 \text{Tr} (\not{s}_1 \gamma^5 \not{p}_2) - m_1^3 \text{Tr} [\not{s}_1 (\gamma^5)^2 \not{p}_2] \end{aligned}$$

Using $\text{Tr}(\gamma^\mu \gamma^5 \gamma^\nu) = 0$, $(\gamma^5)^2 = 1$, and $\text{Tr}(\not{a} \cdot \not{b}) = 4a \cdot b$,

$$\Sigma = 4m_1^2 p_1 \cdot p_2 - 4m_1^3 s_1 \cdot p_2 = 4m_1^3 |\mathbf{p}^*|(1 - \cos \theta).$$

The spin-summed amplitude squared is then

$$|\overline{\mathcal{A}}|^2 = 2G_F^2 f_\pi^2 |V_{ud}|^2 m_1^3 |\mathbf{p}^*|(1 - \cos \theta).$$

Note that amplitude tells us that the neutrino's trajectory is forbidden to be in the $+z$ -direction. Using the two-body decay rate formula (8.12), the decay rate for this process is

$$\Gamma = \frac{|\mathbf{p}^*|}{32\pi^2 m_1^2} \int |\overline{\mathcal{A}}|^2 d\Omega = \frac{1}{4\pi} G_F^2 f_\pi^2 |V_{ud}|^2 m_1 |\mathbf{p}^*|^2.$$

Substituting $|\mathbf{p}^*|$ for the two-body decay with $m_1 = m_\tau$ and $m_2 = 0$,

$$|\mathbf{p}^*| = \frac{\sqrt{[m_1^2 - (m_2 + m_\pi)^2][m_1^2 - (m_2 - m_\pi)^2]}}{2m_1} = \frac{1}{2m_\tau}(m_\tau^2 - m_\pi^2),$$

the decay rate is

$$\Gamma = \frac{1}{16\pi} G_F^2 f_\pi^2 |V_{ud}|^2 m_\tau^3 \left(\frac{m_\tau^2 - m_\pi^2}{m_\tau^2} \right)^2. \quad (\text{A.2})$$

You may wonder whether (A.2) is the correct decay rate since we are assuming that τ is polarized and the differential decay rate has angular dependence. The answer is yes. The decay rate does not depend on the polarization. This is because the decay rate is a Lorentz-invariant quantity, so we can rotate the z -axis by π so that the τ -lepton is polarized in the $-z$ -direction while still having the same decay rate. In general experiments, the τ -lepton is unpolarized. Averaging the results for two polarizations still gives (A.2). It is only when we are interested in the differential decay rate for a particular spin direction, a polarized version calculation is useful. Otherwise, computing the spin-averaged amplitude squared to get the decay rate is much easier.

A.6.2 Chiral Projections of Neutrinos

We claim that a right-handed neutrino contribute nothing to the amplitude, which means a neutrino can never have spin aligned with its momentum. The spin-averaged amplitude squared (not averaging over the initial-state neutrino spin) is

$$\begin{aligned} \sum_{s_1, s_2, s_4} |\overline{\mathcal{A}}|^2 &= \frac{1}{2} G_F^2 \sum_{s_1, s_3, s_4} [\bar{u}_1 \gamma^\mu (1 - \gamma^5) u_4] [\bar{u}_2 \gamma_\mu (1 - \gamma^5) u_3] [\bar{u}_3 \gamma_\nu (1 - \gamma^5) u_2] [\bar{u}_4 \gamma^\nu (1 - \gamma^5) u_1] \\ &= \frac{1}{2} G_F^2 \sum_{s_1, s_3, s_4} [\bar{u}_1 \gamma^\mu (1 - \gamma^5) u_4] [\bar{u}_4 \gamma^\nu (1 - \gamma^5) u_1] [\bar{u}_2 \gamma_\mu (1 - \gamma^5) u_3] [\bar{u}_3 \gamma_\nu (1 - \gamma^5) u_2] \\ &= \frac{1}{2} G_F^2 \text{Tr} \left[\sum_{s_1} u_1 \bar{u}_1 \gamma^\mu (1 - \gamma^5) \sum_{s_4} u_4 \bar{u}_4 \gamma^\nu (1 - \gamma^5) \right] \text{Tr} \left[u_2 \bar{u}_2 \gamma^\mu (1 - \gamma^5) \sum_{s_3} u_3 \bar{u}_3 \gamma^\nu (1 - \gamma^5) \right]. \end{aligned}$$

Let's focus on the second trace since it contains the spinor for the initial-state neutrino u_2 . We know that the polarized outer product of spinors is given by (A.1),

$$u(p, s) \bar{u}(p, s) = (\not{p} + m) \frac{1}{2} (1 + \gamma^5 \not{s}).$$

For a massless or ultra-relativistic particle, this modifies to

$$u(p, s) \bar{u}(p, s) = \not{p} \frac{1}{2} (1 - h \gamma^5), \quad (\text{A.3})$$

where $h = \pm 1$ is the helicity of the particle (+1 if right-handed, -1 if left-handed).

Proof. Let the particle travels in the z -direction. We know that the polarization vector satisfies $s \cdot p = 0$ and $s^2 = -1$ in the particle's rest frame, so these conditions work in any frame. For a particle with 4-momentum $p = (E, 0, 0, p_z)$, its polarization vector can be

$$s = \frac{h}{m} (p_z, 0, 0, E).$$

Compute $u(p, s)\bar{u}(p, s)$ explicitly,

$$\begin{aligned}
u(p, s)\bar{u}(p, s) &= (\not{p} + m)\frac{1}{2}(1 + \gamma^5 \not{s}) \\
&= \frac{1}{2}(E\gamma^0 - p_z\gamma^3 + m) \left[1 - \frac{h}{m}(p_z\gamma^0 - E\gamma_3)\gamma^5 \right] \\
&= \frac{1}{2} \left(E\gamma^0 - \frac{Ep_z h}{m}\gamma^5 - \frac{E^2 h}{m}\gamma^0\gamma^3\gamma^5 - p_z\gamma^3 + \frac{p_z^2 h}{m}\gamma^3\gamma^0\gamma^5 + \frac{Ep_z h}{m}\gamma^5 + m - p_z h\gamma^0\gamma^5 + Eh\gamma^3\gamma^5 \right) \\
&= \frac{1}{2} \left(E\gamma^0 - p_z\gamma^3 + m - \frac{E^2 - p_z^2}{m}h\gamma^0\gamma^3\gamma^5 - p_z h\gamma^0\gamma^5 + Eh\gamma^3\gamma^5 \right) \\
&= \frac{1}{2}(\not{p} + m - mh\gamma^0\gamma^3\gamma^5 - p_z h\gamma^0\gamma^5 + Eh\gamma^3\gamma^5).
\end{aligned}$$

We used the properties $(\gamma^0)^2 = 1$, $(\gamma^3)^2 = -1$, and $\{\gamma^0, \gamma^3\} = 0$ in the steps above. Now take the limit $m \rightarrow 0$ and $E = p_z$ to model neutrinos,

$$u(p, s)\bar{u}(p, s) = \frac{1}{2}[\not{p} - Eh(\gamma^0 - \gamma^3)\gamma^5] = \frac{1}{2}(\not{p} - h\not{p}\gamma^5) = \not{p}\frac{1}{2}(1 - h\gamma^5).$$

□

With the outer product (A.3), we can calculate the second trace in the spin-averaged amplitude squared,

$$\text{Tr} \left[u_2 \bar{u}_2 \gamma^\mu (1 - \gamma^5) \sum_{s_3} u_3 \bar{u}_3 \gamma^\nu (1 - \gamma^5) \right] = \text{Tr} [\not{p}_2 (1 - h\gamma^5) \gamma^\mu (1 - \gamma^5) (\not{p}_3 + m_3) \gamma^\nu (1 - \gamma^5)].$$

If the neutrino is right-handed, $h = +1$, then the trace becomes

$$\text{Tr} [\not{p}_2 (1 - \gamma^5) \gamma^\mu (1 - \gamma^5) (\not{p}_3 + m_3) \gamma^\nu (1 - \gamma^5)] = \text{Tr} [\not{p}_2 \gamma^\mu (1 + \gamma^5) (1 - \gamma^5) (\not{p}_3 + m_3) \gamma^\nu (1 - \gamma^5)].$$

But we know that $(\gamma^5)^2 = 1$, so $(1 + \gamma^5)(1 - \gamma^5) = 0$. The trace is zero, hence is the amplitude.

The calculation above is just an example proof. Nevertheless, in an amplitude squared, there will always be a term that looks like $u\bar{u}\gamma^\mu(1 - \gamma^5)$ for the neutrino. This factor is always zero for right-handed neutrino, $h = +1$:

$$u(p, s)\bar{u}(p, s)\gamma^\mu(1 - \gamma^5) = \not{p}\frac{1}{2}(1 - \gamma^5)\gamma^\mu(1 - \gamma^5) = \frac{1}{2}\not{p}\gamma^\mu(1 + \gamma^5)(1 - \gamma^5) = 0.$$

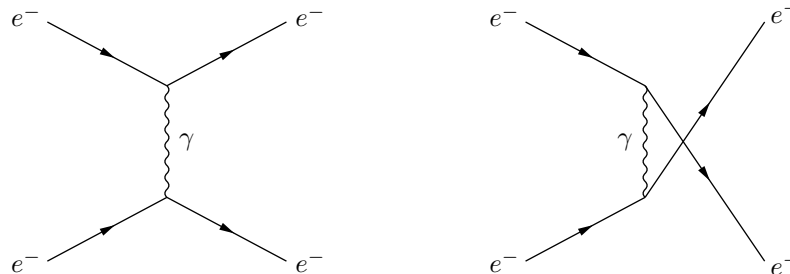
This term can come from the initial-state neutrino or the final-state neutrino, so a right-handed neutrino can never even be produced through weak interactions.

B FEYNMAN DIAGRAMS

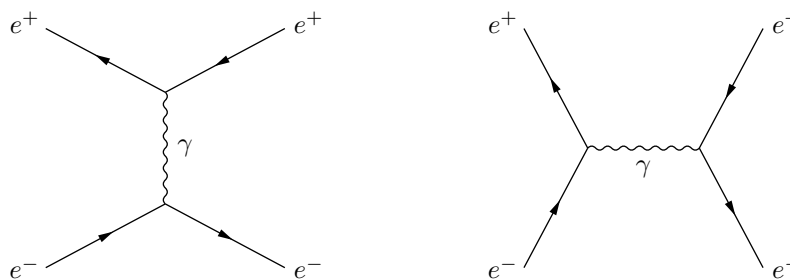
B.1 QED and QCD Processes

B.1.1 Fundamentals

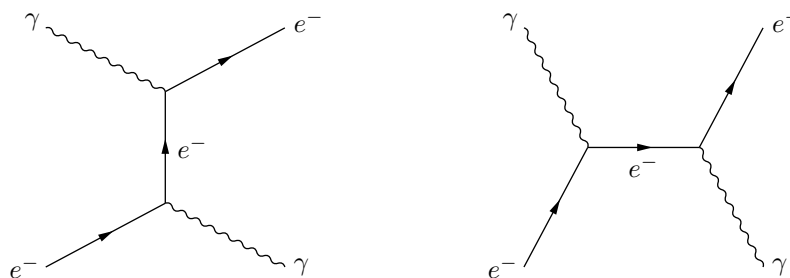
Møller Scattering $e^-e^- \rightarrow e^-e^-$



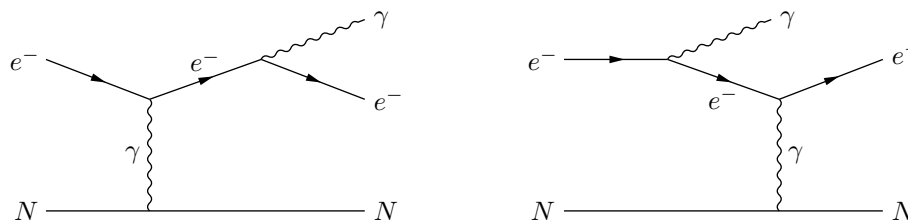
Bhabha Scattering $e^+e^- \rightarrow e^+e^-$



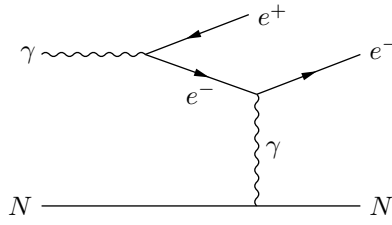
Compton Scattering $\gamma e^- \rightarrow \gamma e^-$



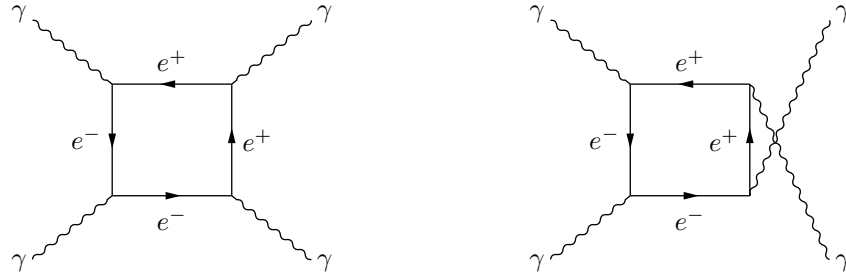
Bremsstrahlung



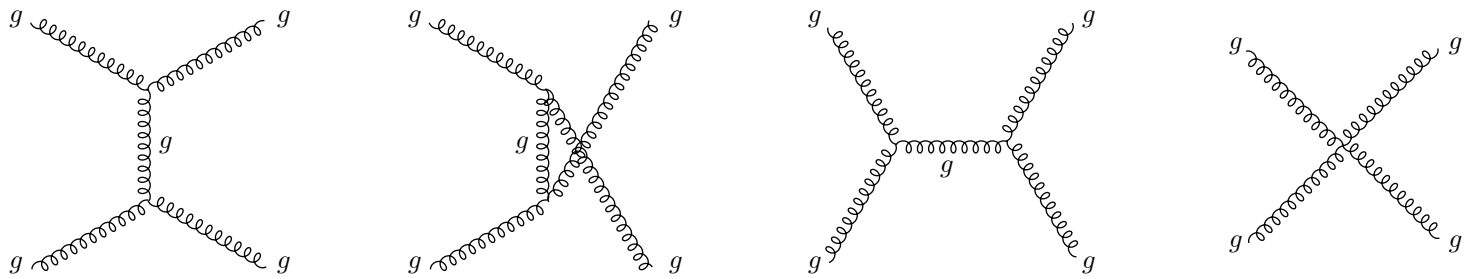
Pair Production



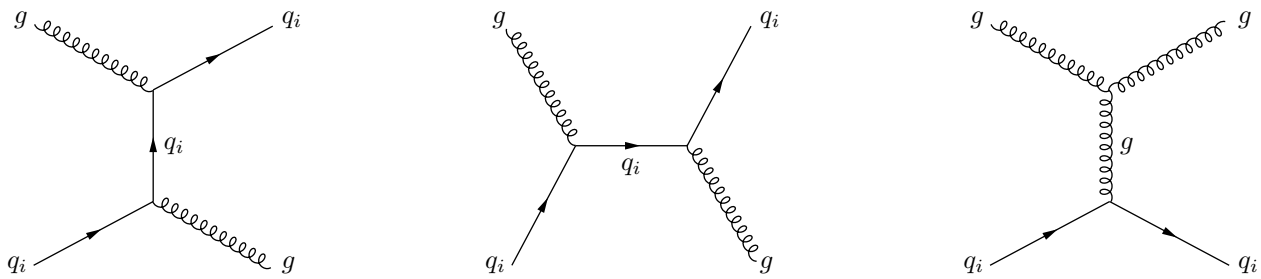
Delbrück Scattering $\gamma\gamma \rightarrow \gamma\gamma$



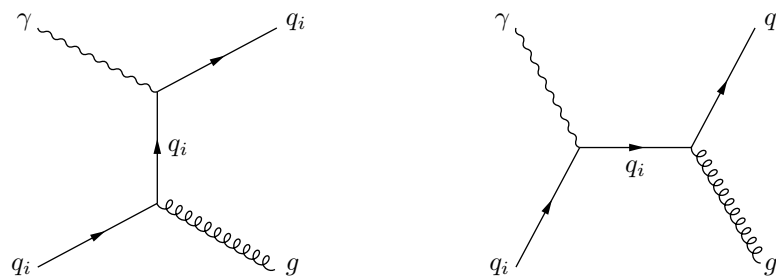
Gluon Scattering $gg \rightarrow gg$



$gq_i \rightarrow gq_i$

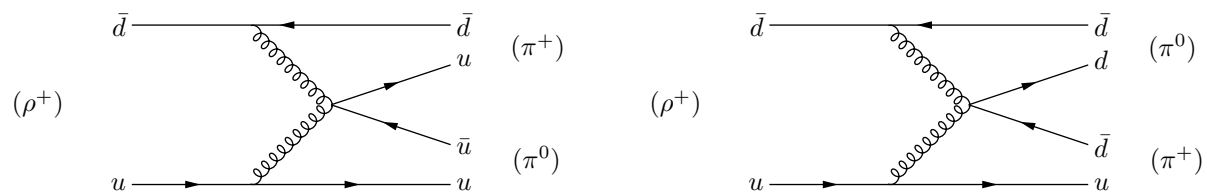


$\gamma q_i \rightarrow gq_i$

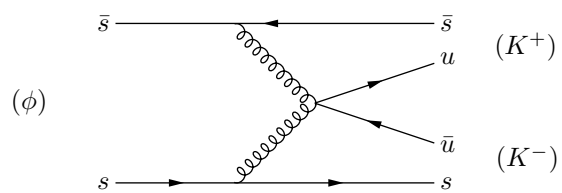


B.1.2 Hadronic Processes

$\rho^+ \rightarrow \pi^+\pi^0$

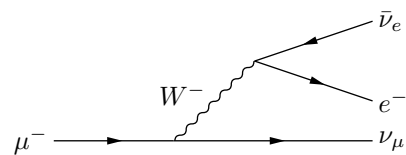
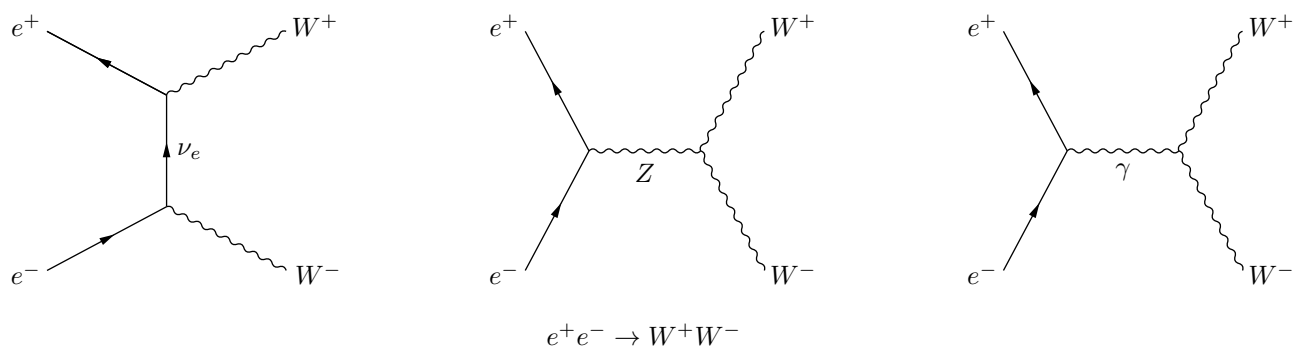
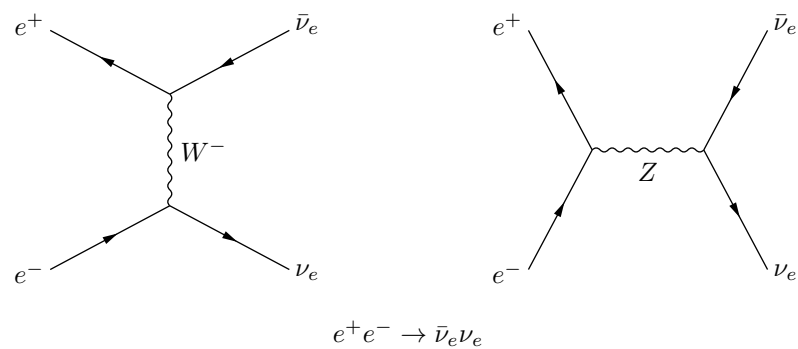


$\phi \rightarrow K^+ K^-$

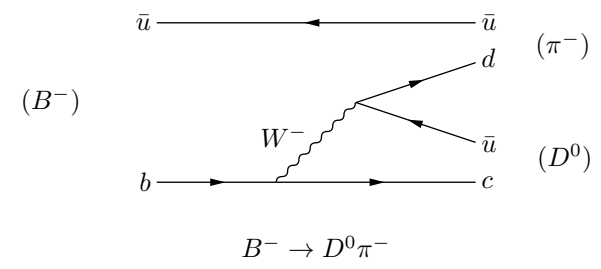
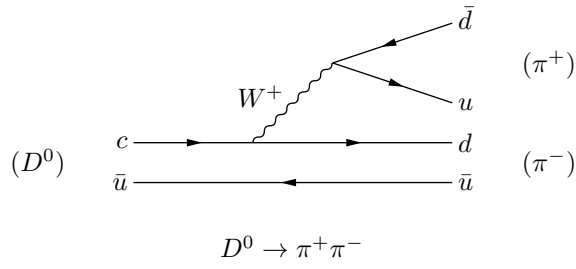
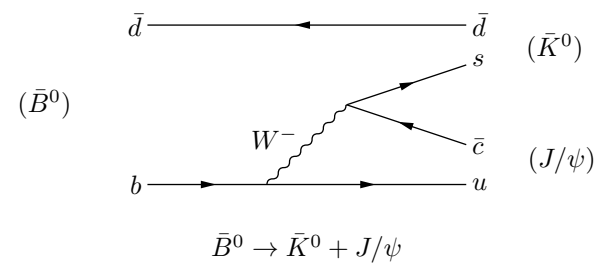
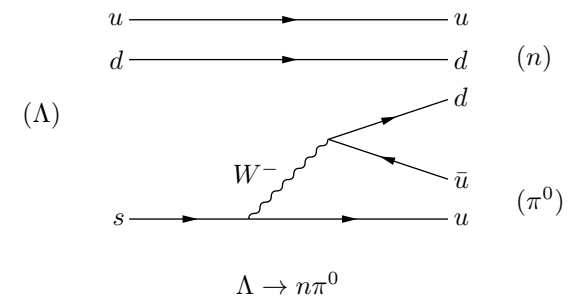


B.2 Feynman Diagrams Involving W and Z

B.2.1 Leptonic



B.2.2 Hadronic



B.2.3 Processes Involving Both Leptons and Quarks

